GameVerse
Mathematics Question Bank

1,000 original multiple-choice questions · 7 areas · 46 subtopics · Easy / Medium / Hard · answer key and worked solutions

Area	Questions	IDs	Easy / Med / Hard
A. Foundations	154	1–154	49 / 56 / 49
B. Calculus and Analysis	176	155–330	56 / 64 / 56
C. Algebra	132	331–462	42 / 48 / 42
D. Geometry and Topology	154	463–616	49 / 56 / 49
E. Discrete Mathematics and Logic	132	617–748	42 / 48 / 42
F. Probability and Statistics	84	749–832	28 / 28 / 28
G. Applied Mathematics	168	833–1000	56 / 56 / 56

How this document is organised

Part I – Questions (grouped by area and subtopic, Easy → Hard).
Part II – Answer key (compact grid).
Part III – Worked solutions.

About correctness

Every answer key was **computed by program** (exact fractions and integer arithmetic), and a large share was cross-checked with SymPy symbolic computation or brute-force enumeration. Each question has exactly four options and exactly one correct option, and the correct letter is randomised. These questions are **not yet reviewed by a human mathematician**: spot-check a sample before using them in reward-bearing or competitive play.

Limits of this edition

Questions are produced from parameterised templates (several different templates per subtopic), so numbers vary but the problem types repeat within a subtopic. Per-option “why this distractor is wrong” explanations, Amharic translations, and proof-evaluation items are not included. Difficulty is relative to the educational level shown for each subtopic.

Difficulty guide: Easy ≈ 20–45 s · Medium ≈ 50–110 s · Hard ≈ 2–5 min.

Part I – Questions

A. Foundations

Arithmetic

Q0001 · Easy · ≈43 s · Middle/High School

1. Compute 24 × 17 + 27.

A. 435
B. 483
C. 1056
D. 381

Q0002 · Easy · ≈43 s · Middle/High School

2. Compute 33 × 5 + 34.

A. 1287
B. 131
C. 203
D. 199

Q0003 · Easy · ≈30 s · Middle/High School

3. Compute 46 × 17 + 39.

A. 743
B. 2576
C. 821
D. 709

Q0004 · Easy · ≈21 s · Middle/High School

4. Compute 18 × 17 + 96.

A. 210
B. 402
C. 2034
D. 1650

Q0005 · Easy · ≈44 s · Middle/High School

5. Compute 28 × 11 + 53.

A. 361
B. 611
C. 255
D. 1792

Q0006 · Easy · ≈35 s · Middle/High School

6. Compute 31 × 15 + 59.

A. 916
B. 524
C. 2294
D. 406

Q0007 · Easy · ≈38 s · Middle/High School

7. Compute 42 × 4 + 77.

A. 3402
B. 245
C. 350
D. 91

Q0008 · Medium · ≈71 s · High School

8. Find gcd(72, 45) + lcm(72, 45).

A. 369
B. 360
C. 9
D. 3240

Q0009 · Medium · ≈76 s · High School

9. Find gcd(110, 66) + lcm(110, 66).

A. 7260
B. 352
C. 330
D. 22

Q0010 · Medium · ≈73 s · High School

10. Find gcd(84, 120) + lcm(84, 120).

A. 852
B. 10080
C. 12
D. 840

Q0011 · Medium · ≈106 s · High School

11. Find gcd(63, 99) + lcm(63, 99).

A. 9
B. 702
C. 693
D. 6237

Q0012 · Medium · ≈75 s · High School

12. Find gcd(45, 40) + lcm(45, 40).

A. 5
B. 365
C. 1800
D. 360

Q0013 · Medium · ≈62 s · High School

13. Find gcd(15, 27) + lcm(15, 27).

A. 138
B. 135
C. 405
D. 3

Q0014 · Medium · ≈54 s · High School

14. Find gcd(110, 70) + lcm(110, 70).

A. 770
B. 7700
C. 780
D. 10

Q0015 · Medium · ≈99 s · High School

15. Find gcd(80, 64) + lcm(80, 64).

A. 320
B. 336
C. 5120
D. 16

Q0016 · Hard · ≈229 s · Advanced High School

16. What is the remainder when 14^{252} is divided by 31?

A. 8
B. 9
C. 28
D. 2

Q0017 · Hard · ≈137 s · Advanced High School

17. What is the remainder when 5^{249} is divided by 9?

A. 8
B. 1
C. 7
D. 4

Q0018 · Hard · ≈219 s · Advanced High School

18. What is the remainder when 3^{239} is divided by 29?

A. 28
B. 20
C. 12
D. 26

Q0019 · Hard · ≈209 s · Advanced High School

19. What is the remainder when 67^{297} is divided by 23?

A. 12
B. 9
C. 2
D. 22

Q0020 · Hard · ≈176 s · Advanced High School

20. What is the remainder when 3^{216} is divided by 13?

A. 1
B. 9
C. 3
D. 2

Q0021 · Hard · ≈273 s · Advanced High School

21. What is the remainder when 70^{228} is divided by 31?

A. 4
B. 8
C. 16
D. 2

Q0022 · Hard · ≈269 s · Advanced High School

22. What is the remainder when 11^{129} is divided by 9?

A. 4
B. 8
C. 7
D. 9

Pre-algebra

Q0023 · Easy · ≈29 s · Middle/High School

23. Evaluate 1/8 + 1/2. Give the answer in lowest terms.

A. 1/5
B. 1/16
C. 5/8
D. 3/8

Q0024 · Easy · ≈20 s · Middle/High School

24. Evaluate 2/8 + 2/3. Give the answer in lowest terms.

A. 11/12
B. 5/12
C. 4/11
D. 1/6

Q0025 · Easy · ≈42 s · Middle/High School

25. Evaluate 2/12 + 1/3. Give the answer in lowest terms.

A. 1/2
B. 1/6
C. 1/5
D. 1/18

Q0026 · Easy · ≈29 s · Middle/High School

26. Evaluate 6/9 + 1/5. Give the answer in lowest terms.

A. 13/15
B. 2/15
C. 1/2
D. 7/15

Q0027 · Easy · ≈25 s · Middle/High School

27. Evaluate 2/3 + 5/9. Give the answer in lowest terms.

A. 10/27
B. 7/12
C. 11/9
D. 1/9

Q0028 · Easy · ≈26 s · Middle/High School

28. Evaluate 2/5 + 2/8. Give the answer in lowest terms.

A. 1/10
B. 3/20
C. 4/13
D. 13/20

Q0029 · Easy · ≈24 s · Middle/High School

29. Evaluate 6/8 + 4/9. Give the answer in lowest terms.

A. 10/17
B. 11/36
C. 43/36
D. 1/3

Q0030 · Medium · ≈108 s · High School

30. A prize of 56 coins is shared in the ratio 9:5. How many coins are in the larger share?

A. 32
B. 36
C. 20
D. 40

Q0031 · Medium · ≈77 s · High School

31. A prize of 48 coins is shared in the ratio 4:2. How many coins are in the larger share?

A. 32
B. 40
C. 16
D. 36

Q0032 · Medium · ≈63 s · High School

32. A prize of 96 coins is shared in the ratio 9:3. How many coins are in the larger share?

A. 72
B. 56
C. 80
D. 24

Q0033 · Medium · ≈68 s · High School

33. A prize of 60 coins is shared in the ratio 2:3. How many coins are in the larger share?

A. 24
B. 36
C. 48
D. 42

Q0034 · Medium · ≈88 s · High School

34. A prize of 156 coins is shared in the ratio 4:9. How many coins are in the larger share?

A. 120
B. 108
C. 90
D. 48

Q0035 · Medium · ≈70 s · High School

35. A prize of 150 coins is shared in the ratio 9:6. How many coins are in the larger share?

A. 85
B. 90
C. 100
D. 60

Q0036 · Medium · ≈106 s · High School

36. A prize of 132 coins is shared in the ratio 5:6. How many coins are in the larger share?

A. 60
B. 72
C. 78
D. 84

Q0037 · Medium · ≈55 s · High School

37. A prize of 60 coins is shared in the ratio 6:9. How many coins are in the larger share?

A. 36
B. 40
C. 34
D. 24

Q0038 · Hard · ≈234 s · Advanced High School

38. The mean of five numbers is 25. After one of them is removed, the mean of the remaining four is 23. What number was removed?

A. 33
B. 10
C. 8
D. 56

Q0039 · Hard · ≈140 s · Advanced High School

39. The mean of five numbers is 28. After one of them is removed, the mean of the remaining four is 19. What number was removed?

A. 64
B. 36
C. 45
D. 83

Q0040 · Hard · ≈159 s · Advanced High School

40. The mean of five numbers is 29. After one of them is removed, the mean of the remaining four is 18. What number was removed?

A. 44
B. 73
C. 91
D. 55

Q0041 · Hard · ≈189 s · Advanced High School

41. The mean of five numbers is 40. After one of them is removed, the mean of the remaining four is 21. What number was removed?

A. 95
B. 137
C. 76
D. 116

Q0042 · Hard · ≈164 s · Advanced High School

42. The mean of five numbers is 39. After one of them is removed, the mean of the remaining four is 24. What number was removed?

A. 60
B. 75
C. 99
D. 123

Q0043 · Hard · ≈152 s · Advanced High School

43. The mean of five numbers is 40. After one of them is removed, the mean of the remaining four is 15. What number was removed?

A. 155
B. 100
C. 125
D. 140

Q0044 · Hard · ≈181 s · Advanced High School

44. The mean of five numbers is 37. After one of them is removed, the mean of the remaining four is 15. What number was removed?

A. 110
B. 140
C. 88
D. 125

Algebra I

Q0045 · Easy · ≈24 s · Middle/High School

45. Solve for x: 9x - 8 = 37.

A. 5
B. 9/2
C. 109/9
D. 29/9

Q0046 · Easy · ≈43 s · Middle/High School

46. Solve for x: 9x + 9 = 99.

A. 2
B. 9
C. 10
D. 12

Q0047 · Easy · ≈35 s · Middle/High School

47. Solve for x: 3x + 12 = 27.

A. 13
B. 5
C. -3
D. 15/4

Q0048 · Easy · ≈33 s · Middle/High School

48. Solve for x: 6x - 5 = -23.

A. -14/3
B. -18/7
C. -3
D. 7/6

Q0049 · Easy · ≈38 s · Middle/High School

49. Solve for x: 4x - 8 = -28.

A. -9
B. -5
C. -4
D. 1

Q0050 · Easy · ≈22 s · Middle/High School

50. Solve for x: 8x - 1 = 55.

A. 27/4
B. 56/9
C. 63/8
D. 7

Q0051 · Easy · ≈33 s · Middle/High School

51. Solve for x: 6x + 6 = 12.

A. 1
B. -4
C. 6/7
D. 3

Q0052 · Medium · ≈78 s · High School

52. The equation x^2 - 5x - 24 = 0 has roots r and s. Find $r^2 + s^2$.

A. 576
B. 73
C. -23
D. 25

Q0053 · Medium · ≈82 s · High School

53. The equation x^2 + 14x + 48 = 0 has roots r and s. Find $r^2 + s^2$.

A. 2304
B. 100
C. 196
D. 292

Q0054 · Medium · ≈55 s · High School

54. The equation x^2 + x - 12 = 0 has roots r and s. Find $r^2 + s^2$.

A. 25
B. 1
C. -23
D. 144

Q0055 · Medium · ≈62 s · High School

55. The equation $x^2 - 5x + 6 = 0$ has roots r and s. Find $r^2 + s^2$.

A. 36
B. 37
C. 13
D. 25

Q0056 · Medium · ≈85 s · High School

56. The equation $x^2 - x - 12 = 0$ has roots r and s. Find $r^2 + s^2$.

A. 1
B. 144
C. -23
D. 25

Q0057 · Medium · ≈99 s · High School

57. The equation $x^2 + x - 42 = 0$ has roots r and s. Find $r^2 + s^2$.

A. 1
B. 85
C. -83
D. 1764

Q0058 · Medium · ≈82 s · High School

58. The equation $x^2 - x - 20 = 0$ has roots r and s. Find $r^2 + s^2$.

A. 41
B. -39
C. 400
D. 1

Q0059 · Medium · ≈68 s · High School

59. The equation $x^2 - 5x + 4 = 0$ has roots r and s. Find $r^2 + s^2$.

A. 17
B. 33
C. 25
D. 16

Q0060 · Hard · ≈170 s · Advanced High School

60. Real numbers x and y satisfy $x + y = -3$ and $xy = -4$. Find $x^3 + y^3$.

A. -39
B. -27
C. 9
D. -63

Q0061 · Hard · ≈147 s · Advanced High School

61. Real numbers x and y satisfy $x + y = -11$ and $xy = 28$. Find $x^3 + y^3$.

A. -1331
B. -2255
C. -1023
D. -407

Q0062 · Hard · ≈146 s · Advanced High School

62. Real numbers x and y satisfy $x + y = -1$ and $xy = -42$. Find $x^3 + y^3$.

A. -1
B. -43
C. -127
D. 125

Q0063 · Hard · ≈266 s · Advanced High School

63. Real numbers x and y satisfy x + y = -5 and xy = 0. Find $x^3 + y^3$.

A. -126
B. -124
C. -123
D. -125

Q0064 · Hard · ≈193 s · Advanced High School

64. Real numbers x and y satisfy x + y = 1 and xy = -30. Find $x^3 + y^3$.

A. 91
B. 1
C. 31
D. -89

Q0065 · Hard · ≈232 s · Advanced High School

65. Real numbers x and y satisfy x + y = 4 and xy = 0. Find $x^3 + y^3$.

A. 66
B. 64
C. 63
D. 65

Q0066 · Hard · ≈254 s · Advanced High School

66. Real numbers x and y satisfy x + y = -7 and xy = 6. Find $x^3 + y^3$.

A. -343
B. -469
C. -301
D. -217

Algebra II

Q0067 · Easy · ≈23 s · Middle/High School

67. If $\log_4 x = 3$, what is x?

A. 64
B. 12
C. 81
D. 16

Q0068 · Easy · ≈32 s · Middle/High School

68. If $\log_4 x = 5$, what is x?

A. 1024
B. 256
C. 20
D. 625

Q0069 · Easy · ≈42 s · Middle/High School

69. If $\log_2 x = 4$, what is x?

A. 16
B. 32
C. 8
D. 17

Q0070 · Easy · ≈25 s · Middle/High School

70. If $\log_2 x = 3$, what is x?

A. 6
B. 8
C. 9
D. 4

Q0071 · Easy · ≈21 s · Middle/High School

71. If $\log_{10} x = 3$, what is x?

A. 59049
B. 100
C. 30
D. 1000

Q0072 · Easy · ≈40 s · Middle/High School

72. If $\log_{3} x = 4$, what is x?

A. 27
B. 81
C. 12
D. 64

Q0073 · Easy · ≈39 s · Middle/High School

73. If $\log_{10} x = 5$, what is x?

A. 9765625
B. 100000
C. 50
D. 10000

Q0074 · Medium · ≈87 s · High School

74. Find the sum of the first 8 terms of the geometric sequence 2, 4, 8, ...

A. 510
B. 508
C. 1022
D. 512

Q0075 · Medium · ≈56 s · High School

75. Find the sum of the first 6 terms of the geometric sequence 1, 2, 4, ...

A. 64
B. 127
C. 62
D. 63

Q0076 · Medium · ≈103 s · High School

76. Find the sum of the first 5 terms of the geometric sequence 2, 4, 8, ...

A. 126
B. 62
C. 60
D. 64

Q0077 · Medium · ≈76 s · High School

77. Find the sum of the first 7 terms of the geometric sequence 2, 6, 18, ...

A. 2186
B. 4374
C. 6560
D. 4372

Q0078 · Medium · ≈57 s · High School

78. Find the sum of the first 6 terms of the geometric sequence 3, 9, 27, ...

A. 3279
B. 1092
C. 2187
D. 2184

Q0079 · Medium · ≈76 s · High School

79. Find the sum of the first 8 terms of the geometric sequence 4, 8, 16, ...

A. 2044
B. 1016
C. 1020
D. 1024

Q0080 · Medium · ≈76 s · High School

80. Find the sum of the first 6 terms of the geometric sequence 1, 3, 9, ...

A. 728
B. 1093
C. 729
D. 364

Q0081 · Medium · ≈89 s · High School

81. Find the sum of the first 7 terms of the geometric sequence 3, 9, 27, ...

A. 9840
B. 3279
C. 6561
D. 6558

Q0082 · Hard · ≈150 s · Advanced High School

82. A polynomial p(x) leaves remainder -3 when divided by x − 1 and remainder -5 when divided by x − 2. What is the remainder when p(x) is divided by (x − 1)(x − 2)?

A. -2x - 1
B. 2x - 7
C. -3x - 2
D. -2x - 3

Q0083 · Hard · ≈227 s · Advanced High School

83. A polynomial p(x) leaves remainder -1 when divided by x − 1 and remainder 5 when divided by x − 2. What is the remainder when p(x) is divided by (x − 1)(x − 2)?

A. 6x - 1
B. 6x - 7
C. -x + 6
D. -6x + 11

Q0084 · Hard · ≈147 s · Advanced High School

84. A polynomial p(x) leaves remainder -8 when divided by x − 1 and remainder 6 when divided by x − 2. What is the remainder when p(x) is divided by (x − 1)(x − 2)?

A. 14x - 22
B. 14x - 8
C. -8x + 14
D. -14x + 20

Q0085 · Hard · ≈258 s · Advanced High School

85. A polynomial p(x) leaves remainder -3 when divided by x − 1 and remainder -9 when divided by x − 2. What is the remainder when p(x) is divided by (x − 1)(x − 2)?

A. -6x - 3
B. 6x - 15
C. -3x - 6
D. -6x + 3

Q0086 · Hard · ≈266 s · Advanced High School

86. A polynomial p(x) leaves remainder 0 when divided by x − 1 and remainder -9 when divided by x − 2. What is the remainder when p(x) is divided by (x − 1)(x − 2)?

A. 9x - 18
B. -9x
C. -9x + 9
D. -9

Q0087 · Hard · ≈196 s · Advanced High School

87. A polynomial p(x) leaves remainder 1 when divided by x − 1 and remainder 7 when divided by x − 2. What is the remainder when p(x) is divided by (x − 1)(x − 2)?

A. -6x + 13
B. 6x - 5
C. x + 6
D. 6x + 1

Q0088 · Hard · ≈161 s · Advanced High School

88. A polynomial p(x) leaves remainder 1 when divided by x − 1 and remainder -5 when divided by x − 2. What is the remainder when p(x) is divided by (x − 1)(x − 2)?

A. 6x - 11
B. -6x + 7
C. -6x + 1
D. x - 6

Geometry

Q0089 · Easy · ≈36 s · Middle/High School

89. A right triangle has legs 9 and 40. What is the length of its hypotenuse?

A. 41
B. 42
C. 40
D. 49

Q0090 · Easy · ≈35 s · Middle/High School

90. A right triangle has legs 18 and 80. What is the length of its hypotenuse?

A. 83
B. 82
C. 81
D. 98

Q0091 · Easy · ≈39 s · Middle/High School

91. A right triangle has legs 7 and 24. What is the length of its hypotenuse?

A. 31
B. 25
C. 24
D. 26

Q0092 · Easy · ≈20 s · Middle/High School

92. A right triangle has legs 10 and 24. What is the length of its hypotenuse?

A. 34
B. 26
C. 25
D. 27

Q0093 · Easy · ≈44 s · Middle/High School

93. A right triangle has legs 40 and 42. What is the length of its hypotenuse?

A. 59
B. 82
C. 58
D. 57

Q0094 · Easy · ≈24 s · Middle/High School

94. A right triangle has legs 6 and 8. What is the length of its hypotenuse?

A. 11
B. 14
C. 9
D. 10

Q0095 · Easy · ≈28 s · Middle/High School

95. A right triangle has legs 15 and 36. What is the length of its hypotenuse?

A. 51
B. 39
C. 40
D. 38

Q0096 · Medium · ≈105 s · High School

96. What is the measure, in degrees, of each interior angle of a regular polygon with 5 sides?

A. 144
B. 72
C. 108
D. 540

Q0097 · Medium · ≈52 s · High School

97. What is the measure, in degrees, of each interior angle of a regular polygon with 9 sides?

A. 160
B. 1260
C. 140
D. 40

Q0098 · Medium · ≈92 s · High School

98. What is the measure, in degrees, of each interior angle of a regular polygon with 15 sides?

A. 156
B. 168
C. 24
D. 2340

Q0099 · Medium · ≈87 s · High School

99. What is the measure, in degrees, of each interior angle of a regular polygon with 12 sides?

A. 150
B. 30
C. 1800
D. 165

Q0100 · Medium · ≈57 s · High School

100. What is the measure, in degrees, of each interior angle of a regular polygon with 20 sides?

A. 171
B. 18
C. 3240
D. 162

Q0101 · Medium · ≈99 s · High School

101. What is the measure, in degrees, of each interior angle of a regular polygon with 8 sides?

A. 135
B. 45
C. 158
D. 1080

Q0102 · Medium · ≈55 s · High School

102. What is the measure, in degrees, of each interior angle of a regular polygon with 18 sides?

A. 160
B. 170
C. 20
D. 2880

Q0103 · Medium · ≈63 s · High School

103. What is the measure, in degrees, of each interior angle of a regular polygon with 10 sides?

A. 144
B. 162
C. 1440
D. 36

Q0104 · Hard · ≈135 s · Advanced High School

104. A triangle has side lengths 5, 5 and 8. What is the radius of its inscribed circle?

A. 8/3
B. 12
C. 2/3
D. 4/3

Q0105 · Hard · ≈250 s · Advanced High School

105. A triangle has side lengths 7, 15 and 20. What is the radius of its inscribed circle?

A. 1
B. 2
C. 42
D. 4

Q0106 · Hard · ≈261 s · Advanced High School

106. A triangle has side lengths 13, 14 and 15. What is the radius of its inscribed circle?

A. 84
B. 8
C. 4
D. 2

Q0107 · Hard · ≈279 s · Advanced High School

107. A triangle has side lengths 11, 13 and 20. What is the radius of its inscribed circle?

A. 3
B. 3/2
C. 66
D. 6

Q0108 · Hard · ≈143 s · Advanced High School

108. A triangle has side lengths 3, 25 and 26. What is the radius of its inscribed circle?

A. 8/3
B. 2/3
C. 36
D. 4/3

Q0109 · Hard · ≈277 s · Advanced High School

109. A triangle has side lengths 10, 17 and 21. What is the radius of its inscribed circle?

A. 7/2
B. 7
C. 7/4
D. 84

Q0110 · Hard · ≈134 s · Advanced High School

110. A triangle has side lengths 10, 13 and 13. What is the radius of its inscribed circle?

A. 10/3
B. 20/3
C. 5/3
D. 60

Trigonometry

Q0111 · Easy · ≈23 s · Middle/High School

111. What is the exact value of tan(60°)?

A. 0
B. √2/2
C. √3/2
D. √3

Q0112 · Easy · ≈31 s · Middle/High School

112. What is the exact value of tan(30°)?

A. √3/2
B. 1/2
C. 1
D. √3/3

Q0113 · Easy · ≈33 s · Middle/High School

113. What is the exact value of cos(0°)?

A. 1
B. 0
C. √3/3
D. √2/2

Q0114 · Easy · ≈44 s · Middle/High School

114. What is the exact value of tan(45°)?

A. √3/3
B. √3
C. 1
D. 0

Q0115 · Easy · ≈21 s · Middle/High School

115. What is the exact value of sin(90°)?

A. √3/3
B. 1
C. √3
D. 0

Q0116 · Easy · ≈40 s · Middle/High School

116. What is the exact value of cos(45°)?

A. √3/3
B. √3
C. √2/2
D. √3/2

Q0117 · Easy · ≈43 s · Middle/High School

117. What is the exact value of sin(45°)?

A. √3/3
B. √2/2
C. 1
D. 0

Q0118 · Medium · ≈79 s · High School

118. If θ is an acute angle with sin θ = 24/25, what is tan θ?

A. 25/7
B. 7/24
C. 24/7
D. 7/25

Q0119 · Medium · ≈80 s · High School

119. If θ is an acute angle with sin θ = 9/41, what is tan θ?

A. 40/9
B. 9/40
C. 41/40
D. 40/41

Q0120 · Medium · ≈83 s · High School

120. If θ is an acute angle with sin θ = 12/13, what is tan θ?

A. 5/12
B. 5/13
C. 13/5
D. 12/5

Q0121 · Medium · ≈79 s · High School

121. If θ is an acute angle with sin θ = 4/5, what is tan θ?

A. 4/3
B. 3/4
C. 5/3
D. 3/5

Q0122 · Medium · ≈80 s · High School

122. If θ is an acute angle with sin θ = 15/17, what is tan θ?

A. 8/15
B. 8/17
C. 15/8
D. 17/8

Q0123 · Medium · ≈99 s · High School

123. If θ is an acute angle with sin θ = 21/29, what is tan θ?

A. 20/29
B. 21/20
C. 20/21
D. 29/20

Q0124 · Medium · ≈72 s · High School

124. If θ is an acute angle with sin θ = 7/25, what is tan θ?

A. 24/7
B. 7/24
C. 25/24
D. 24/25

Q0125 · Medium · ≈54 s · High School

125. If θ is an acute angle with sin θ = 5/13, what is tan θ?

A. 12/5
B. 13/12
C. 12/13
D. 5/12

Q0126 · Hard · ≈245 s · Advanced High School

126. In triangle ABC, AC = 12, BC = 8 and ∠C = 60°. Find the length AB.

A. √304
B. √112
C. 4
D. √208

Q0127 · Hard · ≈275 s · Advanced High School

127. In triangle ABC, AC = 6, BC = 4 and ∠C = 120°. Find the length AB.

A. √76
B. 2
C. √28
D. √52

Q0128 · Hard · ≈146 s · Advanced High School

128. In triangle ABC, AC = 6, BC = 3 and ∠C = 120°. Find the length AB.

A. √45
B. 3
C. √63
D. √27

Q0129 · Hard · ≈275 s · Advanced High School

129. In triangle ABC, AC = 5, BC = 12 and ∠C = 60°. Find the length AB.

A. √229
B. 13
C. 7
D. √109

Q0130 · Hard · ≈144 s · Advanced High School

130. In triangle ABC, AC = 8, BC = 4 and ∠C = 60°. Find the length AB.

A. √80
B. 4
C. √48
D. √112

Q0131 · Hard · ≈268 s · Advanced High School

131. In triangle ABC, AC = 7, BC = 11 and ∠C = 60°. Find the length AB.

A. 4
B. √247
C. √93
D. √170

Q0132 · Hard · ≈214 s · Advanced High School

132. In triangle ABC, AC = 5, BC = 4 and ∠C = 120°. Find the length AB.

A. √61
B. 1
C. √21
D. √41

Precalculus

Q0133 · Easy · ≈40 s · Middle/High School

133. Let f(x) = 2x - 4 and g(x) = $4x^2$ + 2. Find f(g(-2)).

A. 18
B. 10
C. 32
D. 258

Q0134 · Easy · ≈42 s · Middle/High School

134. Let f(x) = 2x + 3 and g(x) = $3x^2$ + 4. Find f(g(-3)).

A. 31
B. 59
C. 65
D. 28

Q0135 · Easy · ≈32 s · Middle/High School

135. Let f(x) = 2x - 1 and g(x) = $1x^2$ + 4. Find f(g(-3)).

A. 13
B. 6
C. 53
D. 25

Q0136 · Easy · ≈35 s · Middle/High School

136. Let f(x) = 3x - 5 and g(x) = $3x^2$ + 3. Find f(g(-1)).

A. -2
B. 13
C. 195
D. 6

Q0137 · Easy · ≈40 s · Middle/High School

137. Let f(x) = 5x + 0 and g(x) = $4x^2$ + 5. Find f(g(1)).

A. 45
B. 105
C. 14
D. 9

Q0138 · Easy · ≈43 s · Middle/High School

138. Let f(x) = 2x - 2 and g(x) = $2x^2$ + 2. Find f(g(0)).

A. 10
B. 0
C. 6
D. 2

Q0139 · Easy · ≈44 s · Middle/High School

139. Let f(x) = 3x - 3 and g(x) = $3x^2$ - 4. Find f(g(2)).

A. 8
B. 23
C. 21
D. 11

Q0140 · Medium · ≈101 s · High School

140. Let f(x) = (4x - 2)/(2x + 0). Find $f^{-1}(1)$.

A. -1
B. 1
C. -1/3
D. -3

Q0141 · Medium · ≈102 s · High School

141. Let f(x) = (2x + 1)/(3x + 1). Find $f^{-1}(1)$.

A. 2/5
B. 0
C. -1/2
D. 3/4

Q0142 · Medium · ≈88 s · High School

142. Let f(x) = (3x + 0)/(3x + 2). Find $f^{-1}(-2)$.

A. -3/4
B. 4/3
C. 3/2
D. -4/9

Q0143 · Medium · ≈110 s · High School

143. Let f(x) = (2x + 0)/(3x - 1). Find $f^{-1}(-2)$.

A. 1/4
B. -1/2
C. 4/7
D. -4/5

Q0144 · Medium · ≈87 s · High School

144. Let f(x) = (4x + 3)/(3x - 1). Find $f^{-1}(-1)$.

A. 4
B. -2/7
C. -7/2
D. 1/4

Q0145 · Medium · ≈97 s · High School

145. Let f(x) = (3x + 3)/(3x + 0). Find $f^{-1}(4)$.

A. 1/3
B. 1/5
C. -3/4
D. 5/4

Q0146 · Medium · ≈110 s · High School

146. Let f(x) = (2x + 2)/(2x + 4). Find $f^{-1}(-1)$.

A. 0
B. 3/2
C. -2/3
D. -3/2

Q0147 · Medium · ≈56 s · High School

147. Let f(x) = (3x - 3)/(3x + 2). Find $f^{-1}(3)$.

A. -12/7
B. 1/4
C. 6/11
D. -3/2

Q0148 · Hard · ≈142 s · Advanced High School

148. Compute $(1 + i)^8$ in the form a + bi.

A. 8 - 8i
B. 17
C. 16
D. 16 + 16i

Q0149 · Hard · ≈160 s · Advanced High School

149. Compute $(1 + i)^6$ in the form a + bi.

A. 8i
B. -8i
C. -4 - 4i
D. 8 - 8i

Q0150 · Hard · ≈250 s · Advanced High School

150. Compute $(1 + i)^9$ in the form a + bi.

A. 16
B. 16 + 16i
C. 16 - 16i
D. 32i

Q0151 · Hard · ≈245 s · Advanced High School

151. Compute $(1 + i)^{11}$ in the form a + bi.

A. -32 - 32i
B. 32i
C. -64
D. -32 + 32i

Q0152 · Hard · ≈234 s · Advanced High School

152. Compute $(1 + i)^{13}$ in the form a + bi.

A. -64 - 64i
B. -64
C. -128i
D. -64 + 64i

Q0153 · Hard · ≈156 s · Advanced High School

153. Compute $(1 + i)^{14}$ in the form a + bi.

A. -64 - 64i
B. 128i
C. 128 - 128i
D. -128i

Q0154 · Hard · ≈233 s · Advanced High School

154. Compute $(1 + i)^{12}$ in the form a + bi.

A. -32 + 32i
B. -64 - 64i
C. 64
D. -64

B. Calculus and Analysis

Single-variable calculus

Q0155 · Easy · ≈25 s · Introductory Undergraduate

155. If $f(x) = 4x^3 - 4x^2 - x + 5$, find $f'(-3)$.

A. 119
B. 131
C. -45
D. -136

Q0156 · Easy · ≈43 s · Introductory Undergraduate

156. If $f(x) = 5x^3 - 4x^2 - 3x - 3$, find $f'(-3)$.

A. 144
B. -165
C. -56
D. 156

Q0157 · Easy · ≈29 s · Introductory Undergraduate

157. If $f(x) = x^3 + 2x^2 - 5x - 2$, find $f'(-1)$.

A. 4
B. -4
C. -5
D. -6

Q0158 · Easy · ≈37 s · Introductory Undergraduate

158. If $f(x) = 2x^3 - x^2 - x + 1$, find $f'(-2)$.

A. -15
B. 27
C. 25
D. -17

Q0159 · Easy · ≈29 s · Introductory Undergraduate

159. If $f(x) = x^3 + 3x^2 + 3x + 4$, find $f'(2)$.

A. 15
B. 27
C. 30
D. 21

Q0160 · Easy · ≈37 s · Introductory Undergraduate

160. If $f(x) = 3x^3 - 2x^2 - 4x - 5$, find $f'(-1)$.

A. -17
B. 7
C. -6
D. 9

Q0161 · Easy · ≈30 s · Introductory Undergraduate

161. If $f(x) = x^3 - x^2 + 4x$, find $f'(2)$.

A. 12
B. 13
C. 14
D. 8

Q0162 · Medium · ≈110 s · Undergraduate

162. Evaluate $\int_0^3 (x^2 + 2)\,dx$.

A. 9
B. 13/2
C. 33
D. 15

Q0163 · Medium · ≈78 s · Undergraduate

163. Evaluate $\int_0^2 (x^2 - x - 2)\,dx$.

A. 2/3
B. -10/3
C. 0
D. -2

Q0164 · Medium · ≈75 s · Undergraduate

164. Evaluate $\int_0^4 (x^2 - 4x - 1)\,dx$.

A. -32/3
B. -44/3
C. -9
D. -4

Q0165 · Medium · ≈70 s · Undergraduate

165. Evaluate $\int_0^4 (2x^2 - 3)\,dx$.

A. 116
B. 128/3
C. 92/3
D. 13

Q0166 · Medium · ≈87 s · Undergraduate

166. Evaluate $\int_0^2 (2x^2 + 2x + 3)\,dx$.

A. 28/3
B. 30
C. 46/3
D. 11

Q0167 · Medium · ≈69 s · Undergraduate

167. Evaluate $\int_0^3 (4x^2 + 3x + 5)\,dx$.

A. 99/2
B. 32
C. 150
D. 129/2

Q0168 · Medium · ≈103 s · Undergraduate

168. Evaluate $\int_0^4 (4x^2 + 3x + 5)\,dx$.

A. 388/3
B. 328/3
C. 324
D. 49

Q0169 · Medium · ≈58 s · Undergraduate

169. Evaluate $\int_0^4 (4x^2 - 2)\,dx$.

A. 30
B. 232/3
C. 248
D. 256/3

Q0170 · Hard · ≈130 s · Upper Undergraduate

170. Evaluate $\lim_{x\to 0} (1 - \cos(3x)) / (x \cdot \sin(1x))$.

A. 9
B. 3/2
C. 3
D. 9/2

Q0171 · Hard · ≈146 s · Upper Undergraduate

171. Evaluate $\lim_{x\to 0} (1 - \cos(2x)) / (x \cdot \sin(5x))$.

A. 2/25
B. 4/5
C. 1/5
D. 2/5

Q0172 · Hard · ≈144 s · Upper Undergraduate

172. Evaluate $\lim_{x\to 0} (1 - \cos(5x)) / (x \cdot \sin(5x))$.

A. 1
B. 5
C. 5/2
D. 1/2

Q0173 · Hard · ≈177 s · Upper Undergraduate

173. Evaluate $\lim_{x\to 0} (1 - \cos(2x)) / (x \cdot \sin(3x))$.

A. 4/3
B. 2/9
C. 1/3
D. 2/3

Q0174 · Hard · ≈214 s · Upper Undergraduate

174. Evaluate $\lim_{x\to 0} (1 - \cos(6x)) / (x \cdot \sin(6x))$.

A. 6
B. 1/2
C. 1
D. 3

Q0175 · Hard · ≈170 s · Upper Undergraduate

175. Evaluate $\lim_{x\to 0} (1 - \cos(6x)) / (x \cdot \sin(5x))$.

A. 18/25
B. 36/5
C. 3/5
D. 18/5

Q0176 · Hard · ≈258 s · Upper Undergraduate

176. Evaluate $\lim_{x\to 0} (1 - \cos(6x)) / (x \cdot \sin(4x))$.

A. 9/2
B. 9/8
C. 9
D. 3/4

Multivariable and vector calculus

Q0177 · Easy · ≈31 s · Introductory Undergraduate

177. Let $f(x, y) = x^4y^4 + xy$. Find $\partial f/\partial y$ at the point (3, 2).

A. 1302
B. 2595
C. 2592
D. 1730

Q0178 · Easy · ≈23 s · Introductory Undergraduate

178. Let $f(x, y) = x^4y^2 + xy$. Find $\partial f/\partial y$ at the point (1, 1).

A. 2
B. 4
C. 5
D. 3

Q0179 · Easy · ≈44 s · Introductory Undergraduate

179. Let $f(x, y) = x^2y^2 + xy$. Find $\partial f/\partial y$ at the point (1, 1).

A. 5
B. 3
C. 4
D. 2

Q0180 · Easy · ≈32 s · Introductory Undergraduate

180. Let f(x, y) = x^4y^2 + xy. Find ∂f/∂y at the point (3, 3).

A. 738
B. 975
C. 489
D. 486

Q0181 · Easy · ≈28 s · Introductory Undergraduate

181. Let f(x, y) = x^4y^2 + xy. Find ∂f/∂y at the point (2, 1).

A. 18
B. 32
C. 33
D. 34

Q0182 · Easy · ≈35 s · Introductory Undergraduate

182. Let f(x, y) = x^4y^3 + xy. Find ∂f/∂y at the point (2, 1).

A. 18
B. 48
C. 33
D. 50

Q0183 · Easy · ≈32 s · Introductory Undergraduate

183. Let f(x, y) = x^4y^2 + xy. Find ∂f/∂y at the point (2, 3).

A. 98
B. 96
C. 291
D. 150

Q0184 · Medium · ≈83 s · Undergraduate

184. Find the directional derivative of f(x, y) = x^2 + (2)xy + $3y^2$ at (0, 1) in the direction of the vector (3, 4).

A. 26/5
B. 6
C. 8
D. 30

Q0185 · Medium · ≈99 s · Undergraduate

185. Find the directional derivative of f(x, y) = x^2 + (-2)xy + $1y^2$ at (1, -1) in the direction of the vector (3, 4).

A. 0
B. 4/5
C. -4/5
D. -4

Q0186 · Medium · ≈101 s · Undergraduate

186. Find the directional derivative of f(x, y) = x^2 + (4)xy + $1y^2$ at (0, 2) in the direction of the vector (3, 4).

A. 40
B. 44/5
C. 8
D. 12

Q0187 · Medium · ≈109 s · Undergraduate

187. Find the directional derivative of $f(x, y) = x^2 + (4)xy + 3y^2$ at (-1, -2) in the direction of the vector (3, 4).

A. -26
B. -88/5
C. -94/5
D. -94

Q0188 · Medium · ≈100 s · Undergraduate

188. Find the directional derivative of $f(x, y) = x^2 + (-3)xy + 3y^2$ at (0, 2) in the direction of the vector (3, 4).

A. 30
B. 6
C. 12/5
D. 6/5

Q0189 · Medium · ≈86 s · Undergraduate

189. Find the directional derivative of $f(x, y) = x^2 + (-1)xy + 1y^2$ at (-2, 0) in the direction of the vector (3, 4).

A. -4
B. -2/5
C. -2
D. -4/5

Q0190 · Medium · ≈87 s · Undergraduate

190. Find the directional derivative of $f(x, y) = x^2 + (0)xy + 3y^2$ at (3, -2) in the direction of the vector (3, 4).

A. -6
B. -12/5
C. -6/5
D. -30

Q0191 · Medium · ≈85 s · Undergraduate

191. Find the directional derivative of $f(x, y) = x^2 + (1)xy + 1y^2$ at (-2, 3) in the direction of the vector (3, 4).

A. 13/5
B. 3
C. 13
D. 8/5

Q0192 · Hard · ≈141 s · Upper Undergraduate

192. Evaluate the double integral of $3x^2 + (1)xy + 1y^2$ over the rectangle $0 \le x \le 2$, $0 \le y \le 2$.

A. 18
B. 76/3
C. 88/3
D. 79/3

Q0193 · Hard · ≈225 s · Upper Undergraduate

193. Evaluate the double integral of $3x^2 + (4)xy + 4y^2$ over the rectangle $0 \le x \le 1$, $0 \le y \le 4$.

A. 364/3
B. 316/3
C. 319/3
D. 46

Q0194 · Hard · ≈250 s · Upper Undergraduate

194. Evaluate the double integral of $3x^2 + (2)xy + 4y^2$ over the rectangle $0 \le x \le 4$, $0 \le y \le 2$.

A. 704/3
B. 88
C. 608/3
D. 611/3

Q0195 · Hard · ≈186 s · Upper Undergraduate

195. Evaluate the double integral of $1x^2 + (3)xy + 3y^2$ over the rectangle $0 \le x \le 2$, $0 \le y \le 3$.

A. 90
B. 89
C. 42
D. 116

Q0196 · Hard · ≈138 s · Upper Undergraduate

196. Evaluate the double integral of $2x^2 + (3)xy + 1y^2$ over the rectangle $0 \le x \le 2$, $0 \le y \le 2$.

A. 18
B. 29
C. 40
D. 28

Q0197 · Hard · ≈236 s · Upper Undergraduate

197. Evaluate the double integral of $2x^2 + (-3)xy + 1y^2$ over the rectangle $0 \le x \le 3$, $0 \le y \le 2$.

A. -10
B. 15
C. 17
D. 18

Q0198 · Hard · ≈278 s · Upper Undergraduate

198. Evaluate the double integral of $4x^2 + (-2)xy + 3y^2$ over the rectangle $0 \le x \le 3$, $0 \le y \le 3$.

A. 297/2
B. 108
C. 299/2
D. 171/2

Ordinary differential equations

Q0199 · Easy · ≈22 s · Introductory Undergraduate

199. Solve y' = 3y with y(0) = 3. What is y(2)?

A. $3e^{6}$
B. $6e^{3}$
C. $3e^{5}$
D. $3e^{3}$

Q0200 · Easy · ≈42 s · Introductory Undergraduate

200. Solve y' = -1y with y(0) = 2. What is y(2)?

A. $4e^{-1}$
B. $2e^{-1}$
C. $2e^{-2}$
D. $2e^{1}$

Q0201 · Easy · ≈39 s · Introductory Undergraduate

201. Solve y' = -3y with y(0) = 5. What is y(2)?

A. $5e^{-1}$
B. $5e^{-6}$
C. $5e^{-3}$
D. $10e^{-3}$

Q0202 · Easy · ≈40 s · Introductory Undergraduate

202. Solve y' = -2y with y(0) = 2. What is y(2)?

A. $2e^{-4}$
B. $4e^{-2}$
C. $2e^{0}$
D. $2e^{-2}$

Q0203 · Easy · ≈44 s · Introductory Undergraduate

203. Solve y' = -2y with y(0) = 6. What is y(2)?

A. $6e^{0}$
B. $12e^{-2}$
C. $6e^{-4}$
D. $6e^{-2}$

Q0204 · Easy · ≈26 s · Introductory Undergraduate

204. Solve y' = 3y with y(0) = 4. What is y(2)?

A. $8e^{3}$
B. $4e^{3}$
C. $4e^{5}$
D. $4e^{6}$

Q0205 · Easy · ≈39 s · Introductory Undergraduate

205. Solve y' = -2y with y(0) = 5. What is y(2)?

A. $5e^{-4}$
B. $5e^{-2}$
C. $10e^{-2}$
D. $5e^{0}$

Q0206 · Medium · ≈86 s · Undergraduate

206. Find the general solution of y'' + 2y' - 8y = 0.

A. $C_1e^{-2x} + C_2e^{4x}$
B. $C_1e^{-4x} + C_2e^{-2x}$
C. $C_1e^{-4x} + C_2e^{2x}$
D. $C_1e^{2x} + C_2e^{4x}$

Q0207 · Medium · ≈88 s · Undergraduate

207. Find the general solution of y'' - 3y' + 2y = 0.

A. $C_1e^{-2x} + C_2e^{-1x}$
B. $C_1e^{1x} + C_2e^{2x}$
C. $C_1e^{-1x} + C_2e^{2x}$
D. $C_1e^{-2x} + C_2e^{1x}$

Q0208 · Medium · ≈50 s · Undergraduate

208. Find the general solution of y'' - 1y' - 2y = 0.

A. $C_1e^{1x} + C_2e^{2x}$
B. $C_1e^{-2x} + C_2e^{1x}$
C. $C_1e^{-1x} + C_2e^{2x}$
D. $C_1e^{-2x} + C_2e^{-1x}$

Q0209 · Medium · ≈60 s · Undergraduate

209. Find the general solution of y'' + 7y' + 12y = 0.

A. $C_1e^{-4x} + C_2e^{3x}$
B. $C_1e^{3x} + C_2e^{4x}$
C. $C_1e^{-4x} + C_2e^{-3x}$
D. $C_1e^{-3x} + C_2e^{4x}$

Q0210 · Medium · ≈104 s · Undergraduate

210. Find the general solution of y'' - 3y' - 4y = 0.

A. $C_1e^{-4x} + C_2e^{-1x}$
B. $C_1e^{-1x} + C_2e^{4x}$
C. $C_1e^{1x} + C_2e^{4x}$
D. $C_1e^{-4x} + C_2e^{1x}$

Q0211 · Medium · ≈50 s · Undergraduate

211. Find the general solution of y'' - 1y' - 6y = 0.

A. $C_1e^{-3x} + C_2e^{-2x}$
B. $C_1e^{2x} + C_2e^{3x}$
C. $C_1e^{-3x} + C_2e^{2x}$
D. $C_1e^{-2x} + C_2e^{3x}$

Q0212 · Medium · ≈99 s · Undergraduate

212. Find the general solution of y'' + 3y' + 2y = 0.

A. $C_1e^{-1x} + C_2e^{2x}$
B. $C_1e^{-2x} + C_2e^{-1x}$
C. $C_1e^{1x} + C_2e^{2x}$
D. $C_1e^{-2x} + C_2e^{1x}$

Q0213 · Medium · ≈57 s · Undergraduate

213. Find the general solution of y'' - 4y' + 3y = 0.

A. $C_1e^{-1x} + C_2e^{3x}$
B. $C_1e^{-3x} + C_2e^{1x}$
C. $C_1e^{-3x} + C_2e^{-1x}$
D. $C_1e^{1x} + C_2e^{3x}$

Q0214 · Hard · ≈254 s · Upper Undergraduate

214. Solve y' + y/x = x^2 for x > 0 with y(1) = 1. What is y(2)?

A. 11/4
B. 9/4
C. 2
D. 19/8

Q0215 · Hard · ≈153 s · Upper Undergraduate

215. Solve y' + y/x = x^8 for x > 0 with y(1) = 1. What is y(2)?

A. 521/10
B. 513/10
C. 256/5
D. 1033/20

Q0216 · Hard · ≈190 s · Upper Undergraduate

216. Solve y' + y/x = x^{13} for x > 0 with y(1) = 1. What is y(2)?

A. 3277/3
B. 16384/15
C. 16391/15
D. 5466/5

Q0217 · Hard · ≈194 s · Upper Undergraduate

217. Solve y' + y/x = x^5 for x > 0 with y(1) = 1. What is y(2)?

A. 10
B. 64/7
C. 65/7
D. 67/7

Q0218 · Hard · ≈279 s · Upper Undergraduate

218. Solve y' + y/x = x^{7} for x > 0 with y(1) = 1. What is y(2)?

A. 257/9
B. 260/9
C. 88/3
D. 256/9

Q0219 · Hard · ≈220 s · Upper Undergraduate

219. Solve y' + y/x = x^{4} for x > 0 with y(1) = 1. What is y(2)?

A. 16/3
B. 37/6
C. 23/4
D. 11/2

Q0220 · Hard · ≈199 s · Upper Undergraduate

220. Solve y' + y/x = x^{11} for x > 0 with y(1) = 1. What is y(2)?

A. 316
B. 4096/13
C. 4097/13
D. 4102/13

Partial differential equations

Q0221 · Easy · ≈32 s · Introductory Undergraduate

221. Classify the PDE $3u_{xx} + (0)u_{xy} + (1)u_{yy} = 0$.

A. Cannot be classified (nonlinear)
B. Hyperbolic
C. Parabolic
D. Elliptic

Q0222 · Easy · ≈43 s · Introductory Undergraduate

222. Classify the PDE $3u_{xx} + (4)u_{xy} + (1)u_{yy} = 0$.

A. Cannot be classified (nonlinear)
B. Hyperbolic
C. Elliptic
D. Parabolic

Q0223 · Easy · ≈41 s · Introductory Undergraduate

223. Classify the PDE $4u_{xx} + (-2)u_{xy} + (4)u_{yy} = 0$.

A. Parabolic
B. Hyperbolic
C. Cannot be classified (nonlinear)
D. Elliptic

Q0224 · Easy · ≈27 s · Introductory Undergraduate

224. Classify the PDE $1u_{xx} + (1)u_{xy} + (2)u_{yy} = 0$.

A. Hyperbolic
B. Cannot be classified (nonlinear)
C. Elliptic
D. Parabolic

Q0225 · Easy · ≈34 s · Introductory Undergraduate

225. Classify the PDE $2u_{xx} + (-1)u_{xy} + (3)u_{yy} = 0$.

A. Parabolic
B. Elliptic
C. Hyperbolic
D. Cannot be classified (nonlinear)

Q0226 · Easy · ≈32 s · Introductory Undergraduate

226. Classify the PDE $2u_{xx} + (0)u_{xy} + (0)u_{yy} = 0$.

A. Elliptic
B. Cannot be classified (nonlinear)
C. Hyperbolic
D. Parabolic

Q0227 · Easy · ≈35 s · Introductory Undergraduate

227. Classify the PDE $3u_{xx} + (4)u_{xy} + (-4)u_{yy} = 0$.

A. Elliptic
B. Parabolic
C. Cannot be classified (nonlinear)
D. Hyperbolic

Q0228 · Medium · ≈83 s · Upper Undergraduate

228. The heat equation $u_t = 2u_{xx}$ on $0 < x < 5$, with $u(0,t) = u(5,t) = 0$, has separated solutions $e^{-\lambda t}\sin(n\pi x/5)$. Find λ for $n = 5$.

A. $10\pi^2$
B. π^2
C. $2\pi^2$
D. $2\pi^2/5$

Q0229 · Medium · ≈67 s · Upper Undergraduate

229. The heat equation $u_t = 5u_{xx}$ on $0 < x < 6$, with $u(0,t) = u(6,t) = 0$, has separated solutions $e^{-\lambda t}\sin(n\pi x/6)$. Find λ for $n = 5$.

A. $125\pi^2$
B. $125\pi^2/36$
C. $25\pi^2/36$
D. $125\pi^2/6$

Q0230 · Medium · ≈78 s · Upper Undergraduate

230. The heat equation $u_t = 5u_{xx}$ on $0 < x < 6$, with $u(0,t) = u(6,t) = 0$, has separated solutions $e^{-\lambda t}\sin(n\pi x/6)$. Find λ for $n = 2$.

A. $5\pi^2/9$
B. $5\pi^2/18$
C. $10\pi^2/3$
D. $\pi^2/9$

Q0231 · Medium · ≈60 s · Upper Undergraduate

231. The heat equation $u_t = 2u_{xx}$ on $0 < x < 3$, with $u(0,t) = u(3,t) = 0$, has separated solutions $e^{-\lambda t}\sin(n\pi x/3)$. Find λ for $n = 5$.

A. $50\pi^2/3$
B. $50\pi^2/9$
C. $25\pi^2/9$
D. $10\pi^2/9$

Q0232 · Medium · ≈50 s · Upper Undergraduate

232. The heat equation $u_t = 2u_{xx}$ on $0 < x < 2$, with $u(0,t) = u(2,t) = 0$, has separated solutions $e^{-\lambda t}\sin(n\pi x/2)$. Find λ for $n = 5$.

A. $5\pi^2/2$
B. $25\pi^2/2$
C. $25\pi^2$
D. $25\pi^2/4$

Q0233 · Medium · ≈85 s · Upper Undergraduate

233. The heat equation $u_t = 2u_{xx}$ on $0 < x < 5$, with $u(0,t) = u(5,t) = 0$, has separated solutions $e^{-\lambda t}\sin(n\pi x/5)$. Find λ for $n = 2$.

A. $8\pi^2/25$
B. $8\pi^2/5$
C. $4\pi^2/25$
D. $8\pi^2$

Q0234 · Medium · ≈99 s · Upper Undergraduate

234. The heat equation $u_t = 3u_{xx}$ on $0 < x < 2$, with $u(0,t) = u(2,t) = 0$, has separated solutions $e^{-\lambda t}\sin(n\pi x/2)$. Find λ for $n = 4$.

A. $3\pi^2$
B. $12\pi^2$
C. $24\pi^2$
D. $4\pi^2$

Q0235 · Medium · ≈102 s · Upper Undergraduate

235. The heat equation $u_t = 2u_{xx}$ on $0 < x < 6$, with $u(0,t) = u(6,t) = 0$, has separated solutions $e^{-\lambda t}\sin(n\pi x/6)$. Find λ for $n = 2$.

A. $\pi^2/9$
B. $8\pi^2$
C. $4\pi^2/3$
D. $2\pi^2/9$

Q0236 · Hard · ≈274 s · Advanced Undergraduate / Early Graduate

236. Solve the wave equation $u_{tt} = 16u_{xx}$ on the real line with $u(x,0) = x^3$ and $u_t(x,0) = 0$. Find $u(4, 2)$.

A. 1664
B. 832
C. 160
D. 320

Q0237 · Hard · ≈199 s · Advanced Undergraduate / Early Graduate

237. Solve the wave equation $u_{tt} = 16u_{xx}$ on the real line with $u(x,0) = x^3$ and $u_t(x,0) = 0$. Find $u(4, 1)$.

A. 112
B. 256
C. 512
D. 128

Q0238 · Hard · ≈224 s · Advanced Undergraduate / Early Graduate

238. Solve the wave equation $u_{tt} = 4u_{xx}$ on the real line with $u(x,0) = x^3$ and $u_t(x,0) = 0$. Find $u(2, 3)$.

A. 224
B. 448
C. 80
D. 44

Q0239 · Hard · ≈236 s · Advanced Undergraduate / Early Graduate

239. Solve the wave equation $u_{tt} = 16u_{xx}$ on the real line with $u(x,0) = x^3$ and $u_t(x,0) = 0$. Find $u(2, 3)$.

A. 296
B. 80
C. 1744
D. 872

Q0240 · Hard · ≈137 s · Advanced Undergraduate / Early Graduate

240. Solve the wave equation $u_tt = 4u_xx$ on the real line with $u(x,0) = x^3$ and $u_t(x,0) = 0$. Find $u(2, 2)$.

A. 208
B. 32
C. 104
D. 40

Q0241 · Hard · ≈210 s · Advanced Undergraduate / Early Graduate

241. Solve the wave equation $u_tt = 9u_xx$ on the real line with $u(x,0) = x^3$ and $u_t(x,0) = 0$. Find $u(3, 1)$.

A. 109
B. 108
C. 216
D. 54

Q0242 · Hard · ≈184 s · Advanced Undergraduate / Early Graduate

242. Solve the wave equation $u_tt = 16u_xx$ on the real line with $u(x,0) = x^3$ and $u_t(x,0) = 0$. Find $u(4, 3)$.

A. 1792
B. 208
C. 3584
D. 640

Real analysis

Q0243 · Easy · ≈28 s · Introductory Undergraduate

243. Find $\lim_{n\to\infty} (8n + 0) / (7n + 6)$.

A. 8/13
B. 8/7
C. 0
D. 4/3

Q0244 · Easy · ≈43 s · Introductory Undergraduate

244. Find $\lim_{n\to\infty} (2n + 2) / (3n - 3)$.

A. 2/3
B. 3/2
C. -2/3
D. 0

Q0245 · Easy · ≈23 s · Introductory Undergraduate

245. Find $\lim_{n\to\infty} (8n + 2) / (6n - 2)$.

A. 5/2
B. 4/3
C. -4
D. -1

Q0246 · Easy · ≈28 s · Introductory Undergraduate

246. Find $\lim_{n\to\infty} (6n - 2) / (8n + 9)$.

A. 2/3
B. 3/4
C. -2/9
D. 4/17

Q0247 · Easy · ≈43 s · Introductory Undergraduate

247. Find $\lim_{n\to\infty} (5n - 1) / (1n + 2)$.

A. -1/2
B. 4/3
C. 5
D. 5/2

Q0248 · Easy · ≈29 s · Introductory Undergraduate

248. Find $\lim_{n\to\infty} (1n + 4) / (2n + 3)$.

A. 1
B. 1/3
C. 4/3
D. 1/2

Q0249 · Easy · ≈24 s · Introductory Undergraduate

249. Find $\lim_{n\to\infty} (5n + 5) / (8n + 0)$.

A. 5/4
B. 8/5
C. 5/8
D. 0

Q0250 · Medium · ≈54 s · Upper Undergraduate

250. Find the radius of convergence of the power series $\sum_{n=1}^{\infty} (n^5 + 1)\, x^n / 9^n$.

A. 1/9
B. 1
C. ∞
D. 9

Q0251 · Medium · ≈67 s · Upper Undergraduate

251. Find the radius of convergence of the power series $\sum_{n=1}^{\infty} (n^1 + 1)\, x^n / 5^n$.

A. ∞
B. 1
C. 1/5
D. 5

Q0252 · Medium · ≈54 s · Upper Undergraduate

252. Find the radius of convergence of the power series $\sum_{n=1}^{\infty} (n^2 + 1)\, x^n / 6^n$.

A. 1/6
B. 6
C. 1
D. ∞

Q0253 · Medium · ≈101 s · Upper Undergraduate

253. Find the radius of convergence of the power series $\sum_{n=1}^{\infty} (n^2 + 1)\, x^n / 8^n$.

A. ∞
B. 1/8
C. 8
D. 1

Q0254 · Medium · ≈60 s · Upper Undergraduate

254. Find the radius of convergence of the power series $\sum_{n=1}^{\infty} (n^1 + 1)\, x^n / 7^n$.

A. 1/7
B. ∞
C. 7
D. 1

Q0255 · Medium · ≈82 s · Upper Undergraduate

255. Find the radius of convergence of the power series $\sum_{n=1}^{\infty} (n^{1} + 1)\, x^{n} / 4^{n}$.

A. 1
B. ∞
C. 1/4
D. 4

Q0256 · Medium · ≈91 s · Upper Undergraduate

256. Find the radius of convergence of the power series $\sum_{n=1}^{\infty} (n^{5} + 1)\, x^{n} / 4^{n}$.

A. ∞
B. 1
C. 1/4
D. 4

Q0257 · Medium · ≈88 s · Upper Undergraduate

257. Find the radius of convergence of the power series $\sum_{n=1}^{\infty} (n^{1} + 1)\, x^{n} / 3^{n}$.

A. 3
B. ∞
C. 1/3
D. 1

Q0258 · Hard · ≈134 s · Advanced Undergraduate / Early Graduate

258. Find the exact value of $\sum_{n=1}^{\infty} 1/(n(n+8))$.

A. 1/8
B. 761/280
C. 1/64
D. 761/2240

Q0259 · Hard · ≈208 s · Advanced Undergraduate / Early Graduate

259. Find the exact value of $\sum_{n=1}^{\infty} 1/(n(n+7))$.

A. 363/980
B. 363/140
C. 1/7
D. 1/49

Q0260 · Hard · ≈182 s · Advanced Undergraduate / Early Graduate

260. Find the exact value of $\sum_{n=1}^{\infty} 1/(n(n+6))$.

A. 49/20
B. 1/6
C. 1/36
D. 49/120

Q0261 · Hard · ≈264 s · Advanced Undergraduate / Early Graduate

261. Find the exact value of $\sum_{n=1}^{\infty} 1/(n(n+4))$.

A. 25/48
B. 25/12
C. 1/16
D. 1/4

Q0262 · Hard · ≈188 s · Advanced Undergraduate / Early Graduate

262. Find the exact value of $\sum_{n=1}^{\infty} 1/(n(n+3))$.

A. 11/6
B. 1/3
C. 11/18
D. 1/9

Q0263 · Hard · ≈241 s · Advanced Undergraduate / Early Graduate

263. Find the exact value of $\sum_{n=1}^{\infty} 1/(n(n+2))$.

A. 1/4
B. 3/2
C. 1/2
D. 3/4

Q0264 · Hard · ≈155 s · Advanced Undergraduate / Early Graduate

264. Find the exact value of $\sum_{n=1}^{\infty} 1/(n(n+9))$.

A. 1/81
B. 7129/22680
C. 7129/2520
D. 1/9

Complex analysis

Q0265 · Easy · ≈44 s · Introductory Undergraduate

265. Find $|z_1/z_2|$ where $z_1 = 8 - 15i$ and $z_2 = 20 + 21i$.

A. 17/29
B. 12
C. 29/17
D. 493

Q0266 · Easy · ≈32 s · Introductory Undergraduate

266. Find $|z_1/z_2|$ where $z_1 = 7 - 24i$ and $z_2 = 8 - 15i$.

A. 8
B. 17/25
C. 425
D. 25/17

Q0267 · Easy · ≈23 s · Introductory Undergraduate

267. Find $|z_1/z_2|$ where $z_1 = 7 - 24i$ and $z_2 = 3 + 4i$.

A. 20
B. 5
C. 1/5
D. 125

Q0268 · Easy · ≈41 s · Introductory Undergraduate

268. Find $|z_1/z_2|$ where $z_1 = 5 - 12i$ and $z_2 = 7 + 24i$.

A. 325
B. 12
C. 13/25
D. 25/13

Q0269 · Easy · ≈21 s · Introductory Undergraduate

269. Find $|z_1/z_2|$ where $z_1 = 8 + 15i$ and $z_2 = 3 - 4i$.

A. 17/5
B. 85
C. 12
D. 5/17

Q0270 · Easy · ≈21 s · Introductory Undergraduate

270. Find $|z_1/z_2|$ where $z_1 = 5 - 12i$ and $z_2 = 20 - 21i$.

A. 13/29
B. 377
C. 16
D. 29/13

Q0271 · Easy · ≈44 s · Introductory Undergraduate

271. Find $|z_1/z_2|$ where $z_1 = 3 + 4i$ and $z_2 = 8 + 15i$.

A. 85
B. 12
C. 17/5
D. 5/17

Q0272 · Medium · ≈53 s · Upper Undergraduate

272. Evaluate $\oint dz / ((z - 2)(z - 4))$ over the circle $|z| = 3$, traversed once counterclockwise.

A. 0
B. πi
C. -πi
D. 2πi

Q0273 · Medium · ≈53 s · Upper Undergraduate

273. Evaluate $\oint dz / ((z)(z - 4))$ over the circle $|z| = 1$, traversed once counterclockwise.

A. 0
B. (1/2)πi
C. (-1/2)πi
D. 2πi

Q0274 · Medium · ≈64 s · Upper Undergraduate

274. Evaluate $\oint dz / ((z + 1)(z - 6))$ over the circle $|z| = 2$, traversed once counterclockwise.

A. 0
B. (-2/7)πi
C. 2πi
D. (2/7)πi

Q0275 · Medium · ≈103 s · Upper Undergraduate

275. Evaluate $\oint dz / ((z + 1)(z - 5))$ over the circle $|z| = 2$, traversed once counterclockwise.

A. 0
B. (-1/3)πi
C. 2πi
D. (1/3)πi

Q0276 · Medium · ≈80 s · Upper Undergraduate

276. Evaluate $\oint dz / ((z - 1)(z - 6))$ over the circle $|z| = 2$, traversed once counterclockwise.

A. 0
B. (-2/5)πi
C. (2/5)πi
D. 2πi

Q0277 · Medium · ≈95 s · Upper Undergraduate

277. Evaluate $\oint dz / ((z - 2)(z + 5))$ over the circle $|z| = 3$, traversed once counterclockwise.

A. 0
B. 2πi
C. (-2/7)πi
D. (2/7)πi

Q0278 · Medium · ≈99 s · Upper Undergraduate

278. Evaluate $\oint dz / ((z + 2)(z - 5))$ over the circle $|z| = 3$, traversed once counterclockwise.

A. 2πi
B. (-2/7)πi
C. (2/7)πi
D. 0

Q0279 · Medium · ≈94 s · Upper Undergraduate

279. Evaluate $\oint dz / ((z-1)(z-3))$ over the circle $|z| = 2$, traversed once counterclockwise.

A. 0
B. πi
C. -πi
D. 2πi

Q0280 · Hard · ≈163 s · Advanced Undergraduate / Early Graduate

280. Compute the residue of e^{2z} / z^3 at $z = 0$.

A. 3
B. 2/3
C. 4/3
D. 2

Q0281 · Hard · ≈182 s · Advanced Undergraduate / Early Graduate

281. Compute the residue of e^{3z} / z^4 at $z = 0$.

A. 9
B. 9/2
C. 27/8
D. 9/8

Q0282 · Hard · ≈196 s · Advanced Undergraduate / Early Graduate

282. Compute the residue of e^{5z} / z^2 at $z = 0$.

A. 125/6
B. 25/2
C. 5/2
D. 5

Q0283 · Hard · ≈246 s · Advanced Undergraduate / Early Graduate

283. Compute the residue of e^{4z} / z^3 at $z = 0$.

A. 8
B. 32/3
C. 8/3
D. 9

Q0284 · Hard · ≈207 s · Advanced Undergraduate / Early Graduate

284. Compute the residue of e^{3z} / z^5 at $z = 0$.

A. 81/4
B. 27/40
C. 81/40
D. 27/8

Q0285 · Hard · ≈224 s · Advanced Undergraduate / Early Graduate

285. Compute the residue of e^{3z} / z^6 at $z = 0$.

A. 27/80
B. 243/5
C. 81/80
D. 81/40

Q0286 · Hard · ≈251 s · Advanced Undergraduate / Early Graduate

286. Compute the residue of e^{5z} / z^4 at $z = 0$.

A. 125/24
B. 125/3
C. 625/24
D. 125/6

Functional analysis

Q0287 · Easy · ≈22 s · Introductory Undergraduate

287. For x = (6, -1, -5) in $\mathbb{R}^3$, compute $\|x\|_1 + \|x\|_\infty$.

A. 18
B. 6
C. 12
D. 19

Q0288 · Easy · ≈32 s · Introductory Undergraduate

288. For x = (-5, -2, -1) in $\mathbb{R}^3$, compute $\|x\|_1 + \|x\|_\infty$.

A. 13
B. 8
C. 5
D. 14

Q0289 · Easy · ≈36 s · Introductory Undergraduate

289. For x = (4, -3, 2) in $\mathbb{R}^3$, compute $\|x\|_1 + \|x\|_\infty$.

A. 9
B. 13
C. 7
D. 4

Q0290 · Easy · ≈40 s · Introductory Undergraduate

290. For x = (-4, 1, 6) in $\mathbb{R}^3$, compute $\|x\|_1 + \|x\|_\infty$.

A. 9
B. 6
C. 17
D. 11

Q0291 · Easy · ≈43 s · Introductory Undergraduate

291. For x = (-3, -5, 2) in $\mathbb{R}^3$, compute $\|x\|_1 + \|x\|_\infty$.

A. 10
B. 11
C. 15
D. 5

Q0292 · Easy · ≈30 s · Introductory Undergraduate

292. For x = (5, 1, -6) in $\mathbb{R}^3$, compute $\|x\|_1 + \|x\|_\infty$.

A. 6
B. 19
C. 12
D. 18

Q0293 · Easy · ≈36 s · Introductory Undergraduate

293. For x = (5, -2, 1) in $\mathbb{R}^3$, compute $\|x\|_1 + \|x\|_\infty$.

A. 9
B. 8
C. 13
D. 5

Q0294 · Medium · ≈105 s · Upper Undergraduate

294. For $1 < p < \infty$, the dual of ℓ^p is ℓ^q with $1/p + 1/q = 1$. If $p = 5/4$, what is q?

A. 4/5
B. 1/5
C. 1/4
D. 5

Q0295 · Medium · ≈72 s · Upper Undergraduate

295. For $1 < p < \infty$, the dual of ℓ^p is ℓ^q with $1/p + 1/q = 1$. If $p = 7/2$, what is q?

A. 5/7
B. 5/2
C. 7/5
D. 2/7

Q0296 · Medium · ≈99 s · Upper Undergraduate

296. For $1 < p < \infty$, the dual of ℓ^p is ℓ^q with $1/p + 1/q = 1$. If $p = 5$, what is q?

A. 4/5
B. 1/5
C. 4
D. 5/4

Q0297 · Medium · ≈78 s · Upper Undergraduate

297. For $1 < p < \infty$, the dual of ℓ^p is ℓ^q with $1/p + 1/q = 1$. If $p = 5/3$, what is q?

A. 5/2
B. 2/5
C. 2/3
D. 3/5

Q0298 · Medium · ≈65 s · Upper Undergraduate

298. For $1 < p < \infty$, the dual of ℓ^p is ℓ^q with $1/p + 1/q = 1$. If $p = 3$, what is q?

A. 2
B. 2/3
C. 3/2
D. 1/3

Q0299 · Medium · ≈67 s · Upper Undergraduate

299. For $1 < p < \infty$, the dual of ℓ^p is ℓ^q with $1/p + 1/q = 1$. If $p = 4$, what is q?

A. 3/4
B. 3
C. 4/3
D. 1/4

Q0300 · Medium · ≈72 s · Upper Undergraduate

300. For $1 < p < \infty$, the dual of ℓ^p is ℓ^q with $1/p + 1/q = 1$. If $p = 6$, what is q?

A. 6/5
B. 5/6
C. 5
D. 1/6

Q0301 · Medium · ≈59 s · Upper Undergraduate

301. For $1 < p < \infty$, the dual of ℓ^p is ℓ^q with $1/p + 1/q = 1$. If $p = 3/2$, what is q?

A. 2/3
B. 3
C. 1/3
D. 1/2

Q0302 · Hard · ≈191 s · Advanced Undergraduate / Early Graduate

302. Let $T: \ell^2 \to \ell^2$ be the diagonal operator $(Tx)_n = 5n/(n^2 + 4) \cdot x_n$. What is $\|T\|$?

A. 5/2
B. 5/4
C. 1
D. 5/8

Q0303 · Hard · ≈204 s · Advanced Undergraduate / Early Graduate

303. Let $T: \ell^2 \to \ell^2$ be the diagonal operator $(Tx)_n = 4n/(n^2 + 4) \cdot x_n$. What is $\|T\|$?

A. 4/5
B. 1
C. 2
D. 1/2

Q0304 · Hard · ≈257 s · Advanced Undergraduate / Early Graduate

304. Let $T: \ell^2 \to \ell^2$ be the diagonal operator $(Tx)_n = 3n/(n^2 + 9) \cdot x_n$. What is $\|T\|$?

A. 1/3
B. 1/2
C. 3/10
D. 1

Q0305 · Hard · ≈219 s · Advanced Undergraduate / Early Graduate

305. Let $T: \ell^2 \to \ell^2$ be the diagonal operator $(Tx)_n = 4n/(n^2 + 25) \cdot x_n$. What is $\|T\|$?

A. 4/25
B. 4/5
C. 2/5
D. 2/13

Q0306 · Hard · ≈201 s · Advanced Undergraduate / Early Graduate

306. Let $T: \ell^2 \to \ell^2$ be the diagonal operator $(Tx)_n = 4n/(n^2 + 36) \cdot x_n$. What is $\|T\|$?

A. 1/3
B. 1/9
C. 4/37
D. 2/3

Q0307 · Hard · ≈165 s · Advanced Undergraduate / Early Graduate

307. Let $T: \ell^2 \to \ell^2$ be the diagonal operator $(Tx)_n = 6n/(n^2 + 16) \cdot x_n$. What is $\|T\|$?

A. 6/17
B. 3/8
C. 3/2
D. 3/4

Q0308 · Hard · ≈266 s · Advanced Undergraduate / Early Graduate

308. Let $T: \ell^2 \to \ell^2$ be the diagonal operator $(Tx)_n = 9n/(n^2 + 16) \cdot x_n$. What is $\|T\|$?

A. 9/4
B. 9/17
C. 9/8
D. 9/16

Measure theory

Q0309 · Easy · ≈32 s · Introductory Undergraduate

309. What is the Lebesgue measure of $[1, 5] \cup (8, 14)$?

A. 24
B. 12
C. 10
D. 13

Q0310 · Easy · ≈24 s · Introductory Undergraduate

310. What is the Lebesgue measure of $[0, 2] \cup (5, 10)$?

A. 7
B. 9
C. 2
D. 10

Q0311 · Easy · ≈32 s · Introductory Undergraduate

311. What is the Lebesgue measure of [0, 5] ∪ (8, 13)?

A. 25
B. 13
C. 10
D. 12

Q0312 · Easy · ≈44 s · Introductory Undergraduate

312. What is the Lebesgue measure of [0, 5] ∪ (7, 9)?

A. 7
B. 8
C. 10
D. 9

Q0313 · Easy · ≈29 s · Introductory Undergraduate

313. What is the Lebesgue measure of [5, 9] ∪ (13, 16)?

A. 12
B. 11
C. 7
D. 10

Q0314 · Easy · ≈45 s · Introductory Undergraduate

314. What is the Lebesgue measure of [3, 8] ∪ (12, 15)?

A. 12
B. 11
C. 8
D. 15

Q0315 · Easy · ≈31 s · Introductory Undergraduate

315. What is the Lebesgue measure of [0, 4] ∪ (8, 10)?

A. 8
B. 10
C. 6
D. 9

Q0316 · Medium · ≈52 s · Upper Undergraduate

316. Start with [0,1]. At each stage, remove the open middle 1/7 (by length) of every remaining interval. What is the total length remaining after 2 stages?

A. 5/7
B. 1/49
C. 12/7
D. 36/49

Q0317 · Medium · ≈76 s · Upper Undergraduate

317. Start with [0,1]. At each stage, remove the open middle 1/5 (by length) of every remaining interval. What is the total length remaining after 3 stages?

A. 64/125
B. 2/5
C. 1/125
D. 12/5

Q0318 · Medium · ≈52 s · Upper Undergraduate

318. Start with [0,1]. At each stage, remove the open middle 1/3 (by length) of every remaining interval. What is the total length remaining after 4 stages?

A. 16/81
B. 8/3
C. 1/81
D. -1/3

Q0319 · Medium · ≈76 s · Upper Undergraduate

319. Start with [0,1]. At each stage, remove the open middle 1/4 (by length) of every remaining interval. What is the total length remaining after 2 stages?

A. 1/16
B. 9/16
C. 1/2
D. 3/2

Q0320 · Medium · ≈81 s · Upper Undergraduate

320. Start with [0,1]. At each stage, remove the open middle 1/4 (by length) of every remaining interval. What is the total length remaining after 3 stages?

A. 1/64
B. 27/64
C. 9/4
D. 1/4

Q0321 · Medium · ≈95 s · Upper Undergraduate

321. Start with [0,1]. At each stage, remove the open middle 1/7 (by length) of every remaining interval. What is the total length remaining after 4 stages?

A. 3/7
B. 1296/2401
C. 24/7
D. 1/2401

Q0322 · Medium · ≈72 s · Upper Undergraduate

322. Start with [0,1]. At each stage, remove the open middle 1/6 (by length) of every remaining interval. What is the total length remaining after 2 stages?

A. 25/36
B. 5/3
C. 2/3
D. 1/36

Q0323 · Medium · ≈52 s · Upper Undergraduate

323. Start with [0,1]. At each stage, remove the open middle 1/6 (by length) of every remaining interval. What is the total length remaining after 4 stages?

A. 1/3
B. 10/3
C. 1/1296
D. 625/1296

Q0324 · Hard · ≈266 s · Advanced Undergraduate / Early Graduate

324. From [0,1], at stage k = 1, 2, 3, … remove 2^(k−1) disjoint open intervals, each of length 9^(−k). The remaining closed set has Lebesgue measure:

A. 7/8
B. 0
C. 8/9
D. 6/7

Q0325 · Hard · ≈147 s · Advanced Undergraduate / Early Graduate

325. From [0,1], at stage k = 1, 2, 3, … remove 2^(k−1) disjoint open intervals, each of length 5^(−k). The remaining closed set has Lebesgue measure:

A. 3/4
B. 0
C. 4/5
D. 2/3

Q0326 · Hard · ≈249 s · Advanced Undergraduate / Early Graduate

326. From [0,1], at stage k = 1, 2, 3, … remove 2^(k−1) disjoint open intervals, each of length 7^(−k). The remaining closed set has Lebesgue measure:

A. 6/7
B. 0
C. 5/6
D. 4/5

Q0327 · Hard · ≈268 s · Advanced Undergraduate / Early Graduate

327. From [0,1], at stage k = 1, 2, 3, … remove 2^(k−1) disjoint open intervals, each of length 10^(−k). The remaining closed set has Lebesgue measure:

A. 7/8
B. 8/9
C. 9/10
D. 0

Q0328 · Hard · ≈179 s · Advanced Undergraduate / Early Graduate

328. From [0,1], at stage k = 1, 2, 3, … remove 2^(k−1) disjoint open intervals, each of length 12^(−k). The remaining closed set has Lebesgue measure:

A. 11/12
B. 10/11
C. 0
D. 9/10

Q0329 · Hard · ≈179 s · Advanced Undergraduate / Early Graduate

329. From [0,1], at stage k = 1, 2, 3, … remove 2^(k−1) disjoint open intervals, each of length 8^(−k). The remaining closed set has Lebesgue measure:

A. 5/6
B. 0
C. 6/7
D. 7/8

Q0330 · Hard · ≈182 s · Advanced Undergraduate / Early Graduate

330. From [0,1], at stage k = 1, 2, 3, … remove 2^(k−1) disjoint open intervals, each of length 11^(−k). The remaining closed set has Lebesgue measure:

A. 9/10
B. 10/11
C. 8/9
D. 0

C. Algebra

Linear algebra

Q0331 · Easy · ≈43 s · Introductory Undergraduate

331. Find the determinant of the matrix A = [[3, -2], [-4, 0]].

A. 3
B. -12
C. -8
D. 8

Q0332 · Easy · ≈36 s · Introductory Undergraduate

332. Find the determinant of the matrix A = [[-4, 1], [0, -1]].

A. 5
B. -5
C. 1
D. 4

Q0333 · Easy · ≈23 s · Introductory Undergraduate

333. Find the determinant of the matrix A = [[-1, 1], [4, -6]].

A. 3
B. -7
C. 10
D. 2

Q0334 · Easy · ≈26 s · Introductory Undergraduate

334. Find the determinant of the matrix A = [[-1, 6], [-6, 7]].

A. 6
B. -43
C. -36
D. 29

Q0335 · Easy · ≈38 s · Introductory Undergraduate

335. Find the determinant of the matrix A = [[3, -1], [-1, -4]].

A. -13
B. -11
C. -1
D. -7

Q0336 · Easy · ≈38 s · Introductory Undergraduate

336. Find the determinant of the matrix A = [[-5, 3], [6, -4]].

A. -18
B. 38
C. 2
D. -9

Q0337 · Easy · ≈25 s · Introductory Undergraduate

337. Find the determinant of the matrix A = [[2, -4], [3, 0]].

A. 6
B. 12
C. 2
D. -12

Q0338 · Medium · ≈98 s · Undergraduate

338. What is the larger eigenvalue of A = [[-5, 2], [6, -1]]?

A. -7
B. 2
C. -6
D. 1

Q0339 · Medium · ≈63 s · Undergraduate

339. What is the larger eigenvalue of A = [[6, 3], [1, 4]]?

A. 10
B. 7
C. 21
D. 3

Q0340 · Medium · ≈87 s · Undergraduate

340. What is the larger eigenvalue of A = [[-4, -4], [3, 3]]?

A. 2
B. 0
C. -1
D. 1

Q0341 · Medium · ≈70 s · Undergraduate

341. What is the larger eigenvalue of A = [[3, 3], [3, -5]]?

A. -2
B. -6
C. 4
D. -24

Q0342 · Medium · ≈58 s · Undergraduate

342. What is the larger eigenvalue of A = [[-2, 3], [-4, 5]]?

A. 4
B. 3
C. 1
D. 2

Q0343 · Medium · ≈65 s · Undergraduate

343. What is the larger eigenvalue of A = [[-4, -3], [-2, -5]]?

A. 14
B. -7
C. -2
D. -9

Q0344 · Medium · ≈106 s · Undergraduate

344. What is the larger eigenvalue of A = [[2, 6], [1, -3]]?

A. -1
B. -12
C. -4
D. 3

Q0345 · Medium · ≈66 s · Undergraduate

345. What is the larger eigenvalue of A = [[-5, 1], [5, -1]]?

A. 0
B. -6
C. -1
D. 1

Q0346 · Hard · ≈272 s · Upper Undergraduate

346. A 3×3 matrix A has eigenvalues 1, 4 and -3. Find det(A^2 + 3A - 2I).

A. 26
B. -128
C. 106
D. -104

Q0347 · Hard · ≈151 s · Upper Undergraduate

347. A 3×3 matrix A has eigenvalues 3, 4 and 1. Find det(A^2 - 2A - 4I).

A. -2
B. -220
C. 20
D. 116

Q0348 · Hard · ≈163 s · Upper Undergraduate

348. A 3×3 matrix A has eigenvalues 1, -1 and -3. Find det(A^2 - 2A + 4I).

A. 147
B. 7
C. 399
D. 29

Q0349 · Hard · ≈207 s · Upper Undergraduate

349. A 3×3 matrix A has eigenvalues 0, 3 and 2. Find det(A^2 - 3A - 2I).

A. -8
B. -16
C. -2
D. -256

Q0350 · Hard · ≈234 s · Upper Undergraduate

350. A 3×3 matrix A has eigenvalues 1, 2 and -3. Find det(A^2 + 0A - 2I).

A. -13
B. -14
C. 34
D. 8

Q0351 · Hard · ≈141 s · Upper Undergraduate

351. A 3×3 matrix A has eigenvalues 0, 2 and -3. Find det(A^2 - 1A + 2I).

A. 20
B. 128
C. 2
D. 112

Q0352 · Hard · ≈277 s · Upper Undergraduate

352. A 3×3 matrix A has eigenvalues 4, 1 and 3. Find det(A^2 - 2A - 2I).

A. 286
B. -18
C. 4
D. 118

Abstract algebra: groups, rings, and fields

Q0353 · Easy · ≈44 s · Introductory Undergraduate

353. In the cyclic group (Z_{16}, +), what is the order of the element 8?

A. 16
B. 2
C. 8
D. 3

Q0354 · Easy · ≈40 s · Introductory Undergraduate

354. In the cyclic group (Z_{32}, +), what is the order of the element 4?

A. 28
B. 8
C. 32
D. 4

Q0355 · Easy · ≈41 s · Introductory Undergraduate

355. In the cyclic group $(Z_{35}, +)$, what is the order of the element 28?

A. 7
B. 6
C. 5
D. 35

Q0356 · Easy · ≈38 s · Introductory Undergraduate

356. In the cyclic group $(Z_{49}, +)$, what is the order of the element 42?

A. 49
B. 8
C. 6
D. 7

Q0357 · Easy · ≈21 s · Introductory Undergraduate

357. In the cyclic group $(Z_{24}, +)$, what is the order of the element 22?

A. 12
B. 2
C. 24
D. 13

Q0358 · Easy · ≈42 s · Introductory Undergraduate

358. In the cyclic group $(Z_{42}, +)$, what is the order of the element 9?

A. 42
B. 14
C. 3
D. 33

Q0359 · Easy · ≈45 s · Introductory Undergraduate

359. In the cyclic group $(Z_{46}, +)$, what is the order of the element 12?

A. 2
B. 46
C. 34
D. 23

Q0360 · Medium · ≈78 s · Undergraduate

360. What is the order of the permutation $\sigma = (1\ 2)(3\ 4\ 5\ 6\ 7\ 8\ 9)$ in S_9?

A. 9
B. 14
C. 7
D. 28

Q0361 · Medium · ≈56 s · Undergraduate

361. What is the order of the permutation $\sigma = (1\ 2)(3\ 4\ 5\ 6)(7\ 8\ 9\ 10\ 11)$ in S_{11}?

A. 11
B. 5
C. 40
D. 20

Q0362 · Medium · ≈107 s · Undergraduate

362. What is the order of the permutation $\sigma = (1\ 2\ 3\ 4\ 5\ 6)(7\ 8\ 9\ 10)$ in S_{10}?

A. 24
B. 6
C. 12
D. 10

Q0363 · Medium · ≈91 s · Undergraduate

363. What is the order of the permutation σ = (1 2 3)(4 5) in S_5?

A. 12
B. 3
C. 6
D. 5

Q0364 · Medium · ≈100 s · Undergraduate

364. What is the order of the permutation σ = (1 2 3 4 5 6 7)(8 9)(10 11 12) in S_{12}?

A. 84
B. 12
C. 7
D. 42

Q0365 · Medium · ≈70 s · Undergraduate

365. What is the order of the permutation σ = (1 2 3 4 5 6 7)(8 9 10 11 12 13)(14 15 16 17 18) in S_{18}?

A. 7
B. 420
C. 210
D. 18

Q0366 · Medium · ≈74 s · Undergraduate

366. What is the order of the permutation σ = (1 2 3 4 5 6)(7 8 9 10 11 12 13)(14 15 16) in S_{16}?

A. 126
B. 42
C. 7
D. 16

Q0367 · Medium · ≈104 s · Undergraduate

367. What is the order of the permutation σ = (1 2 3 4 5)(6 7 8 9 10 11 12)(13 14) in S_{14}?

A. 70
B. 7
C. 140
D. 14

Q0368 · Hard · ≈244 s · Upper Undergraduate

368. How many elements of order 6 does the group $Z_4 \times Z_6$ have?

A. 12
B. 6
C. 4
D. 2

Q0369 · Hard · ≈278 s · Upper Undergraduate

369. How many elements of order 6 does the group $Z_6 \times Z_6$ have?

A. 24
B. 2
C. 6
D. 36

Q0370 · Hard · ≈254 s · Upper Undergraduate

370. How many elements of order 3 does the group $Z_3 \times Z_9$ have?

A. 9
B. 8
C. 7
D. 2

Q0371 · Hard · ≈155 s · Upper Undergraduate

371. How many elements of order 3 does the group $Z_6 \times Z_6$ have?

A. 9
B. 12
C. 8
D. 2

Q0372 · Hard · ≈259 s · Upper Undergraduate

372. How many elements of order 6 does the group $Z_6 \times Z_9$ have?

A. 8
B. 18
C. 2
D. 9

Q0373 · Hard · ≈183 s · Upper Undergraduate

373. How many elements of order 2 does the group $Z_2 \times Z_8$ have?

A. 8
B. 1
C. 4
D. 3

Q0374 · Hard · ≈177 s · Upper Undergraduate

374. How many elements of order 6 does the group $Z_8 \times Z_{12}$ have?

A. 16
B. 12
C. 6
D. 2

Number theory

Q0375 · Easy · ≈23 s · Introductory Undergraduate

375. What is Euler's totient φ(147)?

A. 84
B. 146
C. 73
D. 8

Q0376 · Easy · ≈40 s · Introductory Undergraduate

376. What is Euler's totient φ(74)?

A. 36
B. 37
C. 73
D. 70

Q0377 · Easy · ≈23 s · Introductory Undergraduate

377. What is Euler's totient φ(194)?

A. 190
B. 96
C. 193
D. 97

Q0378 · Easy · ≈25 s · Introductory Undergraduate

378. What is Euler's totient φ(93)?

A. 92
B. 60
C. 32
D. 46

Q0379 · Easy · ≈33 s · Introductory Undergraduate

379. What is Euler's totient φ(55)?

A. 40
B. 14
C. 27
D. 54

Q0380 · Easy · ≈31 s · Introductory Undergraduate

380. What is Euler's totient φ(153)?

A. 96
B. 18
C. 76
D. 152

Q0381 · Easy · ≈21 s · Introductory Undergraduate

381. What is Euler's totient φ(111)?

A. 72
B. 110
C. 55
D. 38

Q0382 · Medium · ≈70 s · Undergraduate

382. Find the multiplicative inverse of 36 modulo 91 (an integer in 1 … 90).

A. 55
B. 43
C. 48
D. 44

Q0383 · Medium · ≈65 s · Undergraduate

383. Find the multiplicative inverse of 67 modulo 73 (an integer in 1 … 72).

A. 13
B. 61
C. 12
D. 6

Q0384 · Medium · ≈78 s · Undergraduate

384. Find the multiplicative inverse of 60 modulo 61 (an integer in 1 … 60).

A. 60
B. 1
C. 2
D. 61

Q0385 · Medium · ≈59 s · Undergraduate

385. Find the multiplicative inverse of 43 modulo 53 (an integer in 1 … 52).

A. 10
B. 38
C. 16
D. 37

Q0386 · Medium · ≈105 s · Undergraduate

386. Find the multiplicative inverse of 37 modulo 67 (an integer in 1 … 66).

A. 29
B. 30
C. 38
D. 37

Q0387 · Medium · ≈88 s · Undergraduate

387. Find the multiplicative inverse of 21 modulo 43 (an integer in 1 … 42).

A. 22
B. 41
C. 42
D. 2

Q0388 · Medium · ≈105 s · Undergraduate

388. Find the multiplicative inverse of 8 modulo 43 (an integer in 1 … 42).

A. 27
B. 28
C. 35
D. 16

Q0389 · Medium · ≈80 s · Undergraduate

389. Find the multiplicative inverse of 32 modulo 33 (an integer in 1 … 32).

A. 32
B. 2
C. 33
D. 1

Q0390 · Hard · ≈137 s · Upper Undergraduate

390. Find the smallest non-negative integer x with $x \equiv 3 \pmod 4$, $x \equiv 2 \pmod 5$, $x \equiv 3 \pmod 9$.

A. 8
B. 33
C. 147
D. 167

Q0391 · Hard · ≈206 s · Upper Undergraduate

391. Find the smallest non-negative integer x with $x \equiv 1 \pmod 3$, $x \equiv 4 \pmod 5$, $x \equiv 0 \pmod 7$.

A. 5
B. 56
C. 49
D. 64

Q0392 · Hard · ≈178 s · Upper Undergraduate

392. Find the smallest non-negative integer x with $x \equiv 4 \pmod 5$, $x \equiv 4 \pmod 7$, $x \equiv 6 \pmod 8$.

A. 214
B. 14
C. 249
D. 66

Q0393 · Hard · ≈213 s · Upper Undergraduate

393. Find the smallest non-negative integer x with $x \equiv 2 \pmod 3$, $x \equiv 0 \pmod 5$, $x \equiv 6 \pmod 7$.

A. 8
B. 20
C. 35
D. 85

Q0394 · Hard · ≈198 s · Upper Undergraduate

394. Find the smallest non-negative integer x with $x \equiv 2 \pmod 4$, $x \equiv 4 \pmod 5$, $x \equiv 2 \pmod 9$.

A. 106
B. 8
C. 74
D. 94

Q0395 · Hard · ≈152 s · Upper Undergraduate

395. Find the smallest non-negative integer x with x ≡ 4 (mod 5), x ≡ 6 (mod 8), x ≡ 3 (mod 9).

A. 13
B. 174
C. 186
D. 214

Q0396 · Hard · ≈244 s · Upper Undergraduate

396. Find the smallest non-negative integer x with x ≡ 0 (mod 3), x ≡ 1 (mod 5), x ≡ 4 (mod 7).

A. 24
B. 5
C. 81
D. 96

Galois theory

Q0397 · Easy · ≈28 s · Introductory Undergraduate

397. How many elements does the finite field F_{2^2} = GF(4) have?

A. 4
B. 2
C. 5
D. 3

Q0398 · Easy · ≈21 s · Introductory Undergraduate

398. How many elements does the finite field F_{2^4} = GF(16) have?

A. 16
B. 8
C. 15
D. 17

Q0399 · Easy · ≈42 s · Introductory Undergraduate

399. How many elements does the finite field F_{7^3} = GF(343) have?

A. 2187
B. 49
C. 343
D. 21

Q0400 · Easy · ≈45 s · Introductory Undergraduate

400. How many elements does the finite field F_{3^4} = GF(81) have?

A. 27
B. 64
C. 81
D. 12

Q0401 · Easy · ≈33 s · Introductory Undergraduate

401. How many elements does the finite field F_{7^4} = GF(2401) have?

A. 16384
B. 2401
C. 28
D. 343

Q0402 · Easy · ≈42 s · Introductory Undergraduate

402. How many elements does the finite field F_{5^4} = GF(625) have?

A. 125
B. 20
C. 1024
D. 625

Q0403 · Easy · ≈22 s · Introductory Undergraduate

403. How many elements does the finite field F_{2^5} = GF(32) have?

A. 10
B. 16
C. 25
D. 32

Q0404 · Medium · ≈69 s · Upper Undergraduate

404. How many subfields (including F_3 itself and the whole field) does the finite field $F_{3^{18}}$ have?

A. 7
B. 18
C. 6
D. 5

Q0405 · Medium · ≈72 s · Upper Undergraduate

405. How many subfields (including F_3 itself and the whole field) does the finite field $F_{3^{16}}$ have?

A. 6
B. 16
C. 4
D. 5

Q0406 · Medium · ≈57 s · Upper Undergraduate

406. How many subfields (including F_2 itself and the whole field) does the finite field $F_{2^{48}}$ have?

A. 48
B. 11
C. 10
D. 9

Q0407 · Medium · ≈86 s · Upper Undergraduate

407. How many subfields (including F_5 itself and the whole field) does the finite field $F_{5^{18}}$ have?

A. 7
B. 6
C. 5
D. 18

Q0408 · Medium · ≈80 s · Upper Undergraduate

408. How many subfields (including F_5 itself and the whole field) does the finite field $F_{5^{36}}$ have?

A. 10
B. 8
C. 9
D. 36

Q0409 · Medium · ≈100 s · Upper Undergraduate

409. How many subfields (including F_3 itself and the whole field) does the finite field $F_{3^{36}}$ have?

A. 36
B. 9
C. 10
D. 8

Q0410 · Medium · ≈97 s · Upper Undergraduate

410. How many subfields (including F_5 itself and the whole field) does the finite field $F_{5^{48}}$ have?

A. 11
B. 10
C. 9
D. 48

Q0411 · Medium · ≈80 s · Upper Undergraduate

411. How many subfields (including F_2 itself and the whole field) does the finite field $F_{2^{20}}$ have?

A. 6
B. 7
C. 20
D. 5

Q0412 · Hard · ≈155 s · Advanced Undergraduate / Early Graduate

412. What is the degree [K : Q] of the splitting field K of $x^{11} - 2$ over Q?

A. 39916800
B. 11
C. 10
D. 110

Q0413 · Hard · ≈276 s · Advanced Undergraduate / Early Graduate

413. What is the degree [K : Q] of the splitting field K of $x^7 - 6$ over Q?

A. 6
B. 5040
C. 42
D. 7

Q0414 · Hard · ≈165 s · Advanced Undergraduate / Early Graduate

414. What is the degree [K : Q] of the splitting field K of $x^3 - 7$ over Q?

A. 3
B. 6
C. 2
D. 9

Q0415 · Hard · ≈193 s · Advanced Undergraduate / Early Graduate

415. What is the degree [K : Q] of the splitting field K of $x^7 - 5$ over Q?

A. 42
B. 5040
C. 6
D. 7

Q0416 · Hard · ≈230 s · Advanced Undergraduate / Early Graduate

416. What is the degree [K : Q] of the splitting field K of $x^5 - 3$ over Q?

A. 120
B. 20
C. 5
D. 4

Q0417 · Hard · ≈202 s · Advanced Undergraduate / Early Graduate

417. What is the degree [K : Q] of the splitting field K of $x^5 - 5$ over Q?

A. 120
B. 5
C. 4
D. 20

Q0418 · Hard · ≈225 s · Advanced Undergraduate / Early Graduate

418. What is the degree [K : Q] of the splitting field K of $x^{11} - 3$ over Q?

A. 11
B. 39916800
C. 10
D. 110

Representation theory

Q0419 · Easy · ≈38 s · Introductory Undergraduate

419. How many non-isomorphic irreducible complex representations does an abelian group of order 29 have?

A. 28
B. 2
C. 1
D. 29

Q0420 · Easy · ≈38 s · Introductory Undergraduate

420. How many non-isomorphic irreducible complex representations does an abelian group of order 27 have?

A. 4
B. 1
C. 27
D. 26

Q0421 · Easy · ≈31 s · Introductory Undergraduate

421. How many non-isomorphic irreducible complex representations does an abelian group of order 24 have?

A. 1
B. 24
C. 23
D. 8

Q0422 · Easy · ≈24 s · Introductory Undergraduate

422. How many non-isomorphic irreducible complex representations does an abelian group of order 37 have?

A. 36
B. 37
C. 2
D. 1

Q0423 · Easy · ≈36 s · Introductory Undergraduate

423. How many non-isomorphic irreducible complex representations does an abelian group of order 7 have?

A. 6
B. 7
C. 1
D. 2

Q0424 · Easy · ≈41 s · Introductory Undergraduate

424. How many non-isomorphic irreducible complex representations does an abelian group of order 38 have?

A. 38
B. 1
C. 4
D. 37

Q0425 · Easy · ≈31 s · Introductory Undergraduate

425. How many non-isomorphic irreducible complex representations does an abelian group of order 12 have?

A. 11
B. 1
C. 6
D. 12

Q0426 · Medium · ≈109 s · Upper Undergraduate

426. How many non-isomorphic irreducible complex representations does the symmetric group S_4 have?

A. 4
B. 6
C. 24
D. 5

Q0427 · Medium · ≈96 s · Upper Undergraduate

427. How many non-isomorphic irreducible complex representations does the symmetric group S_7 have?

A. 7
B. 5040
C. 15
D. 16

Q0428 · Medium · ≈62 s · Upper Undergraduate

428. How many non-isomorphic irreducible complex representations does the symmetric group S_{14} have?

A. 87178291200
B. 136
C. 135
D. 14

Q0429 · Medium · ≈52 s · Upper Undergraduate

429. How many non-isomorphic irreducible complex representations does the symmetric group S_9 have?

A. 30
B. 31
C. 362880
D. 9

Q0430 · Medium · ≈79 s · Upper Undergraduate

430. How many non-isomorphic irreducible complex representations does the symmetric group S_{12} have?

A. 78
B. 77
C. 479001600
D. 12

Q0431 · Medium · ≈72 s · Upper Undergraduate

431. How many non-isomorphic irreducible complex representations does the symmetric group S_5 have?

A. 120
B. 7
C. 5
D. 8

Q0432 · Medium · ≈93 s · Upper Undergraduate

432. How many non-isomorphic irreducible complex representations does the symmetric group S_6 have?

A. 720
B. 12
C. 11
D. 6

Q0433 · Medium · ≈68 s · Upper Undergraduate

433. How many non-isomorphic irreducible complex representations does the symmetric group S_8 have?

A. 8
B. 40320
C. 22
D. 23

Q0434 · Hard · ≈205 s · Advanced Undergraduate / Early Graduate

434. The group S_4 has order 24. Its irreducible complex representations have dimensions 1, 1, 3, 3 and one more irreducible of unknown dimension. What is the missing dimension?

A. 2
B. 4
C. 1
D. 3

Q0435 · Hard · ≈244 s · Advanced Undergraduate / Early Graduate

435. The group A_5 has order 60. Its irreducible complex representations have dimensions 1, 3, 4, 5 and one more irreducible of unknown dimension. What is the missing dimension?

A. 2
B. 3
C. 6
D. 4

Q0436 · Hard · ≈189 s · Advanced Undergraduate / Early Graduate

436. The group SL(2,3) has order 24. Its irreducible complex representations have dimensions 1, 1, 1, 2, 2, 3 and one more irreducible of unknown dimension. What is the missing dimension?

A. 1
B. 4
C. 2
D. 3

Q0437 · Hard · ≈220 s · Advanced Undergraduate / Early Graduate

437. The group S_4 has order 24. Its irreducible complex representations have dimensions 1, 1, 2, 3 and one more irreducible of unknown dimension. What is the missing dimension?

A. 2
B. 3
C. 6
D. 4

Q0438 · Hard · ≈240 s · Advanced Undergraduate / Early Graduate

438. The group A_4 has order 12. Its irreducible complex representations have dimensions 1, 1, 1 and one more irreducible of unknown dimension. What is the missing dimension?

A. 2
B. 3
C. 6
D. 4

Q0439 · Hard · ≈226 s · Advanced Undergraduate / Early Graduate

439. The group A_5 has order 60. Its irreducible complex representations have dimensions 1, 3, 3, 4 and one more irreducible of unknown dimension. What is the missing dimension?

A. 10
B. 4
C. 5
D. 6

Q0440 · Hard · ≈195 s · Advanced Undergraduate / Early Graduate

440. The group S_5 has order 120. Its irreducible complex representations have dimensions 1, 1, 4, 4, 5, 6 and one more irreducible of unknown dimension. What is the missing dimension?

A. 6
B. 10
C. 5
D. 4

Commutative algebra

Q0441 · Easy · ≈31 s · Introductory Undergraduate

441. How many maximal ideals does the ring Z/56Z have?

A. 8
B. 56
C. 2
D. 3

Q0442 · Easy · ≈30 s · Introductory Undergraduate

442. How many maximal ideals does the ring Z/24Z have?

A. 3
B. 8
C. 24
D. 2

Q0443 · Easy · ≈29 s · Introductory Undergraduate

443. How many maximal ideals does the ring Z/324Z have?

A. 3
B. 324
C. 2
D. 15

Q0444 · Easy · ≈22 s · Introductory Undergraduate

444. How many maximal ideals does the ring Z/244Z have?

A. 6
B. 244
C. 3
D. 2

Q0445 · Easy · ≈45 s · Introductory Undergraduate

445. How many maximal ideals does the ring Z/315Z have?

A. 3
B. 315
C. 12
D. 4

Q0446 · Easy · ≈40 s · Introductory Undergraduate

446. How many maximal ideals does the ring Z/99Z have?

A. 6
B. 99
C. 3
D. 2

Q0447 · Easy · ≈29 s · Introductory Undergraduate

447. How many maximal ideals does the ring Z/317Z have?

A. 2
B. 316
C. 1
D. 317

Q0448 · Medium · ≈62 s · Upper Undergraduate

448. Let k be a field. What is $\dim_k$ of the quotient ring k[x, y] / (x^3, y^7)?

A. 20
B. 10
C. 21
D. 7

Q0449 · Medium · ≈101 s · Upper Undergraduate

449. Let k be a field. What is $\dim_k$ of the quotient ring k[x, y] / (x^7, y^3)?

A. 7
B. 21
C. 20
D. 10

Q0450 · Medium · ≈56 s · Upper Undergraduate

450. Let k be a field. What is $\dim_k$ of the quotient ring k[x, y] / (x^5, y^4)?

A. 5
B. 9
C. 20
D. 19

Q0451 · Medium · ≈73 s · Upper Undergraduate

451. Let k be a field. What is $\dim_k$ of the quotient ring k[x, y] / (x^7, y^4)?

A. 28
B. 7
C. 27
D. 11

Q0452 · Medium · ≈99 s · Upper Undergraduate

452. Let k be a field. What is $\dim_k$ of the quotient ring k[x, y] / (x^7, y^2)?

A. 13
B. 9
C. 7
D. 14

Q0453 · Medium · ≈100 s · Upper Undergraduate

453. Let k be a field. What is $\dim_k$ of the quotient ring k[x, y] / (x^7, y^5)?

A. 12
B. 7
C. 35
D. 34

Q0454 · Medium · ≈50 s · Upper Undergraduate

454. Let k be a field. What is $\dim_k$ of the quotient ring k[x, y] / (x^2, y^7)?

A. 7
B. 9
C. 13
D. 14

Q0455 · Medium · ≈109 s · Upper Undergraduate

455. Let k be a field. What is $\dim_k$ of the quotient ring k[x, y] / (x^5, y^2)?

A. 5
B. 9
C. 7
D. 10

Q0456 · Hard · ≈216 s · Advanced Undergraduate / Early Graduate

456. How many nilpotent elements does the ring Z/98Z contain?

A. 56
B. 7
C. 14
D. 98

Q0457 · Hard · ≈162 s · Advanced Undergraduate / Early Graduate

457. How many nilpotent elements does the ring Z/28Z contain?

A. 2
B. 14
C. 16
D. 28

Q0458 · Hard · ≈158 s · Advanced Undergraduate / Early Graduate

458. How many nilpotent elements does the ring Z/112Z contain?

A. 8
B. 64
C. 14
D. 112

Q0459 · Hard · ≈174 s · Advanced Undergraduate / Early Graduate

459. How many nilpotent elements does the ring Z/44Z contain?

A. 22
B. 44
C. 24
D. 2

Q0460 · Hard · ≈209 s · Advanced Undergraduate / Early Graduate

460. How many nilpotent elements does the ring Z/72Z contain?

A. 72
B. 12
C. 48
D. 6

Q0461 · Hard · ≈132 s · Advanced Undergraduate / Early Graduate

461. How many nilpotent elements does the ring Z/100Z contain?

A. 60
B. 100
C. 1
D. 10

Q0462 · Hard · ≈171 s · Advanced Undergraduate / Early Graduate

462. How many nilpotent elements does the ring Z/75Z contain?

A. 35
B. 75
C. 5
D. 15

D. Geometry and Topology

Euclidean geometry

Q0463 · Easy · ≈36 s · Introductory Undergraduate

463. An arc of a circle subtends a central angle of 110°. What angle does it subtend at a point on the remaining part of the circle?

A. 220°
B. 110°
C. 125°
D. 55°

Q0464 · Easy · ≈41 s · Introductory Undergraduate

464. An arc of a circle subtends a central angle of 70°. What angle does it subtend at a point on the remaining part of the circle?

A. 35°
B. 145°
C. 70°
D. 140°

Q0465 · Easy · ≈24 s · Introductory Undergraduate

465. An arc of a circle subtends a central angle of 60°. What angle does it subtend at a point on the remaining part of the circle?

A. 150°
B. 120°
C. 30°
D. 60°

Q0466 · Easy · ≈37 s · Introductory Undergraduate

466. An arc of a circle subtends a central angle of 100°. What angle does it subtend at a point on the remaining part of the circle?

A. 100°
B. 50°
C. 200°
D. 130°

Q0467 · Easy · ≈45 s · Introductory Undergraduate

467. An arc of a circle subtends a central angle of 130°. What angle does it subtend at a point on the remaining part of the circle?

A. 115°
B. 260°
C. 130°
D. 65°

Q0468 · Easy · ≈22 s · Introductory Undergraduate

468. An arc of a circle subtends a central angle of 150°. What angle does it subtend at a point on the remaining part of the circle?

A. 75°
B. 105°
C. 300°
D. 150°

Q0469 · Easy · ≈36 s · Introductory Undergraduate

469. An arc of a circle subtends a central angle of 90°. What angle does it subtend at a point on the remaining part of the circle?

A. 45°
B. 135°
C. 90°
D. 180°

Q0470 · Medium · ≈97 s · Undergraduate

470. Chords AB and CD of a circle meet at an interior point P. If PA = 9, PB = 2 and PC = 3, find PD.

A. 2/3
B. 8
C. 6
D. 18

Q0471 · Medium · ≈81 s · Undergraduate

471. Chords AB and CD of a circle meet at an interior point P. If PA = 8, PB = 10 and PC = 4, find PD.

A. 14
B. 5
C. 80
D. 20

Q0472 · Medium · ≈57 s · Undergraduate

472. Chords AB and CD of a circle meet at an interior point P. If PA = 9, PB = 11 and PC = 3, find PD.

A. 17
B. 99
C. 33
D. 11/3

Q0473 · Medium · ≈73 s · Undergraduate

473. Chords AB and CD of a circle meet at an interior point P. If PA = 10, PB = 4 and PC = 8, find PD.

A. 40
B. 5
C. 16/5
D. 6

Q0474 · Medium · ≈110 s · Undergraduate

474. Chords AB and CD of a circle meet at an interior point P. If PA = 8, PB = 11 and PC = 4, find PD.

A. 15
B. 11/2
C. 22
D. 88

Q0475 · Medium · ≈58 s · Undergraduate

475. Chords AB and CD of a circle meet at an interior point P. If PA = 12, PB = 11 and PC = 4, find PD.

A. 132
B. 19
C. 11/3
D. 33

Q0476 · Medium · ≈99 s · Undergraduate

476. Chords AB and CD of a circle meet at an interior point P. If PA = 2, PB = 6 and PC = 4, find PD.

A. 3
B. 12
C. 2
D. 4

Q0477 · Medium · ≈99 s · Undergraduate

477. Chords AB and CD of a circle meet at an interior point P. If PA = 6, PB = 6 and PC = 4, find PD.

A. 36
B. 9
C. 8
D. 4

Q0478 · Hard · ≈169 s · Upper Undergraduate

478. A triangle has sides 13, 20, 21. What is the radius of its circumscribed circle?

A. 14/3
B. 65/3
C. 65/6
D. 65/12

Q0479 · Hard · ≈214 s · Upper Undergraduate

479. A triangle has sides 17, 25, 26. What is the radius of its circumscribed circle?

A. 325/48
B. 325/24
C. 325/12
D. 6

Q0480 · Hard · ≈224 s · Upper Undergraduate

480. A triangle has sides 11, 13, 20. What is the radius of its circumscribed circle?

A. 3
B. 65/12
C. 65/3
D. 65/6

Q0481 · Hard · ≈263 s · Upper Undergraduate

481. A triangle has sides 5, 5, 6. What is the radius of its circumscribed circle?

A. 25/8
B. 25/16
C. 3/2
D. 25/4

Q0482 · Hard · ≈173 s · Upper Undergraduate

482. A triangle has sides 13, 13, 24. What is the radius of its circumscribed circle?

A. 169/20
B. 169/10
C. 169/5
D. 12/5

Q0483 · Hard · ≈141 s · Upper Undergraduate

483. A triangle has sides 14, 25, 25. What is the radius of its circumscribed circle?

A. 21/4
B. 625/48
C. 625/24
D. 625/96

Q0484 · Hard · ≈207 s · Upper Undergraduate

484. A triangle has sides 4, 13, 15. What is the radius of its circumscribed circle?

A. 3/2
B. 65/8
C. 65/16
D. 65/4

Non-Euclidean geometry

Q0485 · Easy · ≈43 s · Introductory Undergraduate

485. On a sphere of radius 6, two points are separated by a central angle of 60°. What is the length of the shorter great-circle arc between them?

A. π/3
B. 12π
C. 4π
D. 2π

Q0486 · Easy · ≈32 s · Introductory Undergraduate

486. On a sphere of radius 5, two points are separated by a central angle of 120°. What is the length of the shorter great-circle arc between them?

A. 20π/3
B. 2π/3
C. 10π/3
D. 50π/3

Q0487 · Easy · ≈40 s · Introductory Undergraduate

487. On a sphere of radius 7, two points are separated by a central angle of 120°. What is the length of the shorter great-circle arc between them?

A. 98π/3
B. 28π/3
C. 14π/3
D. 2π/3

Q0488 · Easy · ≈22 s · Introductory Undergraduate

488. On a sphere of radius 6, two points are separated by a central angle of 90°. What is the length of the shorter great-circle arc between them?

A. 6π
B. 18π
C. π/2
D. 3π

Q0489 · Easy · ≈33 s · Introductory Undergraduate

489. On a sphere of radius 6, two points are separated by a central angle of 135°. What is the length of the shorter great-circle arc between them?

A. 27π
B. 9π/2
C. 3π/4
D. 9π

Q0490 · Easy · ≈36 s · Introductory Undergraduate

490. On a sphere of radius 4, two points are separated by a central angle of 60°. What is the length of the shorter great-circle arc between them?

A. 4π/3
B. π/3
C. 8π/3
D. 16π/3

Q0491 · Easy · ≈20 s · Introductory Undergraduate

491. On a sphere of radius 3, two points are separated by a central angle of 150°. What is the length of the shorter great-circle arc between them?

A. 5π/6
B. 5π
C. 5π/2
D. 15π/2

Q0492 · Medium · ≈104 s · Undergraduate

492. A spherical triangle on a sphere of radius 3 has angles 113°, 150° and 149°. What is its area?

A. 58π/5
B. 58π/15
C. 58π/45
D. 103π/5

Q0493 · Medium · ≈54 s · Undergraduate

493. A spherical triangle on a sphere of radius 1 has angles 107°, 103° and 122°. What is its area?

A. 76π/45
B. 83π/45
C. 38π/45
D. 7π/45

Q0494 · Medium · ≈84 s · Undergraduate

494. A spherical triangle on a sphere of radius 2 has angles 149°, 113° and 61°. What is its area?

A. 143π/90
B. 143π/45
C. 143π/180
D. 323π/45

Q0495 · Medium · ≈110 s · Undergraduate

495. A spherical triangle on a sphere of radius 5 has angles 133°, 145° and 136°. What is its area?

A. 65π/2
B. 115π/2
C. 13π/10
D. 13π/2

Q0496 · Medium · ≈91 s · Undergraduate

496. A spherical triangle on a sphere of radius 5 has angles 122°, 103° and 66°. What is its area?

A. 485π/12
B. 37π/60
C. 185π/12
D. 37π/12

Q0497 · Medium · ≈90 s · Undergraduate

497. A spherical triangle on a sphere of radius 4 has angles 132°, 74° and 100°. What is its area?

A. 14π/5
B. 7π/10
C. 56π/5
D. 136π/5

Q0498 · Medium · ≈110 s · Undergraduate

498. A spherical triangle on a sphere of radius 2 has angles 146°, 78° and 146°. What is its area?

A. 19π/9
B. 38π/9
C. 19π/18
D. 74π/9

Q0499 · Medium · ≈55 s · Undergraduate

499. A spherical triangle on a sphere of radius 1 has angles 66°, 145° and 93°. What is its area?

A. 76π/45
B. 14π/45
C. 62π/45
D. 31π/45

Q0500 · Hard · ≈238 s · Upper Undergraduate

500. A geodesic triangle in the hyperbolic plane of curvature −1/9 has area 81π/20 and two angles 33° and 54°. Find the third angle (in degrees).

A. 12°
B. 22°
C. 93°
D. 2°

Q0501 · Hard · ≈166 s · Upper Undergraduate

501. A geodesic triangle in the hyperbolic plane of curvature −1/9 has area 63π/20 and two angles 43° and 41°. Find the third angle (in degrees).

A. 33°
B. 96°
C. 23°
D. 43°

Q0502 · Hard · ≈232 s · Upper Undergraduate

502. A geodesic triangle in the hyperbolic plane of curvature −1/4 has area 32π/45 and two angles 65° and 34°. Find the third angle (in degrees).

A. 59°
B. 81°
C. 49°
D. 39°

Q0503 · Hard · ≈255 s · Upper Undergraduate

503. A geodesic triangle in the hyperbolic plane of curvature −1/16 has area 292π/45 and two angles 19° and 57°. Find the third angle (in degrees).

A. 31°
B. 41°
C. 104°
D. 21°

Q0504 · Hard · ≈276 s · Upper Undergraduate

504. A geodesic triangle in the hyperbolic plane of curvature −1/4 has area 82π/45 and two angles 13° and 52°. Find the third angle (in degrees).

A. 43°
B. 115°
C. 23°
D. 33°

Q0505 · Hard · ≈265 s · Upper Undergraduate

505. A geodesic triangle in the hyperbolic plane of curvature −1/4 has area 17π/9 and two angles 36° and 41°. Find the third angle (in degrees).

A. 28°
B. 8°
C. 103°
D. 18°

Q0506 · Hard · ≈196 s · Upper Undergraduate

506. A geodesic triangle in the hyperbolic plane of curvature −1/9 has area 67π/20 and two angles 23° and 18°. Find the third angle (in degrees).

A. 82°
B. 72°
C. 139°
D. 62°

Analytic geometry

Q0507 · Easy · ≈45 s · Introductory Undergraduate

507. Find the distance between (-1, -5) and (8, 35).

A. 49
B. 42
C. 7
D. 41

Q0508 · Easy · ≈45 s · Introductory Undergraduate

508. Find the distance between (3, 0) and (19, 30).

A. 35
B. 46
C. 34
D. 20

Q0509 · Easy · ≈38 s · Introductory Undergraduate

509. Find the distance between (-1, 1) and (59, 64).

A. 88
B. 92
C. 87
D. 123

Q0510 · Easy · ≈20 s · Introductory Undergraduate

510. Find the distance between (1, 2) and (22, 74).

A. 76
B. 48
C. 75
D. 93

Q0511 · Easy · ≈29 s · Introductory Undergraduate

511. Find the distance between (-1, -4) and (2, 0).

A. 6
B. 1
C. 5
D. 7

Q0512 · Easy · ≈36 s · Introductory Undergraduate

512. Find the distance between (3, 2) and (13, 26).

A. 27
B. 34
C. 26
D. 14

Q0513 · Easy · ≈30 s · Introductory Undergraduate

513. Find the distance between (-2, -1) and (18, 20).

A. 30
B. 10
C. 41
D. 29

Q0514 · Medium · ≈54 s · Undergraduate

514. Find the distance from the point (-3, -4) to the line $8x + 15y + 8 = 0$.

A. 76/289
B. 76/17
C. 76
D. 76/23

Q0515 · Medium · ≈109 s · Undergraduate

515. Find the distance from the point (0, -5) to the line 3x + 4y + 1 = 0.

A. 19
B. 19/5
C. 19/7
D. 19/25

Q0516 · Medium · ≈73 s · Undergraduate

516. Find the distance from the point (6, -5) to the line 8x + 15y - 7 = 0.

A. 2/17
B. 34/23
C. 2
D. 34

Q0517 · Medium · ≈100 s · Undergraduate

517. Find the distance from the point (-1, 3) to the line 15x + 8y - 1 = 0.

A. 8/17
B. 8
C. 8/23
D. 8/289

Q0518 · Medium · ≈102 s · Undergraduate

518. Find the distance from the point (-3, 6) to the line 12x + 5y + 2 = 0.

A. 4/169
B. 4/17
C. 4/13
D. 4

Q0519 · Medium · ≈78 s · Undergraduate

519. Find the distance from the point (1, -5) to the line 12x + 5y - 8 = 0.

A. 21
B. 21/17
C. 21/169
D. 21/13

Q0520 · Medium · ≈53 s · Undergraduate

520. Find the distance from the point (-4, 1) to the line 15x + 8y + 6 = 0.

A. 46
B. 46/289
C. 46/17
D. 2

Q0521 · Medium · ≈53 s · Undergraduate

521. Find the distance from the point (0, -2) to the line 15x + 8y + 9 = 0.

A. 7/23
B. 7
C. 7/17
D. 7/289

Q0522 · Hard · ≈143 s · Upper Undergraduate

522. The ellipse $x^2/676 + y^2/100 = 1$ has foci F_1 and F_2 and its minor-axis endpoint B = (0, 10). What is the area of triangle F_1F_2B?

A. 624
B. 120
C. 240
D. 260

Q0523 · Hard · ≈162 s · Upper Undergraduate

523. The ellipse $x^2/6724 + y^2/324 = 1$ has foci F_1 and F_2 and its minor-axis endpoint B = (0, 18). What is the area of triangle F_1F_2B?

A. 1440
B. 720
C. 6560
D. 1476

Q0524 · Hard · ≈137 s · Upper Undergraduate

524. The ellipse $x^2/100 + y^2/64 = 1$ has foci F_1 and F_2 and its minor-axis endpoint B = (0, 8). What is the area of triangle F_1F_2B?

A. 48
B. 24
C. 60
D. 80

Q0525 · Hard · ≈249 s · Upper Undergraduate

525. The ellipse $x^2/289 + y^2/64 = 1$ has foci F_1 and F_2 and its minor-axis endpoint B = (0, 8). What is the area of triangle F_1F_2B?

A. 60
B. 136
C. 120
D. 255

Q0526 · Hard · ≈260 s · Upper Undergraduate

526. The ellipse $x^2/676 + y^2/576 = 1$ has foci F_1 and F_2 and its minor-axis endpoint B = (0, 24). What is the area of triangle F_1F_2B?

A. 240
B. 260
C. 120
D. 624

Q0527 · Hard · ≈246 s · Upper Undergraduate

527. The ellipse $x^2/625 + y^2/49 = 1$ has foci F_1 and F_2 and its minor-axis endpoint B = (0, 7). What is the area of triangle F_1F_2B?

A. 84
B. 600
C. 168
D. 175

Q0528 · Hard · ≈225 s · Upper Undergraduate

528. The ellipse $x^2/3364 + y^2/1600 = 1$ has foci F_1 and F_2 and its minor-axis endpoint B = (0, 40). What is the area of triangle F_1F_2B?

A. 2436
B. 1680
C. 2320
D. 840

Differential geometry

Q0529 · Easy · ≈33 s · Introductory Undergraduate

529. What is the curvature of a circle of radius 4?

A. 1/16
B. 1/4
C. 4
D. 8

Q0530 · Easy · ≈39 s · Introductory Undergraduate

530. What is the curvature of a circle of radius 6?

A. 1/36
B. 6
C. 1/6
D. 12

Q0531 · Easy · ≈27 s · Introductory Undergraduate

531. What is the Gaussian curvature of a sphere of radius 5?

A. 1/5
B. 2/5
C. 25
D. 1/25

Q0532 · Easy · ≈32 s · Introductory Undergraduate

532. What is the curvature of a circle of radius 2?

A. 4
B. 1/2
C. 1/4
D. 2

Q0533 · Easy · ≈28 s · Introductory Undergraduate

533. What is the Gaussian curvature of a sphere of radius 9?

A. 81
B. 1/9
C. 1/81
D. 2/9

Q0534 · Easy · ≈22 s · Introductory Undergraduate

534. What is the curvature of a circle of radius 3?

A. 3
B. 1/3
C. 6
D. 1/9

Q0535 · Easy · ≈44 s · Introductory Undergraduate

535. What is the Gaussian curvature of a sphere of radius 6?

A. 36
B. 1/6
C. 1/36
D. 1/3

Q0536 · Medium · ≈93 s · Upper Undergraduate

536. Find the curvature of the circular helix $\gamma(t) = (3\cos t, 3\sin t, 1t)$.

A. 1/3
B. 1/10
C. 3/4
D. 3/10

Q0537 · Medium · ≈57 s · Upper Undergraduate

537. Find the curvature of the circular helix $\gamma(t) = (2\cos t, 2\sin t, 3t)$.

A. 2/5
B. 2/13
C. 1/2
D. 3/13

Q0538 · Medium · ≈74 s · Upper Undergraduate

538. Find the curvature of the circular helix γ(t) = (4 cos t, 4 sin t, 6t).

A. 1/4
B. 3/26
C. 2/5
D. 1/13

Q0539 · Medium · ≈57 s · Upper Undergraduate

539. Find the curvature of the circular helix γ(t) = (6 cos t, 6 sin t, 1t).

A. 1/37
B. 6/37
C. 6/7
D. 1/6

Q0540 · Medium · ≈80 s · Upper Undergraduate

540. Find the curvature of the circular helix γ(t) = (1 cos t, 1 sin t, 5t).

A. 1
B. 5/26
C. 1/6
D. 1/26

Q0541 · Medium · ≈53 s · Upper Undergraduate

541. Find the curvature of the circular helix γ(t) = (5 cos t, 5 sin t, 1t).

A. 5/6
B. 5/26
C. 1/26
D. 1/5

Q0542 · Medium · ≈76 s · Upper Undergraduate

542. Find the curvature of the circular helix γ(t) = (2 cos t, 2 sin t, 5t).

A. 5/29
B. 1/2
C. 2/7
D. 2/29

Q0543 · Medium · ≈60 s · Upper Undergraduate

543. Find the curvature of the circular helix γ(t) = (3 cos t, 3 sin t, 6t).

A. 1/5
B. 2/15
C. 1/15
D. 1/3

Q0544 · Hard · ≈245 s · Advanced Undergraduate / Early Graduate

544. A closed orientable surface of genus 6 carries a Riemannian metric with Gaussian curvature K. What is ∬K dA?

A. 20π
B. -20π
C. -10π
D. -8π

Q0545 · Hard · ≈174 s · Advanced Undergraduate / Early Graduate

545. Find the curvature of the parabola $y = 3x^2$ at the point where x = 2/9.

A. 54/25
B. 162/125
C. 287/125
D. 324/125

Q0546 · Hard · ≈219 s · Advanced Undergraduate / Early Graduate

546. Find the curvature of the parabola $y = 1x^2$ at the point where $x = 2/3$.

A. 54/125
B. 179/125
C. 108/125
D. 18/25

Q0547 · Hard · ≈139 s · Advanced Undergraduate / Early Graduate

547. A closed orientable surface of genus 3 carries a Riemannian metric with Gaussian curvature K. What is $\iint K\, dA$?

A. 8π
B. -2π
C. -4π
D. -8π

Q0548 · Hard · ≈182 s · Advanced Undergraduate / Early Graduate

548. Find the curvature of the parabola $y = 2x^2$ at the point where $x = 15/32$.

A. 6961/4913
B. 256/289
C. 2048/4913
D. 4096/4913

Q0549 · Hard · ≈149 s · Advanced Undergraduate / Early Graduate

549. Find the curvature of the parabola $y = 2x^2$ at the point where $x = 3/16$.

A. 256/125
B. 64/25
C. 381/125
D. 512/125

Q0550 · Hard · ≈260 s · Advanced Undergraduate / Early Graduate

550. Find the curvature of the parabola $y = 2x^2$ at the point where $x = 3/5$.

A. 100/169
B. 500/2197
C. 2697/2197
D. 1000/2197

Algebraic geometry

Q0551 · Easy · ≈33 s · Introductory Undergraduate

551. Two projective plane curves of degrees 2 and 4 have no common component. How many intersection points do they have over the complex numbers, counted with multiplicity?

A. 7
B. 8
C. 4
D. 6

Q0552 · Easy · ≈20 s · Introductory Undergraduate

552. Two projective plane curves of degrees 2 and 6 have no common component. How many intersection points do they have over the complex numbers, counted with multiplicity?

A. 6
B. 12
C. 11
D. 8

Q0553 · Easy · ≈27 s · Introductory Undergraduate

553. Two projective plane curves of degrees 4 and 6 have no common component. How many intersection points do they have over the complex numbers, counted with multiplicity?

A. 10
B. 24
C. 6
D. 23

Q0554 · Easy · ≈25 s · Introductory Undergraduate

554. Two projective plane curves of degrees 7 and 2 have no common component. How many intersection points do they have over the complex numbers, counted with multiplicity?

A. 13
B. 14
C. 7
D. 9

Q0555 · Easy · ≈31 s · Introductory Undergraduate

555. Two projective plane curves of degrees 4 and 4 have no common component. How many intersection points do they have over the complex numbers, counted with multiplicity?

A. 16
B. 8
C. 4
D. 15

Q0556 · Easy · ≈36 s · Introductory Undergraduate

556. Two projective plane curves of degrees 2 and 7 have no common component. How many intersection points do they have over the complex numbers, counted with multiplicity?

A. 14
B. 9
C. 13
D. 7

Q0557 · Easy · ≈44 s · Introductory Undergraduate

557. Two projective plane curves of degrees 6 and 4 have no common component. How many intersection points do they have over the complex numbers, counted with multiplicity?

A. 23
B. 10
C. 6
D. 24

Q0558 · Medium · ≈95 s · Upper Undergraduate

558. What is the genus of a smooth projective plane curve of degree 3?

A. 2
B. 0
C. 1
D. 3

Q0559 · Medium · ≈85 s · Upper Undergraduate

559. What is the genus of a smooth projective plane curve of degree 8?

A. 42
B. 15
C. 21
D. 28

Q0560 · Medium · ≈77 s · Upper Undergraduate

560. What is the genus of a smooth projective plane curve of degree 4?

A. 1
B. 6
C. 3
D. 4

Q0561 · Medium · ≈87 s · Upper Undergraduate

561. What is the genus of a smooth projective plane curve of degree 5?

A. 3
B. 10
C. 12
D. 6

Q0562 · Medium · ≈70 s · Upper Undergraduate

562. What is the genus of a smooth projective plane curve of degree 6?

A. 15
B. 6
C. 10
D. 20

Q0563 · Medium · ≈57 s · Upper Undergraduate

563. What is the genus of a smooth projective plane curve of degree 9?

A. 21
B. 36
C. 28
D. 56

Q0564 · Medium · ≈54 s · Upper Undergraduate

564. What is the genus of a smooth projective plane curve of degree 10?

A. 36
B. 28
C. 72
D. 45

Q0565 · Medium · ≈70 s · Upper Undergraduate

565. What is the genus of a smooth projective plane curve of degree 7?

A. 21
B. 10
C. 15
D. 30

Q0566 · Hard · ≈133 s · Advanced Undergraduate / Early Graduate

566. How many points does the elliptic curve $y^2 = x^3 + 4x + 4$ have over F_5, including the point at infinity?

A. 7
B. 6
C. 8
D. 9

Q0567 · Hard · ≈173 s · Advanced Undergraduate / Early Graduate

567. How many points does the elliptic curve $y^2 = x^3 + 5x + 10$ have over F_{11}, including the point at infinity?

A. 14
B. 11
C. 12
D. 13

Q0568 · Hard · ≈189 s · Advanced Undergraduate / Early Graduate

568. How many points does the elliptic curve $y^2 = x^3 + 0x + 5$ have over F_7, including the point at infinity?

A. 7
B. 9
C. 8
D. 6

Q0569 · Hard · ≈157 s · Advanced Undergraduate / Early Graduate

569. How many points does the elliptic curve $y^2 = x^3 + 5x + 7$ have over F_{11}, including the point at infinity?

A. 15
B. 16
C. 12
D. 17

Q0570 · Hard · ≈137 s · Advanced Undergraduate / Early Graduate

570. How many points does the elliptic curve $y^2 = x^3 + 6x + 5$ have over F_{11}, including the point at infinity?

A. 12
B. 16
C. 17
D. 18

Q0571 · Hard · ≈253 s · Advanced Undergraduate / Early Graduate

571. How many points does the elliptic curve $y^2 = x^3 + 5x + 0$ have over F_7, including the point at infinity?

A. 10
B. 7
C. 8
D. 9

Q0572 · Hard · ≈166 s · Advanced Undergraduate / Early Graduate

572. How many points does the elliptic curve $y^2 = x^3 + 4x + 1$ have over F_5, including the point at infinity?

A. 9
B. 8
C. 7
D. 6

General topology

Q0573 · Easy · ≈28 s · Introductory Undergraduate

573. Let $X = \{1, \ldots, 6\}$ with open sets $\varnothing$ and $\{1, \ldots, i\}$ for $i = 1, \ldots, 6$. What is the closure of $A = \{1, 5\}$?

A. $\{1, 2, 3, 4, 5, 6\}$
B. $\{1, 5\}$
C. $\{1, 2, 3, 4, 5\}$
D. $\{1\}$

Q0574 · Easy · ≈24 s · Introductory Undergraduate

574. Let $X = \{1, \ldots, 9\}$ with open sets $\varnothing$ and $\{1, \ldots, i\}$ for $i = 1, \ldots, 9$. What is the closure of $A = \{2, 7\}$?

A. $\{2, 3, 4, 5, 6, 7\}$
B. $\{2, 7\}$
C. $\{2, 3, 4, 5, 6, 7, 8, 9\}$
D. $\{1, 2, 3, 4, 5, 6, 7\}$

Q0575 · Easy · ≈23 s · Introductory Undergraduate

575. Let X = {1, …, 9} with open sets ∅ and {1, …, i} for i = 1, …, 9. What is the closure of A = {3, 4, 9}?

A. {1, 2, 3}
B. {3, 4, 9}
C. {3, 4, 5, 6, 7, 8, 9}
D. {1, 2, 3, 4, 5, 6, 7, 8, 9}

Q0576 · Easy · ≈24 s · Introductory Undergraduate

576. Let X = {1, …, 9} with open sets ∅ and {1, …, i} for i = 1, …, 9. What is the closure of A = {3, 6}?

A. {3, 6}
B. {3, 4, 5, 6, 7, 8, 9}
C. {3, 4, 5, 6}
D. {1, 2, 3, 4, 5, 6}

Q0577 · Easy · ≈22 s · Introductory Undergraduate

577. Let X = {1, …, 7} with open sets ∅ and {1, …, i} for i = 1, …, 7. What is the closure of A = {4, 5, 6}?

A. {1, 2, 3, 4, 5, 6}
B. {4, 5, 6, 7}
C. {4, 5, 6}
D. {1, 2, 3, 4}

Q0578 · Easy · ≈30 s · Introductory Undergraduate

578. Let X = {1, …, 7} with open sets ∅ and {1, …, i} for i = 1, …, 7. What is the closure of A = {3, 4, 7}?

A. {3, 4, 5, 6, 7}
B. {1, 2, 3, 4, 5, 6, 7}
C. {3, 4, 7}
D. {1, 2, 3}

Q0579 · Easy · ≈24 s · Introductory Undergraduate

579. Let X = {1, …, 7} with open sets ∅ and {1, …, i} for i = 1, …, 7. What is the closure of A = {2, 6}?

A. {2, 6}
B. {2, 3, 4, 5, 6, 7}
C. {2, 3, 4, 5, 6}
D. {1, 2, 3, 4, 5, 6}

Q0580 · Medium · ≈76 s · Upper Undergraduate

580. A set X with 15 points is partitioned into 6 non-empty blocks of sizes 4, 3, 4, 1, 1, 2. The open sets are exactly the unions of blocks (including ∅). How many open sets are there?

A. 64
B. 63
C. 6
D. 32768

Q0581 · Medium · ≈60 s · Upper Undergraduate

581. A set X with 7 points is partitioned into 3 non-empty blocks of sizes 2, 4, 1. The open sets are exactly the unions of blocks (including ∅). How many open sets are there?

A. 3
B. 7
C. 8
D. 128

Q0582 · Medium · ≈85 s · Upper Undergraduate

582. A set X with 8 points is partitioned into 3 non-empty blocks of sizes 2, 4, 2. The open sets are exactly the unions of blocks (including ∅). How many open sets are there?

A. 256
B. 3
C. 7
D. 8

Q0583 · Medium · ≈56 s · Upper Undergraduate

583. A set X with 16 points is partitioned into 6 non-empty blocks of sizes 4, 2, 1, 4, 1, 4. The open sets are exactly the unions of blocks (including ∅). How many open sets are there?

A. 65536
B. 6
C. 64
D. 63

Q0584 · Medium · ≈70 s · Upper Undergraduate

584. A set X with 11 points is partitioned into 5 non-empty blocks of sizes 1, 3, 4, 1, 2. The open sets are exactly the unions of blocks (including ∅). How many open sets are there?

A. 31
B. 2048
C. 32
D. 5

Q0585 · Medium · ≈101 s · Upper Undergraduate

585. A set X with 14 points is partitioned into 6 non-empty blocks of sizes 1, 4, 3, 2, 2, 2. The open sets are exactly the unions of blocks (including ∅). How many open sets are there?

A. 6
B. 16384
C. 64
D. 63

Q0586 · Medium · ≈75 s · Upper Undergraduate

586. A set X with 13 points is partitioned into 5 non-empty blocks of sizes 3, 2, 4, 3, 1. The open sets are exactly the unions of blocks (including ∅). How many open sets are there?

A. 31
B. 32
C. 8192
D. 5

Q0587 · Medium · ≈56 s · Upper Undergraduate

587. A set X with 8 points is partitioned into 3 non-empty blocks of sizes 3, 4, 1. The open sets are exactly the unions of blocks (including ∅). How many open sets are there?

A. 7
B. 256
C. 8
D. 3

Q0588 · Hard · ≈219 s · Advanced Undergraduate / Early Graduate

588. On X = {1, …, 4}, let τ be the smallest topology containing the subsets {1,3,4}, {1,2,3}, {1}. How many open sets does τ have (including ∅ and X)?

A. 8
B. 6
C. 5
D. 7

Q0589 · Hard · ≈149 s · Advanced Undergraduate / Early Graduate

589. Let X = {1, …, 3} with open sets ∅ and {1, …, i} for i = 1, …, 3. How many continuous maps f: X → X are there?

A. 10
B. 27
C. 4
D. 6

Q0590 · Hard · ≈224 s · Advanced Undergraduate / Early Graduate

590. On X = {1, …, 5}, let τ be the smallest topology containing the subsets {1,2,4}, {2,3,5}, {1,2,3,5}. How many open sets does τ have (including ∅ and X)?

A. 8
B. 6
C. 7
D. 9

Q0591 · Hard · ≈265 s · Advanced Undergraduate / Early Graduate

591. On X = {1, …, 5}, let τ be the smallest topology containing the subsets {2,4,5}, {1,3}, {1,4,5}. How many open sets does τ have (including ∅ and X)?

A. 9
B. 8
C. 7
D. 10

Q0592 · Hard · ≈133 s · Advanced Undergraduate / Early Graduate

592. Let X = {1, …, 4} with open sets ∅ and {1, …, i} for i = 1, …, 4. How many continuous maps f: X → X are there?

A. 35
B. 24
C. 256
D. 5

Q0593 · Hard · ≈190 s · Advanced Undergraduate / Early Graduate

593. On X = {1, …, 6}, let τ be the smallest topology containing the subsets {3,6}, {1,3,5}, {4,6}. How many open sets does τ have (including ∅ and X)?

A. 10
B. 11
C. 9
D. 8

Q0594 · Hard · ≈241 s · Advanced Undergraduate / Early Graduate

594. Let X = {1, …, 5} with open sets ∅ and {1, …, i} for i = 1, …, 5. How many continuous maps f: X → X are there?

A. 3125
B. 126
C. 120
D. 6

Algebraic topology

Q0595 · Easy · ≈25 s · Introductory Undergraduate

595. What is the Euler characteristic of the closed orientable surface of genus 0?

A. 1
B. -2
C. 3
D. 2

Q0596 · Easy · ≈23 s · Introductory Undergraduate

596. What is the Euler characteristic of the closed orientable surface of genus 1?

A. 0
B. -1
C. 4
D. 1

Q0597 · Easy · ≈43 s · Introductory Undergraduate

597. What is the Euler characteristic of the closed orientable surface of genus 3?

A. -1
B. 4
C. -2
D. -4

Q0598 · Easy · ≈21 s · Introductory Undergraduate

598. What is the Euler characteristic of the closed orientable surface of genus 6?

A. -10
B. 10
C. -4
D. -5

Q0599 · Easy · ≈39 s · Introductory Undergraduate

599. What is the Euler characteristic of the closed orientable surface of genus 5?

A. -3
B. -4
C. 8
D. -8

Q0600 · Easy · ≈38 s · Introductory Undergraduate

600. What is the Euler characteristic of the closed orientable surface of genus 4?

A. -3
B. -6
C. -2
D. 6

Q0601 · Easy · ≈41 s · Introductory Undergraduate

601. What is the Euler characteristic of the closed orientable surface of genus 7?

A. -5
B. -12
C. -6
D. 12

Q0602 · Medium · ≈80 s · Upper Undergraduate

602. A connected graph has 12 vertices and 19 edges. What is the rank of its first homology group $H_1(\Gamma;\mathbb{Z})$?

A. 19
B. 8
C. 7
D. 9

Q0603 · Medium · ≈68 s · Upper Undergraduate

603. A connected graph has 7 vertices and 10 edges. What is the rank of its first homology group $H_1(\Gamma;\mathbb{Z})$?

A. 10
B. 5
C. 4
D. 3

Q0604 · Medium · ≈64 s · Upper Undergraduate

604. A connected graph has 9 vertices and 10 edges. What is the rank of its first homology group $H_1(\Gamma;\mathbb{Z})$?

A. 3
B. 2
C. 10
D. 1

Q0605 · Medium · ≈79 s · Upper Undergraduate

605. A connected graph has 5 vertices and 6 edges. What is the rank of its first homology group $H_1(\Gamma;\mathbb{Z})$?

A. 2
B. 3
C. 1
D. 6

Q0606 · Medium · ≈89 s · Upper Undergraduate

606. A connected graph has 7 vertices and 14 edges. What is the rank of its first homology group $H_1(\Gamma;\mathbb{Z})$?

A. 9
B. 14
C. 7
D. 8

Q0607 · Medium · ≈98 s · Upper Undergraduate

607. A connected graph has 7 vertices and 15 edges. What is the rank of its first homology group $H_1(\Gamma;\mathbb{Z})$?

A. 15
B. 9
C. 10
D. 8

Q0608 · Medium · ≈91 s · Upper Undergraduate

608. A connected graph has 12 vertices and 16 edges. What is the rank of its first homology group $H_1(\Gamma;\mathbb{Z})$?

A. 5
B. 4
C. 16
D. 6

Q0609 · Medium · ≈55 s · Upper Undergraduate

609. A connected graph has 5 vertices and 13 edges. What is the rank of its first homology group $H_1(\Gamma;\mathbb{Z})$?

A. 8
B. 10
C. 9
D. 13

Q0610 · Hard · ≈229 s · Advanced Undergraduate / Early Graduate

610. A connected 2-sheeted covering space of the closed orientable surface of genus 5 is itself a closed orientable surface. What is its genus?

A. 9
B. 10
C. 8
D. 6

Q0611 · Hard · ≈196 s · Advanced Undergraduate / Early Graduate

611. What is the Euler characteristic of $\Sigma_4 \times S^2$, where Σ_4 is the closed orientable surface of genus 4?

A. -6
B. -12
C. -4
D. 6

Q0612 · Hard · ≈223 s · Advanced Undergraduate / Early Graduate

612. What is the Euler characteristic of $\Sigma_6 \times S^4$, where Σ_6 is the closed orientable surface of genus 6?

A. -10
B. -40
C. -20
D. -8

Q0613 · Hard · ≈233 s · Advanced Undergraduate / Early Graduate

613. What is the Euler characteristic of $\Sigma_2 \times S^2$, where Σ_2 is the closed orientable surface of genus 2?

A. 0
B. 2
C. -2
D. -4

Q0614 · Hard · ≈280 s · Advanced Undergraduate / Early Graduate

614. A connected 4-sheeted covering space of the closed orientable surface of genus 2 is itself a closed orientable surface. What is its genus?

A. 8
B. 3
C. 5
D. 4

Q0615 · Hard · ≈274 s · Advanced Undergraduate / Early Graduate

615. What is the Euler characteristic of $\Sigma_2 \times S^6$, where Σ_2 is the closed orientable surface of genus 2?

A. 0
B. -12
C. -2
D. -4

Q0616 · Hard · ≈270 s · Advanced Undergraduate / Early Graduate

616. A connected 3-sheeted covering space of the closed orientable surface of genus 5 is itself a closed orientable surface. What is its genus?

A. 7
B. 12
C. 15
D. 13

E. Discrete Mathematics and Logic

Combinatorics

Q0617 · Easy · ≈30 s · Introductory Undergraduate

617. In how many ways can a committee of 3 be chosen from 6 people?

A. 120
B. 18
C. 20
D. 26

Q0618 · Easy · ≈44 s · Introductory Undergraduate

618. In how many ways can a committee of 4 be chosen from 10 people?

A. 220
B. 210
C. 5040
D. 40

Q0619 · Easy · ≈29 s · Introductory Undergraduate

619. In how many ways can a committee of 2 be chosen from 7 people?

A. 14
B. 42
C. 21
D. 28

Q0620 · Easy · ≈35 s · Introductory Undergraduate

620. In how many ways can a committee of 5 be chosen from 12 people?

A. 95040
B. 792
C. 804
D. 60

Q0621 · Easy · ≈41 s · Introductory Undergraduate

621. In how many ways can a committee of 2 be chosen from 9 people?

A. 36
B. 45
C. 18
D. 72

Q0622 · Easy · ≈31 s · Introductory Undergraduate

622. In how many ways can a committee of 3 be chosen from 14 people?

A. 42
B. 2184
C. 364
D. 378

Q0623 · Easy · ≈40 s · Introductory Undergraduate

623. In how many ways can a committee of 4 be chosen from 11 people?

A. 341
B. 7920
C. 44
D. 330

Q0624 · Medium · ≈85 s · Undergraduate

624. How many distinct arrangements are there of the letters of the multiset AAABCCD?

A. 840
B. 5040
C. 421
D. 420

Q0625 · Medium · ≈108 s · Undergraduate

625. How many distinct arrangements are there of the letters of the multiset ABBBBCCCDD?

A. 3628800
B. 151200
C. 12601
D. 12600

Q0626 · Medium · ≈87 s · Undergraduate

626. How many distinct arrangements are there of the letters of the multiset AAABCCCD?

A. 1121
B. 1120
C. 6720
D. 40320

Q0627 · Medium · ≈90 s · Undergraduate

627. How many distinct arrangements are there of the letters of the multiset AAABC?

A. 21
B. 20
C. 12
D. 120

Q0628 · Medium · ≈71 s · Undergraduate

628. How many distinct arrangements are there of the letters of the multiset AABBBBCCDD?

A. 3628800
B. 18901
C. 18900
D. 151200

Q0629 · Medium · ≈54 s · Undergraduate

629. How many distinct arrangements are there of the letters of the multiset AAABCCCC?

A. 1680
B. 280
C. 40320
D. 281

Q0630 · Medium · ≈74 s · Undergraduate

630. How many distinct arrangements are there of the letters of the multiset AAABBCC?

A. 211
B. 210
C. 840
D. 5040

Q0631 · Medium · ≈77 s · Undergraduate

631. How many distinct arrangements are there of the letters of the multiset AAABBCCDDDD?

A. 1663200
B. 69300
C. 69301
D. 39916800

Q0632 · Hard · ≈221 s · Upper Undergraduate

632. How many surjective functions are there from a 6-element set onto a 3-element set?

A. 729
B. 120
C. 10
D. 540

Q0633 · Hard · ≈274 s · Upper Undergraduate

633. How many surjective functions are there from a 4-element set onto a 4-element set?

A. 48
B. 24
C. 256
D. 1

Q0634 · Hard · ≈159 s · Upper Undergraduate

634. How many surjective functions are there from a 4-element set onto a 2-element set?

A. 16
B. 14
C. 12
D. 3

Q0635 · Hard · ≈266 s · Upper Undergraduate

635. How many surjective functions are there from a 6-element set onto a 4-element set?

A. 1560
B. 360
C. 10
D. 4096

Q0636 · Hard · ≈149 s · Upper Undergraduate

636. How many surjective functions are there from a 7-element set onto a 2-element set?

A. 42
B. 6
C. 126
D. 128

Q0637 · Hard · ≈158 s · Upper Undergraduate

637. How many surjective functions are there from a 4-element set onto a 3-element set?

A. 81
B. 3
C. 24
D. 36

Q0638 · Hard · ≈133 s · Upper Undergraduate

638. How many surjective functions are there from a 5-element set onto a 4-element set?

A. 240
B. 1024
C. 120
D. 4

Graph theory

Q0639 · Easy · ≈43 s · Introductory Undergraduate

639. How many edges does the complete graph K_{10} have?

A. 10
B. 45
C. 90
D. 55

Q0640 · Easy · ≈31 s · Introductory Undergraduate

640. How many edges does the complete graph K_{12} have?

A. 132
B. 12
C. 66
D. 78

Q0641 · Easy · ≈25 s · Introductory Undergraduate

641. How many edges does the complete graph K_{14} have?

A. 105
B. 91
C. 182
D. 14

Q0642 · Easy · ≈23 s · Introductory Undergraduate

642. How many edges does the complete graph K_{13} have?

A. 91
B. 78
C. 13
D. 156

Q0643 · Easy · ≈20 s · Introductory Undergraduate

643. How many edges does the complete graph K_{11} have?

A. 55
B. 11
C. 66
D. 110

Q0644 · Easy · ≈20 s · Introductory Undergraduate

644. How many edges does the complete graph K_{8} have?

A. 8
B. 36
C. 56
D. 28

Q0645 · Easy · ≈21 s · Introductory Undergraduate

645. How many edges does the complete graph K_{9} have?

A. 9
B. 36
C. 72
D. 45

Q0646 · Medium · ≈59 s · Undergraduate

646. A connected planar graph drawn in the plane has 7 vertices and 9 edges. How many faces (including the outer face) does it have?

A. 2
B. 14
C. 4
D. 3

Q0647 · Medium · ≈50 s · Undergraduate

647. A connected planar graph drawn in the plane has 7 vertices and 14 edges. How many faces (including the outer face) does it have?

A. 19
B. 7
C. 8
D. 9

Q0648 · Medium · ≈66 s · Undergraduate

648. How many spanning trees does the complete graph K_{5} have?

A. 625
B. 125
C. 10
D. 24

Q0649 · Medium · ≈92 s · Undergraduate

649. How many spanning trees does the complete graph K_{7} have?

A. 16807
B. 21
C. 117649
D. 720

Q0650 · Medium · ≈108 s · Undergraduate

650. A connected planar graph drawn in the plane has 15 vertices and 17 edges. How many faces (including the outer face) does it have?

A. 30
B. 4
C. 2
D. 3

Q0651 · Medium · ≈57 s · Undergraduate

651. A connected planar graph drawn in the plane has 11 vertices and 18 edges. How many faces (including the outer face) does it have?

A. 7
B. 8
C. 27
D. 9

Q0652 · Medium · ≈85 s · Undergraduate

652. How many spanning trees does the complete graph K_8 have?

A. 5040
B. 28
C. 262144
D. 2097152

Q0653 · Medium · ≈85 s · Undergraduate

653. A connected planar graph drawn in the plane has 9 vertices and 13 edges. How many faces (including the outer face) does it have?

A. 20
B. 5
C. 6
D. 4

Q0654 · Hard · ≈168 s · Upper Undergraduate

654. In how many ways can the vertices of the cycle C_4 be properly coloured using 4 available colours?

A. 78
B. 81
C. 84
D. 108

Q0655 · Hard · ≈179 s · Upper Undergraduate

655. In how many ways can the vertices of the cycle C_8 be properly coloured using 4 available colours?

A. 8748
B. 6558
C. 6564
D. 6561

Q0656 · Hard · ≈270 s · Upper Undergraduate

656. In how many ways can the vertices of the cycle C_5 be properly coloured using 4 available colours?

A. 1012
B. 324
C. 240
D. 243

Q0657 · Hard · ≈199 s · Upper Undergraduate

657. In how many ways can the vertices of the cycle C_6 be properly coloured using 3 available colours?

A. 66
B. 64
C. 96
D. 62

Q0658 · Hard · ≈251 s · Upper Undergraduate

658. In how many ways can the vertices of the cycle C_6 be properly coloured using 4 available colours?

A. 729
B. 972
C. 732
D. 726

Q0659 · Hard · ≈153 s · Upper Undergraduate

659. In how many ways can the vertices of the cycle C_4 be properly coloured using 3 available colours?

A. 16
B. 14
C. 18
D. 24

Q0660 · Hard · ≈164 s · Upper Undergraduate

660. In how many ways can the vertices of the cycle C_7 be properly coloured using 3 available colours?

A. 126
B. 128
C. 2181
D. 192

Set theory

Q0661 · Easy · ≈40 s · Introductory Undergraduate

661. If $|A| = 22$, $|B| = 38$ and $|A \cap B| = 8$, what is $|A \cup B|$?

A. 60
B. 68
C. 44
D. 52

Q0662 · Easy · ≈24 s · Introductory Undergraduate

662. If $|A| = 12$, $|B| = 27$ and $|A \cap B| = 6$, what is $|A \cup B|$?

A. 33
B. 27
C. 39
D. 45

Q0663 · Easy · ≈22 s · Introductory Undergraduate

663. If $|A| = 20$, $|B| = 32$ and $|A \cap B| = 7$, what is $|A \cup B|$?

A. 59
B. 38
C. 45
D. 52

Q0664 · Easy · ≈44 s · Introductory Undergraduate

664. If |A| = 37, |B| = 11 and |A ∩ B| = 9, what is |A ∪ B|?

A. 39
B. 48
C. 30
D. 57

Q0665 · Easy · ≈25 s · Introductory Undergraduate

665. If |A| = 13, |B| = 12 and |A ∩ B| = 2, what is |A ∪ B|?

A. 25
B. 23
C. 21
D. 27

Q0666 · Easy · ≈22 s · Introductory Undergraduate

666. If |A| = 13, |B| = 12 and |A ∩ B| = 9, what is |A ∪ B|?

A. 34
B. 7
C. 16
D. 25

Q0667 · Easy · ≈37 s · Introductory Undergraduate

667. If |A| = 27, |B| = 34 and |A ∩ B| = 5, what is |A ∪ B|?

A. 66
B. 51
C. 56
D. 61

Q0668 · Medium · ≈66 s · Undergraduate

668. How many different equivalence relations exist on a set with 9 elements?

A. 21147
B. 512
C. 4140
D. 362880

Q0669 · Medium · ≈81 s · Undergraduate

669. How many different equivalence relations exist on a set with 7 elements?

A. 877
B. 5040
C. 128
D. 203

Q0670 · Medium · ≈53 s · Undergraduate

670. How many different equivalence relations exist on a set with 4 elements?

A. 5
B. 15
C. 24
D. 16

Q0671 · Medium · ≈51 s · Undergraduate

671. How many different equivalence relations exist on a set with 10 elements?

A. 1024
B. 3628800
C. 115975
D. 21147

Q0672 · Medium · ≈95 s · Undergraduate

672. How many different equivalence relations exist on a set with 6 elements?

A. 720
B. 203
C. 52
D. 64

Q0673 · Medium · ≈100 s · Undergraduate

673. How many different equivalence relations exist on a set with 3 elements?

A. 6
B. 2
C. 5
D. 8

Q0674 · Medium · ≈93 s · Undergraduate

674. How many different equivalence relations exist on a set with 11 elements?

A. 678570
B. 115975
C. 39916800
D. 2048

Q0675 · Medium · ≈86 s · Undergraduate

675. How many different equivalence relations exist on a set with 5 elements?

A. 32
B. 120
C. 52
D. 15

Q0676 · Hard · ≈168 s · Upper Undergraduate

676. How many binary relations on an 6-element set are symmetric?

A. 32768
B. 2097152
C. 68719476736
D. 1073741824

Q0677 · Hard · ≈210 s · Upper Undergraduate

677. How many binary relations on an 4-element set are symmetric?

A. 4096
B. 64
C. 65536
D. 1024

Q0678 · Hard · ≈234 s · Upper Undergraduate

678. How many binary relations on an 4-element set are both reflexive and symmetric?

A. 65536
B. 4096
C. 1024
D. 64

Q0679 · Hard · ≈257 s · Upper Undergraduate

679. How many binary relations on an 3-element set are antisymmetric?

A. 64
B. 8
C. 216
D. 512

Q0680 · Hard · ≈202 s · Upper Undergraduate

680. How many binary relations on an 5-element set are antisymmetric?

A. 1889568
B. 33554432
C. 1024
D. 32768

Q0681 · Hard · ≈230 s · Upper Undergraduate

681. How many binary relations on an 6-element set are antisymmetric?

A. 918330048
B. 32768
C. 68719476736
D. 2097152

Q0682 · Hard · ≈185 s · Upper Undergraduate

682. How many binary relations on an 3-element set are symmetric?

A. 8
B. 64
C. 512
D. 27

Mathematical logic

Q0683 · Easy · ≈22 s · Introductory Undergraduate

683. For how many of the 8 truth assignments to (p, q, r) is the formula (p → q) ∧ (q → r) true?

A. 4
B. 3
C. 6
D. 5

Q0684 · Easy · ≈36 s · Introductory Undergraduate

684. For how many of the 8 truth assignments to (p, q, r) is the formula p ∧ (q ∨ r) true?

A. 2
B. 4
C. 3
D. 5

Q0685 · Easy · ≈38 s · Introductory Undergraduate

685. For how many of the 8 truth assignments to (p, q, r) is the formula (p ⊕ q) ∧ r true?

A. 6
B. 1
C. 2
D. 3

Q0686 · Easy · ≈40 s · Introductory Undergraduate

686. For how many of the 8 truth assignments to (p, q, r) is the formula ¬(p ∨ q) ∨ r true?

A. 3
B. 5
C. 6
D. 4

Q0687 · Easy · ≈20 s · Introductory Undergraduate

687. For how many of the 8 truth assignments to (p, q, r) is the formula (p ∧ ¬q) → r true?

A. 1
B. 6
C. 8
D. 7

Q0688 · Easy · ≈41 s · Introductory Undergraduate

688. For how many of the 8 truth assignments to (p, q, r) is the formula (p ∨ q) → r true?

A. 3
B. 4
C. 5
D. 6

Q0689 · Easy · ≈23 s · Introductory Undergraduate

689. For how many of the 8 truth assignments to (p, q, r) is the formula p ↔ (q ∧ r) true?

A. 6
B. 5
C. 4
D. 3

Q0690 · Medium · ≈68 s · Undergraduate

690. Which of the following formulas is logically equivalent to (p ∨ q) ∧ ¬(p ∧ q)?

A. p ∨ q
B. ¬p ∧ ¬q
C. p → q
D. ¬(p ↔ q)

Q0691 · Medium · ≈91 s · Undergraduate

691. Which of the following formulas is logically equivalent to ¬p ∨ q?

A. p ∧ ¬q
B. ¬q → ¬p
C. q → p
D. ¬(p ↔ q)

Q0692 · Medium · ≈96 s · Undergraduate

692. Which of the following formulas is logically equivalent to ¬(p ∨ q)?

A. ¬p ∧ ¬q
B. p ∨ q
C. ¬(p ↔ q)
D. ¬p ∨ ¬q

Q0693 · Medium · ≈51 s · Undergraduate

693. Which of the following formulas is logically equivalent to (p → q) ∧ (q → p)?

A. q
B. p
C. p ↔ q
D. p ∧ q

Q0694 · Medium · ≈64 s · Undergraduate

694. Which of the following formulas is logically equivalent to ¬p ∧ ¬q?

A. ¬p
B. ¬(p ∨ q)
C. ¬(p ↔ q)
D. p ∨ q

Q0695 · Medium · ≈64 s · Undergraduate

695. Which of the following formulas is logically equivalent to ¬(p ↔ q)?

A. ¬p ∧ ¬q
B. ¬p ∨ ¬q
C. p ∨ q
D. (p ∨ q) ∧ ¬(p ∧ q)

Q0696 · Medium · ≈90 s · Undergraduate

696. Which of the following formulas is logically equivalent to ¬(p ∧ ¬q)?

A. p ∧ q
B. ¬(p ↔ q)
C. ¬q → ¬p
D. ¬p ∨ ¬q

Q0697 · Medium · ≈87 s · Undergraduate

697. Which of the following formulas is logically equivalent to ¬(p → q)?

A. p ∧ q
B. p
C. p ∧ ¬q
D. q

Q0698 · Hard · ≈148 s · Upper Undergraduate

698. Work in $\mathbb{Z}_{11} = \{0, 1, \ldots, 10\}$ with arithmetic mod 11 and let c = 0. Exactly one of the four listed statements is true. Which one?

A. $\forall x \neq 0\ \exists y\ (x \cdot y \equiv 1)$
B. $\exists x \notin \{0, 1\}\ (x^2 \equiv x)$
C. $\forall x\ \exists y\ (y^2 \equiv x)$
D. $\exists x \neq 0\ \exists y \neq 0\ (x \cdot y \equiv 0)$

Q0699 · Hard · ≈135 s · Upper Undergraduate

699. Work in $\mathbb{Z}_{18} = \{0, 1, \ldots, 17\}$ with arithmetic mod 18 and let c = 4. Exactly one of the four listed statements is true. Which one?

A. $\exists x \notin \{0, 1\}\ (x^2 \equiv x)$
B. $\forall x \neq 0\ \exists y\ (x \cdot y \equiv 1)$
C. $\forall x\ \exists y\ (y^2 \equiv x)$
D. $\forall x \neq 0\ \forall y\ \forall y'\ (x \cdot y \equiv x \cdot y' \Longrightarrow y = y')$

Q0700 · Hard · ≈272 s · Upper Undergraduate

700. Work in $\mathbb{Z}_{22} = \{0, 1, \ldots, 21\}$ with arithmetic mod 22 and let c = 4. Exactly one of the four listed statements is true. Which one?

A. $\forall x \neq 0\ \forall y\ \forall y'\ (x \cdot y \equiv x \cdot y' \Longrightarrow y = y')$
B. $\forall x\ \exists y\ (y^2 \equiv x)$
C. $\forall x \neq 0\ \exists y\ (x \cdot y \equiv 1)$
D. $\exists x \neq 0\ \exists y \neq 0\ (x \cdot y \equiv 0)$

Q0701 · Hard · ≈251 s · Upper Undergraduate

701. Work in $\mathbb{Z}_{37} = \{0, 1, \ldots, 36\}$ with arithmetic mod 37 and let c = 21. Exactly one of the four listed statements is true. Which one?

A. $\forall x\ \exists y\ (y^2 \equiv x)$
B. $\exists x\ (2x \equiv c)$
C. $\forall x\ \exists y\ (y^3 \equiv x)$
D. $\exists x\ (x^3 \equiv c)$

Q0702 · Hard · ≈231 s · Upper Undergraduate

702. Work in $\mathbb{Z}_{20} = \{0, 1, \ldots, 19\}$ with arithmetic mod 20 and let c = 3. Exactly one of the four listed statements is true. Which one?

A. $\exists x\ (x^2 \equiv c)$
B. $\forall x\ \exists y\ (y^3 \equiv x)$
C. $\forall x\ \exists y\ (y^2 \equiv x)$
D. $\exists x\ (x^3 \equiv c)$

Q0703 · Hard · ≈277 s · Upper Undergraduate

703. Work in $\mathbb{Z}_{29} = \{0, 1, \ldots, 28\}$ with arithmetic mod 29 and let c = 4. Exactly one of the four listed statements is true. Which one?

A. $\exists x\ (x^2 \equiv c)$
B. $\exists x \neq 0\ \exists y \neq 0\ (x \cdot y \equiv 0)$
C. $\forall x\ \exists y\ (y^2 \equiv x)$
D. $\exists x \notin \{0, 1\}\ (x^2 \equiv x)$

Q0704 · Hard · ≈141 s · Upper Undergraduate

704. Work in $\mathbb{Z}_{34} = \{0, 1, \ldots, 33\}$ with arithmetic mod 34 and let c = 9. Exactly one of the four listed statements is true. Which one?

A. $\forall x \neq 0\ \exists y\ (x \cdot y \equiv 1)$
B. $\forall x\ \exists y\ (y^2 \equiv x)$
C. $\forall x\ \exists y\ (y^3 \equiv x)$
D. $\forall x \neq 0\ \forall y\ \forall y'\ (x \cdot y \equiv x \cdot y' \Longrightarrow y = y')$

Cryptography

Q0705 · Easy · ≈27 s · Introductory Undergraduate

705. A Caesar cipher shifted every letter forward by 5 positions to produce the ciphertext HNUMJW. What was the plaintext?

A. CIPHER
B. BHOGDQ
C. DJQIFS
D. AGNFCP

Q0706 · Easy · ≈35 s · Introductory Undergraduate

706. A Caesar cipher shifted every letter forward by 3 positions to produce the ciphertext ORJLF. What was the plaintext?

A. LOGIC
B. MPHJD
C. KNFHB
D. JMEGA

Q0707 · Easy · ≈32 s · Introductory Undergraduate

707. A Caesar cipher shifted every letter forward by 9 positions to produce the ciphertext CARP. What was the plaintext?

A. RPGE
B. USJH
C. TRIG
D. SQHF

Q0708 · Easy · ≈31 s · Introductory Undergraduate

708. A Caesar cipher shifted every letter forward by 9 positions to produce the ciphertext YARVN. What was the plaintext?

A. PRIME
B. NPGKC
C. OQHLD
D. QSJNF

Q0709 · Easy · ≈27 s · Introductory Undergraduate

709. A Caesar cipher shifted every letter forward by 6 positions to produce the ciphertext MXGVN. What was the plaintext?

A. EPYNF
B. GRAPH
C. HSBQI
D. FQZOG

Q0710 · Easy · ≈35 s · Introductory Undergraduate

710. A Caesar cipher shifted every letter forward by 6 positions to produce the ciphertext TASHKX. What was the plaintext?

A. NUMBER
B. OVNCFS
C. MTLADQ
D. LSKZCP

Q0711 · Easy · ≈43 s · Introductory Undergraduate

711. A Caesar cipher shifted every letter forward by 8 positions to produce the ciphertext VCUJMZ. What was the plaintext?

A. OVNCFS
B. NUMBER
C. LSKZCP
D. MTLADQ

Q0712 · Medium · ≈53 s · Undergraduate

712. In an RSA setup, p = 13, q = 17 and the public exponent is e = 13. What is the private exponent d (smallest positive)?

A. 13
B. 134
C. 133
D. 59

Q0713 · Medium · ≈99 s · Undergraduate

713. In an RSA setup, p = 23, q = 13 and the public exponent is e = 5. What is the private exponent d (smallest positive)?

A. 60
B. 53
C. 5
D. 211

Q0714 · Medium · ≈50 s · Undergraduate

714. In an RSA setup, p = 19, q = 5 and the public exponent is e = 5. What is the private exponent d (smallest positive)?

A. 5
B. 29
C. 43
D. 30

Q0715 · Medium · ≈106 s · Undergraduate

715. In an RSA setup, p = 11, q = 23 and the public exponent is e = 13. What is the private exponent d (smallest positive)?

A. 39
B. 203
C. 13
D. 17

Q0716 · Medium · ≈103 s · Undergraduate

716. In an RSA setup, p = 19, q = 5 and the public exponent is e = 7. What is the private exponent d (smallest positive)?

A. 31
B. 68
C. 7
D. 41

Q0717 · Medium · ≈97 s · Undergraduate

717. In an RSA setup, p = 13, q = 11 and the public exponent is e = 17. What is the private exponent d (smallest positive)?

A. 101
B. 7
C. 113
D. 17

Q0718 · Medium · ≈96 s · Undergraduate

718. In an RSA setup, p = 11, q = 5 and the public exponent is e = 9. What is the private exponent d (smallest positive)?

A. 49
B. 31
C. 9
D. 10

Q0719 · Medium · ≈72 s · Undergraduate

719. In an RSA setup, p = 5, q = 23 and the public exponent is e = 3. What is the private exponent d (smallest positive)?

A. 3
B. 59
C. 29
D. 77

Q0720 · Hard · ≈213 s · Upper Undergraduate

720. In Diffie–Hellman with p = 41 and g = 2, Alice's public value is A = 21 and Bob's secret exponent is b = 8. What is the shared secret key?

A. 5
B. 10
C. 37
D. 31

Q0721 · Hard · ≈261 s · Upper Undergraduate

721. In Diffie–Hellman with p = 37 and g = 3, Alice's public value is A = 9 and Bob's secret exponent is b = 9. What is the shared secret key?

A. 1
B. 36
C. 28
D. 7

Q0722 · Hard · ≈134 s · Upper Undergraduate

722. In Diffie–Hellman with p = 31 and g = 3, Alice's public value is A = 27 and Bob's secret exponent is b = 4. What is the shared secret key?

A. 19
B. 17
C. 16
D. 8

Q0723 · Hard · ≈223 s · Upper Undergraduate

723. In Diffie–Hellman with p = 31 and g = 5, Alice's public value is A = 5 and Bob's secret exponent is b = 23. What is the shared secret key?

A. 1
B. 5
C. 26
D. 25

Q0724 · Hard · ≈228 s · Upper Undergraduate

724. In Diffie–Hellman with p = 37 and g = 3, Alice's public value is A = 12 and Bob's secret exponent is b = 32. What is the shared secret key?

A. 33
B. 7
C. 8
D. 16

Q0725 · Hard · ≈197 s · Upper Undergraduate

725. In Diffie–Hellman with p = 41 and g = 3, Alice's public value is A = 40 and Bob's secret exponent is b = 23. What is the shared secret key?

A. 40
B. 1
C. 14
D. 27

Q0726 · Hard · ≈188 s · Upper Undergraduate

726. In Diffie–Hellman with p = 41 and g = 2, Alice's public value is A = 23 and Bob's secret exponent is b = 23. What is the shared secret key?

A. 37
B. 20
C. 8
D. 31

Discrete mathematics

Q0727 · Easy · ≈44 s · Introductory Undergraduate

727. Evaluate $1^2 + 2^2 + 3^2 + \ldots + 9^2$.

A. 45
B. 2025
C. 570
D. 285

Q0728 · Easy · ≈35 s · Introductory Undergraduate

728. Evaluate $1^2 + 2^2 + 3^2 + \ldots + 10^2$.

A. 55
B. 770
C. 385
D. 3025

Q0729 · Easy · ≈25 s · Introductory Undergraduate

729. Evaluate $1^2 + 2^2 + 3^2 + \ldots + 12^2$.

A. 1300
B. 78
C. 650
D. 6084

Q0730 · Easy · ≈21 s · Introductory Undergraduate

730. Evaluate $1^2 + 2^2 + 3^2 + \ldots + 26^2$.

A. 123201
B. 351
C. 6201
D. 12402

Q0731 · Easy · ≈38 s · Introductory Undergraduate

731. Evaluate $1^2 + 2^2 + 3^2 + \ldots + 23^2$.

A. 8648
B. 276
C. 4324
D. 76176

Q0732 · Easy · ≈30 s · Introductory Undergraduate

732. Evaluate $1^2 + 2^2 + 3^2 + \ldots + 14^2$.

A. 105
B. 11025
C. 2030
D. 1015

Q0733 · Easy · ≈34 s · Introductory Undergraduate

733. Evaluate $1^2 + 2^2 + 3^2 + \ldots + 22^2$.

A. 64009
B. 253
C. 7590
D. 3795

Q0734 · Medium · ≈76 s · Undergraduate

734. A sequence satisfies $a_n = 3a_{n-1} + 2a_{n-2}$ with $a_0 = 3$, $a_1 = 1$. Find a_9.

A. 60039
B. 16857
C. 55304
D. 60037

Q0735 · Medium · ≈73 s · Undergraduate

735. A sequence satisfies $a_n = 1a_{n-1} + 2a_{n-2}$ with $a_0 = 0$, $a_1 = 2$. Find a_6.

A. 42
B. 22
C. 32
D. 44

Q0736 · Medium · ≈62 s · Undergraduate

736. A sequence satisfies $a_n = 2a_{n-1} + 1a_{n-2}$ with $a_0 = 0$, $a_1 = 1$. Find a_7.

A. 169
B. 70
C. 140
D. 170

Q0737 · Medium · ≈78 s · Undergraduate

737. A sequence satisfies $a_n = 2a_{n-1} + 2a_{n-2}$ with $a_0 = 3$, $a_1 = 2$. Find a_7.

A. 1376
B. 504
C. 1378
D. 1192

Q0738 · Medium · ≈95 s · Undergraduate

738. A sequence satisfies $a_n = 1a_{n-1} + 1a_{n-2}$ with $a_0 = 0$, $a_1 = 1$. Find a_7.

A. 8
B. 13
C. 14
D. 12

Q0739 · Medium · ≈52 s · Undergraduate

739. A sequence satisfies $a_n = 1a_{n-1} + 1a_{n-2}$ with $a_0 = 0$, $a_1 = 3$. Find a_8.

A. 62
B. 39
C. 64
D. 63

Q0740 · Medium · ≈53 s · Undergraduate

740. A sequence satisfies $a_n = 1a_{n-1} + 3a_{n-2}$ with $a_0 = 2$, $a_1 = 2$. Find a_6.

A. 156
B. 118
C. 194
D. 80

Q0741 · Medium · ≈53 s · Undergraduate

741. A sequence satisfies $a_n = 2a_{n-1} + 3a_{n-2}$ with $a_0 = 3$, $a_1 = 3$. Find a_6.

A. 363
B. 1095
C. 849
D. 972

Q0742 · Hard · ≈280 s · Upper Undergraduate

742. How many binary strings of length 10 contain no three consecutive 1s?

A. 520
B. 1014
C. 423
D. 504

Q0743 · Hard · ≈162 s · Upper Undergraduate

743. How many binary strings of length 11 contain no three consecutive 1s?

A. 1121
B. 778
C. 927
D. 2037

Q0744 · Hard · ≈191 s · Upper Undergraduate

744. How many binary strings of length 9 contain no three consecutive 1s?

A. 230
B. 274
C. 238
D. 503

Q0745 · Hard · ≈182 s · Upper Undergraduate

745. How many binary strings of length 13 contain no three consecutive 1s?

A. 2632
B. 3136
C. 8179
D. 5056

Q0746 · Hard · ≈228 s · Upper Undergraduate

746. How many binary strings of length 12 contain no three consecutive 1s?

A. 1431
B. 1705
C. 4084
D. 2391

Q0747 · Hard · ≈147 s · Upper Undergraduate

747. How many binary strings of length 14 contain no three consecutive 1s?

A. 5768
B. 4841
C. 16370
D. 10616

Q0748 · Hard · ≈203 s · Upper Undergraduate

748. How many binary strings of length 8 contain no three consecutive 1s?

A. 107
B. 125
C. 149
D. 248

F. Probability and Statistics

Probability theory

Q0749 · Easy · ≈40 s · Introductory Undergraduate

749. Two fair six-sided dice are rolled. What is the probability that the sum equals 9?

A. 1/3
B. 1/11
C. 1/12
D. 1/9

Q0750 · Easy · ≈42 s · Introductory Undergraduate

750. Two fair six-sided dice are rolled. What is the probability that the sum equals 3?

A. 1/12
B. 1/6
C. 1/18
D. 1/11

Q0751 · Easy · ≈27 s · Introductory Undergraduate

751. Two fair six-sided dice are rolled. What is the probability that the sum equals 4?

A. 1/4
B. 1/6
C. 1/11
D. 1/12

Q0752 · Easy · ≈30 s · Introductory Undergraduate

752. Two fair six-sided dice are rolled. What is the probability that the sum equals 6?

A. 1/11
B. 1/12
C. 5/36
D. 5/12

Q0753 · Easy · ≈35 s · Introductory Undergraduate

753. Two fair six-sided dice are rolled. What is the probability that the sum equals 10?

A. 1/11
B. 1/4
C. 1/12
D. 1/6

Q0754 · Easy · ≈36 s · Introductory Undergraduate

754. Two fair six-sided dice are rolled. What is the probability that the sum equals 8?

A. 5/36
B. 1/11
C. 5/12
D. 1/12

Q0755 · Easy · ≈37 s · Introductory Undergraduate

755. Two fair six-sided dice are rolled. What is the probability that the sum equals 5?

A. 1/11
B. 1/9
C. 1/3
D. 1/12

Q0756 · Medium · ≈58 s · Undergraduate

756. An urn has 3 red and 6 blue balls. Two balls are drawn without replacement. What is the probability that exactly one is red?

A. 4/9
B. 1/12
C. 1/2
D. 1/4

Q0757 · Medium · ≈56 s · Undergraduate

757. An urn has 8 red and 3 blue balls. Two balls are drawn without replacement. What is the probability that exactly one is red?

A. 28/55
B. 12/55
C. 48/121
D. 24/55

Q0758 · Medium · ≈56 s · Undergraduate

758. An urn has 8 red and 6 blue balls. Two balls are drawn without replacement. What is the probability that exactly one is red?

A. 24/91
B. 4/13
C. 24/49
D. 48/91

Q0759 · Medium · ≈102 s · Undergraduate

759. An urn has 3 red and 7 blue balls. Two balls are drawn without replacement. What is the probability that exactly one is red?

A. 1/15
B. 7/30
C. 7/15
D. 21/50

Q0760 · Medium · ≈104 s · Undergraduate

760. An urn has 4 red and 8 blue balls. Two balls are drawn without replacement. What is the probability that exactly one is red?

A. 1/11
B. 4/9
C. 16/33
D. 8/33

Q0761 · Medium · ≈52 s · Undergraduate

761. An urn has 6 red and 8 blue balls. Two balls are drawn without replacement. What is the probability that exactly one is red?

A. 24/91
B. 24/49
C. 15/91
D. 48/91

Q0762 · Medium · ≈66 s · Undergraduate

762. An urn has 8 red and 4 blue balls. Two balls are drawn without replacement. What is the probability that exactly one is red?

A. 14/33
B. 8/33
C. 4/9
D. 16/33

Q0763 · Hard · ≈237 s · Upper Undergraduate

763. 4 fair 8-sided dice (faces 1 to 8) are rolled. What is the expected value of the largest face shown?

A. 9/2
B. 32/5
C. 18
D. 7023/1024

Q0764 · Hard · ≈168 s · Upper Undergraduate

764. 2 fair 8-sided dice (faces 1 to 8) are rolled. What is the expected value of the largest face shown?

A. 9
B. 9/2
C. 16/3
D. 93/16

Q0765 · Hard · ≈169 s · Upper Undergraduate

765. 4 fair 4-sided dice (faces 1 to 4) are rolled. What is the expected value of the largest face shown?

A. 5/2
B. 463/128
C. 16/5
D. 10

Q0766 · Hard · ≈172 s · Upper Undergraduate

766. 4 fair 6-sided dice (faces 1 to 6) are rolled. What is the expected value of the largest face shown?

A. 6797/1296
B. 14
C. 24/5
D. 7/2

Q0767 · Hard · ≈130 s · Upper Undergraduate

767. 3 fair 4-sided dice (faces 1 to 4) are rolled. What is the expected value of the largest face shown?

A. 15/2
B. 5/2
C. 55/16
D. 3

Q0768 · Hard · ≈130 s · Upper Undergraduate

768. 3 fair 8-sided dice (faces 1 to 8) are rolled. What is the expected value of the largest face shown?

A. 207/32
B. 6
C. 9/2
D. 27/2

Q0769 · Hard · ≈254 s · Upper Undergraduate

769. 3 fair 6-sided dice (faces 1 to 6) are rolled. What is the expected value of the largest face shown?

A. 119/24
B. 9/2
C. 7/2
D. 21/2

Statistics

Q0770 · Easy · ≈42 s · Introductory Undergraduate

770. For the data set 10, 34, 16, 29, 18, what is the mean minus the median?

A. 35/4
B. 17/5
C. 107/5
D. -17/5

Q0771 · Easy · ≈29 s · Introductory Undergraduate

771. For the data set 10, 34, 31, 30, 7, what is the mean minus the median?

A. 38/5
B. -38/5
C. 112/5
D. -2

Q0772 · Easy · ≈35 s · Introductory Undergraduate

772. For the data set 37, 12, 7, 3, 5, what is the mean minus the median?

A. -29/5
B. 64/5
C. 9
D. 29/5

Q0773 · Easy · ≈37 s · Introductory Undergraduate

773. For the data set 18, 24, 3, 15, 7, what is the mean minus the median?

A. 67/5
B. -8/5
C. 8/5
D. 7/4

Q0774 · Easy · ≈41 s · Introductory Undergraduate

774. For the data set 23, 32, 35, 22, 29, what is the mean minus the median?

A. 4/5
B. -4/5
C. 25/4
D. 141/5

Q0775 · Easy · ≈41 s · Introductory Undergraduate

775. For the data set 35, 7, 18, 6, 14, what is the mean minus the median?

A. 6
B. 16
C. -2
D. 2

Q0776 · Easy · ≈31 s · Introductory Undergraduate

776. For the data set 24, 27, 19, 9, 34, what is the mean minus the median?

A. 17/4
B. 113/5
C. -7/5
D. 7/5

Q0777 · Medium · ≈76 s · Undergraduate

777. Find the sample variance (with denominator $n-1$) of the data 12, 8, 14, 15, 14.

A. 26/5
B. 39/5
C. 156/5
D. 156/25

Q0778 · Medium · ≈85 s · Undergraduate

778. Find the sample variance (with denominator $n-1$) of the data 2, 8, 9, 15, 5, 8.

A. 569/6
B. 569/30
C. 569/42
D. 569/36

Q0779 · Medium · ≈56 s · Undergraduate

779. Find the sample variance (with denominator $n-1$) of the data 2, 15, 9, 6, 14.

A. 297/10
B. 594/5
C. 99/5
D. 594/25

Q0780 · Medium · ≈60 s · Undergraduate

780. Find the sample variance (with denominator $n-1$) of the data 6, 10, 7, 10, 7, 12.

A. 82/3
B. 41/9
C. 82/15
D. 82/21

Q0781 · Medium · ≈62 s · Undergraduate

781. Find the sample variance (with denominator $n-1$) of the data 10, 10, 6, 2, 2.

A. 16
B. 64/5
C. 64
D. 32/3

Q0782 · Medium · ≈67 s · Undergraduate

782. Find the sample variance (with denominator $n-1$) of the data 12, 10, 4, 5, 13, 8, 14.

A. 321/28
B. 107/7
C. 642/49
D. 642/7

Q0783 · Medium · ≈79 s · Undergraduate

783. Find the sample variance (with denominator n − 1) of the data 5, 9, 15, 11, 7, 4, 5.

A. 47/4
B. 94/7
C. 94
D. 47/3

Q0784 · Hard · ≈155 s · Upper Undergraduate

784. The least-squares line y = a + bx is fitted to the points (1, 12), (7, 13), (5, 10), (3, 12), (11, 5), (6, 5). Find the slope b.

A. 79/119
B. -119/79
C. -79/119
D. -79/131

Q0785 · Hard · ≈151 s · Upper Undergraduate

785. The least-squares line y = a + bx is fitted to the points (10, 6), (5, 14), (9, 14), (6, 12), (2, 4). Find the slope b.

A. -40/103
B. 2/11
C. 40/103
D. 103/40

Q0786 · Hard · ≈267 s · Upper Undergraduate

786. The least-squares line y = a + bx is fitted to the points (6, 12), (1, 12), (9, 5), (11, 12). Find the slope b.

A. -3/7
B. 63/227
C. -63/227
D. -227/63

Q0787 · Hard · ≈198 s · Upper Undergraduate

787. The least-squares line y = a + bx is fitted to the points (1, 11), (3, 4), (2, 7), (9, 12), (6, 13). Find the slope b.

A. 64/143
B. 64/107
C. 107/64
D. -64/107

Q0788 · Hard · ≈181 s · Upper Undergraduate

788. The least-squares line y = a + bx is fitted to the points (9, 6), (1, 9), (7, 13), (8, 6), (4, 10). Find the slope b.

A. -33/107
B. -11/29
C. 33/107
D. -107/33

Q0789 · Hard · ≈155 s · Upper Undergraduate

789. The least-squares line y = a + bx is fitted to the points (8, 3), (11, 6), (10, 15), (9, 1), (1, 15), (4, 9). Find the slope b.

A. 307/449
B. -307/449
C. -449/307
D. -307/1061

Q0790 · Hard · ≈271 s · Upper Undergraduate

790. The least-squares line y = a + bx is fitted to the points (1, 1), (9, 11), (3, 8), (6, 9). Find the slope b.

A. 161/227
B. 23/21
C. 21/23
D. -23/21

Stochastic processes

Q0791 · Easy · ≈23 s · Introductory Undergraduate

791. Arrivals follow a Poisson process with rate 2 per hour. What is the expected number of arrivals in 4 hours?

A. 32
B. 8
C. 6
D. 1/2

Q0792 · Easy · ≈25 s · Introductory Undergraduate

792. Arrivals follow a Poisson process with rate 3 per hour. What is the expected number of arrivals in 4 hours?

A. 3/4
B. 48
C. 12
D. 7

Q0793 · Easy · ≈28 s · Introductory Undergraduate

793. Arrivals follow a Poisson process with rate 3 per hour. What is the expected number of arrivals in 2 hours?

A. 5
B. 12
C. 6
D. 3/2

Q0794 · Easy · ≈45 s · Introductory Undergraduate

794. Arrivals follow a Poisson process with rate 5 per hour. What is the expected number of arrivals in 3 hours?

A. 8
B. 45
C. 5/3
D. 15

Q0795 · Easy · ≈20 s · Introductory Undergraduate

795. Arrivals follow a Poisson process with rate 4 per hour. What is the expected number of arrivals in 5 hours?

A. 100
B. 9
C. 20
D. 4/5

Q0796 · Easy · ≈27 s · Introductory Undergraduate

796. Arrivals follow a Poisson process with rate 5 per hour. What is the expected number of arrivals in 2 hours?

A. 10
B. 20
C. 5/2
D. 7

Q0797 · Easy · ≈21 s · Introductory Undergraduate

797. Arrivals follow a Poisson process with rate 3 per hour. What is the expected number of arrivals in 3 hours?

A. 1
B. 9
C. 6
D. 27

Q0798 · Medium · ≈59 s · Upper Undergraduate

798. A two-state Markov chain has P(1→2) = 1/5 and P(2→1) = 1/10. What is the stationary probability of state 1?

A. 2/3
B. 1/3
C. 1/2
D. 1/10

Q0799 · Medium · ≈88 s · Upper Undergraduate

799. A two-state Markov chain has P(1→2) = 3/10 and P(2→1) = 2/5. What is the stationary probability of state 1?

A. 2/5
B. 3/7
C. 4/7
D. 1/2

Q0800 · Medium · ≈104 s · Upper Undergraduate

800. A two-state Markov chain has P(1→2) = 1/5 and P(2→1) = 3/10. What is the stationary probability of state 1?

A. 2/5
B. 3/10
C. 1/2
D. 3/5

Q0801 · Medium · ≈81 s · Upper Undergraduate

801. A two-state Markov chain has P(1→2) = 1/10 and P(2→1) = 3/10. What is the stationary probability of state 1?

A. 1/4
B. 3/10
C. 3/4
D. 1/2

Q0802 · Medium · ≈101 s · Upper Undergraduate

802. A two-state Markov chain has P(1→2) = 1/5 and P(2→1) = 2/5. What is the stationary probability of state 1?

A. 1/3
B. 2/3
C. 2/5
D. 1/2

Q0803 · Medium · ≈80 s · Upper Undergraduate

803. A two-state Markov chain has P(1→2) = 1/10 and P(2→1) = 1/5. What is the stationary probability of state 1?

A. 1/2
B. 1/3
C. 2/3
D. 1/5

Q0804 · Medium · ≈76 s · Upper Undergraduate

804. A two-state Markov chain has P(1→2) = 2/5 and P(2→1) = 1/5. What is the stationary probability of state 1?

A. 1/3
B. 1/5
C. 2/3
D. 1/2

Q0805 · Hard · ≈143 s · Advanced Undergraduate / Early Graduate

805. A gambler starts with 2 units and plays rounds winning 1 unit with probability 1/3 and losing 1 unit with probability 2/3, until reaching 8 units or going broke. What is the probability of reaching 8?

A. 84/85
B. 1/85
C. 21/85
D. 1/4

Q0806 · Hard · ≈280 s · Advanced Undergraduate / Early Graduate

806. A gambler starts with 4 units and plays rounds winning 1 unit with probability 1/4 and losing 1 unit with probability 3/4, until reaching 5 units or going broke. What is the probability of reaching 5?

A. 1/121
B. 4/5
C. 81/121
D. 40/121

Q0807 · Hard · ≈208 s · Advanced Undergraduate / Early Graduate

807. A gambler starts with 1 units and plays rounds winning 1 unit with probability 3/4 and losing 1 unit with probability 1/4, until reaching 5 units or going broke. What is the probability of reaching 5?

A. 1/5
B. 81/121
C. 120/121
D. 40/121

Q0808 · Hard · ≈209 s · Advanced Undergraduate / Early Graduate

808. A gambler starts with 3 units and plays rounds winning 1 unit with probability 2/3 and losing 1 unit with probability 1/3, until reaching 8 units or going broke. What is the probability of reaching 8?

A. 224/255
B. 248/255
C. 31/255
D. 3/8

Q0809 · Hard · ≈202 s · Advanced Undergraduate / Early Graduate

809. A gambler starts with 3 units and plays rounds winning 1 unit with probability 1/3 and losing 1 unit with probability 2/3, until reaching 4 units or going broke. What is the probability of reaching 4?

A. 1/15
B. 7/15
C. 8/15
D. 3/4

Q0810 · Hard · ≈135 s · Advanced Undergraduate / Early Graduate

810. A gambler starts with 3 units and plays rounds winning 1 unit with probability 1/4 and losing 1 unit with probability 3/4, until reaching 5 units or going broke. What is the probability of reaching 5?

A. 3/5
B. 108/121
C. 4/121
D. 13/121

Q0811 · Hard · ≈234 s · Advanced Undergraduate / Early Graduate

811. A gambler starts with 3 units and plays rounds winning 1 unit with probability 2/3 and losing 1 unit with probability 1/3, until reaching 4 units or going broke. What is the probability of reaching 4?

A. 3/4
B. 14/15
C. 1/15
D. 8/15

Bayesian statistics

Q0812 · Easy · ≈39 s · Introductory Undergraduate

812. A disease has prevalence 1/10. A test is positive with probability 4/5 for sick people and 1/20 for healthy people. Given a positive test, what is the probability of disease?

A. 16/25
B. 2/25
C. 4/5
D. 19/20

Q0813 · Easy · ≈42 s · Introductory Undergraduate

813. A disease has prevalence 1/10. A test is positive with probability 9/10 for sick people and 1/25 for healthy people. Given a positive test, what is the probability of disease?

A. 5/7
B. 9/100
C. 9/10
D. 24/25

Q0814 · Easy · ≈31 s · Introductory Undergraduate

814. A disease has prevalence 1/20. A test is positive with probability 9/10 for sick people and 1/20 for healthy people. Given a positive test, what is the probability of disease?

A. 18/37
B. 9/200
C. 9/10
D. 19/20

Q0815 · Easy · ≈43 s · Introductory Undergraduate

815. A disease has prevalence 1/10. A test is positive with probability 19/20 for sick people and 1/25 for healthy people. Given a positive test, what is the probability of disease?

A. 24/25
B. 19/200
C. 19/20
D. 95/131

Q0816 · Easy · ≈33 s · Introductory Undergraduate

816. A disease has prevalence 1/100. A test is positive with probability 9/10 for sick people and 1/20 for healthy people. Given a positive test, what is the probability of disease?

A. 9/10
B. 19/20
C. 2/13
D. 9/1000

Q0817 · Easy · ≈20 s · Introductory Undergraduate

817. A disease has prevalence 1/20. A test is positive with probability 4/5 for sick people and 1/10 for healthy people. Given a positive test, what is the probability of disease?

A. 1/25
B. 9/10
C. 8/27
D. 4/5

Q0818 · Easy · ≈42 s · Introductory Undergraduate

818. A disease has prevalence 1/50. A test is positive with probability 19/20 for sick people and 1/20 for healthy people. Given a positive test, what is the probability of disease?

A. 361/400
B. 19/20
C. 19/1000
D. 19/68

Q0819 · Medium · ≈54 s · Undergraduate

819. A coin's heads probability has a Beta(2, 3) prior. After observing 2 heads and 5 tails, what is the posterior mean?

A. 2/5
B. 2/7
C. 1/3
D. 4/13

Q0820 · Medium · ≈103 s · Undergraduate

820. A coin's heads probability has a Beta(2, 5) prior. After observing 9 heads and 12 tails, what is the posterior mean?

A. 11/29
B. 2/7
C. 11/28
D. 3/7

Q0821 · Medium · ≈106 s · Undergraduate

821. A coin's heads probability has a Beta(3, 2) prior. After observing 11 heads and 7 tails, what is the posterior mean?

A. 14/23
B. 7/12
C. 11/18
D. 3/5

Q0822 · Medium · ≈106 s · Undergraduate

822. A coin's heads probability has a Beta(2, 1) prior. After observing 12 heads and 3 tails, what is the posterior mean?

A. 4/5
B. 14/19
C. 7/9
D. 2/3

Q0823 · Medium · ≈78 s · Undergraduate

823. A coin's heads probability has a Beta(2, 2) prior. After observing 10 heads and 2 tails, what is the posterior mean?

A. 3/4
B. 1/2
C. 5/6
D. 12/17

Q0824 · Medium · ≈66 s · Undergraduate

824. A coin's heads probability has a Beta(3, 2) prior. After observing 8 heads and 11 tails, what is the posterior mean?

A. 11/25
B. 11/24
C. 3/5
D. 8/19

Q0825 · Medium · ≈95 s · Undergraduate

825. A coin's heads probability has a Beta(2, 5) prior. After observing 2 heads and 4 tails, what is the posterior mean?

A. 4/13
B. 2/7
C. 3/11
D. 1/3

Q0826 · Hard · ≈215 s · Upper Undergraduate

826. Data are 4 independent observations from N(θ, σ² = 4) with sample mean 1. The prior is θ ~ N(6, 4). What is the posterior mean of θ?

A. 6/5
B. 2
C. 7/2
D. 1

Q0827 · Hard · ≈140 s · Upper Undergraduate

827. Data are 4 independent observations from N(θ, σ² = 16) with sample mean 9. The prior is θ ~ N(3, 4). What is the posterior mean of θ?

A. 9
B. 39/5
C. 6
D. 3/2

Q0828 · Hard · ≈168 s · Upper Undergraduate

828. Data are 9 independent observations from N(θ, σ² = 16) with sample mean 1. The prior is θ ~ N(10, 9). What is the posterior mean of θ?

A. 241/97
B. 11/2
C. 19/10
D. 1

Q0829 · Hard · ≈227 s · Upper Undergraduate

829. Data are 8 independent observations from N(θ, σ² = 1) with sample mean 11. The prior is θ ~ N(1, 4). What is the posterior mean of θ?

A. 6
B. 353/33
C. 11
D. 89/9

Q0830 · Hard · ≈194 s · Upper Undergraduate

830. Data are 4 independent observations from N(θ, σ² = 16) with sample mean 14. The prior is θ ~ N(2, 9). What is the posterior mean of θ?

A. 134/13
B. 14
C. 8
D. 58/5

Q0831 · Hard · ≈249 s · Upper Undergraduate

831. Data are 8 independent observations from N(θ, σ² = 16) with sample mean 10. The prior is θ ~ N(3, 4). What is the posterior mean of θ?

A. 83/9
B. 10
C. 13/2
D. 23/3

Q0832 · Hard · ≈256 s · Upper Undergraduate

832. Data are 5 independent observations from $N(\theta, \sigma^2 = 1)$ with sample mean 3. The prior is $\theta \sim N(7, 9)$. What is the posterior mean of θ?

A. 11/3
B. 71/23
C. 5
D. 3

G. Applied Mathematics

Numerical analysis

Q0833 · Easy · ≈25 s · Introductory Undergraduate

833. Apply one Newton step to $f(x) = x^2 - 10$ starting from $x_0 = 4$. What is x_1?

A. 7
B. -2
C. 13/4
D. 5/2

Q0834 · Easy · ≈44 s · Introductory Undergraduate

834. Apply one Newton step to $f(x) = x^2 - 10$ starting from $x_0 = 2$. What is x_1?

A. 6
B. 5
C. 8
D. 7/2

Q0835 · Easy · ≈36 s · Introductory Undergraduate

835. Apply one Newton step to $f(x) = x^2 - 5$ starting from $x_0 = 1$. What is x_1?

A. 5
B. 0
C. 3
D. 4

Q0836 · Easy · ≈34 s · Introductory Undergraduate

836. Apply one Newton step to $f(x) = x^2 - 7$ starting from $x_0 = 1$. What is x_1?

A. -1
B. 5
C. 7
D. 4

Q0837 · Easy · ≈21 s · Introductory Undergraduate

837. Apply one Newton step to $f(x) = x^2 - 5$ starting from $x_0 = 2$. What is x_1?

A. 3
B. 5/2
C. 7/2
D. 9/4

Q0838 · Easy · ≈33 s · Introductory Undergraduate

838. Apply one Newton step to $f(x) = x^2 - 11$ starting from $x_0 = 2$. What is x_1?

A. 15/4
B. 9
C. 11/2
D. 13/2

Q0839 · Easy · ≈45 s · Introductory Undergraduate

839. Apply one Newton step to $f(x) = x^2 - 2$ starting from $x_0 = 3$. What is x_1?

A. -4
B. 2/3
C. 5/2
D. 11/6

Q0840 · Medium · ≈54 s · Undergraduate

840. Approximate $\int_0^2 x^2\,dx$ with the trapezoidal rule using 4 equal subintervals.

A. 15/4
B. 11/4
C. 7/4
D. 8/3

Q0841 · Medium · ≈103 s · Undergraduate

841. Approximate $\int_0^4 x^2\,dx$ with the trapezoidal rule using 5 equal subintervals.

A. 544/25
B. 704/25
C. 64/3
D. 384/25

Q0842 · Medium · ≈80 s · Undergraduate

842. Approximate $\int_0^2 x^2\,dx$ with the trapezoidal rule using 2 equal subintervals.

A. 1
B. 3
C. 8/3
D. 5

Q0843 · Medium · ≈84 s · Undergraduate

843. Approximate $\int_0^3 x^2\,dx$ with the trapezoidal rule using 2 equal subintervals.

A. 27/8
B. 81/8
C. 9
D. 135/8

Q0844 · Medium · ≈84 s · Undergraduate

844. Approximate $\int_0^5 x^2\,dx$ with the trapezoidal rule using 4 equal subintervals.

A. 875/32
B. 1375/32
C. 125/3
D. 1875/32

Q0845 · Medium · ≈94 s · Undergraduate

845. Approximate $\int_0^4 x^2\,dx$ with the trapezoidal rule using 2 equal subintervals.

A. 24
B. 8
C. 64/3
D. 40

Q0846 · Medium · ≈69 s · Undergraduate

846. Approximate $\int_0^4 x^2\,dx$ with the trapezoidal rule using 4 equal subintervals.

A. 30
B. 64/3
C. 14
D. 22

Q0847 · Hard · ≈242 s · Upper Undergraduate

847. Use one step of the classical fourth-order Runge–Kutta method with h = 1/4 for $y' = -4y$, $y(0) = 3$. What is y_1?

A. 3/2
B. 0
C. 9/8
D. 1

Q0848 · Hard · ≈222 s · Upper Undergraduate

848. Use one step of the classical fourth-order Runge–Kutta method with h = 1/2 for $y' = -3y$, $y(0) = 3$. What is y_1?

A. 3/16
B. 15/8
C. -3/2
D. 105/128

Q0849 · Hard · ≈278 s · Upper Undergraduate

849. Use one step of the classical fourth-order Runge–Kutta method with h = 1/4 for $y' = -1y$, $y(0) = 4$. What is y_1?

A. 3
B. 1595/512
C. 25/8
D. 299/96

Q0850 · Hard · ≈223 s · Upper Undergraduate

850. Use one step of the classical fourth-order Runge–Kutta method with h = 1/5 for $y' = -4y$, $y(0) = 5$. What is y_1?

A. 13/5
B. 1
C. 163/75
D. 847/375

Q0851 · Hard · ≈161 s · Upper Undergraduate

851. Use one step of the classical fourth-order Runge–Kutta method with h = 1/4 for $y' = -3y$, $y(0) = 1$. What is y_1?

A. 17/32
B. 1/4
C. 971/2048
D. 59/128

Q0852 · Hard · ≈132 s · Upper Undergraduate

852. Use one step of the classical fourth-order Runge–Kutta method with h = 1/4 for $y' = -4y$, $y(0) = 1$. What is y_1?

A. 0
B. 3/8
C. 1/2
D. 1/3

Q0853 · Hard · ≈149 s · Upper Undergraduate

853. Use one step of the classical fourth-order Runge–Kutta method with h = 1/5 for $y' = -1y$, $y(0) = 2$. What is y_1?

A. 12281/7500
B. 41/25
C. 614/375
D. 8/5

Optimization

Q0854 · Easy · ≈20 s · Introductory Undergraduate

854. At what x does f(x) = $3x^2$ - 7x - 3 attain its minimum?

A. 7/3
B. -7/6
C. 7/6
D. 7/2

Q0855 · Easy · ≈29 s · Introductory Undergraduate

855. At what x does f(x) = x^2 + 8x - 4 attain its minimum?

A. 4
B. -4
C. -3
D. -8

Q0856 · Easy · ≈44 s · Introductory Undergraduate

856. At what x does f(x) = $4x^2$ - 2x + 3 attain its minimum?

A. 1/4
B. -1/4
C. 1
D. 1/2

Q0857 · Easy · ≈31 s · Introductory Undergraduate

857. At what x does f(x) = $5x^2$ - x + 9 attain its minimum?

A. 1/5
B. -1/10
C. 1/10
D. 1/2

Q0858 · Easy · ≈33 s · Introductory Undergraduate

858. At what x does f(x) = $4x^2$ + 7x - 5 attain its minimum?

A. 7/8
B. -7/4
C. -7/8
D. -7/2

Q0859 · Easy · ≈27 s · Introductory Undergraduate

859. At what x does f(x) = $3x^2$ + 5x - 2 attain its minimum?

A. -5/3
B. -5/6
C. 5/6
D. -5/2

Q0860 · Easy · ≈44 s · Introductory Undergraduate

860. At what x does f(x) = $4x^2$ + 2 attain its minimum?

A. -1
B. 2
C. 1
D. 0

Q0861 · Medium · ≈98 s · Undergraduate

861. Minimise $x^2 + y^2$ subject to 4x + 2y = 4. What is the minimum value?

A. 2/3
B. 1/5
C. 4/5
D. 8/3

Q0862 · Medium · ≈106 s · Undergraduate

862. Minimise $x^2 + y^2$ subject to $3x + 3y = 3$. What is the minimum value?

A. 3/2
B. 1/4
C. 1/2
D. 1/6

Q0863 · Medium · ≈50 s · Undergraduate

863. Minimise $x^2 + y^2$ subject to $5x + 1y = 14$. What is the minimum value?

A. 7/3
B. 98/3
C. 7/13
D. 98/13

Q0864 · Medium · ≈77 s · Undergraduate

864. Minimise $x^2 + y^2$ subject to $5x + 2y = 8$. What is the minimum value?

A. 8/29
B. 64/29
C. 8/7
D. 64/7

Q0865 · Medium · ≈97 s · Undergraduate

865. Minimise $x^2 + y^2$ subject to $1x + 4y = 3$. What is the minimum value?

A. 9/17
B. 3/17
C. 3/5
D. 9/5

Q0866 · Medium · ≈62 s · Undergraduate

866. Minimise $x^2 + y^2$ subject to $2x + 5y = 18$. What is the minimum value?

A. 324/7
B. 324/29
C. 18/29
D. 18/7

Q0867 · Medium · ≈86 s · Undergraduate

867. Minimise $x^2 + y^2$ subject to $2x + 3y = 18$. What is the minimum value?

A. 18/13
B. 324/5
C. 324/13
D. 18/5

Q0868 · Hard · ≈151 s · Upper Undergraduate

868. Maximise $3x + 3y$ subject to $1x + 1y \le 14$, $1x + 3y \le 18$, $2x + 1y \le 15$, $x \ge 0$, $y \ge 0$. What is the maximum value?

A. 139/5
B. 45/2
C. 144/5
D. 149/5

Q0869 · Hard · ≈130 s · Upper Undergraduate

869. Maximise $1x + 3y$ subject to $1x + 1y \le 8$, $1x + 2y \le 13$, $3x + 1y \le 16$, $x \ge 0$, $y \ge 0$. What is the maximum value?

A. 8
B. 39/2
C. 37/2
D. 41/2

Q0870 · Hard · ≈189 s · Upper Undergraduate

870. Maximise 3x + 1y subject to 1x + 1y ≤ 7, 1x + 3y ≤ 10, 2x + 1y ≤ 18, x ≥ 0, y ≥ 0. What is the maximum value?

A. 21
B. 20
C. 22
D. 23

Q0871 · Hard · ≈137 s · Upper Undergraduate

871. Maximise 3x + 1y subject to 1x + 1y ≤ 13, 1x + 3y ≤ 10, 3x + 1y ≤ 19, x ≥ 0, y ≥ 0. What is the maximum value?

A. 18
B. 39
C. 19
D. 20

Q0872 · Hard · ≈254 s · Upper Undergraduate

872. Maximise 5x + 5y subject to 1x + 1y ≤ 7, 1x + 2y ≤ 20, 3x + 1y ≤ 20, x ≥ 0, y ≥ 0. What is the maximum value?

A. 37
B. 35
C. 36
D. 34

Q0873 · Hard · ≈185 s · Upper Undergraduate

873. Maximise 5x + 4y subject to 1x + 1y ≤ 12, 1x + 2y ≤ 17, 3x + 1y ≤ 12, x ≥ 0, y ≥ 0. What is the maximum value?

A. 191/5
B. 34
C. 186/5
D. 196/5

Q0874 · Hard · ≈221 s · Upper Undergraduate

874. Maximise 5x + 3y subject to 1x + 1y ≤ 13, 1x + 2y ≤ 18, 3x + 1y ≤ 14, x ≥ 0, y ≥ 0. What is the maximum value?

A. 34
B. 27
C. 33
D. 35

Operations research

Q0875 · Easy · ≈29 s · Introductory Undergraduate

875. The EOQ model has annual demand D = 1000, ordering cost S = 25 per order and holding cost H = 5 per unit per year. What is the economic order quantity √(2DS/H)?

A. 10000
B. 100
C. 50
D. 200

Q0876 · Easy · ≈37 s · Introductory Undergraduate

876. The EOQ model has annual demand D = 250, ordering cost S = 25 per order and holding cost H = 5 per unit per year. What is the economic order quantity √(2DS/H)?

A. 25
B. 100
C. 2500
D. 50

Q0877 · Easy · ≈26 s · Introductory Undergraduate

877. The EOQ model has annual demand D = 128, ordering cost S = 25 per order and holding cost H = 4 per unit per year. What is the economic order quantity √(2DS/H)?

A. 20
B. 40
C. 80
D. 1600

Q0878 · Easy · ≈22 s · Introductory Undergraduate

878. The EOQ model has annual demand D = 40, ordering cost S = 20 per order and holding cost H = 1 per unit per year. What is the economic order quantity √(2DS/H)?

A. 20
B. 80
C. 1600
D. 40

Q0879 · Easy · ≈34 s · Introductory Undergraduate

879. The EOQ model has annual demand D = 200, ordering cost S = 50 per order and holding cost H = 2 per unit per year. What is the economic order quantity √(2DS/H)?

A. 100
B. 200
C. 10000
D. 50

Q0880 · Easy · ≈40 s · Introductory Undergraduate

880. The EOQ model has annual demand D = 1250, ordering cost S = 20 per order and holding cost H = 5 per unit per year. What is the economic order quantity √(2DS/H)?

A. 10000
B. 100
C. 200
D. 50

Q0881 · Easy · ≈29 s · Introductory Undergraduate

881. The EOQ model has annual demand D = 80, ordering cost S = 10 per order and holding cost H = 1 per unit per year. What is the economic order quantity √(2DS/H)?

A. 40
B. 80
C. 20
D. 1600

Q0882 · Medium · ≈108 s · Undergraduate

882. In an M/M/1 queue, customers arrive at rate λ = 10 per minute and are served at rate μ = 11 per minute. What is the expected number of customers in the system?

A. 10/11
B. 1
C. 10
D. 100/11

Q0883 · Medium · ≈84 s · Undergraduate

883. In an M/M/1 queue, customers arrive at rate λ = 1 per minute and are served at rate μ = 4 per minute. What is the expected number of customers in the system?

A. 1/12
B. 4/3
C. 1/3
D. 1/4

Q0884 · Medium · ≈108 s · Undergraduate

884. In an M/M/1 queue, customers arrive at rate λ = 9 per minute and are served at rate μ = 10 per minute. What is the expected number of customers in the system?

A. 81/10
B. 1
C. 9
D. 9/10

Q0885 · Medium · ≈109 s · Undergraduate

885. In an M/M/1 queue, customers arrive at rate λ = 1 per minute and are served at rate μ = 5 per minute. What is the expected number of customers in the system?

A. 1/20
B. 5/4
C. 1/5
D. 1/4

Q0886 · Medium · ≈71 s · Undergraduate

886. In an M/M/1 queue, customers arrive at rate λ = 4 per minute and are served at rate μ = 11 per minute. What is the expected number of customers in the system?

A. 16/77
B. 4/11
C. 1/7
D. 4/7

Q0887 · Medium · ≈105 s · Undergraduate

887. In an M/M/1 queue, customers arrive at rate λ = 5 per minute and are served at rate μ = 6 per minute. What is the expected number of customers in the system?

A. 5/6
B. 1
C. 5
D. 25/6

Q0888 · Medium · ≈106 s · Undergraduate

888. In an M/M/1 queue, customers arrive at rate λ = 5 per minute and are served at rate μ = 8 per minute. What is the expected number of customers in the system?

A. 25/24
B. 1/3
C. 5/8
D. 5/3

Q0889 · Hard · ≈157 s · Upper Undergraduate

889. Four workers must each be assigned to a different one of four jobs. The cost matrix (rows = workers, columns = jobs) is [15, 10, 17, 20]; [4, 10, 7, 10]; [11, 16, 14, 5]; [16, 13, 8, 18]. What is the minimum total cost?

A. 57
B. 27
C. 26
D. 29

Q0890 · Hard · ≈156 s · Upper Undergraduate

890. Four workers must each be assigned to a different one of four jobs. The cost matrix (rows = workers, columns = jobs) is [2, 11, 10, 14]; [7, 18, 18, 6]; [9, 17, 15, 5]; [6, 9, 16, 7]. What is the minimum total cost?

A. 42
B. 32
C. 31
D. 33

Q0891 · Hard · ≈214 s · Upper Undergraduate

891. Four workers must each be assigned to a different one of four jobs. The cost matrix (rows = workers, columns = jobs) is [7, 7, 5, 15]; [12, 7, 12, 3]; [10, 16, 19, 14]; [12, 17, 16, 16]. What is the minimum total cost?

A. 34
B. 37
C. 49
D. 35

Q0892 · Hard · ≈279 s · Upper Undergraduate

892. Four workers must each be assigned to a different one of four jobs. The cost matrix (rows = workers, columns = jobs) is [5, 11, 10, 11]; [14, 5, 17, 13]; [17, 9, 4, 9]; [8, 17, 8, 5]. What is the minimum total cost?

A. 18
B. 19
C. 20
D. 21

Q0893 · Hard · ≈192 s · Upper Undergraduate

893. Four workers must each be assigned to a different one of four jobs. The cost matrix (rows = workers, columns = jobs) is [14, 3, 12, 14]; [13, 15, 13, 11]; [14, 16, 11, 15]; [5, 2, 16, 15]. What is the minimum total cost?

A. 55
B. 29
C. 32
D. 30

Q0894 · Hard · ≈273 s · Upper Undergraduate

894. Four workers must each be assigned to a different one of four jobs. The cost matrix (rows = workers, columns = jobs) is [7, 15, 4, 19]; [3, 8, 13, 3]; [6, 7, 16, 15]; [8, 11, 16, 15]. What is the minimum total cost?

A. 24
B. 46
C. 29
D. 22

Q0895 · Hard · ≈272 s · Upper Undergraduate

895. Four workers must each be assigned to a different one of four jobs. The cost matrix (rows = workers, columns = jobs) is [20, 15, 11, 4]; [14, 14, 7, 20]; [6, 8, 3, 7]; [2, 9, 18, 5]. What is the minimum total cost?

A. 23
B. 42
C. 26
D. 21

Mathematical modeling

Q0896 · Easy · ≈39 s · Introductory Undergraduate

896. A sample of 320 g of a radioactive isotope has half-life 4 years. How many grams remain after 12 years?

A. 40
B. 308
C. 160/3
D. 320/3

Q0897 · Easy · ≈28 s · Introductory Undergraduate

897. A bacterial culture of 720 cells doubles every 5 hours. How many cells are there after 15 hours?

A. 10800
B. 5760
C. 23592960
D. 4320

Q0898 · Easy · ≈36 s · Introductory Undergraduate

898. A bacterial culture of 530 cells doubles every 5 hours. How many cells are there after 15 hours?

A. 17367040
B. 3180
C. 4240
D. 7950

Q0899 · Easy · ≈20 s · Introductory Undergraduate

899. A bacterial culture of 486 cells doubles every 4 hours. How many cells are there after 20 hours?

A. 509607936
B. 9720
C. 4860
D. 15552

Q0900 · Easy · ≈37 s · Introductory Undergraduate

900. A bacterial culture of 418 cells doubles every 3 hours. How many cells are there after 9 hours?

A. 2508
B. 214016
C. 3762
D. 3344

Q0901 · Easy · ≈23 s · Introductory Undergraduate

901. A bacterial culture of 581 cells doubles every 4 hours. How many cells are there after 8 hours?

A. 148736
B. 585
C. 2324
D. 4648

Q0902 · Easy · ≈44 s · Introductory Undergraduate

902. A bacterial culture of 245 cells doubles every 2 hours. How many cells are there after 8 hours?

A. 62720
B. 1960
C. 3920
D. 261

Q0903 · Medium · ≈88 s · Undergraduate

903. A population obeys $dP/dt = 4P(1 - P/400)$. What is the maximum possible growth rate dP/dt?

A. 800
B. 400
C. 1600
D. 200

Q0904 · Medium · ≈89 s · Undergraduate

904. A population obeys $dP/dt = 3P(1 - P/400)$. What is the maximum possible growth rate dP/dt?

A. 300
B. 200
C. 600
D. 1200

Q0905 · Medium · ≈61 s · Undergraduate

905. A population obeys dP/dt = 6P(1 − P/400). What is the maximum possible growth rate dP/dt?

A. 1200
B. 600
C. 200
D. 2400

Q0906 · Medium · ≈65 s · Undergraduate

906. A population obeys dP/dt = 6P(1 − P/800). What is the maximum possible growth rate dP/dt?

A. 1200
B. 4800
C. 400
D. 2400

Q0907 · Medium · ≈64 s · Undergraduate

907. A population obeys dP/dt = 6P(1 − P/1000). What is the maximum possible growth rate dP/dt?

A. 1500
B. 3000
C. 500
D. 6000

Q0908 · Medium · ≈53 s · Undergraduate

908. A population obeys dP/dt = 1P(1 − P/100). What is the maximum possible growth rate dP/dt?

A. 100
B. 25
C. 50
D. 2500

Q0909 · Medium · ≈76 s · Undergraduate

909. A population obeys dP/dt = 2P(1 − P/1000). What is the maximum possible growth rate dP/dt?

A. 2000
B. 500000
C. 1000
D. 500

Q0910 · Hard · ≈270 s · Upper Undergraduate

910. In an SIR model the transmission rate is $\beta = 3$ per week and the recovery rate is $\gamma = 1$ per week. What fraction of the population must be immune to reach the herd-immunity threshold $1 - 1/R_0$?

A. 1/3
B. 2/3
C. 3
D. 2

Q0911 · Hard · ≈243 s · Upper Undergraduate

911. In an SIR model the transmission rate is $\beta = 5$ per week and the recovery rate is $\gamma = 2$ per week. What fraction of the population must be immune to reach the herd-immunity threshold $1 - 1/R_0$?

A. 2/5
B. 3/5
C. 3/2
D. 5/2

Q0912 · Hard · ≈157 s · Upper Undergraduate

912. In an SIR model the transmission rate is $\beta = 4$ per week and the recovery rate is $\gamma = 1$ per week. What fraction of the population must be immune to reach the herd-immunity threshold $1 - 1/R_0$?

A. 3
B. 1/4
C. 4
D. 3/4

Q0913 · Hard · ≈144 s · Upper Undergraduate

913. In an SIR model the transmission rate is $\beta = 9$ per week and the recovery rate is $\gamma = 3$ per week. What fraction of the population must be immune to reach the herd-immunity threshold $1 - 1/R_0$?

A. 2/3
B. 3
C. 2
D. 1/3

Q0914 · Hard · ≈188 s · Upper Undergraduate

914. In an SIR model the transmission rate is $\beta = 8$ per week and the recovery rate is $\gamma = 5$ per week. What fraction of the population must be immune to reach the herd-immunity threshold $1 - 1/R_0$?

A. 5/8
B. 8/5
C. 3/8
D. 3/5

Q0915 · Hard · ≈142 s · Upper Undergraduate

915. In an SIR model the transmission rate is $\beta = 6$ per week and the recovery rate is $\gamma = 2$ per week. What fraction of the population must be immune to reach the herd-immunity threshold $1 - 1/R_0$?

A. 2/3
B. 2
C. 1/3
D. 3

Q0916 · Hard · ≈190 s · Upper Undergraduate

916. In an SIR model the transmission rate is $\beta = 7$ per week and the recovery rate is $\gamma = 2$ per week. What fraction of the population must be immune to reach the herd-immunity threshold $1 - 1/R_0$?

A. 2/7
B. 5/2
C. 5/7
D. 7/2

Game theory

Q0917 · Easy · ≈35 s · Introductory Undergraduate

917. In a 2×2 game, the payoff pairs (Row, Column) are [(4,8), (0,2)]; [(8,3), (7,5)]. Which outcome is the unique pure-strategy Nash equilibrium?

A. (Row 2, Col 1)
B. (Row 1, Col 2)
C. (Row 1, Col 1)
D. (Row 2, Col 2)

Q0918 · Easy · ≈27 s · Introductory Undergraduate

918. In a 2×2 game, the payoff pairs (Row, Column) are [(4,5), (6,2)]; [(2,8), (3,9)]. Which outcome is the unique pure-strategy Nash equilibrium?

A. (Row 1, Col 1)
B. (Row 2, Col 1)
C. (Row 2, Col 2)
D. (Row 1, Col 2)

Q0919 · Easy · ≈39 s · Introductory Undergraduate

919. In a 2×2 game, the payoff pairs (Row, Column) are [(5,1), (5,5)]; [(3,0), (6,4)]. Which outcome is the unique pure-strategy Nash equilibrium?

A. (Row 2, Col 1)
B. (Row 1, Col 2)
C. (Row 2, Col 2)
D. (Row 1, Col 1)

Q0920 · Easy · ≈31 s · Introductory Undergraduate

920. In a 2×2 game, the payoff pairs (Row, Column) are [(6,3), (9,2)]; [(8,9), (9,5)]. Which outcome is the unique pure-strategy Nash equilibrium?

A. (Row 1, Col 2)
B. (Row 1, Col 1)
C. (Row 2, Col 2)
D. (Row 2, Col 1)

Q0921 · Easy · ≈43 s · Introductory Undergraduate

921. In a 2×2 game, the payoff pairs (Row, Column) are [(7,4), (0,9)]; [(5,0), (7,6)]. Which outcome is the unique pure-strategy Nash equilibrium?

A. (Row 2, Col 2)
B. (Row 2, Col 1)
C. (Row 1, Col 2)
D. (Row 1, Col 1)

Q0922 · Easy · ≈23 s · Introductory Undergraduate

922. In a 2×2 game, the payoff pairs (Row, Column) are [(6,1), (4,2)]; [(2,5), (3,4)]. Which outcome is the unique pure-strategy Nash equilibrium?

A. (Row 1, Col 1)
B. (Row 2, Col 2)
C. (Row 2, Col 1)
D. (Row 1, Col 2)

Q0923 · Easy · ≈21 s · Introductory Undergraduate

923. In a 2×2 game, the payoff pairs (Row, Column) are [(4,4), (5,1)]; [(9,4), (8,8)]. Which outcome is the unique pure-strategy Nash equilibrium?

A. (Row 2, Col 2)
B. (Row 1, Col 2)
C. (Row 2, Col 1)
D. (Row 1, Col 1)

Q0924 · Medium · ≈80 s · Undergraduate

924. In a zero-sum game the Row player's payoff matrix is [[5, 9], [8, 3]]. Row plays row 1 with probability p and row 2 with probability 1 − p. For which p is Row's expected payoff the same against either column?

A. 3/8
B. 2/3
C. 4/9
D. 5/9

Q0925 · Medium · ≈89 s · Undergraduate

925. In a zero-sum game the Row player's payoff matrix is [[7, 9], [4, 3]]. Row plays row 1 with probability p and row 2 with probability 1 − p. For which p is Row's expected payoff the same against either column?

A. 1/3
B. 3/10
C. 2
D. 2/3

Q0926 · Medium · ≈109 s · Undergraduate

926. In a zero-sum game the Row player's payoff matrix is [[7, 9], [5, 3]]. Row plays row 1 with probability p and row 2 with probability 1 − p. For which p is Row's expected payoff the same against either column?

A. 3/2
B. 1/2
C. 3/10
D. -1/2

Q0927 · Medium · ≈58 s · Undergraduate

927. In a zero-sum game the Row player's payoff matrix is [[0, 6], [9, 3]]. Row plays row 1 with probability p and row 2 with probability 1 − p. For which p is Row's expected payoff the same against either column?

A. 1/2
B. 1/4
C. 3/4
D. 1

Q0928 · Medium · ≈56 s · Undergraduate

928. In a zero-sum game the Row player's payoff matrix is [[6, 4], [2, 7]]. Row plays row 1 with probability p and row 2 with probability 1 − p. For which p is Row's expected payoff the same against either column?

A. 5/7
B. 2/7
C. 3/7
D. 7/13

Q0929 · Medium · ≈70 s · Undergraduate

929. In a zero-sum game the Row player's payoff matrix is [[2, 8], [2, 0]]. Row plays row 1 with probability p and row 2 with probability 1 − p. For which p is Row's expected payoff the same against either column?

A. 1
B. 0
C. 3/4
D. 1/4

Q0930 · Medium · ≈59 s · Undergraduate

930. In a zero-sum game the Row player's payoff matrix is [[2, 8], [9, 4]]. Row plays row 1 with probability p and row 2 with probability 1 − p. For which p is Row's expected payoff the same against either column?

A. 4/11
B. 2/3
C. 6/11
D. 5/11

Q0931 · Hard · ≈221 s · Upper Undergraduate

931. A zero-sum game has Row payoff matrix [[1, 12], [6, 4]] (Row maximises). It has no saddle point. What is the value of the game?

A. 4
B. 23/4
C. -76/13
D. 68/13

Q0932 · Hard · ≈257 s · Upper Undergraduate

932. A zero-sum game has Row payoff matrix [[12, 2], [4, 7]] (Row maximises). It has no saddle point. What is the value of the game?

A. 25/4
B. 76/13
C. 92/13
D. 4

Q0933 · Hard · ≈231 s · Upper Undergraduate

933. A zero-sum game has Row payoff matrix [[2, 1], [1, 10]] (Row maximises). It has no saddle point. What is the value of the game?

A. 1
B. 19/10
C. 7/2
D. 21/10

Q0934 · Hard · ≈179 s · Upper Undergraduate

934. A zero-sum game has Row payoff matrix [[12, 8], [11, 12]] (Row maximises). It has no saddle point. What is the value of the game?

A. 43/4
B. 232/5
C. 11
D. 56/5

Q0935 · Hard · ≈189 s · Upper Undergraduate

935. A zero-sum game has Row payoff matrix [[9, 8], [0, 11]] (Row maximises). It has no saddle point. What is the value of the game?

A. 7
B. 33/4
C. 8
D. 9

Q0936 · Hard · ≈226 s · Upper Undergraduate

936. A zero-sum game has Row payoff matrix [[2, 11], [7, 2]] (Row maximises). It has no saddle point. What is the value of the game?

A. 73/14
B. -81/14
C. 2
D. 11/2

Q0937 · Hard · ≈195 s · Upper Undergraduate

937. A zero-sum game has Row payoff matrix [[4, 0], [1, 9]] (Row maximises). It has no saddle point. What is the value of the game?

A. 1
B. 7/2
C. 4
D. 3

Dynamical systems and chaos theory

Q0938 · Easy · ≈22 s · Introductory Undergraduate

938. Find the non-zero fixed point of the logistic map $x_{n+1} = 5x_n(1 - x_n)$.

A. 5/6
B. 4/5
C. 1/5
D. 3/5

Q0939 · Easy · ≈23 s · Introductory Undergraduate

939. Find the non-zero fixed point of the logistic map $x_{n+1} = 10x_n(1 - x_n)$.

A. 4/5
B. 1/10
C. 9/10
D. 10/11

Q0940 · Easy · ≈33 s · Introductory Undergraduate

940. Find the non-zero fixed point of the logistic map $x_{n+1} = 8x_n(1 - x_n)$.

A. 1/8
B. 3/4
C. 8/9
D. 7/8

Q0941 · Easy · ≈30 s · Introductory Undergraduate

941. Find the non-zero fixed point of the logistic map $x_{n+1} = 11x_n(1 - x_n)$.

A. 1/11
B. 11/12
C. 9/11
D. 10/11

Q0942 · Easy · ≈28 s · Introductory Undergraduate

942. Find the non-zero fixed point of the logistic map $x_{n+1} = 6x_n(1 - x_n)$.

A. 6/7
B. 2/3
C. 5/6
D. 1/6

Q0943 · Easy · ≈37 s · Introductory Undergraduate

943. Find the non-zero fixed point of the logistic map $x_{n+1} = 7x_n(1 - x_n)$.

A. 5/7
B. 6/7
C. 1/7
D. 7/8

Q0944 · Easy · ≈27 s · Introductory Undergraduate

944. Find the non-zero fixed point of the logistic map $x_{n+1} = 13x_n(1 - x_n)$.

A. 1/13
B. 13/14
C. 12/13
D. 11/13

Q0945 · Medium · ≈107 s · Undergraduate

945. Classify the equilibrium at the origin of the linear system x′ = Ax with A = [[-4, -4], [2, -2]].

A. Saddle
B. Stable spiral
C. Unstable node
D. Unstable spiral

Q0946 · Medium · ≈92 s · Undergraduate

946. Classify the equilibrium at the origin of the linear system x′ = Ax with A = [[0, -1], [3, -1]].

A. Stable spiral
B. Unstable node
C. Stable node
D. Saddle

Q0947 · Medium · ≈57 s · Undergraduate

947. Classify the equilibrium at the origin of the linear system $x' = Ax$ with $A = [[3, -2], [1, -1]]$.

A. Saddle
B. Unstable spiral
C. Stable spiral
D. Stable node

Q0948 · Medium · ≈64 s · Undergraduate

948. Classify the equilibrium at the origin of the linear system $x' = Ax$ with $A = [[0, -4], [-3, -1]]$.

A. Saddle
B. Unstable node
C. Stable node
D. Unstable spiral

Q0949 · Medium · ≈55 s · Undergraduate

949. Classify the equilibrium at the origin of the linear system $x' = Ax$ with $A = [[5, 4], [4, 1]]$.

A. Unstable spiral
B. Saddle
C. Unstable node
D. Stable spiral

Q0950 · Medium · ≈87 s · Undergraduate

950. Classify the equilibrium at the origin of the linear system $x' = Ax$ with $A = [[0, -3], [-5, -4]]$.

A. Unstable spiral
B. Stable node
C. Saddle
D. Unstable node

Q0951 · Medium · ≈51 s · Undergraduate

951. Classify the equilibrium at the origin of the linear system $x' = Ax$ with $A = [[-5, -5], [1, 2]]$.

A. Stable node
B. Unstable spiral
C. Saddle
D. Stable spiral

Q0952 · Hard · ≈233 s · Upper Undergraduate

952. For the logistic map $x_{n+1} = (11/3)\cdot x_n(1 - x_n)$, the map has a period-2 orbit $\{x_1, x_2\}$. What is $x_1 + x_2$?

A. 8/11
B. 16/11
C. 11/14
D. 14/11

Q0953 · Hard · ≈133 s · Upper Undergraduate

953. For the logistic map $x_{n+1} = (10/3)\cdot x_n(1 - x_n)$, the map has a period-2 orbit $\{x_1, x_2\}$. What is $x_1 + x_2$?

A. 7/10
B. 10/13
C. 7/5
D. 13/10

Q0954 · Hard · ≈207 s · Upper Undergraduate

954. For the logistic map $x_{n+1} = (17/5)\cdot x_n(1 - x_n)$, the map has a period-2 orbit $\{x_1, x_2\}$. What is $x_1 + x_2$?

A. 22/17
B. 17/22
C. 24/17
D. 12/17

Q0955 · Hard · ≈278 s · Upper Undergraduate

955. For the logistic map $x_{n+1} = 4\cdot x_n(1 - x_n)$, the map has a period-2 orbit $\{x_1, x_2\}$. What is $x_1 + x_2$?

A. 3/4
B. 3/2
C. 5/4
D. 4/5

Q0956 · Hard · ≈180 s · Upper Undergraduate

956. For the logistic map $x_{n+1} = (16/5)\cdot x_n(1 - x_n)$, the map has a period-2 orbit $\{x_1, x_2\}$. What is $x_1 + x_2$?

A. 16/21
B. 11/8
C. 11/16
D. 21/16

Q0957 · Hard · ≈254 s · Upper Undergraduate

957. For the logistic map $x_{n+1} = (19/5)\cdot x_n(1 - x_n)$, the map has a period-2 orbit $\{x_1, x_2\}$. What is $x_1 + x_2$?

A. 14/19
B. 28/19
C. 24/19
D. 19/24

Q0958 · Hard · ≈280 s · Upper Undergraduate

958. For the logistic map $x_{n+1} = (13/4)\cdot x_n(1 - x_n)$, the map has a period-2 orbit $\{x_1, x_2\}$. What is $x_1 + x_2$?

A. 17/13
B. 18/13
C. 13/17
D. 9/13

Information theory

Q0959 · Easy · ≈40 s · Introductory Undergraduate

959. A source emits symbols with probabilities 1/4, 1/4, 1/4, 1/4. What is its entropy in bits?

A. 2
B. 4
C. 5/2
D. 7/4

Q0960 · Easy · ≈23 s · Introductory Undergraduate

960. A source emits symbols with probabilities 1/16, 1/16, 1/8, 1/2, 1/8, 1/8. What is its entropy in bits?

A. 6
B. 17/8
C. 21/8
D. 3

Q0961 · Easy · ≈26 s · Introductory Undergraduate

961. A source emits symbols with probabilities 1/4, 1/4, 1/2. What is its entropy in bits?

A. 5/4
B. 3/2
C. 2
D. 3

Q0962 · Easy · ≈40 s · Introductory Undergraduate

962. A source emits symbols with probabilities 1/16, 1/16, 1/8, 1/4, 1/2. What is its entropy in bits?

A. 5
B. 19/8
C. 15/8
D. 5/2

Q0963 · Easy · ≈44 s · Introductory Undergraduate

963. A source emits symbols with probabilities 1/8, 1/4, 1/4, 1/16, 1/4, 1/16. What is its entropy in bits?

A. 19/8
B. 23/8
C. 6
D. 3

Q0964 · Easy · ≈42 s · Introductory Undergraduate

964. A source emits symbols with probabilities 1/8, 1/8, 1/8, 1/8, 1/4, 1/4. What is its entropy in bits?

A. 5/2
B. 3
C. 9/4
D. 6

Q0965 · Easy · ≈33 s · Introductory Undergraduate

965. A source emits symbols with probabilities 1/2, 1/4, 1/8, 1/8. What is its entropy in bits?

A. 2
B. 4
C. 7/4
D. 9/4

Q0966 · Medium · ≈88 s · Undergraduate

966. A source has symbol frequencies 8, 9, 12, 12, 12. What is the average codeword length (bits per symbol) of an optimal Huffman code?

A. 3
B. 124/53
C. 122/53
D. 123/53

Q0967 · Medium · ≈93 s · Undergraduate

967. A source has symbol frequencies 11, 3, 1, 5, 9, 7. What is the average codeword length (bits per symbol) of an optimal Huffman code?

A. 85/36
B. 7/3
C. 3
D. 43/18

Q0968 · Medium · ≈90 s · Undergraduate

968. A source has symbol frequencies 12, 2, 11, 8. What is the average codeword length (bits per symbol) of an optimal Huffman code?

A. 65/33
B. 2
C. 21/11
D. 64/33

Q0969 · Medium · ≈90 s · Undergraduate

969. A source has symbol frequencies 4, 5, 9, 5, 2, 3. What is the average codeword length (bits per symbol) of an optimal Huffman code?

A. 5/2
B. 71/28
C. 69/28
D. 3

Q0970 · Medium · ≈97 s · Undergraduate

970. A source has symbol frequencies 2, 11, 6, 4, 5. What is the average codeword length (bits per symbol) of an optimal Huffman code?

A. 31/14
B. 61/28
C. 3
D. 9/4

Q0971 · Medium · ≈73 s · Undergraduate

971. A source has symbol frequencies 10, 12, 3, 10, 2, 4. What is the average codeword length (bits per symbol) of an optimal Huffman code?

A. 3
B. 96/41
C. 97/41
D. 95/41

Q0972 · Medium · ≈84 s · Undergraduate

972. A source has symbol frequencies 5, 3, 2, 2, 9. What is the average codeword length (bits per symbol) of an optimal Huffman code?

A. 3
B. 43/21
C. 15/7
D. 44/21

Q0973 · Hard · ≈167 s · Upper Undergraduate

973. What is the minimum number of parity bits needed for a single-error-correcting Hamming-type code that protects 26 data bits?

A. 5
B. 4
C. 13
D. 6

Q0974 · Hard · ≈193 s · Upper Undergraduate

974. What is the minimum number of parity bits needed for a single-error-correcting Hamming-type code that protects 57 data bits?

A. 6
B. 5
C. 7
D. 28

Q0975 · Hard · ≈140 s · Upper Undergraduate

975. What is the minimum number of parity bits needed for a single-error-correcting Hamming-type code that protects 8 data bits?

A. 5
B. 4
C. 3
D. 6

Q0976 · Hard · ≈269 s · Upper Undergraduate

976. What is the minimum number of parity bits needed for a single-error-correcting Hamming-type code that protects 30 data bits?

A. 5
B. 6
C. 7
D. 15

Q0977 · Hard · ≈268 s · Upper Undergraduate

977. What is the minimum number of parity bits needed for a single-error-correcting Hamming-type code that protects 4 data bits?

A. 5
B. 2
C. 3
D. 4

Q0978 · Hard · ≈242 s · Upper Undergraduate

978. What is the minimum number of parity bits needed for a single-error-correcting Hamming-type code that protects 20 data bits?

A. 5
B. 6
C. 10
D. 4

Q0979 · Hard · ≈198 s · Upper Undergraduate

979. What is the minimum number of parity bits needed for a single-error-correcting Hamming-type code that protects 11 data bits?

A. 5
B. 3
C. 6
D. 4

Financial mathematics

Q0980 · Easy · ≈25 s · Introductory Undergraduate

980. An investment of 500 earns 8% interest per year, compounded annually. What is its value after 3 years (rounded to 2 decimals)?

A. 583.20
B. 620.00
C. 680.24
D. 629.86

Q0981 · Easy · ≈39 s · Introductory Undergraduate

981. An investment of 1000 earns 8% interest per year, compounded annually. What is its value after 2 years (rounded to 2 decimals)?

A. 1259.71
B. 1166.40
C. 1160.00
D. 1080.00

Q0982 · Easy · ≈45 s · Introductory Undergraduate

982. An investment of 5000 earns 4% interest per year, compounded annually. What is its value after 3 years (rounded to 2 decimals)?

A. 5849.29
B. 5624.32
C. 5408.00
D. 5600.00

Q0983 · Easy · ≈35 s · Introductory Undergraduate

983. An investment of 1000 earns 4% interest per year, compounded annually. What is its value after 5 years (rounded to 2 decimals)?

A. 1265.32
B. 1200.00
C. 1216.65
D. 1169.86

Q0984 · Easy · ≈35 s · Introductory Undergraduate

984. An investment of 1000 earns 10% interest per year, compounded annually. What is its value after 2 years (rounded to 2 decimals)?

A. 1200.00
B. 1100.00
C. 1331.00
D. 1210.00

Q0985 · Easy · ≈35 s · Introductory Undergraduate

985. An investment of 5000 earns 8% interest per year, compounded annually. What is its value after 5 years (rounded to 2 decimals)?

A. 6802.44
B. 7000.00
C. 7346.64
D. 7934.37

Q0986 · Easy · ≈28 s · Introductory Undergraduate

986. An investment of 1000 earns 10% interest per year, compounded annually. What is its value after 5 years (rounded to 2 decimals)?

A. 1771.56
B. 1610.51
C. 1500.00
D. 1464.10

Q0987 · Medium · ≈105 s · Undergraduate

987. What is the present value of an ordinary annuity paying 200 at the end of each year for 11 years, at an annual discount rate of 5%? (Round to 2 decimals.)

A. 2841.36
B. 1661.28
C. 4000.00
D. 2200.00

Q0988 · Medium · ≈66 s · Undergraduate

988. What is the present value of an ordinary annuity paying 100 at the end of each year for 5 years, at an annual discount rate of 5%? (Round to 2 decimals.)

A. 552.56
B. 500.00
C. 432.95
D. 2000.00

Q0989 · Medium · ≈66 s · Undergraduate

989. What is the present value of an ordinary annuity paying 1000 at the end of each year for 8 years, at an annual discount rate of 8%? (Round to 2 decimals.)

A. 10636.63
B. 5746.64
C. 12500.00
D. 8000.00

Q0990 · Medium · ≈54 s · Undergraduate

990. What is the present value of an ordinary annuity paying 200 at the end of each year for 12 years, at an annual discount rate of 10%? (Round to 2 decimals.)

A. 4276.86
B. 2000.00
C. 2400.00
D. 1362.74

Q0991 · Medium · ≈86 s · Undergraduate

991. What is the present value of an ordinary annuity paying 500 at the end of each year for 12 years, at an annual discount rate of 8%? (Round to 2 decimals.)

A. 6000.00
B. 3768.04
C. 9488.56
D. 6250.00

Q0992 · Medium · ≈96 s · Undergraduate

992. What is the present value of an ordinary annuity paying 200 at the end of each year for 7 years, at an annual discount rate of 8%? (Round to 2 decimals.)

A. 1784.56
B. 2500.00
C. 1400.00
D. 1041.27

Q0993 · Medium · ≈58 s · Undergraduate

993. What is the present value of an ordinary annuity paying 100 at the end of each year for 6 years, at an annual discount rate of 5%? (Round to 2 decimals.)

A. 2000.00
B. 507.57
C. 600.00
D. 680.19

Q0994 · Hard · ≈246 s · Upper Undergraduate

994. One-period binomial model: $S_0 = 50$, the stock goes to 60 (up) or 40 (down), and the risk-free rate is 1/25 per period. What is the no-arbitrage price of a call option with strike 52? (Round to 2 decimals.)

A. 4.80
B. 8.00
C. 3.85
D. 4.62

Q0995 · Hard · ≈141 s · Upper Undergraduate

995. One-period binomial model: $S_0 = 80$, the stock goes to 100 (up) or 64 (down), and the risk-free rate is 1/20 per period. What is the no-arbitrage price of a call option with strike 84? (Round to 2 decimals.)

A. 16.00
B. 8.47
C. 8.89
D. 7.62

Q0996 · Hard · ≈210 s · Upper Undergraduate

996. One-period binomial model: $S_0 = 50$, the stock goes to 60 (up) or 40 (down), and the risk-free rate is 1/25 per period. What is the no-arbitrage price of a call option with strike 45? (Round to 2 decimals.)

A. 15.00
B. 9.00
C. 8.65
D. 7.21

Q0997 · Hard · ≈188 s · Upper Undergraduate

997. One-period binomial model: $S_0 = 100$, the stock goes to 120 (up) or 90 (down), and the risk-free rate is 1/20 per period. What is the no-arbitrage price of a call option with strike 100? (Round to 2 decimals.)

A. 20.00
B. 9.52
C. 10.02
D. 10.00

Q0998 · Hard · ≈142 s · Upper Undergraduate

998. One-period binomial model: $S_0 = 120$, the stock goes to 150 (up) or 108 (down), and the risk-free rate is 1/10 per period. What is the no-arbitrage price of a call option with strike 120? (Round to 2 decimals.)

A. 30.00
B. 17.14
C. 15.58
D. 13.64

Q0999 · Hard · ≈274 s · Upper Undergraduate

999. One-period binomial model: $S_0 = 120$, the stock goes to 150 (up) or 108 (down), and the risk-free rate is 1/10 per period. What is the no-arbitrage price of a call option with strike 126? (Round to 2 decimals.)

A. 13.71
B. 24.00
C. 10.91
D. 12.47

Q1000 · Hard · ≈252 s · Upper Undergraduate

1000. One-period binomial model: $S_0 = 80$, the stock goes to 100 (up) or 64 (down), and the risk-free rate is 1/20 per period. What is the no-arbitrage price of a call option with strike 72? (Round to 2 decimals.)

A. 14.81
B. 15.56
C. 28.00
D. 13.33

Part II – Answer key

1 A	**2** D	**3** C	**4** B	**5** A	**6** B	**7** B	**8** A	**9** B	**10** A
11 B	**12** B	**13** A	**14** C	**15** B	**16** D	**17** A	**18** D	**19** D	**20** A
21 C	**22** B	**23** C	**24** A	**25** A	**26** A	**27** C	**28** D	**29** C	**30** B
31 A	**32** A	**33** B	**34** B	**35** B	**36** B	**37** A	**38** A	**39** A	**40** B
41 D	**42** C	**43** D	**44** D	**45** A	**46** C	**47** B	**48** C	**49** B	**50** D
51 A	**52** B	**53** B	**54** A	**55** C	**56** D	**57** B	**58** A	**59** A	**60** D
61 D	**62** C	**63** D	**64** A	**65** B	**66** D	**67** A	**68** A	**69** A	**70** B
71 D	**72** B	**73** B	**74** A	**75** D	**76** B	**77** A	**78** B	**79** C	**80** D
81 B	**82** A	**83** B	**84** A	**85** D	**86** C	**87** B	**88** B	**89** A	**90** B
91 B	**92** B	**93** C	**94** D	**95** B	**96** C	**97** C	**98** A	**99** A	**100** D
101 A	**102** A	**103** A	**104** D	**105** B	**106** C	**107** A	**108** D	**109** A	**110** A
111 D	**112** D	**113** A	**114** C	**115** B	**116** C	**117** B	**118** C	**119** B	**120** D
121 A	**122** C	**123** B	**124** B	**125** D	**126** B	**127** A	**128** C	**129** D	**130** C
131 C	**132** A	**133** C	**134** C	**135** D	**136** B	**137** A	**138** D	**139** C	**140** B
141 B	**142** D	**143** A	**144** B	**145** A	**146** D	**147** D	**148** C	**149** B	**150** B
151 D	**152** A	**153** D	**154** D	**155** B	**156** D	**157** D	**158** B	**159** B	**160** D
161 A	**162** D	**163** B	**164** B	**165** C	**166** C	**167** D	**168** A	**169** B	**170** D
171 D	**172** C	**173** D	**174** D	**175** D	**176** A	**177** B	**178** D	**179** B	**180** C
181 D	**182** D	**183** A	**184** B	**185** C	**186** C	**187** C	**188** B	**189** D	**190** A
191 A	**192** B	**193** B	**194** C	**195** B	**196** D	**197** C	**198** A	**199** A	**200** C
201 B	**202** A	**203** C	**204** D	**205** A	**206** C	**207** B	**208** C	**209** C	**210** B
211 D	**212** B	**213** D	**214** D	**215** D	**216** C	**217** D	**218** B	**219** C	**220** D
221 D	**222** B	**223** D	**224** C	**225** B	**226** D	**227** D	**228** C	**229** B	**230** A
231 B	**232** B	**233** A	**234** B	**235** D	**236** B	**237** B	**238** A	**239** D	**240** C
241 B	**242** A	**243** B	**244** A	**245** B	**246** B	**247** C	**248** D	**249** C	**250** D
251 D	**252** B	**253** C	**254** C	**255** D	**256** D	**257** A	**258** D	**259** A	**260** D
261 A	**262** C	**263** D	**264** B	**265** A	**266** D	**267** B	**268** C	**269** A	**270** A
271 D	**272** C	**273** C	**274** B	**275** B	**276** B	**277** D	**278** B	**279** C	**280** D
281 B	**282** D	**283** A	**284** D	**285** D	**286** D	**287** A	**288** A	**289** B	**290** C
291 C	**292** D	**293** C	**294** D	**295** C	**296** D	**297** A	**298** C	**299** C	**300** A
301 B	**302** B	**303** B	**304** B	**305** C	**306** A	**307** D	**308** C	**309** C	**310** A
311 C	**312** A	**313** C	**314** C	**315** C	**316** D	**317** A	**318** A	**319** B	**320** B
321 B	**322** A	**323** D	**324** D	**325** D	**326** D	**327** A	**328** D	**329** A	**330** C
331 C	**332** D	**333** D	**334** D	**335** A	**336** C	**337** B	**338** D	**339** B	**340** B
341 C	**342** D	**343** C	**344** D	**345** A	**346** D	**347** C	**348** C	**349** B	**350** B
351 D	**352** B	**353** B	**354** B	**355** C	**356** D	**357** A	**358** B	**359** D	**360** B
361 D	**362** C	**363** C	**364** D	**365** C	**366** B	**367** A	**368** B	**369** A	**370** B
371 C	**372** A	**373** D	**374** C	**375** A	**376** A	**377** B	**378** B	**379** A	**380** A
381 A	**382** B	**383** C	**384** A	**385** D	**386** A	**387** B	**388** A	**389** A	**390** C
391 C	**392** A	**393** B	**394** C	**395** B	**396** C	**397** A	**398** A	**399** C	**400** C
401 B	**402** D	**403** D	**404** C	**405** D	**406** C	**407** B	**408** C	**409** B	**410** B
411 A	**412** D	**413** C	**414** B	**415** A	**416** B	**417** D	**418** D	**419** D	**420** C
421 B	**422** B	**423** B	**424** A	**425** D	**426** D	**427** C	**428** C	**429** A	**430** B
431 B	**432** C	**433** C	**434** A	**435** B	**436** C	**437** B	**438** B	**439** C	**440** C
441 C	**442** D	**443** C	**444** D	**445** A	**446** D	**447** C	**448** C	**449** B	**450** C
451 A	**452** D	**453** C	**454** D	**455** D	**456** B	**457** A	**458** A	**459** D	**460** B
461 D	**462** C	**463** D	**464** A	**465** C	**466** B	**467** D	**468** A	**469** A	**470** C
471 D	**472** C	**473** B	**474** C	**475** D	**476** A	**477** B	**478** C	**479** B	**480** D
481 A	**482** B	**483** B	**484** B	**485** D	**486** C	**487** C	**488** D	**489** B	**490** A

491 C	**492** A	**493** C	**494** B	**495** A	**496** C	**497** C	**498** B	**499** D	**500** A
501 A	**502** C	**503** A	**504** D	**505** D	**506** B	**507** D	**508** C	**509** C	**510** C
511 C	**512** C	**513** D	**514** B	**515** B	**516** C	**517** A	**518** C	**519** D	**520** C
521 C	**522** C	**523** A	**524** A	**525** C	**526** A	**527** C	**528** B	**529** B	**530** C
531 D	**532** B	**533** C	**534** B	**535** C	**536** D	**537** B	**538** D	**539** B	**540** D
541 B	**542** D	**543** C	**544** B	**545** B	**546** A	**547** D	**548** C	**549** A	**550** B
551 B	**552** B	**553** B	**554** B	**555** A	**556** A	**557** D	**558** C	**559** C	**560** C
561 D	**562** C	**563** C	**564** A	**565** C	**566** C	**567** D	**568** A	**569** B	**570** C
571 C	**572** B	**573** A	**574** C	**575** C	**576** B	**577** B	**578** A	**579** B	**580** A
581 C	**582** D	**583** C	**584** C	**585** C	**586** B	**587** C	**588** B	**589** A	**590** C
591 A	**592** A	**593** A	**594** B	**595** D	**596** A	**597** D	**598** A	**599** D	**600** B
601 B	**602** B	**603** C	**604** B	**605** A	**606** D	**607** B	**608** A	**609** C	**610** A
611 B	**612** C	**613** D	**614** C	**615** D	**616** D	**617** C	**618** B	**619** C	**620** B
621 A	**622** C	**623** D	**624** D	**625** D	**626** B	**627** B	**628** C	**629** B	**630** B
631 B	**632** D	**633** B	**634** B	**635** A	**636** C	**637** D	**638** A	**639** B	**640** C
641 B	**642** B	**643** A	**644** D	**645** B	**646** C	**647** D	**648** B	**649** A	**650** B
651 D	**652** C	**653** C	**654** C	**655** C	**656** C	**657** A	**658** C	**659** C	**660** A
661 D	**662** A	**663** C	**664** A	**665** B	**666** C	**667** C	**668** A	**669** A	**670** B
671 C	**672** B	**673** C	**674** A	**675** C	**676** B	**677** D	**678** D	**679** C	**680** A
681 A	**682** B	**683** A	**684** C	**685** C	**686** B	**687** D	**688** C	**689** C	**690** D
691 B	**692** A	**693** C	**694** B	**695** D	**696** C	**697** C	**698** A	**699** A	**700** D
701 B	**702** D	**703** A	**704** C	**705** A	**706** A	**707** C	**708** A	**709** B	**710** A
711 B	**712** C	**713** B	**714** B	**715** D	**716** A	**717** C	**718** C	**719** B	**720** C
721 A	**722** D	**723** D	**724** B	**725** A	**726** D	**727** D	**728** C	**729** C	**730** C
731 C	**732** D	**733** D	**734** D	**735** A	**736** A	**737** A	**738** B	**739** D	**740** C
741 B	**742** D	**743** C	**744** B	**745** B	**746** B	**747** A	**748** C	**749** D	**750** C
751 D	**752** C	**753** C	**754** A	**755** B	**756** C	**757** D	**758** D	**759** C	**760** C
761 D	**762** D	**763** D	**764** D	**765** B	**766** A	**767** C	**768** A	**769** A	**770** B
771 B	**772** D	**773** B	**774** B	**775** D	**776** C	**777** B	**778** B	**779** A	**780** C
781 A	**782** B	**783** D	**784** C	**785** C	**786** C	**787** B	**788** A	**789** B	**790** B
791 B	**792** C	**793** C	**794** D	**795** C	**796** A	**797** B	**798** B	**799** C	**800** D
801 C	**802** B	**803** C	**804** A	**805** B	**806** D	**807** B	**808** A	**809** B	**810** D
811 B	**812** A	**813** A	**814** A	**815** D	**816** C	**817** C	**818** D	**819** C	**820** C
821 A	**822** C	**823** A	**824** B	**825** A	**826** B	**827** C	**828** A	**829** B	**830** A
831 D	**832** B	**833** C	**834** D	**835** C	**836** D	**837** D	**838** A	**839** D	**840** B
841 A	**842** B	**843** B	**844** B	**845** A	**846** D	**847** C	**848** D	**849** B	**850** D
851 C	**852** B	**853** A	**854** C	**855** B	**856** A	**857** C	**858** C	**859** B	**860** D
861 C	**862** C	**863** D	**864** B	**865** A	**866** B	**867** C	**868** C	**869** B	**870** A
871 C	**872** B	**873** A	**874** A	**875** B	**876** D	**877** B	**878** D	**879** A	**880** B
881 A	**882** C	**883** C	**884** C	**885** D	**886** D	**887** C	**888** D	**889** B	**890** C
891 D	**892** B	**893** D	**894** D	**895** D	**896** A	**897** B	**898** C	**899** D	**900** D
901 C	**902** C	**903** B	**904** A	**905** B	**906** A	**907** A	**908** B	**909** D	**910** B
911 B	**912** D	**913** A	**914** C	**915** A	**916** C	**917** D	**918** A	**919** C	**920** D
921 A	**922** D	**923** A	**924** D	**925** A	**926** B	**927** A	**928** A	**929** D	**930** D
931 D	**932** B	**933** B	**934** D	**935** B	**936** A	**937** D	**938** B	**939** C	**940** D
941 D	**942** C	**943** B	**944** C	**945** B	**946** A	**947** A	**948** A	**949** B	**950** C
951 C	**952** D	**953** D	**954** A	**955** C	**956** D	**957** C	**958** A	**959** A	**960** B
961 B	**962** C	**963** A	**964** A	**965** C	**966** D	**967** A	**968** D	**969** A	**970** A
971 B	**972** D	**973** A	**974** A	**975** B	**976** B	**977** C	**978** A	**979** D	**980** D
981 B	**982** B	**983** C	**984** D	**985** C	**986** B	**987** B	**988** C	**989** B	**990** D
991 B	**992** D	**993** B	**994** D	**995** B	**996** C	**997** B	**998** C	**999** D	**1000** A

Part III – Worked solutions

1. Answer A Arithmetic · Easy

Multiplication precedes addition: 24 × 17 = 408, and 408 + 27 = 435.

2. Answer D Arithmetic · Easy

Multiplication precedes addition: 33 × 5 = 165, and 165 + 34 = 199.

3. Answer C Arithmetic · Easy

Multiplication precedes addition: 46 × 17 = 782, and 782 + 39 = 821.

4. Answer B Arithmetic · Easy

Multiplication precedes addition: 18 × 17 = 306, and 306 + 96 = 402.

5. Answer A Arithmetic · Easy

Multiplication precedes addition: 28 × 11 = 308, and 308 + 53 = 361.

6. Answer B Arithmetic · Easy

Multiplication precedes addition: 31 × 15 = 465, and 465 + 59 = 524.

7. Answer B Arithmetic · Easy

Multiplication precedes addition: 42 × 4 = 168, and 168 + 77 = 245.

8. Answer A Arithmetic · Medium

gcd(72,45) = 9; lcm = 72·45/9 = 360. Their sum is 369.

9. Answer B Arithmetic · Medium

gcd(110,66) = 22; lcm = 110·66/22 = 330. Their sum is 352.

10. Answer A Arithmetic · Medium

gcd(84,120) = 12; lcm = 84·120/12 = 840. Their sum is 852.

11. Answer B Arithmetic · Medium

gcd(63,99) = 9; lcm = 63·99/9 = 693. Their sum is 702.

12. Answer B Arithmetic · Medium

gcd(45,40) = 5; lcm = 45·40/5 = 360. Their sum is 365.

13. Answer A Arithmetic · Medium

gcd(15,27) = 3; lcm = 15·27/3 = 135. Their sum is 138.

14. Answer C Arithmetic · Medium

gcd(110,70) = 10; lcm = 110·70/10 = 770. Their sum is 780.

15. Answer B Arithmetic · Medium

gcd(80,64) = 16; lcm = 80·64/16 = 320. Their sum is 336.

16. Answer D Arithmetic · Hard

The order of 14 modulo 31 is 15 (since $14^{15} \equiv 1$). Because 252 = 16·15 + 12, $14^{252} \equiv 14^{12} \equiv 2 \pmod{31}$.

17. Answer A Arithmetic · Hard

The order of 5 modulo 9 is 6 (since $5^{6} \equiv 1$). Because 249 = 41·6 + 3, $5^{249} \equiv 5^{3} \equiv 8 \pmod{9}$.

18. Answer D Arithmetic · Hard

The order of 3 modulo 29 is 28 (since $3^{28} \equiv 1$). Because 239 = 8·28 + 15, $3^{239} \equiv 3^{15} \equiv 26 \pmod{29}$.

19. Answer D Arithmetic · Hard

The order of 67 modulo 23 is 22 (since $67^{22} \equiv 1$). Because 297 = 13·22 + 11, $67^{297} \equiv 67^{11} \equiv 22 \pmod{23}$.

20. Answer A Arithmetic · Hard

The order of 3 modulo 13 is 3 (since $3^{3} \equiv 1$). Because 216 = 72·3 + 0, $3^{216} \equiv 3^{0} \equiv 1 \pmod{13}$.

21. Answer C Arithmetic · Hard

The order of 70 modulo 31 is 5 (since $70^5 \equiv 1$). Because $228 = 45 \cdot 5 + 3$, $70^{228} \equiv 70^3 \equiv 16 \pmod{31}$.

22. Answer B Arithmetic · Hard

The order of 11 modulo 9 is 6 (since $11^6 \equiv 1$). Because $129 = 21 \cdot 6 + 3$, $11^{129} \equiv 11^3 \equiv 8 \pmod{9}$.

23. Answer C Pre-algebra · Easy

Use the common denominator 16: (2 + 8)/16 = 5/8 in lowest terms.

24. Answer A Pre-algebra · Easy

Use the common denominator 24: (6 + 16)/24 = 11/12 in lowest terms.

25. Answer A Pre-algebra · Easy

Use the common denominator 36: (6 + 12)/36 = 1/2 in lowest terms.

26. Answer A Pre-algebra · Easy

Use the common denominator 45: (30 + 9)/45 = 13/15 in lowest terms.

27. Answer C Pre-algebra · Easy

Use the common denominator 27: (18 + 15)/27 = 11/9 in lowest terms.

28. Answer D Pre-algebra · Easy

Use the common denominator 40: (16 + 10)/40 = 13/20 in lowest terms.

29. Answer C Pre-algebra · Easy

Use the common denominator 72: (54 + 32)/72 = 43/36 in lowest terms.

30. Answer B Pre-algebra · Medium

There are 9+5 = 14 parts, so one part is 56/14 = 4. The larger share is 9 × 4 = 36.

31. Answer A Pre-algebra · Medium

There are 4+2 = 6 parts, so one part is 48/6 = 8. The larger share is 4 × 8 = 32.

32. Answer A Pre-algebra · Medium

There are 9+3 = 12 parts, so one part is 96/12 = 8. The larger share is 9 × 8 = 72.

33. Answer B Pre-algebra · Medium

There are 2+3 = 5 parts, so one part is 60/5 = 12. The larger share is 3 × 12 = 36.

34. Answer B Pre-algebra · Medium

There are 4+9 = 13 parts, so one part is 156/13 = 12. The larger share is 9 × 12 = 108.

35. Answer B Pre-algebra · Medium

There are 9+6 = 15 parts, so one part is 150/15 = 10. The larger share is 9 × 10 = 90.

36. Answer B Pre-algebra · Medium

There are 5+6 = 11 parts, so one part is 132/11 = 12. The larger share is 6 × 12 = 72.

37. Answer A Pre-algebra · Medium

There are 6+9 = 15 parts, so one part is 60/15 = 4. The larger share is 9 × 4 = 36.

38. Answer A Pre-algebra · Hard

The sum of five numbers is 5·25 = 125; the sum of four is 4·23 = 92. The removed number is 125 − 92 = 33.

39. Answer A Pre-algebra · Hard

The sum of five numbers is 5·28 = 140; the sum of four is 4·19 = 76. The removed number is 140 − 76 = 64.

40. Answer B Pre-algebra · Hard

The sum of five numbers is 5·29 = 145; the sum of four is 4·18 = 72. The removed number is 145 − 72 = 73.

41. Answer D Pre-algebra · Hard

The sum of five numbers is 5·40 = 200; the sum of four is 4·21 = 84. The removed number is 200 − 84 = 116.

42. Answer C Pre-algebra · Hard

The sum of five numbers is 5·39 = 195; the sum of four is 4·24 = 96. The removed number is 195 − 96 = 99.

43. Answer D Pre-algebra · Hard

The sum of five numbers is 5·40 = 200; the sum of four is 4·15 = 60. The removed number is 200 − 60 = 140.

44. Answer D Pre-algebra · Hard

The sum of five numbers is 5·37 = 185; the sum of four is 4·15 = 60. The removed number is 185 − 60 = 125.

45. Answer A Algebra I · Easy

Subtract (-8) from both sides: 9x = 45. Divide by 9: x = 5.

46. Answer C Algebra I · Easy

Subtract (9) from both sides: 9x = 90. Divide by 9: x = 10.

47. Answer B Algebra I · Easy

Subtract (12) from both sides: 3x = 15. Divide by 3: x = 5.

48. Answer C Algebra I · Easy

Subtract (-5) from both sides: 6x = -18. Divide by 6: x = -3.

49. Answer B Algebra I · Easy

Subtract (-8) from both sides: 4x = -20. Divide by 4: x = -5.

50. Answer D Algebra I · Easy

Subtract (-1) from both sides: 8x = 56. Divide by 8: x = 7.

51. Answer A Algebra I · Easy

Subtract (6) from both sides: 6x = 6. Divide by 6: x = 1.

52. Answer B Algebra I · Medium

Vieta: r+s = 5, rs = -24. Then $r^2+s^2 = (r+s)^2 - 2rs = 25 - -48 = 73$.

53. Answer B Algebra I · Medium

Vieta: r+s = -14, rs = 48. Then $r^2+s^2 = (r+s)^2 - 2rs = 196 - 96 = 100$.

54. Answer A Algebra I · Medium

Vieta: r+s = -1, rs = -12. Then $r^2+s^2 = (r+s)^2 - 2rs = 1 - -24 = 25$.

55. Answer C Algebra I · Medium

Vieta: r+s = 5, rs = 6. Then $r^2+s^2 = (r+s)^2 - 2rs = 25 - 12 = 13$.

56. Answer D Algebra I · Medium

Vieta: r+s = 1, rs = -12. Then $r^2+s^2 = (r+s)^2 - 2rs = 1 - -24 = 25$.

57. Answer B Algebra I · Medium

Vieta: r+s = -1, rs = -42. Then $r^2+s^2 = (r+s)^2 - 2rs = 1 - -84 = 85$.

58. Answer A Algebra I · Medium

Vieta: r+s = 1, rs = -20. Then $r^2+s^2 = (r+s)^2 - 2rs = 1 - -40 = 41$.

59. Answer A Algebra I · Medium

Vieta: r+s = 5, rs = 4. Then $r^2+s^2 = (r+s)^2 - 2rs = 25 - 8 = 17$.

60. Answer D Algebra I · Hard

$x^3+y^3 = (x+y)^3 - 3xy(x+y) = -27 - 3\cdot-4\cdot-3 = -63$.

61. Answer D Algebra I · Hard

$x^3+y^3 = (x+y)^3 - 3xy(x+y) = -1331 - 3\cdot28\cdot-11 = -407$.

62. Answer C Algebra I · Hard

$x^3+y^3 = (x+y)^3 - 3xy(x+y) = -1 - 3\cdot-42\cdot-1 = -127$.

63. Answer D Algebra I · Hard

$x^3+y^3 = (x+y)^3 - 3xy(x+y) = -125 - 3\cdot0\cdot-5 = -125$.

64. Answer A Algebra I · Hard

$x^3+y^3 = (x+y)^3 - 3xy(x+y) = 1 - 3\cdot-30\cdot1 = 91$.

65. Answer B Algebra I · Hard

$x^3+y^3 = (x+y)^3 - 3xy(x+y) = 64 - 3\cdot0\cdot4 = 64$.

66. Answer D Algebra I · Hard

$x^3+y^3 = (x+y)^3 - 3xy(x+y) = -343 - 3\cdot6\cdot-7 = -217$.

67. Answer A Algebra II · Easy

By definition, $x = 4^3 = 64$.

68. Answer A Algebra II · Easy

By definition, $x = 4^5 = 1024$.

69. Answer A Algebra II · Easy

By definition, $x = 2^4 = 16$.

70. Answer B Algebra II · Easy

By definition, $x = 2^3 = 8$.

71. Answer D Algebra II · Easy

By definition, $x = 10^3 = 1000$.

72. Answer B Algebra II · Easy

By definition, $x = 3^4 = 81$.

73. Answer B Algebra II · Easy

By definition, $x = 10^5 = 100000$.

74. Answer A Algebra II · Medium

$S_n = a(r^n - 1)/(r - 1) = 2(2^8 - 1)/1 = 510$.

75. Answer D Algebra II · Medium

$S_n = a(r^n - 1)/(r - 1) = 1(2^6 - 1)/1 = 63$.

76. Answer B Algebra II · Medium

$S_n = a(r^n - 1)/(r - 1) = 2(2^5 - 1)/1 = 62$.

77. Answer A Algebra II · Medium

$S_n = a(r^n - 1)/(r - 1) = 2(3^7 - 1)/2 = 2186$.

78. Answer B Algebra II · Medium

$S_n = a(r^n - 1)/(r - 1) = 3(3^6 - 1)/2 = 1092$.

79. Answer C Algebra II · Medium

$S_n = a(r^n - 1)/(r - 1) = 4(2^8 - 1)/1 = 1020$.

80. Answer D Algebra II · Medium

$S_n = a(r^n - 1)/(r - 1) = 1(3^6 - 1)/2 = 364$.

81. Answer B Algebra II · Medium

$S_n = a(r^n - 1)/(r - 1) = 3(3^7 - 1)/2 = 3279$.

82. Answer A Algebra II · Hard

The remainder has the form mx + c with m+c = p(1) = -3 and 2m+c = p(2) = -5. So m = -2, c = -1.

83. Answer B Algebra II · Hard

The remainder has the form mx + c with m+c = p(1) = -1 and 2m+c = p(2) = 5. So m = 6, c = -7.

84. Answer A Algebra II · Hard

The remainder has the form mx + c with m+c = p(1) = -8 and 2m+c = p(2) = 6. So m = 14, c = -22.

85. Answer D Algebra II · Hard

The remainder has the form mx + c with m+c = p(1) = -3 and 2m+c = p(2) = -9. So m = -6, c = 3.

86. Answer C Algebra II · Hard

The remainder has the form mx + c with m+c = p(1) = 0 and 2m+c = p(2) = -9. So m = -9, c = 9.

87. Answer B Algebra II · Hard

The remainder has the form mx + c with m+c = p(1) = 1 and 2m+c = p(2) = 7. So m = 6, c = -5.

88. Answer B Algebra II · Hard

The remainder has the form mx + c with m+c = p(1) = 1 and 2m+c = p(2) = -5. So m = -6, c = 7.

89. Answer A Geometry · Easy

Pythagoras: $\sqrt{(9^2 + 40^2)} = \sqrt{1681} = 41$.

90. Answer B Geometry · Easy

Pythagoras: $\sqrt{(18^2 + 80^2)} = \sqrt{6724} = 82$.

91. Answer B Geometry · Easy

Pythagoras: $\sqrt{(7^2 + 24^2)} = \sqrt{625} = 25$.

92. Answer B Geometry · Easy

Pythagoras: $\sqrt{(10^2 + 24^2)} = \sqrt{676} = 26$.

93. Answer C Geometry · Easy

Pythagoras: $\sqrt{(40^2 + 42^2)} = \sqrt{3364} = 58$.

94. Answer D Geometry · Easy

Pythagoras: $\sqrt{(6^2 + 8^2)} = \sqrt{100} = 10$.

95. Answer B Geometry · Easy

Pythagoras: $\sqrt{(15^2 + 36^2)} = \sqrt{1521} = 39$.

96. Answer C Geometry · Medium

Interior angle = $(n - 2)\cdot 180°/n = 3\cdot 180/5 = 108°$.

97. Answer C Geometry · Medium

Interior angle = $(n - 2)\cdot 180°/n = 7\cdot 180/9 = 140°$.

98. Answer A Geometry · Medium

Interior angle = $(n - 2)\cdot 180°/n = 13\cdot 180/15 = 156°$.

99. Answer A Geometry · Medium

Interior angle = $(n - 2)\cdot 180°/n = 10\cdot 180/12 = 150°$.

100. Answer D Geometry · Medium

Interior angle = $(n - 2)\cdot 180°/n = 18\cdot 180/20 = 162°$.

101. Answer A Geometry · Medium

Interior angle = $(n - 2)\cdot 180°/n = 6\cdot 180/8 = 135°$.

102. Answer A Geometry · Medium

Interior angle = $(n - 2)\cdot 180°/n = 16\cdot 180/18 = 160°$.

103. Answer A Geometry · Medium

Interior angle = $(n - 2)\cdot 180°/n = 8\cdot 180/10 = 144°$.

104. Answer D Geometry · Hard

Heron: s = 9, area = $\sqrt{(s(s-a)(s-b)(s-c))} = 12$. Inradius r = Area/s = 4/3.

105. Answer B Geometry · Hard

Heron: s = 21, area = √(s(s−a)(s−b)(s−c)) = 42. Inradius r = Area/s = 2.

106. Answer C Geometry · Hard

Heron: s = 21, area = √(s(s−a)(s−b)(s−c)) = 84. Inradius r = Area/s = 4.

107. Answer A Geometry · Hard

Heron: s = 22, area = √(s(s−a)(s−b)(s−c)) = 66. Inradius r = Area/s = 3.

108. Answer D Geometry · Hard

Heron: s = 27, area = √(s(s−a)(s−b)(s−c)) = 36. Inradius r = Area/s = 4/3.

109. Answer A Geometry · Hard

Heron: s = 24, area = √(s(s−a)(s−b)(s−c)) = 84. Inradius r = Area/s = 7/2.

110. Answer A Geometry · Hard

Heron: s = 18, area = √(s(s−a)(s−b)(s−c)) = 60. Inradius r = Area/s = 10/3.

111. Answer D Trigonometry · Easy

This is a standard special-angle value: tan(60°) = √3.

112. Answer D Trigonometry · Easy

This is a standard special-angle value: tan(30°) = √3/3.

113. Answer A Trigonometry · Easy

This is a standard special-angle value: cos(0°) = 1.

114. Answer C Trigonometry · Easy

This is a standard special-angle value: tan(45°) = 1.

115. Answer B Trigonometry · Easy

This is a standard special-angle value: sin(90°) = 1.

116. Answer C Trigonometry · Easy

This is a standard special-angle value: cos(45°) = √2/2.

117. Answer B Trigonometry · Easy

This is a standard special-angle value: sin(45°) = √2/2.

118. Answer C Trigonometry · Medium

The opposite side is 24 and the hypotenuse is 25, so the adjacent side is $\sqrt{25^2-24^2} = 7$. tan θ = 24/7.

119. Answer B Trigonometry · Medium

The opposite side is 9 and the hypotenuse is 41, so the adjacent side is $\sqrt{41^2-9^2} = 40$. tan θ = 9/40.

120. Answer D Trigonometry · Medium

The opposite side is 12 and the hypotenuse is 13, so the adjacent side is $\sqrt{13^2-12^2} = 5$. tan θ = 12/5.

121. Answer A Trigonometry · Medium

The opposite side is 4 and the hypotenuse is 5, so the adjacent side is $\sqrt{5^2-4^2} = 3$. tan θ = 4/3.

122. Answer C Trigonometry · Medium

The opposite side is 15 and the hypotenuse is 17, so the adjacent side is $\sqrt{17^2-15^2} = 8$. tan θ = 15/8.

123. Answer B Trigonometry · Medium

The opposite side is 21 and the hypotenuse is 29, so the adjacent side is $\sqrt{29^2-21^2} = 20$. tan θ = 21/20.

124. Answer B Trigonometry · Medium

The opposite side is 7 and the hypotenuse is 25, so the adjacent side is $\sqrt{25^2-7^2} = 24$. tan θ = 7/24.

125. Answer D Trigonometry · Medium

The opposite side is 5 and the hypotenuse is 13, so the adjacent side is $\sqrt{13^2-5^2} = 12$. tan θ = 5/12.

126. Answer B Trigonometry · Hard

Law of cosines: $AB^2 = 12^2 + 8^2 - 2 \cdot 12 \cdot 8 \cdot \cos 60° = 112$. So $AB = \sqrt{112}$.

127. Answer A Trigonometry · Hard

Law of cosines: $AB^2 = 6^2 + 4^2 - 2 \cdot 6 \cdot 4 \cdot \cos 120° = 76$. So $AB = \sqrt{76}$.

128. Answer C Trigonometry · Hard

Law of cosines: $AB^2 = 6^2 + 3^2 - 2 \cdot 6 \cdot 3 \cdot \cos 120° = 63$. So $AB = \sqrt{63}$.

129. Answer D Trigonometry · Hard

Law of cosines: $AB^2 = 5^2 + 12^2 - 2 \cdot 5 \cdot 12 \cdot \cos 60° = 109$. So $AB = \sqrt{109}$.

130. Answer C Trigonometry · Hard

Law of cosines: $AB^2 = 8^2 + 4^2 - 2 \cdot 8 \cdot 4 \cdot \cos 60° = 48$. So $AB = \sqrt{48}$.

131. Answer C Trigonometry · Hard

Law of cosines: $AB^2 = 7^2 + 11^2 - 2 \cdot 7 \cdot 11 \cdot \cos 60° = 93$. So $AB = \sqrt{93}$.

132. Answer A Trigonometry · Hard

Law of cosines: $AB^2 = 5^2 + 4^2 - 2 \cdot 5 \cdot 4 \cdot \cos 120° = 61$. So $AB = \sqrt{61}$.

133. Answer C Precalculus · Easy

g(-2) = 18, then f(18) = 2·18 - 4 = 32.

134. Answer C Precalculus · Easy

g(-3) = 31, then f(31) = 2·31 + 3 = 65.

135. Answer D Precalculus · Easy

g(-3) = 13, then f(13) = 2·13 - 1 = 25.

136. Answer B Precalculus · Easy

g(-1) = 6, then f(6) = 3·6 - 5 = 13.

137. Answer A Precalculus · Easy

g(1) = 9, then f(9) = 5·9 + 0 = 45.

138. Answer D Precalculus · Easy

g(0) = 2, then f(2) = 2·2 - 2 = 2.

139. Answer C Precalculus · Easy

g(2) = 8, then f(8) = 3·8 - 3 = 21.

140. Answer B Precalculus · Medium

Solve (4x - 2)/(2x + 0) = 1: x(4 − 2·1) = 0·1 − (-2), giving x = 1.

141. Answer B Precalculus · Medium

Solve (2x + 1)/(3x + 1) = 1: x(2 − 3·1) = 1·1 − (1), giving x = 0.

142. Answer D Precalculus · Medium

Solve (3x + 0)/(3x + 2) = -2: x(3 − 3·-2) = 2·-2 − (0), giving x = -4/9.

143. Answer A Precalculus · Medium

Solve (2x + 0)/(3x - 1) = -2: x(2 − 3·-2) = -1·-2 − (0), giving x = 1/4.

144. Answer B Precalculus · Medium

Solve (4x + 3)/(3x - 1) = -1: x(4 − 3·-1) = -1·-1 − (3), giving x = -2/7.

145. Answer A Precalculus · Medium

Solve (3x + 3)/(3x + 0) = 4: x(3 − 3·4) = 0·4 − (3), giving x = 1/3.

146. Answer D Precalculus · Medium

Solve (2x + 2)/(2x + 4) = -1: x(2 − 2·-1) = 4·-1 − (2), giving x = -3/2.

147. Answer D Precalculus · Medium

Solve (3x - 3)/(3x + 2) = 3: x(3 − 3·3) = 2·3 − (-3), giving x = -3/2.

148. Answer C Precalculus · Hard

$1+i = \sqrt{2}\cdot e^{i\pi/4}$, so $(1+i)^8 = 2^{8/2}\cdot e^{i8\pi/4}$, which equals 16. (Check by repeated squaring: $(1+i)^2 = 2i$.)

149. Answer B Precalculus · Hard

$1+i = \sqrt{2}\cdot e^{i\pi/4}$, so $(1+i)^6 = 2^{6/2}\cdot e^{i6\pi/4}$, which equals -8i. (Check by repeated squaring: $(1+i)^2 = 2i$.)

150. Answer B Precalculus · Hard

$1+i = \sqrt{2}\cdot e^{i\pi/4}$, so $(1+i)^9 = 2^{9/2}\cdot e^{i9\pi/4}$, which equals 16 + 16i. (Check by repeated squaring: $(1+i)^2 = 2i$.)

151. Answer D Precalculus · Hard

$1+i = \sqrt{2}\cdot e^{i\pi/4}$, so $(1+i)^{11} = 2^{11/2}\cdot e^{i11\pi/4}$, which equals -32 + 32i. (Check by repeated squaring: $(1+i)^2 = 2i$.)

152. Answer A Precalculus · Hard

$1+i = \sqrt{2}\cdot e^{i\pi/4}$, so $(1+i)^{13} = 2^{13/2}\cdot e^{i13\pi/4}$, which equals -64 - 64i. (Check by repeated squaring: $(1+i)^2 = 2i$.)

153. Answer D Precalculus · Hard

$1+i = \sqrt{2}\cdot e^{i\pi/4}$, so $(1+i)^{14} = 2^{14/2}\cdot e^{i14\pi/4}$, which equals -128i. (Check by repeated squaring: $(1+i)^2 = 2i$.)

154. Answer D Precalculus · Hard

$1+i = \sqrt{2}\cdot e^{i\pi/4}$, so $(1+i)^{12} = 2^{12/2}\cdot e^{i12\pi/4}$, which equals -64. (Check by repeated squaring: $(1+i)^2 = 2i$.)

155. Answer B Single-variable calculus · Easy

$f'(x) = 12x^2 - 8x - 1$. Substituting x = -3 gives 131.

156. Answer D Single-variable calculus · Easy

$f'(x) = 15x^2 - 8x - 3$. Substituting x = -3 gives 156.

157. Answer D Single-variable calculus · Easy

$f'(x) = 3x^2 + 4x - 5$. Substituting x = -1 gives -6.

158. Answer B Single-variable calculus · Easy

$f'(x) = 6x^2 - 2x - 1$. Substituting x = -2 gives 27.

159. Answer B Single-variable calculus · Easy

$f'(x) = 3x^2 + 6x + 3$. Substituting x = 2 gives 27.

160. Answer D Single-variable calculus · Easy

$f'(x) = 9x^2 - 4x - 4$. Substituting x = -1 gives 9.

161. Answer A Single-variable calculus · Easy

$f'(x) = 3x^2 - 2x + 4$. Substituting x = 2 gives 12.

162. Answer D Single-variable calculus · Medium

Antiderivative: $1x^3/3 + 0x^2/2 + 2x$. Evaluate at 3 (it is 0 at 0): 15.

163. Answer B Single-variable calculus · Medium

Antiderivative: $1x^3/3 + -1x^2/2 + -2x$. Evaluate at 2 (it is 0 at 0): -10/3.

164. Answer B Single-variable calculus · Medium

Antiderivative: $1x^3/3 + -4x^2/2 + -1x$. Evaluate at 4 (it is 0 at 0): -44/3.

165. Answer C Single-variable calculus · Medium

Antiderivative: $2x^3/3 + 0x^2/2 + -3x$. Evaluate at 4 (it is 0 at 0): 92/3.

166. Answer C Single-variable calculus · Medium

Antiderivative: $2x^3/3 + 2x^2/2 + 3x$. Evaluate at 2 (it is 0 at 0): 46/3.

167. Answer D Single-variable calculus · Medium

Antiderivative: $4x^3/3 + 3x^2/2 + 5x$. Evaluate at 3 (it is 0 at 0): 129/2.

168. Answer A Single-variable calculus · Medium

Antiderivative: $4x^3/3 + 3x^2/2 + 5x$. Evaluate at 4 (it is 0 at 0): 388/3.

169. Answer B Single-variable calculus · Medium

Antiderivative: $4x^3/3 + 0x^2/2 + -2x$. Evaluate at 4 (it is 0 at 0): 232/3.

170. Answer D Single-variable calculus · Hard

Near 0: $1 - \cos(3x) \approx 3^2x^2/2$ and $x \cdot \sin(1x) \approx 1x^2$. The ratio tends to $3^2/(2 \cdot 1) = 9/2$.

171. Answer D Single-variable calculus · Hard

Near 0: $1 - \cos(2x) \approx 2^2x^2/2$ and $x \cdot \sin(5x) \approx 5x^2$. The ratio tends to $2^2/(2 \cdot 5) = 2/5$.

172. Answer C Single-variable calculus · Hard

Near 0: $1 - \cos(5x) \approx 5^2x^2/2$ and $x \cdot \sin(5x) \approx 5x^2$. The ratio tends to $5^2/(2 \cdot 5) = 5/2$.

173. Answer D Single-variable calculus · Hard

Near 0: $1 - \cos(2x) \approx 2^2x^2/2$ and $x \cdot \sin(3x) \approx 3x^2$. The ratio tends to $2^2/(2 \cdot 3) = 2/3$.

174. Answer D Single-variable calculus · Hard

Near 0: $1 - \cos(6x) \approx 6^2x^2/2$ and $x \cdot \sin(6x) \approx 6x^2$. The ratio tends to $6^2/(2 \cdot 6) = 3$.

175. Answer D Single-variable calculus · Hard

Near 0: $1 - \cos(6x) \approx 6^2x^2/2$ and $x \cdot \sin(5x) \approx 5x^2$. The ratio tends to $6^2/(2 \cdot 5) = 18/5$.

176. Answer A Single-variable calculus · Hard

Near 0: $1 - \cos(6x) \approx 6^2x^2/2$ and $x \cdot \sin(4x) \approx 4x^2$. The ratio tends to $6^2/(2 \cdot 4) = 9/2$.

177. Answer B Multivariable and vector calculus · Easy

$\partial f/\partial y = 4x^4y^3 + x$. At (3,2) this equals 2595.

178. Answer D Multivariable and vector calculus · Easy

$\partial f/\partial y = 2x^4y^1 + x$. At (1,1) this equals 3.

179. Answer B Multivariable and vector calculus · Easy

$\partial f/\partial y = 2x^2y^1 + x$. At (1,1) this equals 3.

180. Answer C Multivariable and vector calculus · Easy

$\partial f/\partial y = 2x^4y^1 + x$. At (3,3) this equals 489.

181. Answer D Multivariable and vector calculus · Easy

$\partial f/\partial y = 2x^4y^1 + x$. At (2,1) this equals 34.

182. Answer D Multivariable and vector calculus · Easy

$\partial f/\partial y = 3x^4y^2 + x$. At (2,1) this equals 50.

183. Answer A Multivariable and vector calculus · Easy

$\partial f/\partial y = 2x^4y^1 + x$. At (2,3) this equals 98.

184. Answer B Multivariable and vector calculus · Medium

$\nabla f = (2x + 2y, 2x + 6y) = (2, 6)$ at (0,1). The unit vector is (3/5, 4/5), so $D_u f = (6 + 24)/5 = 6$.

185. Answer C Multivariable and vector calculus · Medium

$\nabla f = (2x + -2y, -2x + 2y) = (4, -4)$ at (1,-1). The unit vector is (3/5, 4/5), so $D_u f = (12 + -16)/5 = -4/5$.

186. Answer C Multivariable and vector calculus · Medium

$\nabla f = (2x + 4y, 4x + 2y) = (8, 4)$ at (0,2). The unit vector is (3/5, 4/5), so $D_u f = (24 + 16)/5 = 8$.

187. Answer C Multivariable and vector calculus · Medium

$\nabla f = (2x + 4y, 4x + 6y) = (-10, -16)$ at (-1,-2). The unit vector is (3/5, 4/5), so $D_u f = (-30 + -64)/5 = -94/5$.

188. Answer B Multivariable and vector calculus · Medium

$\nabla f = (2x + -3y, -3x + 6y) = (-6, 12)$ at (0,2). The unit vector is (3/5, 4/5), so $D_u f = (-18 + 48)/5 = 6$.

189. Answer D Multivariable and vector calculus · Medium

$\nabla f = (2x + -1y, -1x + 2y) = (-4, 2)$ at (-2,0). The unit vector is (3/5, 4/5), so $D_u f = (-12 + 8)/5 = -4/5$.

190. Answer A Multivariable and vector calculus · Medium

$\nabla f = (2x + 0y, 0x + 6y) = (6, -12)$ at (3,-2). The unit vector is (3/5, 4/5), so $D_u f = (18 + -48)/5 = -6$.

191. Answer A Multivariable and vector calculus · Medium

$\nabla f = (2x + 1y, 1x + 2y) = (-1, 4)$ at (-2,3). The unit vector is (3/5, 4/5), so $D_u f = (-3 + 16)/5 = 13/5$.

192. Answer B Multivariable and vector calculus · Hard

Integrate termwise: $3 \cdot a^3b/3 + 1 \cdot a^2b^2/4 + 1 \cdot ab^3/3$ with a=2, b=2, giving 76/3.

193. Answer B Multivariable and vector calculus · Hard

Integrate termwise: $3 \cdot a^3b/3 + 4 \cdot a^2b^2/4 + 4 \cdot ab^3/3$ with a=1, b=4, giving 316/3.

194. Answer C Multivariable and vector calculus · Hard

Integrate termwise: $3 \cdot a^3b/3 + 2 \cdot a^2b^2/4 + 4 \cdot ab^3/3$ with a=4, b=2, giving 608/3.

195. Answer B Multivariable and vector calculus · Hard

Integrate termwise: $1 \cdot a^3b/3 + 3 \cdot a^2b^2/4 + 3 \cdot ab^3/3$ with a=2, b=3, giving 89.

196. Answer D Multivariable and vector calculus · Hard

Integrate termwise: $2 \cdot a^3b/3 + 3 \cdot a^2b^2/4 + 1 \cdot ab^3/3$ with a=2, b=2, giving 28.

197. Answer C Multivariable and vector calculus · Hard

Integrate termwise: $2 \cdot a^3b/3 + -3 \cdot a^2b^2/4 + 1 \cdot ab^3/3$ with a=3, b=2, giving 17.

198. Answer A Multivariable and vector calculus · Hard

Integrate termwise: $4 \cdot a^3b/3 + -2 \cdot a^2b^2/4 + 3 \cdot ab^3/3$ with a=3, b=3, giving 297/2.

199. Answer A Ordinary differential equations · Easy

The solution is $y = 3e^{3t}$, so $y(2) = 3e^{6}$.

200. Answer C Ordinary differential equations · Easy

The solution is $y = 2e^{-1t}$, so $y(2) = 2e^{-2}$.

201. Answer B Ordinary differential equations · Easy

The solution is $y = 5e^{-3t}$, so $y(2) = 5e^{-6}$.

202. Answer A Ordinary differential equations · Easy

The solution is $y = 2e^{-2t}$, so $y(2) = 2e^{-4}$.

203. Answer C Ordinary differential equations · Easy

The solution is $y = 6e^{-2t}$, so $y(2) = 6e^{-4}$.

204. Answer D Ordinary differential equations · Easy

The solution is $y = 4e^{3t}$, so $y(2) = 4e^{6}$.

205. Answer A Ordinary differential equations · Easy

The solution is $y = 5e^{-2t}$, so $y(2) = 5e^{-4}$.

206. Answer C Ordinary differential equations · Medium

The characteristic equation $\lambda^2 + 2\lambda - 8 = 0$ has roots -4 and 2.

207. Answer B Ordinary differential equations · Medium

The characteristic equation $\lambda^2 - 3\lambda + 2 = 0$ has roots 1 and 2.

208. Answer C Ordinary differential equations · Medium

The characteristic equation $\lambda^2 - 1\lambda - 2 = 0$ has roots -1 and 2.

209. Answer C Ordinary differential equations · Medium

The characteristic equation $\lambda^2 + 7\lambda + 12 = 0$ has roots -4 and -3.

210. Answer B Ordinary differential equations · Medium

The characteristic equation $\lambda^2 - 3\lambda - 4 = 0$ has roots -1 and 4.

211. Answer D Ordinary differential equations · Medium

The characteristic equation $\lambda^2 - 1\lambda - 6 = 0$ has roots -2 and 3.

212. Answer B Ordinary differential equations · Medium

The characteristic equation $\lambda^2 + 3\lambda + 2 = 0$ has roots -2 and -1.

213. Answer D Ordinary differential equations · Medium

The characteristic equation $\lambda^2 - 4\lambda + 3 = 0$ has roots 1 and 3.

214. Answer D Ordinary differential equations · Hard

Multiply by the integrating factor x: $(xy)' = x^3$, so $y = x^3/4 + C/x$. $y(1) = 1$ gives $C = 3/4$. Then $y(2) = 19/8$.

215. Answer D Ordinary differential equations · Hard

Multiply by the integrating factor x: $(xy)' = x^9$, so $y = x^9/10 + C/x$. $y(1) = 1$ gives $C = 9/10$. Then $y(2) = 1033/20$.

216. Answer C Ordinary differential equations · Hard

Multiply by the integrating factor x: $(xy)' = x^{14}$, so $y = x^{14}/15 + C/x$. $y(1) = 1$ gives $C = 14/15$. Then $y(2) = 16391/15$.

217. Answer D Ordinary differential equations · Hard

Multiply by the integrating factor x: $(xy)' = x^6$, so $y = x^6/7 + C/x$. $y(1) = 1$ gives $C = 6/7$. Then $y(2) = 67/7$.

218. Answer B Ordinary differential equations · Hard

Multiply by the integrating factor x: $(xy)' = x^8$, so $y = x^8/9 + C/x$. $y(1) = 1$ gives $C = 8/9$. Then $y(2) = 260/9$.

219. Answer C Ordinary differential equations · Hard

Multiply by the integrating factor x: $(xy)' = x^5$, so $y = x^5/6 + C/x$. $y(1) = 1$ gives $C = 5/6$. Then $y(2) = 23/4$.

220. Answer D Ordinary differential equations · Hard

Multiply by the integrating factor x: $(xy)' = x^{12}$, so $y = x^{12}/13 + C/x$. $y(1) = 1$ gives $C = 12/13$. Then $y(2) = 4102/13$.

221. Answer D Partial differential equations · Easy

The discriminant is $B^2 - 4AC = 0 - 12 = -12$; negative → elliptic.

222. Answer B Partial differential equations · Easy

The discriminant is $B^2 - 4AC = 16 - 12 = 4$; positive → hyperbolic.

223. Answer D Partial differential equations · Easy

The discriminant is $B^2 - 4AC = 4 - 64 = -60$; negative → elliptic.

224. Answer C Partial differential equations · Easy

The discriminant is $B^2 - 4AC = 1 - 8 = -7$; negative → elliptic.

225. Answer B Partial differential equations · Easy

The discriminant is $B^2 - 4AC = 1 - 24 = -23$; negative → elliptic.

226. Answer D Partial differential equations · Easy

The discriminant is $B^2 - 4AC = 0 - 0 = 0$; zero → parabolic.

227. Answer D Partial differential equations · Easy

The discriminant is $B^2 - 4AC = 16 - -48 = 64$; positive → hyperbolic.

228. Answer C Partial differential equations · Medium

Substituting gives $-\lambda = 2\cdot(-(n\pi/5)^2)$, so $\lambda = 2n^2\pi^2/5^2 = 2\pi^2$ for $n = 5$.

229. Answer B Partial differential equations · Medium

Substituting gives $-\lambda = 5\cdot(-(n\pi/6)^2)$, so $\lambda = 5n^2\pi^2/6^2 = 125\pi^2/36$ for $n = 5$.

230. Answer A Partial differential equations · Medium

Substituting gives $-\lambda = 5\cdot(-(n\pi/6)^2)$, so $\lambda = 5n^2\pi^2/6^2 = 5\pi^2/9$ for n = 2.

231. Answer B Partial differential equations · Medium

Substituting gives $-\lambda = 2\cdot(-(n\pi/3)^2)$, so $\lambda = 2n^2\pi^2/3^2 = 50\pi^2/9$ for n = 5.

232. Answer B Partial differential equations · Medium

Substituting gives $-\lambda = 2\cdot(-(n\pi/2)^2)$, so $\lambda = 2n^2\pi^2/2^2 = 25\pi^2/2$ for n = 5.

233. Answer A Partial differential equations · Medium

Substituting gives $-\lambda = 2\cdot(-(n\pi/5)^2)$, so $\lambda = 2n^2\pi^2/5^2 = 8\pi^2/25$ for n = 2.

234. Answer B Partial differential equations · Medium

Substituting gives $-\lambda = 3\cdot(-(n\pi/2)^2)$, so $\lambda = 3n^2\pi^2/2^2 = 12\pi^2$ for n = 4.

235. Answer D Partial differential equations · Medium

Substituting gives $-\lambda = 2\cdot(-(n\pi/6)^2)$, so $\lambda = 2n^2\pi^2/6^2 = 2\pi^2/9$ for n = 2.

236. Answer B Partial differential equations · Hard

d'Alembert: $u = [f(x+ct)+f(x-ct)]/2$ with $f(x)=x^3$, which simplifies to $x^3 + 3c^2xt^2$. At (4,2) with c=4: 832.

237. Answer B Partial differential equations · Hard

d'Alembert: $u = [f(x+ct)+f(x-ct)]/2$ with $f(x)=x^3$, which simplifies to $x^3 + 3c^2xt^2$. At (4,1) with c=4: 256.

238. Answer A Partial differential equations · Hard

d'Alembert: $u = [f(x+ct)+f(x-ct)]/2$ with $f(x)=x^3$, which simplifies to $x^3 + 3c^2xt^2$. At (2,3) with c=2: 224.

239. Answer D Partial differential equations · Hard

d'Alembert: $u = [f(x+ct)+f(x-ct)]/2$ with $f(x)=x^3$, which simplifies to $x^3 + 3c^2xt^2$. At (2,3) with c=4: 872.

240. Answer C Partial differential equations · Hard

d'Alembert: $u = [f(x+ct)+f(x-ct)]/2$ with $f(x)=x^3$, which simplifies to $x^3 + 3c^2xt^2$. At (2,2) with c=2: 104.

241. Answer B Partial differential equations · Hard

d'Alembert: $u = [f(x+ct)+f(x-ct)]/2$ with $f(x)=x^3$, which simplifies to $x^3 + 3c^2xt^2$. At (3,1) with c=3: 108.

242. Answer A Partial differential equations · Hard

d'Alembert: $u = [f(x+ct)+f(x-ct)]/2$ with $f(x)=x^3$, which simplifies to $x^3 + 3c^2xt^2$. At (4,3) with c=4: 1792.

243. Answer B Real analysis · Easy

Divide numerator and denominator by n: the limit is the ratio of leading coefficients, 8/7.

244. Answer A Real analysis · Easy

Divide numerator and denominator by n: the limit is the ratio of leading coefficients, 2/3.

245. Answer B Real analysis · Easy

Divide numerator and denominator by n: the limit is the ratio of leading coefficients, 8/6 = 4/3.

246. Answer B Real analysis · Easy

Divide numerator and denominator by n: the limit is the ratio of leading coefficients, 6/8 = 3/4.

247. Answer C Real analysis · Easy

Divide numerator and denominator by n: the limit is the ratio of leading coefficients, 5/1 = 5.

248. Answer D Real analysis · Easy

Divide numerator and denominator by n: the limit is the ratio of leading coefficients, 1/2.

249. Answer C Real analysis · Easy

Divide numerator and denominator by n: the limit is the ratio of leading coefficients, 5/8.

250. Answer D Real analysis · Medium

Ratio test: $|a_{n+1}/a_n| \to 1/9$, because the polynomial factor tends to ratio 1. Hence R = 9.

251. Answer D Real analysis · Medium

Ratio test: $|a_{n+1}/a_n| \to 1/5$, because the polynomial factor tends to ratio 1. Hence R = 5.

252. Answer B Real analysis · Medium

Ratio test: $|a_{n+1}/a_n| \to 1/6$, because the polynomial factor tends to ratio 1. Hence R = 6.

253. Answer C Real analysis · Medium

Ratio test: $|a_{n+1}/a_n| \to 1/8$, because the polynomial factor tends to ratio 1. Hence R = 8.

254. Answer C Real analysis · Medium

Ratio test: $|a_{n+1}/a_n| \to 1/7$, because the polynomial factor tends to ratio 1. Hence R = 7.

255. Answer D Real analysis · Medium

Ratio test: $|a_{n+1}/a_n| \to 1/4$, because the polynomial factor tends to ratio 1. Hence R = 4.

256. Answer D Real analysis · Medium

Ratio test: $|a_{n+1}/a_n| \to 1/4$, because the polynomial factor tends to ratio 1. Hence R = 4.

257. Answer A Real analysis · Medium

Ratio test: $|a_{n+1}/a_n| \to 1/3$, because the polynomial factor tends to ratio 1. Hence R = 3.

258. Answer D Real analysis · Hard

Partial fractions: 1/(n(n+8)) = (1/8)(1/n − 1/(n+8)). The series telescopes to (1/8)(1 + 1/2 + ... + 1/8) = 761/2240.

259. Answer A Real analysis · Hard

Partial fractions: 1/(n(n+7)) = (1/7)(1/n − 1/(n+7)). The series telescopes to (1/7)(1 + 1/2 + ... + 1/7) = 363/980.

260. Answer D Real analysis · Hard

Partial fractions: 1/(n(n+6)) = (1/6)(1/n − 1/(n+6)). The series telescopes to (1/6)(1 + 1/2 + ... + 1/6) = 49/120.

261. Answer A Real analysis · Hard

Partial fractions: 1/(n(n+4)) = (1/4)(1/n − 1/(n+4)). The series telescopes to (1/4)(1 + 1/2 + ... + 1/4) = 25/48.

262. Answer C Real analysis · Hard

Partial fractions: 1/(n(n+3)) = (1/3)(1/n − 1/(n+3)). The series telescopes to (1/3)(1 + 1/2 + ... + 1/3) = 11/18.

263. Answer D Real analysis · Hard

Partial fractions: 1/(n(n+2)) = (1/2)(1/n − 1/(n+2)). The series telescopes to (1/2)(1 + 1/2 + ... + 1/2) = 3/4.

264. Answer B Real analysis · Hard

Partial fractions: 1/(n(n+9)) = (1/9)(1/n − 1/(n+9)). The series telescopes to (1/9)(1 + 1/2 + ... + 1/9) = 7129/22680.

265. Answer A Complex analysis · Easy

$|z_1| = \sqrt{(8^2+15^2)} = 17$, $|z_2| = \sqrt{(20^2+21^2)} = 29$. The modulus of a quotient is the quotient of moduli: 17/29.

266. Answer D Complex analysis · Easy

$|z_1| = \sqrt{(7^2+24^2)} = 25$, $|z_2| = \sqrt{(8^2+15^2)} = 17$. The modulus of a quotient is the quotient of moduli: 25/17.

267. Answer B Complex analysis · Easy

$|z_1| = \sqrt{(7^2+24^2)} = 25$, $|z_2| = \sqrt{(3^2+4^2)} = 5$. The modulus of a quotient is the quotient of moduli: 25/5.

268. Answer C Complex analysis · Easy

$|z_1| = \sqrt{(5^2+12^2)} = 13$, $|z_2| = \sqrt{(7^2+24^2)} = 25$. The modulus of a quotient is the quotient of moduli: 13/25.

269. Answer A Complex analysis · Easy

$|z_1| = \sqrt{(8^2+15^2)} = 17$, $|z_2| = \sqrt{(3^2+4^2)} = 5$. The modulus of a quotient is the quotient of moduli: 17/5.

270. Answer A Complex analysis · Easy

$|z_1| = \sqrt{(5^2+12^2)} = 13$, $|z_2| = \sqrt{(20^2+21^2)} = 29$. The modulus of a quotient is the quotient of moduli: 13/29.

271. Answer D Complex analysis · Easy

$|z_1| = \sqrt{(3^2+4^2)} = 5$, $|z_2| = \sqrt{(8^2+15^2)} = 17$. The modulus of a quotient is the quotient of moduli: 5/17.

272. Answer C Complex analysis · Medium

Only the pole z = 2 lies inside |z| = 3 (|4| > 3). Its residue is 1/(2 − (4)) = -1/2, so the integral is 2πi · -1/2 = -πi.

273. Answer C Complex analysis · Medium

Only the pole z = 0 lies inside |z| = 1 (|4| > 1). Its residue is 1/(0 − (4)) = -1/4, so the integral is 2πi · -1/4 = (-1/2)πi.

274. Answer B Complex analysis · Medium

Only the pole z = -1 lies inside |z| = 2 (|6| > 2). Its residue is 1/(-1 − (6)) = -1/7, so the integral is 2πi · -1/7 = (-2/7)πi.

275. Answer B Complex analysis · Medium

Only the pole z = -1 lies inside |z| = 2 (|5| > 2). Its residue is 1/(-1 − (5)) = -1/6, so the integral is 2πi · -1/6 = (-1/3)πi.

276. Answer B Complex analysis · Medium

Only the pole z = 1 lies inside |z| = 2 (|6| > 2). Its residue is 1/(1 − (6)) = -1/5, so the integral is 2πi · -1/5 = (-2/5)πi.

277. Answer D Complex analysis · Medium

Only the pole z = 2 lies inside |z| = 3 (|-5| > 3). Its residue is 1/(2 − (-5)) = 1/7, so the integral is 2πi · 1/7 = (2/7)πi.

278. Answer B Complex analysis · Medium

Only the pole z = -2 lies inside |z| = 3 (|5| > 3). Its residue is 1/(-2 − (5)) = -1/7, so the integral is 2πi · -1/7 = (-2/7)πi.

279. Answer C Complex analysis · Medium

Only the pole z = 1 lies inside |z| = 2 (|3| > 2). Its residue is 1/(1 − (3)) = -1/2, so the integral is 2πi · -1/2 = -πi.

280. Answer D Complex analysis · Hard

The residue is the coefficient of z^{-1}: the term z^2 of e^{2z} is $2^2z^2/2!$, so the residue is 2.

281. Answer B Complex analysis · Hard

The residue is the coefficient of z^{-1}: the term z^3 of e^{3z} is $3^3z^3/3!$, so the residue is 9/2.

282. Answer D Complex analysis · Hard

The residue is the coefficient of z^{-1}: the term z^1 of e^{5z} is $5^1z^1/1!$, so the residue is 5.

283. Answer A Complex analysis · Hard

The residue is the coefficient of z^{-1}: the term z^2 of e^{4z} is $4^2z^2/2!$, so the residue is 8.

284. Answer D Complex analysis · Hard

The residue is the coefficient of z^{-1}: the term z^4 of e^{3z} is $3^4z^4/4!$, so the residue is 27/8.

285. Answer D Complex analysis · Hard

The residue is the coefficient of z^{-1}: the term z^5 of e^{3z} is $3^5z^5/5!$, so the residue is 81/40.

286. Answer D Complex analysis · Hard

The residue is the coefficient of z^{-1}: the term z^3 of e^{5z} is $5^3z^3/3!$, so the residue is 125/6.

287. Answer A Functional analysis · Easy

$\|x\|_1 = \Sigma|x_i| = 12$ and $\|x\|_\infty = \max|x_i| = 6$. The sum is 18.

288. Answer A Functional analysis · Easy

$\|x\|_1 = \Sigma|x_i| = 8$ and $\|x\|_\infty = \max|x_i| = 5$. The sum is 13.

289. Answer B Functional analysis · Easy

$\|x\|_1 = \Sigma|x_i| = 9$ and $\|x\|_\infty = \max|x_i| = 4$. The sum is 13.

290. Answer C Functional analysis · Easy

$\|x\|_1 = \Sigma|x_i| = 11$ and $\|x\|_\infty = \max|x_i| = 6$. The sum is 17.

291. Answer C Functional analysis · Easy

$\|x\|_1 = \Sigma|x_i| = 10$ and $\|x\|_\infty = \max|x_i| = 5$. The sum is 15.

292. Answer D Functional analysis · Easy

$\|x\|_1 = \Sigma|x_i| = 12$ and $\|x\|_\infty = \max|x_i| = 6$. The sum is 18.

293. Answer C Functional analysis · Easy

$\|x\|_1 = \Sigma|x_i| = 8$ and $\|x\|_\infty = \max|x_i| = 5$. The sum is 13.

294. Answer D Functional analysis · Medium

$1/q = 1 - 1/p = 1/5$, so $q = 5$.

295. Answer C Functional analysis · Medium

$1/q = 1 - 1/p = 5/7$, so $q = 7/5$.

296. Answer D Functional analysis · Medium

$1/q = 1 - 1/p = 4/5$, so $q = 5/4$.

297. Answer A Functional analysis · Medium

$1/q = 1 - 1/p = 2/5$, so $q = 5/2$.

298. Answer C Functional analysis · Medium

$1/q = 1 - 1/p = 2/3$, so $q = 3/2$.

299. Answer C Functional analysis · Medium

$1/q = 1 - 1/p = 3/4$, so $q = 4/3$.

300. Answer A Functional analysis · Medium

$1/q = 1 - 1/p = 5/6$, so $q = 6/5$.

301. Answer B Functional analysis · Medium

$1/q = 1 - 1/p = 1/3$, so $q = 3$.

302. Answer B Functional analysis · Hard

The norm of a diagonal operator is $\sup|d_n|$. The function $5t/(t^2+4)$ peaks at $t = 2$ with value $5/4$; $n = 2$ is an integer, so $\|T\| = 5/4$.

303. Answer B Functional analysis · Hard

The norm of a diagonal operator is $\sup|d_n|$. The function $4t/(t^2+4)$ peaks at $t = 2$ with value $4/4$; $n = 2$ is an integer, so $\|T\| = 1$.

304. Answer B Functional analysis · Hard

The norm of a diagonal operator is $\sup|d_n|$. The function $3t/(t^2+9)$ peaks at $t = 3$ with value $3/6$; $n = 3$ is an integer, so $\|T\| = 1/2$.

305. Answer C Functional analysis · Hard

The norm of a diagonal operator is $\sup|d_n|$. The function $4t/(t^2+25)$ peaks at $t = 5$ with value $4/10$; $n = 5$ is an integer, so $\|T\| = 2/5$.

306. Answer A Functional analysis · Hard

The norm of a diagonal operator is $\sup|d_n|$. The function $4t/(t^2+36)$ peaks at $t = 6$ with value $4/12$; $n = 6$ is an integer, so $\|T\| = 1/3$.

307. Answer D Functional analysis · Hard

The norm of a diagonal operator is sup|d_n|. The function $6t/(t^2+16)$ peaks at t = 4 with value 6/8; n = 4 is an integer, so ‖T‖ = 3/4.

308. Answer C Functional analysis · Hard

The norm of a diagonal operator is sup|d_n|. The function $9t/(t^2+16)$ peaks at t = 4 with value 9/8; n = 4 is an integer, so ‖T‖ = 9/8.

309. Answer C Measure theory · Easy

The intervals are disjoint, so measure adds: (5−1) + (14−8) = 10.

310. Answer A Measure theory · Easy

The intervals are disjoint, so measure adds: (2−0) + (10−5) = 7.

311. Answer C Measure theory · Easy

The intervals are disjoint, so measure adds: (5−0) + (13−8) = 10.

312. Answer A Measure theory · Easy

The intervals are disjoint, so measure adds: (5−0) + (9−7) = 7.

313. Answer C Measure theory · Easy

The intervals are disjoint, so measure adds: (9−5) + (16−13) = 7.

314. Answer C Measure theory · Easy

The intervals are disjoint, so measure adds: (8−3) + (15−12) = 8.

315. Answer C Measure theory · Easy

The intervals are disjoint, so measure adds: (4−0) + (10−8) = 6.

316. Answer D Measure theory · Medium

Each stage multiplies the total length by 1 − 1/7 = 6/7. After 2 stages: $(6/7)^2$ = 36/49.

317. Answer A Measure theory · Medium

Each stage multiplies the total length by 1 − 1/5 = 4/5. After 3 stages: $(4/5)^3$ = 64/125.

318. Answer A Measure theory · Medium

Each stage multiplies the total length by 1 − 1/3 = 2/3. After 4 stages: $(2/3)^4$ = 16/81.

319. Answer B Measure theory · Medium

Each stage multiplies the total length by 1 − 1/4 = 3/4. After 2 stages: $(3/4)^2$ = 9/16.

320. Answer B Measure theory · Medium

Each stage multiplies the total length by 1 − 1/4 = 3/4. After 3 stages: $(3/4)^3$ = 27/64.

321. Answer B Measure theory · Medium

Each stage multiplies the total length by 1 − 1/7 = 6/7. After 4 stages: $(6/7)^4$ = 1296/2401.

322. Answer A Measure theory · Medium

Each stage multiplies the total length by 1 − 1/6 = 5/6. After 2 stages: $(5/6)^2$ = 25/36.

323. Answer D Measure theory · Medium

Each stage multiplies the total length by 1 − 1/6 = 5/6. After 4 stages: $(5/6)^4$ = 625/1296.

324. Answer D Measure theory · Hard

Total removed = $\Sigma\ 2^{k-1}/9^k$ = (1/9)/(1 − 2/9) = 1/7. Measure = 1 − 1/7 = 6/7.

325. Answer D Measure theory · Hard

Total removed = $\Sigma\ 2^{k-1}/5^k$ = (1/5)/(1 − 2/5) = 1/3. Measure = 1 − 1/3 = 2/3.

326. Answer D Measure theory · Hard

Total removed = $\Sigma\ 2^{k-1}/7^k$ = (1/7)/(1 − 2/7) = 1/5. Measure = 1 − 1/5 = 4/5.

327. Answer A Measure theory · Hard

Total removed = $\Sigma\ 2^{k-1}/10^k$ = (1/10)/(1 − 2/10) = 1/8. Measure = 1 − 1/8 = 7/8.

328. Answer D Measure theory · Hard

Total removed = $\Sigma\ 2^{k-1}/12^k = (1/12)/(1 - 2/12) = 1/10$. Measure = $1 - 1/10 = 9/10$.

329. Answer A Measure theory · Hard

Total removed = $\Sigma\ 2^{k-1}/8^k = (1/8)/(1 - 2/8) = 1/6$. Measure = $1 - 1/6 = 5/6$.

330. Answer C Measure theory · Hard

Total removed = $\Sigma\ 2^{k-1}/11^k = (1/11)/(1 - 2/11) = 1/9$. Measure = $1 - 1/9 = 8/9$.

331. Answer C Linear algebra · Easy

For a 2×2 matrix, det = ad − bc = (3)·(0) − (-2)·(-4) = -8.

332. Answer D Linear algebra · Easy

For a 2×2 matrix, det = ad − bc = (-4)·(-1) − (1)·(0) = 4.

333. Answer D Linear algebra · Easy

For a 2×2 matrix, det = ad − bc = (-1)·(-6) − (1)·(4) = 2.

334. Answer D Linear algebra · Easy

For a 2×2 matrix, det = ad − bc = (-1)·(7) − (6)·(-6) = 29.

335. Answer A Linear algebra · Easy

For a 2×2 matrix, det = ad − bc = (3)·(-4) − (-1)·(-1) = -13.

336. Answer C Linear algebra · Easy

For a 2×2 matrix, det = ad − bc = (-5)·(-4) − (3)·(6) = 2.

337. Answer B Linear algebra · Easy

For a 2×2 matrix, det = ad − bc = (2)·(0) − (-4)·(3) = 12.

338. Answer D Linear algebra · Medium

Characteristic polynomial: $\lambda^2 + 6\lambda - 7 = 0$, discriminant 64. Roots: 1 and -7.

339. Answer B Linear algebra · Medium

Characteristic polynomial: $\lambda^2 - 10\lambda + 21 = 0$, discriminant 16. Roots: 7 and 3.

340. Answer B Linear algebra · Medium

Characteristic polynomial: $\lambda^2 + 1\lambda + 0 = 0$, discriminant 1. Roots: 0 and -1.

341. Answer C Linear algebra · Medium

Characteristic polynomial: $\lambda^2 + 2\lambda - 24 = 0$, discriminant 100. Roots: 4 and -6.

342. Answer D Linear algebra · Medium

Characteristic polynomial: $\lambda^2 - 3\lambda + 2 = 0$, discriminant 1. Roots: 2 and 1.

343. Answer C Linear algebra · Medium

Characteristic polynomial: $\lambda^2 + 9\lambda + 14 = 0$, discriminant 25. Roots: -2 and -7.

344. Answer D Linear algebra · Medium

Characteristic polynomial: $\lambda^2 + 1\lambda - 12 = 0$, discriminant 49. Roots: 3 and -4.

345. Answer A Linear algebra · Medium

Characteristic polynomial: $\lambda^2 + 6\lambda + 0 = 0$, discriminant 36. Roots: 0 and -6.

346. Answer D Linear algebra · Hard

The eigenvalues of p(A) are p(λ) for each eigenvalue λ, and det is their product: (2) · (26) · (-2) = -104.

347. Answer C Linear algebra · Hard

The eigenvalues of p(A) are p(λ) for each eigenvalue λ, and det is their product: (-1) · (4) · (-5) = 20.

348. Answer C Linear algebra · Hard

The eigenvalues of p(A) are p(λ) for each eigenvalue λ, and det is their product: (3) · (7) · (19) = 399.

349. Answer B Linear algebra · Hard

The eigenvalues of p(A) are p(λ) for each eigenvalue λ, and det is their product: (-2) · (-2) · (-4) = -16.

350. Answer B Linear algebra · Hard

The eigenvalues of p(A) are p(λ) for each eigenvalue λ, and det is their product: (-1) · (2) · (7) = -14.

351. Answer D Linear algebra · Hard

The eigenvalues of p(A) are p(λ) for each eigenvalue λ, and det is their product: (2) · (4) · (14) = 112.

352. Answer B Linear algebra · Hard

The eigenvalues of p(A) are p(λ) for each eigenvalue λ, and det is their product: (6) · (-3) · (1) = -18.

353. Answer B Abstract algebra: groups, rings, and fields · Easy

The order of k in Z_n is n/gcd(n,k) = 16/8 = 2.

354. Answer B Abstract algebra: groups, rings, and fields · Easy

The order of k in Z_n is n/gcd(n,k) = 32/4 = 8.

355. Answer C Abstract algebra: groups, rings, and fields · Easy

The order of k in Z_n is n/gcd(n,k) = 35/7 = 5.

356. Answer D Abstract algebra: groups, rings, and fields · Easy

The order of k in Z_n is n/gcd(n,k) = 49/7 = 7.

357. Answer A Abstract algebra: groups, rings, and fields · Easy

The order of k in Z_n is n/gcd(n,k) = 24/2 = 12.

358. Answer B Abstract algebra: groups, rings, and fields · Easy

The order of k in Z_n is n/gcd(n,k) = 42/3 = 14.

359. Answer D Abstract algebra: groups, rings, and fields · Easy

The order of k in Z_n is n/gcd(n,k) = 46/2 = 23.

360. Answer B Abstract algebra: groups, rings, and fields · Medium

The order of a product of disjoint cycles is the lcm of their lengths: lcm(2, 7) = 14.

361. Answer D Abstract algebra: groups, rings, and fields · Medium

The order of a product of disjoint cycles is the lcm of their lengths: lcm(2, 4, 5) = 20.

362. Answer C Abstract algebra: groups, rings, and fields · Medium

The order of a product of disjoint cycles is the lcm of their lengths: lcm(6, 4) = 12.

363. Answer C Abstract algebra: groups, rings, and fields · Medium

The order of a product of disjoint cycles is the lcm of their lengths: lcm(3, 2) = 6.

364. Answer D Abstract algebra: groups, rings, and fields · Medium

The order of a product of disjoint cycles is the lcm of their lengths: lcm(7, 2, 3) = 42.

365. Answer C Abstract algebra: groups, rings, and fields · Medium

The order of a product of disjoint cycles is the lcm of their lengths: lcm(7, 6, 5) = 210.

366. Answer B Abstract algebra: groups, rings, and fields · Medium

The order of a product of disjoint cycles is the lcm of their lengths: lcm(6, 7, 3) = 42.

367. Answer A Abstract algebra: groups, rings, and fields · Medium

The order of a product of disjoint cycles is the lcm of their lengths: lcm(5, 7, 2) = 70.

368. Answer B Abstract algebra: groups, rings, and fields · Hard

An element (x,y) has order lcm(ord x, ord y). Counting all 24 elements by order gives exactly 6 of order 6.

369. Answer A Abstract algebra: groups, rings, and fields · Hard

An element (x,y) has order lcm(ord x, ord y). Counting all 36 elements by order gives exactly 24 of order 6.

370. Answer B Abstract algebra: groups, rings, and fields · Hard

An element (x,y) has order lcm(ord x, ord y). Counting all 27 elements by order gives exactly 8 of order 3.

371. Answer C Abstract algebra: groups, rings, and fields · Hard

An element (x,y) has order lcm(ord x, ord y). Counting all 36 elements by order gives exactly 8 of order 3.

372. Answer A Abstract algebra: groups, rings, and fields · Hard

An element (x,y) has order lcm(ord x, ord y). Counting all 54 elements by order gives exactly 8 of order 6.

373. Answer D Abstract algebra: groups, rings, and fields · Hard

An element (x,y) has order lcm(ord x, ord y). Counting all 16 elements by order gives exactly 3 of order 2.

374. Answer C Abstract algebra: groups, rings, and fields · Hard

An element (x,y) has order lcm(ord x, ord y). Counting all 96 elements by order gives exactly 6 of order 6.

375. Answer A Number theory · Easy

$\varphi(n) = n \cdot \prod(1 - 1/p)$ over primes $p \mid n$. Here $147 = 3^1 \cdot 7^2$, so $\varphi(147) = 84$.

376. Answer A Number theory · Easy

$\varphi(n) = n \cdot \prod(1 - 1/p)$ over primes $p \mid n$. Here $74 = 2^1 \cdot 37^1$, so $\varphi(74) = 36$.

377. Answer B Number theory · Easy

$\varphi(n) = n \cdot \prod(1 - 1/p)$ over primes $p \mid n$. Here $194 = 2^1 \cdot 97^1$, so $\varphi(194) = 96$.

378. Answer B Number theory · Easy

$\varphi(n) = n \cdot \prod(1 - 1/p)$ over primes $p \mid n$. Here $93 = 3^1 \cdot 31^1$, so $\varphi(93) = 60$.

379. Answer A Number theory · Easy

$\varphi(n) = n \cdot \prod(1 - 1/p)$ over primes $p \mid n$. Here $55 = 5^1 \cdot 11^1$, so $\varphi(55) = 40$.

380. Answer A Number theory · Easy

$\varphi(n) = n \cdot \prod(1 - 1/p)$ over primes $p \mid n$. Here $153 = 3^2 \cdot 17^1$, so $\varphi(153) = 96$.

381. Answer A Number theory · Easy

$\varphi(n) = n \cdot \prod(1 - 1/p)$ over primes $p \mid n$. Here $111 = 3^1 \cdot 37^1$, so $\varphi(111) = 72$.

382. Answer B Number theory · Medium

The extended Euclidean algorithm gives $36 \cdot 43 = 1548 \equiv 1 \pmod{91}$, since $1548 = 17 \cdot 91 + 1$.

383. Answer C Number theory · Medium

The extended Euclidean algorithm gives $67 \cdot 12 = 804 \equiv 1 \pmod{73}$, since $804 = 11 \cdot 73 + 1$.

384. Answer A Number theory · Medium

The extended Euclidean algorithm gives $60 \cdot 60 = 3600 \equiv 1 \pmod{61}$, since $3600 = 59 \cdot 61 + 1$.

385. Answer D Number theory · Medium

The extended Euclidean algorithm gives $43 \cdot 37 = 1591 \equiv 1 \pmod{53}$, since $1591 = 30 \cdot 53 + 1$.

386. Answer A Number theory · Medium

The extended Euclidean algorithm gives $37 \cdot 29 = 1073 \equiv 1 \pmod{67}$, since $1073 = 16 \cdot 67 + 1$.

387. Answer B Number theory · Medium

The extended Euclidean algorithm gives $21 \cdot 41 = 861 \equiv 1 \pmod{43}$, since $861 = 20 \cdot 43 + 1$.

388. Answer A Number theory · Medium

The extended Euclidean algorithm gives $8 \cdot 27 = 216 \equiv 1 \pmod{43}$, since $216 = 5 \cdot 43 + 1$.

389. Answer A Number theory · Medium

The extended Euclidean algorithm gives $32 \cdot 32 = 1024 \equiv 1 \pmod{33}$, since $1024 = 31 \cdot 33 + 1$.

390. Answer C Number theory · Hard

The moduli are pairwise coprime, so the Chinese Remainder Theorem gives a unique solution modulo 180. Checking: 147 leaves remainders 3, 2, 3.

391. Answer C Number theory · Hard

The moduli are pairwise coprime, so the Chinese Remainder Theorem gives a unique solution modulo 105. Checking: 49 leaves remainders 1, 4, 0.

392. Answer A Number theory · Hard

The moduli are pairwise coprime, so the Chinese Remainder Theorem gives a unique solution modulo 280. Checking: 214 leaves remainders 4, 4, 6.

393. Answer B Number theory · Hard

The moduli are pairwise coprime, so the Chinese Remainder Theorem gives a unique solution modulo 105. Checking: 20 leaves remainders 2, 0, 6.

394. Answer C Number theory · Hard

The moduli are pairwise coprime, so the Chinese Remainder Theorem gives a unique solution modulo 180. Checking: 74 leaves remainders 2, 4, 2.

395. Answer B Number theory · Hard

The moduli are pairwise coprime, so the Chinese Remainder Theorem gives a unique solution modulo 360. Checking: 174 leaves remainders 4, 6, 3.

396. Answer C Number theory · Hard

The moduli are pairwise coprime, so the Chinese Remainder Theorem gives a unique solution modulo 105. Checking: 81 leaves remainders 0, 1, 4.

397. Answer A Galois theory · Easy

A field of order p^n has exactly $p^n = 4$ elements.

398. Answer A Galois theory · Easy

A field of order p^n has exactly $p^n = 16$ elements.

399. Answer C Galois theory · Easy

A field of order p^n has exactly $p^n = 343$ elements.

400. Answer C Galois theory · Easy

A field of order p^n has exactly $p^n = 81$ elements.

401. Answer B Galois theory · Easy

A field of order p^n has exactly $p^n = 2401$ elements.

402. Answer D Galois theory · Easy

A field of order p^n has exactly $p^n = 625$ elements.

403. Answer D Galois theory · Easy

A field of order p^n has exactly $p^n = 32$ elements.

404. Answer C Galois theory · Medium

Subfields of F_{p^n} correspond to divisors of n. 18 has $\tau(18) = 6$ divisors.

405. Answer D Galois theory · Medium

Subfields of F_{p^n} correspond to divisors of n. 16 has $\tau(16) = 5$ divisors.

406. Answer C Galois theory · Medium

Subfields of F_{p^n} correspond to divisors of n. 48 has $\tau(48) = 10$ divisors.

407. Answer B Galois theory · Medium

Subfields of F_{p^n} correspond to divisors of n. 18 has $\tau(18) = 6$ divisors.

408. Answer C Galois theory · Medium

Subfields of F_{p^n} correspond to divisors of n. 36 has $\tau(36) = 9$ divisors.

409. Answer B Galois theory · Medium

Subfields of F_{p^n} correspond to divisors of n. 36 has $\tau(36) = 9$ divisors.

410. Answer B Galois theory · Medium

Subfields of F_{p^n} correspond to divisors of n. 48 has $\tau(48) = 10$ divisors.

411. Answer A Galois theory · Medium

Subfields of F_{p^n} correspond to divisors of n. 20 has $\tau(20) = 6$ divisors.

412. Answer D Galois theory · Hard

$K = Q(\zeta_{11}, 2^{1/11})$. The degrees 10 and 11 are coprime, and both subextensions are as large as possible ($x^{11}-2$ is irreducible), so $[K:Q] = 11 \cdot 10 = 110$.

413. Answer C Galois theory · Hard

$K = Q(\zeta_7, 6^{1/7})$. The degrees 6 and 7 are coprime, and both subextensions are as large as possible (x^7-6 is irreducible), so $[K:Q] = 7 \cdot 6 = 42$.

414. Answer B Galois theory · Hard

$K = Q(\zeta_3, 7^{1/3})$. The degrees 2 and 3 are coprime, and both subextensions are as large as possible (x^3-7 is irreducible), so $[K:Q] = 3 \cdot 2 = 6$.

415. Answer A Galois theory · Hard

$K = Q(\zeta_7, 5^{1/7})$. The degrees 6 and 7 are coprime, and both subextensions are as large as possible (x^7-5 is irreducible), so $[K:Q] = 7 \cdot 6 = 42$.

416. Answer B Galois theory · Hard

$K = Q(\zeta_5, 3^{1/5})$. The degrees 4 and 5 are coprime, and both subextensions are as large as possible (x^5-3 is irreducible), so $[K:Q] = 5 \cdot 4 = 20$.

417. Answer D Galois theory · Hard

$K = Q(\zeta_5, 5^{1/5})$. The degrees 4 and 5 are coprime, and both subextensions are as large as possible (x^5-5 is irreducible), so $[K:Q] = 5 \cdot 4 = 20$.

418. Answer D Galois theory · Hard

$K = Q(\zeta_{11}, 3^{1/11})$. The degrees 10 and 11 are coprime, and both subextensions are as large as possible ($x^{11}-3$ is irreducible), so $[K:Q] = 11 \cdot 10 = 110$.

419. Answer D Representation theory · Easy

For abelian groups every irreducible complex representation is 1-dimensional, and the number of irreps equals the number of conjugacy classes = $|G| = 29$.

420. Answer C Representation theory · Easy

For abelian groups every irreducible complex representation is 1-dimensional, and the number of irreps equals the number of conjugacy classes = $|G| = 27$.

421. Answer B Representation theory · Easy

For abelian groups every irreducible complex representation is 1-dimensional, and the number of irreps equals the number of conjugacy classes = $|G| = 24$.

422. Answer B Representation theory · Easy

For abelian groups every irreducible complex representation is 1-dimensional, and the number of irreps equals the number of conjugacy classes = $|G| = 37$.

423. Answer B Representation theory · Easy

For abelian groups every irreducible complex representation is 1-dimensional, and the number of irreps equals the number of conjugacy classes = $|G| = 7$.

424. Answer A Representation theory · Easy

For abelian groups every irreducible complex representation is 1-dimensional, and the number of irreps equals the number of conjugacy classes = $|G| = 38$.

425. Answer D Representation theory · Easy

For abelian groups every irreducible complex representation is 1-dimensional, and the number of irreps equals the number of conjugacy classes = $|G| = 12$.

426. Answer D Representation theory · Medium

Irreps of S_n correspond to conjugacy classes, which correspond to partitions of n. p(4) = 5.

427. Answer C Representation theory · Medium

Irreps of S_n correspond to conjugacy classes, which correspond to partitions of n. p(7) = 15.

428. Answer C Representation theory · Medium

Irreps of S_n correspond to conjugacy classes, which correspond to partitions of n. p(14) = 135.

429. Answer A Representation theory · Medium

Irreps of S_n correspond to conjugacy classes, which correspond to partitions of n. p(9) = 30.

430. Answer B Representation theory · Medium

Irreps of S_n correspond to conjugacy classes, which correspond to partitions of n. p(12) = 77.

431. Answer B Representation theory · Medium

Irreps of S_n correspond to conjugacy classes, which correspond to partitions of n. p(5) = 7.

432. Answer C Representation theory · Medium

Irreps of S_n correspond to conjugacy classes, which correspond to partitions of n. p(6) = 11.

433. Answer C Representation theory · Medium

Irreps of S_n correspond to conjugacy classes, which correspond to partitions of n. p(8) = 22.

434. Answer A Representation theory · Hard

The sum of squares of irreducible dimensions equals |G|. Known squares sum to 20, so the missing one satisfies $d^2 = 24 - 20 = 4$, hence d = 2.

435. Answer B Representation theory · Hard

The sum of squares of irreducible dimensions equals |G|. Known squares sum to 51, so the missing one satisfies $d^2 = 60 - 51 = 9$, hence d = 3.

436. Answer C Representation theory · Hard

The sum of squares of irreducible dimensions equals |G|. Known squares sum to 20, so the missing one satisfies $d^2 = 24 - 20 = 4$, hence d = 2.

437. Answer B Representation theory · Hard

The sum of squares of irreducible dimensions equals |G|. Known squares sum to 15, so the missing one satisfies $d^2 = 24 - 15 = 9$, hence d = 3.

438. Answer B Representation theory · Hard

The sum of squares of irreducible dimensions equals |G|. Known squares sum to 3, so the missing one satisfies $d^2 = 12 - 3 = 9$, hence d = 3.

439. Answer C Representation theory · Hard

The sum of squares of irreducible dimensions equals |G|. Known squares sum to 35, so the missing one satisfies $d^2 = 60 - 35 = 25$, hence d = 5.

440. Answer C Representation theory · Hard

The sum of squares of irreducible dimensions equals |G|. Known squares sum to 95, so the missing one satisfies $d^2 = 120 - 95 = 25$, hence d = 5.

441. Answer C Commutative algebra · Easy

Maximal ideals of Z/nZ correspond to the distinct primes dividing n. $56 = 2^3 \cdot 7$, so there are 2.

442. Answer D Commutative algebra · Easy

Maximal ideals of Z/nZ correspond to the distinct primes dividing n. $24 = 2^3 \cdot 3$, so there are 2.

443. Answer C Commutative algebra · Easy

Maximal ideals of Z/nZ correspond to the distinct primes dividing n. $324 = 2^2 \cdot 3^4$, so there are 2.

444. Answer D Commutative algebra · Easy

Maximal ideals of Z/nZ correspond to the distinct primes dividing n. $244 = 2^2 \cdot 61$, so there are 2.

445. Answer A Commutative algebra · Easy

Maximal ideals of Z/nZ correspond to the distinct primes dividing n. $315 = 3^2 \cdot 5 \cdot 7$, so there are 3.

446. Answer D Commutative algebra · Easy

Maximal ideals of Z/nZ correspond to the distinct primes dividing n. $99 = 3^2 \cdot 11$, so there are 2.

447. Answer C Commutative algebra · Easy

Maximal ideals of Z/nZ correspond to the distinct primes dividing n. 317 = 317, so there are 1.

448. Answer C Commutative algebra · Medium

A basis is $x^i y^j$ with $0 \le i < 3$ and $0 \le j < 7$, giving $3 \cdot 7 = 21$ basis elements.

449. Answer B Commutative algebra · Medium

A basis is $x^i y^j$ with $0 \le i < 7$ and $0 \le j < 3$, giving $7 \cdot 3 = 21$ basis elements.

450. Answer C Commutative algebra · Medium

A basis is $x^i y^j$ with $0 \le i < 5$ and $0 \le j < 4$, giving $5 \cdot 4 = 20$ basis elements.

451. Answer A Commutative algebra · Medium

A basis is $x^i y^j$ with $0 \le i < 7$ and $0 \le j < 4$, giving $7 \cdot 4 = 28$ basis elements.

452. Answer D Commutative algebra · Medium

A basis is $x^i y^j$ with $0 \le i < 7$ and $0 \le j < 2$, giving $7 \cdot 2 = 14$ basis elements.

453. Answer C Commutative algebra · Medium

A basis is $x^i y^j$ with $0 \le i < 7$ and $0 \le j < 5$, giving $7 \cdot 5 = 35$ basis elements.

454. Answer D Commutative algebra · Medium

A basis is $x^i y^j$ with $0 \le i < 2$ and $0 \le j < 7$, giving $2 \cdot 7 = 14$ basis elements.

455. Answer D Commutative algebra · Medium

A basis is $x^i y^j$ with $0 \le i < 5$ and $0 \le j < 2$, giving $5 \cdot 2 = 10$ basis elements.

456. Answer B Commutative algebra · Hard

x is nilpotent mod n iff every prime dividing n divides x, i.e. rad(n) = 14 divides x. The multiples of 14 in Z/98Z number 98/14 = 7.

457. Answer A Commutative algebra · Hard

x is nilpotent mod n iff every prime dividing n divides x, i.e. rad(n) = 14 divides x. The multiples of 14 in Z/28Z number 28/14 = 2.

458. Answer A Commutative algebra · Hard

x is nilpotent mod n iff every prime dividing n divides x, i.e. rad(n) = 14 divides x. The multiples of 14 in Z/112Z number 112/14 = 8.

459. Answer D Commutative algebra · Hard

x is nilpotent mod n iff every prime dividing n divides x, i.e. rad(n) = 22 divides x. The multiples of 22 in Z/44Z number 44/22 = 2.

460. Answer B Commutative algebra · Hard

x is nilpotent mod n iff every prime dividing n divides x, i.e. rad(n) = 6 divides x. The multiples of 6 in Z/72Z number 72/6 = 12.

461. Answer D Commutative algebra · Hard

x is nilpotent mod n iff every prime dividing n divides x, i.e. rad(n) = 10 divides x. The multiples of 10 in Z/100Z number 100/10 = 10.

462. Answer C Commutative algebra · Hard

x is nilpotent mod n iff every prime dividing n divides x, i.e. rad(n) = 15 divides x. The multiples of 15 in Z/75Z number 75/15 = 5.

463. Answer D Euclidean geometry · Easy

By the inscribed angle theorem, the inscribed angle is half the central angle: 110°/2 = 55°.

464. Answer A Euclidean geometry · Easy

By the inscribed angle theorem, the inscribed angle is half the central angle: 70°/2 = 35°.

465. Answer C Euclidean geometry · Easy

By the inscribed angle theorem, the inscribed angle is half the central angle: 60°/2 = 30°.

466. Answer B Euclidean geometry · Easy

By the inscribed angle theorem, the inscribed angle is half the central angle: 100°/2 = 50°.

467. Answer D Euclidean geometry · Easy

By the inscribed angle theorem, the inscribed angle is half the central angle: 130°/2 = 65°.

468. Answer A Euclidean geometry · Easy

By the inscribed angle theorem, the inscribed angle is half the central angle: 150°/2 = 75°.

469. Answer A Euclidean geometry · Easy

By the inscribed angle theorem, the inscribed angle is half the central angle: 90°/2 = 45°.

470. Answer C Euclidean geometry · Medium

Intersecting chords theorem: PA·PB = PC·PD, so PD = 9·2/3 = 6.

471. Answer D Euclidean geometry · Medium

Intersecting chords theorem: PA·PB = PC·PD, so PD = 8·10/4 = 20.

472. Answer C Euclidean geometry · Medium

Intersecting chords theorem: PA·PB = PC·PD, so PD = 9·11/3 = 33.

473. Answer B Euclidean geometry · Medium

Intersecting chords theorem: PA·PB = PC·PD, so PD = 10·4/8 = 5.

474. Answer C Euclidean geometry · Medium

Intersecting chords theorem: PA·PB = PC·PD, so PD = 8·11/4 = 22.

475. Answer D Euclidean geometry · Medium

Intersecting chords theorem: PA·PB = PC·PD, so PD = 12·11/4 = 33.

476. Answer A Euclidean geometry · Medium

Intersecting chords theorem: PA·PB = PC·PD, so PD = 2·6/4 = 3.

477. Answer B Euclidean geometry · Medium

Intersecting chords theorem: PA·PB = PC·PD, so PD = 6·6/4 = 9.

478. Answer C Euclidean geometry · Hard

Area K = 126 (Heron). Circumradius R = abc/(4K) = 5460/(4·126) = 65/6.

479. Answer B Euclidean geometry · Hard

Area K = 204 (Heron). Circumradius R = abc/(4K) = 11050/(4·204) = 325/24.

480. Answer D Euclidean geometry · Hard

Area K = 66 (Heron). Circumradius R = abc/(4K) = 2860/(4·66) = 65/6.

481. Answer A Euclidean geometry · Hard

Area K = 12 (Heron). Circumradius R = abc/(4K) = 150/(4·12) = 25/8.

482. Answer B Euclidean geometry · Hard

Area K = 60 (Heron). Circumradius R = abc/(4K) = 4056/(4·60) = 169/10.

483. Answer B Euclidean geometry · Hard

Area K = 168 (Heron). Circumradius R = abc/(4K) = 8750/(4·168) = 625/48.

484. Answer B Euclidean geometry · Hard

Area K = 24 (Heron). Circumradius R = abc/(4K) = 780/(4·24) = 65/8.

485. Answer D Non-Euclidean geometry · Easy

Arc length = R·θ with θ in radians: 6·(60π/180) = 2π.

486. Answer C Non-Euclidean geometry · Easy

Arc length = R·θ with θ in radians: 5·(120π/180) = 10π/3.

487. Answer C Non-Euclidean geometry · Easy

Arc length = R·θ with θ in radians: 7·(120π/180) = 14π/3.

488. Answer D Non-Euclidean geometry · Easy

Arc length = R·θ with θ in radians: 6·(90π/180) = 3π.

489. Answer B Non-Euclidean geometry · Easy

Arc length = R·θ with θ in radians: 6·(135π/180) = 9π/2.

490. Answer A Non-Euclidean geometry · Easy

Arc length = R·θ with θ in radians: 4·(60π/180) = 4π/3.

491. Answer C Non-Euclidean geometry · Easy

Arc length = R·θ with θ in radians: 3·(150π/180) = 5π/2.

492. Answer A Non-Euclidean geometry · Medium

Girard's theorem: Area = $R^2(A + B + C - \pi) = 3^2 \cdot (412° - 180°) = 9 \cdot 232°$ in radians = 58π/5.

493. Answer C Non-Euclidean geometry · Medium

Girard's theorem: Area = $R^2(A + B + C - \pi) = 1^2 \cdot (332° - 180°) = 1 \cdot 152°$ in radians = 38π/45.

494. Answer B Non-Euclidean geometry · Medium

Girard's theorem: Area = $R^2(A + B + C - \pi) = 2^2 \cdot (323° - 180°) = 4 \cdot 143°$ in radians = 143π/45.

495. Answer A Non-Euclidean geometry · Medium

Girard's theorem: Area = $R^2(A + B + C - \pi) = 5^2 \cdot (414° - 180°) = 25 \cdot 234°$ in radians = 65π/2.

496. Answer C Non-Euclidean geometry · Medium

Girard's theorem: Area = $R^2(A + B + C - \pi) = 5^2 \cdot (291° - 180°) = 25 \cdot 111°$ in radians = 185π/12.

497. Answer C Non-Euclidean geometry · Medium

Girard's theorem: Area = $R^2(A + B + C - \pi) = 4^2 \cdot (306° - 180°) = 16 \cdot 126°$ in radians = 56π/5.

498. Answer B Non-Euclidean geometry · Medium

Girard's theorem: Area = $R^2(A + B + C - \pi) = 2^2 \cdot (370° - 180°) = 4 \cdot 190°$ in radians = 38π/9.

499. Answer D Non-Euclidean geometry · Medium

Girard's theorem: Area = $R^2(A + B + C - \pi) = 1^2 \cdot (304° - 180°) = 1 \cdot 124°$ in radians = 31π/45.

500. Answer A Non-Euclidean geometry · Hard

For curvature $-1/R^2$, Area = $R^2(\pi -$ sum of angles). Here $R^2 = 9$, so $\pi -$ sum = 9/20π, i.e. the angle sum is 99°. Third angle = 99° − 33° − 54° = 12°.

501. Answer A Non-Euclidean geometry · Hard

For curvature $-1/R^2$, Area = $R^2(\pi -$ sum of angles). Here $R^2 = 9$, so $\pi -$ sum = 7/20π, i.e. the angle sum is 117°. Third angle = 117° − 43° − 41° = 33°.

502. Answer C Non-Euclidean geometry · Hard

For curvature $-1/R^2$, Area = $R^2(\pi -$ sum of angles). Here $R^2 = 4$, so $\pi -$ sum = 8/45π, i.e. the angle sum is 148°. Third angle = 148° − 65° − 34° = 49°.

503. Answer A Non-Euclidean geometry · Hard

For curvature $-1/R^2$, Area = $R^2(\pi -$ sum of angles). Here $R^2 = 16$, so $\pi -$ sum = 73/180π, i.e. the angle sum is 107°. Third angle = 107° − 19° − 57° = 31°.

504. Answer D Non-Euclidean geometry · Hard

For curvature $-1/R^2$, Area = $R^2(\pi -$ sum of angles). Here $R^2 = 4$, so $\pi -$ sum = 41/90π, i.e. the angle sum is 98°. Third angle = 98° − 13° − 52° = 33°.

505. Answer D Non-Euclidean geometry · Hard

For curvature $-1/R^2$, Area = $R^2(\pi - \text{sum of angles})$. Here $R^2 = 4$, so $\pi - \text{sum} = 17/36\pi$, i.e. the angle sum is 95°. Third angle = 95° − 36° − 41° = 18°.

506. Answer B Non-Euclidean geometry · Hard

For curvature $-1/R^2$, Area = $R^2(\pi - \text{sum of angles})$. Here $R^2 = 9$, so $\pi - \text{sum} = 67/180\pi$, i.e. the angle sum is 113°. Third angle = 113° − 23° − 18° = 72°.

507. Answer D Analytic geometry · Easy

Distance = $\sqrt{((9)^2 + (40)^2)} = 41$.

508. Answer C Analytic geometry · Easy

Distance = $\sqrt{((16)^2 + (30)^2)} = 34$.

509. Answer C Analytic geometry · Easy

Distance = $\sqrt{((60)^2 + (63)^2)} = 87$.

510. Answer C Analytic geometry · Easy

Distance = $\sqrt{((21)^2 + (72)^2)} = 75$.

511. Answer C Analytic geometry · Easy

Distance = $\sqrt{((3)^2 + (4)^2)} = 5$.

512. Answer C Analytic geometry · Easy

Distance = $\sqrt{((10)^2 + (24)^2)} = 26$.

513. Answer D Analytic geometry · Easy

Distance = $\sqrt{((20)^2 + (21)^2)} = 29$.

514. Answer B Analytic geometry · Medium

Distance = $|ax_0 + by_0 + c|/\sqrt{(a^2+b^2)} = |-76|/17 = 76/17$.

515. Answer B Analytic geometry · Medium

Distance = $|ax_0 + by_0 + c|/\sqrt{(a^2+b^2)} = |-19|/5 = 19/5$.

516. Answer C Analytic geometry · Medium

Distance = $|ax_0 + by_0 + c|/\sqrt{(a^2+b^2)} = |-34|/17 = 2$.

517. Answer A Analytic geometry · Medium

Distance = $|ax_0 + by_0 + c|/\sqrt{(a^2+b^2)} = |8|/17 = 8/17$.

518. Answer C Analytic geometry · Medium

Distance = $|ax_0 + by_0 + c|/\sqrt{(a^2+b^2)} = |-4|/13 = 4/13$.

519. Answer D Analytic geometry · Medium

Distance = $|ax_0 + by_0 + c|/\sqrt{(a^2+b^2)} = |-21|/13 = 21/13$.

520. Answer C Analytic geometry · Medium

Distance = $|ax_0 + by_0 + c|/\sqrt{(a^2+b^2)} = |-46|/17 = 46/17$.

521. Answer C Analytic geometry · Medium

Distance = $|ax_0 + by_0 + c|/\sqrt{(a^2+b^2)} = |-7|/17 = 7/17$.

522. Answer C Analytic geometry · Hard

Here $c^2 = a^2 - b^2 = 676 - 100$, so c = 24 and the foci are (±24, 0). The triangle has base 2c = 48 and height b = 10, so area = ½·48·10 = 240.

523. Answer A Analytic geometry · Hard

Here $c^2 = a^2 - b^2 = 6724 - 324$, so c = 80 and the foci are (±80, 0). The triangle has base 2c = 160 and height b = 18, so area = ½·160·18 = 1440.

524. Answer A Analytic geometry · Hard

Here $c^2 = a^2 - b^2 = 100 - 64$, so c = 6 and the foci are (±6, 0). The triangle has base 2c = 12 and height b = 8, so area = ½·12·8 = 48.

525. Answer C Analytic geometry · Hard

Here $c^2 = a^2 - b^2 = 289 - 64$, so $c = 15$ and the foci are $(\pm 15, 0)$. The triangle has base $2c = 30$ and height $b = 8$, so area $= \tfrac{1}{2} \cdot 30 \cdot 8 = 120$.

526. Answer A Analytic geometry · Hard

Here $c^2 = a^2 - b^2 = 676 - 576$, so $c = 10$ and the foci are $(\pm 10, 0)$. The triangle has base $2c = 20$ and height $b = 24$, so area $= \tfrac{1}{2} \cdot 20 \cdot 24 = 240$.

527. Answer C Analytic geometry · Hard

Here $c^2 = a^2 - b^2 = 625 - 49$, so $c = 24$ and the foci are $(\pm 24, 0)$. The triangle has base $2c = 48$ and height $b = 7$, so area $= \tfrac{1}{2} \cdot 48 \cdot 7 = 168$.

528. Answer B Analytic geometry · Hard

Here $c^2 = a^2 - b^2 = 3364 - 1600$, so $c = 42$ and the foci are $(\pm 42, 0)$. The triangle has base $2c = 84$ and height $b = 40$, so area $= \tfrac{1}{2} \cdot 84 \cdot 40 = 1680$.

529. Answer B Differential geometry · Easy

Curvature of a circle is 1/radius = 1/4.

530. Answer C Differential geometry · Easy

Curvature of a circle is 1/radius = 1/6.

531. Answer D Differential geometry · Easy

The principal curvatures are both 1/5; their product is 1/25.

532. Answer B Differential geometry · Easy

Curvature of a circle is 1/radius = 1/2.

533. Answer C Differential geometry · Easy

The principal curvatures are both 1/9; their product is 1/81.

534. Answer B Differential geometry · Easy

Curvature of a circle is 1/radius = 1/3.

535. Answer C Differential geometry · Easy

The principal curvatures are both 1/6; their product is 1/36.

536. Answer D Differential geometry · Medium

For a helix of radius a and pitch parameter b, $\kappa = a/(a^2 + b^2) = 3/(9+1) = 3/10$.

537. Answer B Differential geometry · Medium

For a helix of radius a and pitch parameter b, $\kappa = a/(a^2 + b^2) = 2/(4+9) = 2/13$.

538. Answer D Differential geometry · Medium

For a helix of radius a and pitch parameter b, $\kappa = a/(a^2 + b^2) = 4/(16+36) = 1/13$.

539. Answer B Differential geometry · Medium

For a helix of radius a and pitch parameter b, $\kappa = a/(a^2 + b^2) = 6/(36+1) = 6/37$.

540. Answer D Differential geometry · Medium

For a helix of radius a and pitch parameter b, $\kappa = a/(a^2 + b^2) = 1/(1+25) = 1/26$.

541. Answer B Differential geometry · Medium

For a helix of radius a and pitch parameter b, $\kappa = a/(a^2 + b^2) = 5/(25+1) = 5/26$.

542. Answer D Differential geometry · Medium

For a helix of radius a and pitch parameter b, $\kappa = a/(a^2 + b^2) = 2/(4+25) = 2/29$.

543. Answer C Differential geometry · Medium

For a helix of radius a and pitch parameter b, $\kappa = a/(a^2 + b^2) = 3/(9+36) = 1/15$.

544. Answer B Differential geometry · Hard

Gauss–Bonnet: $\iint K \, dA = 2\pi\chi = 2\pi(2 - 2g) = -20\pi$.

545. Answer B Differential geometry · Hard

$\kappa = |y''|/(1 + y'^2)^{3/2}$. With $y'' = 6$ and $y' = 4/3$, $1 + y'^2 = 25/9$, so $\kappa = 6/(5/3)^3 = 162/125$.

546. Answer A Differential geometry · Hard

$\kappa = |y''|/(1 + y'^2)^{3/2}$. With $y'' = 2$ and $y' = 4/3$, $1 + y'^2 = 25/9$, so $\kappa = 2/(5/3)^3 = 54/125$.

547. Answer D Differential geometry · Hard

Gauss–Bonnet: $\iint K\, dA = 2\pi\chi = 2\pi(2 - 2g) = -8\pi$.

548. Answer C Differential geometry · Hard

$\kappa = |y''|/(1 + y'^2)^{3/2}$. With $y'' = 4$ and $y' = 15/8$, $1 + y'^2 = 289/64$, so $\kappa = 4/(17/8)^3 = 2048/4913$.

549. Answer A Differential geometry · Hard

$\kappa = |y''|/(1 + y'^2)^{3/2}$. With $y'' = 4$ and $y' = 3/4$, $1 + y'^2 = 25/16$, so $\kappa = 4/(5/4)^3 = 256/125$.

550. Answer B Differential geometry · Hard

$\kappa = |y''|/(1 + y'^2)^{3/2}$. With $y'' = 4$ and $y' = 12/5$, $1 + y'^2 = 169/25$, so $\kappa = 4/(13/5)^3 = 500/2197$.

551. Answer B Algebraic geometry · Easy

Bézout's theorem: the number is $2 \cdot 4 = 8$.

552. Answer B Algebraic geometry · Easy

Bézout's theorem: the number is $2 \cdot 6 = 12$.

553. Answer B Algebraic geometry · Easy

Bézout's theorem: the number is $4 \cdot 6 = 24$.

554. Answer B Algebraic geometry · Easy

Bézout's theorem: the number is $7 \cdot 2 = 14$.

555. Answer A Algebraic geometry · Easy

Bézout's theorem: the number is $4 \cdot 4 = 16$.

556. Answer A Algebraic geometry · Easy

Bézout's theorem: the number is $2 \cdot 7 = 14$.

557. Answer D Algebraic geometry · Easy

Bézout's theorem: the number is $6 \cdot 4 = 24$.

558. Answer C Algebraic geometry · Medium

Genus formula: $(d - 1)(d - 2)/2 = 2 \cdot 1/2 = 1$.

559. Answer C Algebraic geometry · Medium

Genus formula: $(d - 1)(d - 2)/2 = 7 \cdot 6/2 = 21$.

560. Answer C Algebraic geometry · Medium

Genus formula: $(d - 1)(d - 2)/2 = 3 \cdot 2/2 = 3$.

561. Answer D Algebraic geometry · Medium

Genus formula: $(d - 1)(d - 2)/2 = 4 \cdot 3/2 = 6$.

562. Answer C Algebraic geometry · Medium

Genus formula: $(d - 1)(d - 2)/2 = 5 \cdot 4/2 = 10$.

563. Answer C Algebraic geometry · Medium

Genus formula: $(d - 1)(d - 2)/2 = 8 \cdot 7/2 = 28$.

564. Answer A Algebraic geometry · Medium

Genus formula: $(d - 1)(d - 2)/2 = 9 \cdot 8/2 = 36$.

565. Answer C Algebraic geometry · Medium

Genus formula: $(d - 1)(d - 2)/2 = 6 \cdot 5/2 = 15$.

566. Answer C Algebraic geometry · Hard

Count affine solutions $(x,y) \in \mathbb{F}_5^2$ by checking each x: 7 solutions. Add the point at infinity: 8. (Hasse bound check: $|8 - 6| \le 2\sqrt{5}$.)

567. Answer D Algebraic geometry · Hard

Count affine solutions $(x,y) \in \mathbb{F}_{11}^2$ by checking each x: 12 solutions. Add the point at infinity: 13. (Hasse bound check: $|13 - 12| \le 2\sqrt{11}$.)

568. Answer A Algebraic geometry · Hard

Count affine solutions $(x,y) \in \mathbb{F}_7^2$ by checking each x: 6 solutions. Add the point at infinity: 7. (Hasse bound check: $|7 - 8| \le 2\sqrt{7}$.)

569. Answer B Algebraic geometry · Hard

Count affine solutions $(x,y) \in \mathbb{F}_{11}^2$ by checking each x: 15 solutions. Add the point at infinity: 16. (Hasse bound check: $|16 - 12| \le 2\sqrt{11}$.)

570. Answer C Algebraic geometry · Hard

Count affine solutions $(x,y) \in \mathbb{F}_{11}^2$ by checking each x: 16 solutions. Add the point at infinity: 17. (Hasse bound check: $|17 - 12| \le 2\sqrt{11}$.)

571. Answer C Algebraic geometry · Hard

Count affine solutions $(x,y) \in \mathbb{F}_7^2$ by checking each x: 7 solutions. Add the point at infinity: 8. (Hasse bound check: $|8 - 8| \le 2\sqrt{7}$.)

572. Answer B Algebraic geometry · Hard

Count affine solutions $(x,y) \in \mathbb{F}_5^2$ by checking each x: 7 solutions. Add the point at infinity: 8. (Hasse bound check: $|8 - 6| \le 2\sqrt{5}$.)

573. Answer A General topology · Easy

Closed sets are the complements of open sets: ∅ and {j, ..., 6}. The smallest closed set containing A starts at min A = 1, so cl(A) = {1, 2, 3, 4, 5, 6}.

574. Answer C General topology · Easy

Closed sets are the complements of open sets: ∅ and {j, ..., 9}. The smallest closed set containing A starts at min A = 2, so cl(A) = {2, 3, 4, 5, 6, 7, 8, 9}.

575. Answer C General topology · Easy

Closed sets are the complements of open sets: ∅ and {j, ..., 9}. The smallest closed set containing A starts at min A = 3, so cl(A) = {3, 4, 5, 6, 7, 8, 9}.

576. Answer B General topology · Easy

Closed sets are the complements of open sets: ∅ and {j, ..., 9}. The smallest closed set containing A starts at min A = 3, so cl(A) = {3, 4, 5, 6, 7, 8, 9}.

577. Answer B General topology · Easy

Closed sets are the complements of open sets: ∅ and {j, ..., 7}. The smallest closed set containing A starts at min A = 4, so cl(A) = {4, 5, 6, 7}.

578. Answer A General topology · Easy

Closed sets are the complements of open sets: ∅ and {j, ..., 7}. The smallest closed set containing A starts at min A = 3, so cl(A) = {3, 4, 5, 6, 7}.

579. Answer B General topology · Easy

Closed sets are the complements of open sets: ∅ and {j, ..., 7}. The smallest closed set containing A starts at min A = 2, so cl(A) = {2, 3, 4, 5, 6, 7}.

580. Answer A General topology · Medium

An open set is determined by which of the 6 blocks it contains, giving $2^6 = 64$ open sets.

581. Answer C General topology · Medium

An open set is determined by which of the 3 blocks it contains, giving $2^3 = 8$ open sets.

582. Answer D General topology · Medium

An open set is determined by which of the 3 blocks it contains, giving $2^3 = 8$ open sets.

583. Answer C General topology · Medium

An open set is determined by which of the 6 blocks it contains, giving $2^6 = 64$ open sets.

584. Answer C General topology · Medium

An open set is determined by which of the 5 blocks it contains, giving $2^5 = 32$ open sets.

585. Answer C General topology · Medium

An open set is determined by which of the 6 blocks it contains, giving $2^6 = 64$ open sets.

586. Answer B General topology · Medium

An open set is determined by which of the 5 blocks it contains, giving $2^5 = 32$ open sets.

587. Answer C General topology · Medium

An open set is determined by which of the 3 blocks it contains, giving $2^3 = 8$ open sets.

588. Answer B General topology · Hard

Close the family under finite intersections to get a base of 5 sets (including X), then take all unions of base sets, including the empty union. This yields 6 distinct open sets.

589. Answer A General topology · Hard

A map is continuous iff every preimage of an open initial segment is an initial segment (or ∅). Enumerating all $3^3 = 27$ maps and testing this condition gives 10.

590. Answer C General topology · Hard

Close the family under finite intersections to get a base of 6 sets (including X), then take all unions of base sets, including the empty union. This yields 7 distinct open sets.

591. Answer A General topology · Hard

Close the family under finite intersections to get a base of 7 sets (including X), then take all unions of base sets, including the empty union. This yields 9 distinct open sets.

592. Answer A General topology · Hard

A map is continuous iff every preimage of an open initial segment is an initial segment (or ∅). Enumerating all $4^4 = 256$ maps and testing this condition gives 35.

593. Answer A General topology · Hard

Close the family under finite intersections to get a base of 7 sets (including X), then take all unions of base sets, including the empty union. This yields 10 distinct open sets.

594. Answer B General topology · Hard

A map is continuous iff every preimage of an open initial segment is an initial segment (or ∅). Enumerating all $5^5 = 3125$ maps and testing this condition gives 126.

595. Answer D Algebraic topology · Easy

$\chi = 2 - 2g = 2 - 0 = 2$.

596. Answer A Algebraic topology · Easy

$\chi = 2 - 2g = 2 - 2 = 0$.

597. Answer D Algebraic topology · Easy

$\chi = 2 - 2g = 2 - 6 = -4$.

598. Answer A Algebraic topology · Easy

$\chi = 2 - 2g = 2 - 12 = -10$.

599. Answer D Algebraic topology · Easy

$\chi = 2 - 2g = 2 - 10 = -8$.

600. Answer B Algebraic topology · Easy

$\chi = 2 - 2g = 2 - 8 = -6$.

601. Answer B Algebraic topology · Easy

$\chi = 2 - 2g = 2 - 14 = -12$.

602. Answer B Algebraic topology · Medium

A connected graph is homotopy equivalent to a wedge of $E - V + 1$ circles, so rank $H_1 = 19 - 12 + 1 = 8$.

603. Answer C Algebraic topology · Medium

A connected graph is homotopy equivalent to a wedge of $E - V + 1$ circles, so rank $H_1 = 10 - 7 + 1 = 4$.

604. Answer B Algebraic topology · Medium

A connected graph is homotopy equivalent to a wedge of $E - V + 1$ circles, so rank $H_1 = 10 - 9 + 1 = 2$.

605. Answer A Algebraic topology · Medium

A connected graph is homotopy equivalent to a wedge of $E - V + 1$ circles, so rank $H_1 = 6 - 5 + 1 = 2$.

606. Answer D Algebraic topology · Medium

A connected graph is homotopy equivalent to a wedge of $E - V + 1$ circles, so rank $H_1 = 14 - 7 + 1 = 8$.

607. Answer B Algebraic topology · Medium

A connected graph is homotopy equivalent to a wedge of $E - V + 1$ circles, so rank $H_1 = 15 - 7 + 1 = 9$.

608. Answer A Algebraic topology · Medium

A connected graph is homotopy equivalent to a wedge of $E - V + 1$ circles, so rank $H_1 = 16 - 12 + 1 = 5$.

609. Answer C Algebraic topology · Medium

A connected graph is homotopy equivalent to a wedge of $E - V + 1$ circles, so rank $H_1 = 13 - 5 + 1 = 9$.

610. Answer A Algebraic topology · Hard

Euler characteristic multiplies by the number of sheets: $2 - 2g' = 2(2 - 2\cdot5)$, so $g' = 2(5-1) + 1 = 9$.

611. Answer B Algebraic topology · Hard

χ is multiplicative for products: $\chi(\Sigma_4)\cdot\chi(S^2) = (-6)\cdot2 = -12$.

612. Answer C Algebraic topology · Hard

χ is multiplicative for products: $\chi(\Sigma_6)\cdot\chi(S^4) = (-10)\cdot2 = -20$.

613. Answer D Algebraic topology · Hard

χ is multiplicative for products: $\chi(\Sigma_2)\cdot\chi(S^2) = (-2)\cdot2 = -4$.

614. Answer C Algebraic topology · Hard

Euler characteristic multiplies by the number of sheets: $2 - 2g' = 4(2 - 2\cdot2)$, so $g' = 4(2-1) + 1 = 5$.

615. Answer D Algebraic topology · Hard

χ is multiplicative for products: $\chi(\Sigma_2)\cdot\chi(S^6) = (-2)\cdot2 = -4$.

616. Answer D Algebraic topology · Hard

Euler characteristic multiplies by the number of sheets: $2 - 2g' = 3(2 - 2\cdot5)$, so $g' = 3(5-1) + 1 = 13$.

617. Answer C Combinatorics · Easy

$C(6,3) = 6!/(3!\cdot3!) = 20$.

618. Answer B Combinatorics · Easy

$C(10,4) = 10!/(4!\cdot6!) = 210$.

619. Answer C Combinatorics · Easy

$C(7,2) = 7!/(2!\cdot5!) = 21$.

620. Answer B Combinatorics · Easy

$C(12,5) = 12!/(5!\cdot7!) = 792$.

621. Answer A Combinatorics · Easy

$C(9,2) = 9!/(2!\cdot7!) = 36$.

622. Answer C Combinatorics · Easy

C(14,3) = 14!/(3!·11!) = 364.

623. Answer D Combinatorics · Easy

C(11,4) = 11!/(4!·7!) = 330.

624. Answer D Combinatorics · Medium

Divide 7! by the factorials of repeat counts: 7!/(3!·1!·2!·1!) = 420.

625. Answer D Combinatorics · Medium

Divide 10! by the factorials of repeat counts: 10!/(1!·4!·3!·2!) = 12600.

626. Answer B Combinatorics · Medium

Divide 8! by the factorials of repeat counts: 8!/(3!·1!·3!·1!) = 1120.

627. Answer B Combinatorics · Medium

Divide 5! by the factorials of repeat counts: 5!/(3!·1!·1!) = 20.

628. Answer C Combinatorics · Medium

Divide 10! by the factorials of repeat counts: 10!/(2!·4!·2!·2!) = 18900.

629. Answer B Combinatorics · Medium

Divide 8! by the factorials of repeat counts: 8!/(3!·1!·4!) = 280.

630. Answer B Combinatorics · Medium

Divide 7! by the factorials of repeat counts: 7!/(3!·2!·2!) = 210.

631. Answer B Combinatorics · Medium

Divide 11! by the factorials of repeat counts: 11!/(3!·2!·2!·4!) = 69300.

632. Answer D Combinatorics · Hard

Inclusion–exclusion: $\Sigma_j (-1)^j C(3,j)(3-j)^6 = 540$ (confirmed by direct enumeration).

633. Answer B Combinatorics · Hard

Inclusion–exclusion: $\Sigma_j (-1)^j C(4,j)(4-j)^4 = 24$ (confirmed by direct enumeration).

634. Answer B Combinatorics · Hard

Inclusion–exclusion: $\Sigma_j (-1)^j C(2,j)(2-j)^4 = 14$ (confirmed by direct enumeration).

635. Answer A Combinatorics · Hard

Inclusion–exclusion: $\Sigma_j (-1)^j C(4,j)(4-j)^6 = 1560$ (confirmed by direct enumeration).

636. Answer C Combinatorics · Hard

Inclusion–exclusion: $\Sigma_j (-1)^j C(2,j)(2-j)^7 = 126$ (confirmed by direct enumeration).

637. Answer D Combinatorics · Hard

Inclusion–exclusion: $\Sigma_j (-1)^j C(3,j)(3-j)^4 = 36$ (confirmed by direct enumeration).

638. Answer A Combinatorics · Hard

Inclusion–exclusion: $\Sigma_j (-1)^j C(4,j)(4-j)^5 = 240$ (confirmed by direct enumeration).

639. Answer B Graph theory · Easy

Each pair of vertices is joined once: C(10,2) = 45.

640. Answer C Graph theory · Easy

Each pair of vertices is joined once: C(12,2) = 66.

641. Answer B Graph theory · Easy

Each pair of vertices is joined once: C(14,2) = 91.

642. Answer B Graph theory · Easy

Each pair of vertices is joined once: C(13,2) = 78.

643. Answer A Graph theory · Easy

Each pair of vertices is joined once: C(11,2) = 55.

644. Answer D Graph theory · Easy

Each pair of vertices is joined once: C(8,2) = 28.

645. Answer B Graph theory · Easy

Each pair of vertices is joined once: C(9,2) = 36.

646. Answer C Graph theory · Medium

Euler's formula V − E + F = 2 gives F = 9 − 7 + 2 = 4.

647. Answer D Graph theory · Medium

Euler's formula V − E + F = 2 gives F = 14 − 7 + 2 = 9.

648. Answer B Graph theory · Medium

Cayley's formula: n^(n−2) = 5^3 = 125.

649. Answer A Graph theory · Medium

Cayley's formula: n^(n−2) = 7^5 = 16807.

650. Answer B Graph theory · Medium

Euler's formula V − E + F = 2 gives F = 17 − 15 + 2 = 4.

651. Answer D Graph theory · Medium

Euler's formula V − E + F = 2 gives F = 18 − 11 + 2 = 9.

652. Answer C Graph theory · Medium

Cayley's formula: n^(n−2) = 8^6 = 262144.

653. Answer C Graph theory · Medium

Euler's formula V − E + F = 2 gives F = 13 − 9 + 2 = 6.

654. Answer C Graph theory · Hard

The chromatic polynomial of C_n is $(k-1)^n + (-1)^n(k-1)$. With n=4, k=4: 84 (confirmed by enumeration).

655. Answer C Graph theory · Hard

The chromatic polynomial of C_n is $(k-1)^n + (-1)^n(k-1)$. With n=8, k=4: 6564 (confirmed by enumeration).

656. Answer C Graph theory · Hard

The chromatic polynomial of C_n is $(k-1)^n + (-1)^n(k-1)$. With n=5, k=4: 240 (confirmed by enumeration).

657. Answer A Graph theory · Hard

The chromatic polynomial of C_n is $(k-1)^n + (-1)^n(k-1)$. With n=6, k=3: 66 (confirmed by enumeration).

658. Answer C Graph theory · Hard

The chromatic polynomial of C_n is $(k-1)^n + (-1)^n(k-1)$. With n=6, k=4: 732 (confirmed by enumeration).

659. Answer C Graph theory · Hard

The chromatic polynomial of C_n is $(k-1)^n + (-1)^n(k-1)$. With n=4, k=3: 18 (confirmed by enumeration).

660. Answer A Graph theory · Hard

The chromatic polynomial of C_n is $(k-1)^n + (-1)^n(k-1)$. With n=7, k=3: 126 (confirmed by enumeration).

661. Answer D Set theory · Easy

Inclusion–exclusion: 22 + 38 − 8 = 52.

662. Answer A Set theory · Easy

Inclusion–exclusion: 12 + 27 − 6 = 33.

663. Answer C Set theory · Easy

Inclusion–exclusion: 20 + 32 − 7 = 45.

664. Answer A Set theory · Easy

Inclusion–exclusion: 37 + 11 − 9 = 39.

665. Answer B Set theory · Easy

Inclusion–exclusion: 13 + 12 − 2 = 23.

666. Answer C Set theory · Easy

Inclusion–exclusion: 13 + 12 − 9 = 16.

667. Answer C Set theory · Easy

Inclusion–exclusion: 27 + 34 − 5 = 56.

668. Answer A Set theory · Medium

Equivalence relations correspond to set partitions; the Bell number $B_9 = 21147$.

669. Answer A Set theory · Medium

Equivalence relations correspond to set partitions; the Bell number $B_7 = 877$.

670. Answer B Set theory · Medium

Equivalence relations correspond to set partitions; the Bell number $B_4 = 15$.

671. Answer C Set theory · Medium

Equivalence relations correspond to set partitions; the Bell number $B_10 = 115975$.

672. Answer B Set theory · Medium

Equivalence relations correspond to set partitions; the Bell number $B_6 = 203$.

673. Answer C Set theory · Medium

Equivalence relations correspond to set partitions; the Bell number $B_3 = 5$.

674. Answer A Set theory · Medium

Equivalence relations correspond to set partitions; the Bell number $B_11 = 678570$.

675. Answer C Set theory · Medium

Equivalence relations correspond to set partitions; the Bell number $B_5 = 52$.

676. Answer B Set theory · Hard

Count independent choices: the formula for 'symmetric' relations gives 2097152 (the formula was checked by brute force for n = 3).

677. Answer D Set theory · Hard

Count independent choices: the formula for 'symmetric' relations gives 1024 (the formula was checked by brute force for n = 3).

678. Answer D Set theory · Hard

Count independent choices: the formula for 'both reflexive and symmetric' relations gives 64 (the formula was checked by brute force for n = 3).

679. Answer C Set theory · Hard

Count independent choices: the formula for 'antisymmetric' relations gives 216 (the formula was checked by brute force for n = 3).

680. Answer A Set theory · Hard

Count independent choices: the formula for 'antisymmetric' relations gives 1889568 (the formula was checked by brute force for n = 3).

681. Answer A Set theory · Hard

Count independent choices: the formula for 'antisymmetric' relations gives 918330048 (the formula was checked by brute force for n = 3).

682. Answer B Set theory · Hard

Count independent choices: the formula for 'symmetric' relations gives 64 (the formula was checked by brute force for n = 3).

683. Answer A Mathematical logic · Easy

Build the 8-row truth table and count the rows evaluating to true: 4.

684. Answer C Mathematical logic · Easy

Build the 8-row truth table and count the rows evaluating to true: 3.

685. Answer C Mathematical logic · Easy

Build the 8-row truth table and count the rows evaluating to true: 2.

686. Answer B Mathematical logic · Easy

Build the 8-row truth table and count the rows evaluating to true: 5.

687. Answer D Mathematical logic · Easy

Build the 8-row truth table and count the rows evaluating to true: 7.

688. Answer C Mathematical logic · Easy

Build the 8-row truth table and count the rows evaluating to true: 5.

689. Answer C Mathematical logic · Easy

Build the 8-row truth table and count the rows evaluating to true: 4.

690. Answer D Mathematical logic · Medium

Compare truth tables: $(p \vee q) \wedge \neg(p \wedge q)$ and $\neg(p \leftrightarrow q)$ have identical truth values for all four assignments of (p, q); each other option differs on at least one row.

691. Answer B Mathematical logic · Medium

Compare truth tables: $\neg p \vee q$ and $\neg q \to \neg p$ have identical truth values for all four assignments of (p, q); each other option differs on at least one row.

692. Answer A Mathematical logic · Medium

Compare truth tables: $\neg(p \vee q)$ and $\neg p \wedge \neg q$ have identical truth values for all four assignments of (p, q); each other option differs on at least one row.

693. Answer C Mathematical logic · Medium

Compare truth tables: $(p \to q) \wedge (q \to p)$ and $p \leftrightarrow q$ have identical truth values for all four assignments of (p, q); each other option differs on at least one row.

694. Answer B Mathematical logic · Medium

Compare truth tables: $\neg p \wedge \neg q$ and $\neg(p \vee q)$ have identical truth values for all four assignments of (p, q); each other option differs on at least one row.

695. Answer D Mathematical logic · Medium

Compare truth tables: $\neg(p \leftrightarrow q)$ and $(p \vee q) \wedge \neg(p \wedge q)$ have identical truth values for all four assignments of (p, q); each other option differs on at least one row.

696. Answer C Mathematical logic · Medium

Compare truth tables: $\neg(p \wedge \neg q)$ and $\neg q \to \neg p$ have identical truth values for all four assignments of (p, q); each other option differs on at least one row.

697. Answer C Mathematical logic · Medium

Compare truth tables: $\neg(p \to q)$ and $p \wedge \neg q$ have identical truth values for all four assignments of (p, q); each other option differs on at least one row.

698. Answer A Mathematical logic · Hard

Check each statement by exhaustive search over $\mathbb{Z}_{11}$; only this one holds.

699. Answer A Mathematical logic · Hard

Check each statement by exhaustive search over $\mathbb{Z}_{18}$; only this one holds.

700. Answer D Mathematical logic · Hard

Check each statement by exhaustive search over $\mathbb{Z}_{22}$; only this one holds.

701. Answer B Mathematical logic · Hard

Check each statement by exhaustive search over $\mathbb{Z}_{37}$; only this one holds.

702. Answer D Mathematical logic · Hard

Check each statement by exhaustive search over $\mathbb{Z}_{20}$; only this one holds.

703. Answer A Mathematical logic · Hard

Check each statement by exhaustive search over $\mathbb{Z}_{29}$; only this one holds.

704. Answer C Mathematical logic · Hard

Check each statement by exhaustive search over $\mathbb{Z}_{34}$; only this one holds.

705. Answer A Cryptography · Easy

Shift each ciphertext letter back by 5: HNUMJW → CIPHER.

706. Answer A Cryptography · Easy

Shift each ciphertext letter back by 3: ORJLF → LOGIC.

707. Answer C Cryptography · Easy

Shift each ciphertext letter back by 9: CARP → TRIG.

708. Answer A Cryptography · Easy

Shift each ciphertext letter back by 9: YARVN → PRIME.

709. Answer B Cryptography · Easy

Shift each ciphertext letter back by 6: MXGVN → GRAPH.

710. Answer A Cryptography · Easy

Shift each ciphertext letter back by 6: TASHKX → NUMBER.

711. Answer B Cryptography · Easy

Shift each ciphertext letter back by 8: VCUJMZ → NUMBER.

712. Answer C Cryptography · Medium

$\varphi(n) = (13-1)(17-1) = 192$. We need $d \cdot 13 \equiv 1 \pmod{192}$; $d = 133$ since $133 \cdot 13 = 1729 = 9 \cdot 192 + 1$.

713. Answer B Cryptography · Medium

$\varphi(n) = (23-1)(13-1) = 264$. We need $d \cdot 5 \equiv 1 \pmod{264}$; $d = 53$ since $53 \cdot 5 = 265 = 1 \cdot 264 + 1$.

714. Answer B Cryptography · Medium

$\varphi(n) = (19-1)(5-1) = 72$. We need $d \cdot 5 \equiv 1 \pmod{72}$; $d = 29$ since $29 \cdot 5 = 145 = 2 \cdot 72 + 1$.

715. Answer D Cryptography · Medium

$\varphi(n) = (11-1)(23-1) = 220$. We need $d \cdot 13 \equiv 1 \pmod{220}$; $d = 17$ since $17 \cdot 13 = 221 = 1 \cdot 220 + 1$.

716. Answer A Cryptography · Medium

$\varphi(n) = (19-1)(5-1) = 72$. We need $d \cdot 7 \equiv 1 \pmod{72}$; $d = 31$ since $31 \cdot 7 = 217 = 3 \cdot 72 + 1$.

717. Answer C Cryptography · Medium

$\varphi(n) = (13-1)(11-1) = 120$. We need $d \cdot 17 \equiv 1 \pmod{120}$; $d = 113$ since $113 \cdot 17 = 1921 = 16 \cdot 120 + 1$.

718. Answer C Cryptography · Medium

$\varphi(n) = (11-1)(5-1) = 40$. We need $d \cdot 9 \equiv 1 \pmod{40}$; $d = 9$ since $9 \cdot 9 = 81 = 2 \cdot 40 + 1$.

719. Answer B Cryptography · Medium

$\varphi(n) = (5-1)(23-1) = 88$. We need $d \cdot 3 \equiv 1 \pmod{88}$; $d = 59$ since $59 \cdot 3 = 177 = 2 \cdot 88 + 1$.

720. Answer C Cryptography · Hard

The key is $A^b \bmod p = 21^8 \bmod 41 = 37$ (equivalently $g^{ab} \bmod p$).

721. Answer A Cryptography · Hard

The key is $A^b \bmod p = 9^9 \bmod 37 = 1$ (equivalently $g^{ab} \bmod p$).

722. Answer D Cryptography · Hard

The key is $A^b \bmod p = 27^4 \bmod 31 = 8$ (equivalently $g^{ab} \bmod p$).

723. Answer D Cryptography · Hard

The key is $A^b \bmod p = 5^{23} \bmod 31 = 25$ (equivalently $g^{ab} \bmod p$).

724. Answer B Cryptography · Hard

The key is $A^b \bmod p = 12^{32} \bmod 37 = 7$ (equivalently $g^{ab} \bmod p$).

725. Answer A Cryptography · Hard

The key is $A^b \bmod p = 40^{23} \bmod 41 = 40$ (equivalently $g^{ab} \bmod p$).

726. Answer D Cryptography · Hard

The key is $A^b \bmod p = 23^{23} \bmod 41 = 31$ (equivalently $g^{ab} \bmod p$).

727. Answer D Discrete mathematics · Easy

Use $n(n+1)(2n+1)/6 = 9 \cdot 10 \cdot 19/6 = 285$.

728. Answer C Discrete mathematics · Easy

Use $n(n+1)(2n+1)/6 = 10 \cdot 11 \cdot 21/6 = 385$.

729. Answer C Discrete mathematics · Easy

Use $n(n+1)(2n+1)/6 = 12 \cdot 13 \cdot 25/6 = 650$.

730. Answer C Discrete mathematics · Easy

Use $n(n+1)(2n+1)/6 = 26 \cdot 27 \cdot 53/6 = 6201$.

731. Answer C Discrete mathematics · Easy

Use $n(n+1)(2n+1)/6 = 23 \cdot 24 \cdot 47/6 = 4324$.

732. Answer D Discrete mathematics · Easy

Use $n(n+1)(2n+1)/6 = 14 \cdot 15 \cdot 29/6 = 1015$.

733. Answer D Discrete mathematics · Easy

Use $n(n+1)(2n+1)/6 = 22 \cdot 23 \cdot 45/6 = 3795$.

734. Answer D Discrete mathematics · Medium

Compute iteratively: 3, 1, 9, 29, 105, 373, 1329, 4733, 16857, 60037.

735. Answer A Discrete mathematics · Medium

Compute iteratively: 0, 2, 2, 6, 10, 22, 42.

736. Answer A Discrete mathematics · Medium

Compute iteratively: 0, 1, 2, 5, 12, 29, 70, 169.

737. Answer A Discrete mathematics · Medium

Compute iteratively: 3, 2, 10, 24, 68, 184, 504, 1376.

738. Answer B Discrete mathematics · Medium

Compute iteratively: 0, 1, 1, 2, 3, 5, 8, 13.

739. Answer D Discrete mathematics · Medium

Compute iteratively: 0, 3, 3, 6, 9, 15, 24, 39, 63.

740. Answer C Discrete mathematics · Medium

Compute iteratively: 2, 2, 8, 14, 38, 80, 194.

741. Answer B Discrete mathematics · Medium

Compute iteratively: 3, 3, 15, 39, 123, 363, 1095.

742. Answer D Discrete mathematics · Hard

Let a_n count such strings; $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ with $a_0=1$, $a_1=2$, $a_2=4$. This gives $a_{10} = 504$ (confirmed by enumerating all 2^{10} strings).

743. Answer C Discrete mathematics · Hard

Let a_n count such strings; $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ with $a_0=1$, $a_1=2$, $a_2=4$. This gives $a_{11} = 927$ (confirmed by enumerating all 2^{11} strings).

744. Answer B Discrete mathematics · Hard

Let a_n count such strings; $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ with $a_0=1$, $a_1=2$, $a_2=4$. This gives $a_9 = 274$ (confirmed by enumerating all 2^9 strings).

745. Answer B Discrete mathematics · Hard

Let a_n count such strings; $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ with $a_0=1$, $a_1=2$, $a_2=4$. This gives $a_{13} = 3136$ (confirmed by enumerating all 2^{13} strings).

746. Answer B Discrete mathematics · Hard

Let a_n count such strings; $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ with $a_0=1$, $a_1=2$, $a_2=4$. This gives $a_{12} = 1705$ (confirmed by enumerating all 2^{12} strings).

747. Answer A Discrete mathematics · Hard

Let a_n count such strings; $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ with $a_0=1$, $a_1=2$, $a_2=4$. This gives $a_{14} = 5768$ (confirmed by enumerating all 2^{14} strings).

748. Answer C Discrete mathematics · Hard

Let a_n count such strings; $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ with $a_0=1$, $a_1=2$, $a_2=4$. This gives $a_8 = 149$ (confirmed by enumerating all 2^8 strings).

749. Answer D Probability theory · Easy

There are 4 outcomes with sum 9 out of 36 equally likely outcomes: 1/9.

750. Answer C Probability theory · Easy

There are 2 outcomes with sum 3 out of 36 equally likely outcomes: 1/18.

751. Answer D Probability theory · Easy

There are 3 outcomes with sum 4 out of 36 equally likely outcomes: 1/12.

752. Answer C Probability theory · Easy

There are 5 outcomes with sum 6 out of 36 equally likely outcomes: 5/36.

753. Answer C Probability theory · Easy

There are 3 outcomes with sum 10 out of 36 equally likely outcomes: 1/12.

754. Answer A Probability theory · Easy

There are 5 outcomes with sum 8 out of 36 equally likely outcomes: 5/36.

755. Answer B Probability theory · Easy

There are 4 outcomes with sum 5 out of 36 equally likely outcomes: 1/9.

756. Answer C Probability theory · Medium

$P = 2\cdot(3/9)\cdot(6/8) = 1/2$.

757. Answer D Probability theory · Medium

$P = 2\cdot(8/11)\cdot(3/10) = 24/55$.

758. Answer D Probability theory · Medium

$P = 2\cdot(8/14)\cdot(6/13) = 48/91$.

759. Answer C Probability theory · Medium

$P = 2\cdot(3/10)\cdot(7/9) = 7/15$.

760. Answer C Probability theory · Medium

$P = 2\cdot(4/12)\cdot(8/11) = 16/33$.

761. Answer D Probability theory · Medium

$P = 2\cdot(6/14)\cdot(8/13) = 48/91$.

762. Answer D Probability theory · Medium

P = 2·(8/12)·(4/11) = 16/33.

763. Answer D Probability theory · Hard

P(max = k) = ($k^4 - (k-1)^4$)/8^4. Summing k·P(max = k) over k = 1…8 gives 7023/1024 (confirmed by enumerating all 4096 outcomes).

764. Answer D Probability theory · Hard

P(max = k) = ($k^2 - (k-1)^2$)/8^2. Summing k·P(max = k) over k = 1…8 gives 93/16 (confirmed by enumerating all 64 outcomes).

765. Answer B Probability theory · Hard

P(max = k) = ($k^4 - (k-1)^4$)/4^4. Summing k·P(max = k) over k = 1…4 gives 463/128 (confirmed by enumerating all 256 outcomes).

766. Answer A Probability theory · Hard

P(max = k) = ($k^4 - (k-1)^4$)/6^4. Summing k·P(max = k) over k = 1…6 gives 6797/1296 (confirmed by enumerating all 1296 outcomes).

767. Answer C Probability theory · Hard

P(max = k) = ($k^3 - (k-1)^3$)/4^3. Summing k·P(max = k) over k = 1…4 gives 55/16 (confirmed by enumerating all 64 outcomes).

768. Answer A Probability theory · Hard

P(max = k) = ($k^3 - (k-1)^3$)/8^3. Summing k·P(max = k) over k = 1…8 gives 207/32 (confirmed by enumerating all 512 outcomes).

769. Answer A Probability theory · Hard

P(max = k) = ($k^3 - (k-1)^3$)/6^3. Summing k·P(max = k) over k = 1…6 gives 119/24 (confirmed by enumerating all 216 outcomes).

770. Answer B Statistics · Easy

Sorted: 10, 16, 18, 29, 34. Median = 18. Mean = 107/5. Difference = 17/5.

771. Answer B Statistics · Easy

Sorted: 7, 10, 30, 31, 34. Median = 30. Mean = 112/5. Difference = -38/5.

772. Answer D Statistics · Easy

Sorted: 3, 5, 7, 12, 37. Median = 7. Mean = 64/5. Difference = 29/5.

773. Answer B Statistics · Easy

Sorted: 3, 7, 15, 18, 24. Median = 15. Mean = 67/5. Difference = -8/5.

774. Answer B Statistics · Easy

Sorted: 22, 23, 29, 32, 35. Median = 29. Mean = 141/5. Difference = -4/5.

775. Answer D Statistics · Easy

Sorted: 6, 7, 14, 18, 35. Median = 14. Mean = 16. Difference = 2.

776. Answer C Statistics · Easy

Sorted: 9, 19, 24, 27, 34. Median = 24. Mean = 113/5. Difference = -7/5.

777. Answer B Statistics · Medium

Mean = 63/5. Sum of squared deviations = 156/5. Divide by n − 1 = 4: 39/5.

778. Answer B Statistics · Medium

Mean = 47/6. Sum of squared deviations = 569/6. Divide by n − 1 = 5: 569/30.

779. Answer A Statistics · Medium

Mean = 46/5. Sum of squared deviations = 594/5. Divide by n − 1 = 4: 297/10.

780. Answer C Statistics · Medium

Mean = 26/3. Sum of squared deviations = 82/3. Divide by n − 1 = 5: 82/15.

781. Answer A Statistics · Medium

Mean = 6. Sum of squared deviations = 64. Divide by n − 1 = 4: 16.

782. Answer B Statistics · Medium

Mean = 66/7. Sum of squared deviations = 642/7. Divide by n − 1 = 6: 107/7.

783. Answer D Statistics · Medium

Mean = 8. Sum of squared deviations = 94. Divide by n − 1 = 6: 47/3.

784. Answer C Statistics · Hard

$\bar{x} = 11/2$, $\bar{y} = 19/2$, Sxy = -79/2, Sxx = 119/2. Slope b = Sxy/Sxx = -79/119.

785. Answer C Statistics · Hard

$\bar{x} = 32/5$, $\bar{y} = 10$, Sxy = 16, Sxx = 206/5. Slope b = Sxy/Sxx = 40/103.

786. Answer C Statistics · Hard

$\bar{x} = 27/4$, $\bar{y} = 41/4$, Sxy = -63/4, Sxx = 227/4. Slope b = Sxy/Sxx = -63/227.

787. Answer B Statistics · Hard

$\bar{x} = 21/5$, $\bar{y} = 47/5$, Sxy = 128/5, Sxx = 214/5. Slope b = Sxy/Sxx = 64/107.

788. Answer A Statistics · Hard

$\bar{x} = 29/5$, $\bar{y} = 44/5$, Sxy = -66/5, Sxx = 214/5. Slope b = Sxy/Sxx = -33/107.

789. Answer B Statistics · Hard

$\bar{x} = 43/6$, $\bar{y} = 49/6$, Sxy = -307/6, Sxx = 449/6. Slope b = Sxy/Sxx = -307/449.

790. Answer B Statistics · Hard

$\bar{x} = 19/4$, $\bar{y} = 29/4$, Sxy = 161/4, Sxx = 147/4. Slope b = Sxy/Sxx = 23/21.

791. Answer B Stochastic processes · Easy

$E[N(t)] = \lambda t = 2 \cdot 4 = 8$.

792. Answer C Stochastic processes · Easy

$E[N(t)] = \lambda t = 3 \cdot 4 = 12$.

793. Answer C Stochastic processes · Easy

$E[N(t)] = \lambda t = 3 \cdot 2 = 6$.

794. Answer D Stochastic processes · Easy

$E[N(t)] = \lambda t = 5 \cdot 3 = 15$.

795. Answer C Stochastic processes · Easy

$E[N(t)] = \lambda t = 4 \cdot 5 = 20$.

796. Answer A Stochastic processes · Easy

$E[N(t)] = \lambda t = 5 \cdot 2 = 10$.

797. Answer B Stochastic processes · Easy

$E[N(t)] = \lambda t = 3 \cdot 3 = 9$.

798. Answer B Stochastic processes · Medium

Balance equation $\pi_1 \cdot 1/5 = \pi_2 \cdot 1/10$ with $\pi_1 + \pi_2 = 1$ gives $\pi_1 = 1/10/(1/5+1/10) = 1/3$.

799. Answer C Stochastic processes · Medium

Balance equation $\pi_1 \cdot 3/10 = \pi_2 \cdot 2/5$ with $\pi_1 + \pi_2 = 1$ gives $\pi_1 = 2/5/(3/10+2/5) = 4/7$.

800. Answer D Stochastic processes · Medium

Balance equation $\pi_1 \cdot 1/5 = \pi_2 \cdot 3/10$ with $\pi_1 + \pi_2 = 1$ gives $\pi_1 = 3/10/(1/5+3/10) = 3/5$.

801. Answer C Stochastic processes · Medium

Balance equation $\pi_1 \cdot 1/10 = \pi_2 \cdot 3/10$ with $\pi_1 + \pi_2 = 1$ gives $\pi_1 = 3/10/(1/10+3/10) = 3/4$.

802. Answer B Stochastic processes · Medium

Balance equation $\pi_1 \cdot 1/5 = \pi_2 \cdot 2/5$ with $\pi_1 + \pi_2 = 1$ gives $\pi_1 = 2/5/(1/5+2/5) = 2/3$.

803. Answer C Stochastic processes · Medium

Balance equation $\pi_1 \cdot 1/10 = \pi_2 \cdot 1/5$ with $\pi_1 + \pi_2 = 1$ gives $\pi_1 = 1/5/(1/10+1/5) = 2/3$.

804. Answer A Stochastic processes · Medium

Balance equation $\pi_1 \cdot 2/5 = \pi_2 \cdot 1/5$ with $\pi_1 + \pi_2 = 1$ gives $\pi_1 = 1/5/(2/5+1/5) = 1/3$.

805. Answer B Stochastic processes · Hard

With $\rho = q/p = 2$: $P = (1 - \rho^2)/(1 - \rho^8) = 1/85$ (confirmed by solving the linear system exactly).

806. Answer D Stochastic processes · Hard

With $\rho = q/p = 3$: $P = (1 - \rho^4)/(1 - \rho^5) = 40/121$ (confirmed by solving the linear system exactly).

807. Answer B Stochastic processes · Hard

With $\rho = q/p = 1/3$: $P = (1 - \rho^1)/(1 - \rho^5) = 81/121$ (confirmed by solving the linear system exactly).

808. Answer A Stochastic processes · Hard

With $\rho = q/p = 1/2$: $P = (1 - \rho^3)/(1 - \rho^8) = 224/255$ (confirmed by solving the linear system exactly).

809. Answer B Stochastic processes · Hard

With $\rho = q/p = 2$: $P = (1 - \rho^3)/(1 - \rho^4) = 7/15$ (confirmed by solving the linear system exactly).

810. Answer D Stochastic processes · Hard

With $\rho = q/p = 3$: $P = (1 - \rho^3)/(1 - \rho^5) = 13/121$ (confirmed by solving the linear system exactly).

811. Answer B Stochastic processes · Hard

With $\rho = q/p = 1/2$: $P = (1 - \rho^3)/(1 - \rho^4) = 14/15$ (confirmed by solving the linear system exactly).

812. Answer A Bayesian statistics · Easy

Bayes: $P(D|+) = P(+|D)P(D)/P(+) = (4/5 \cdot 1/10)/(1/8) = 16/25$.

813. Answer A Bayesian statistics · Easy

Bayes: $P(D|+) = P(+|D)P(D)/P(+) = (9/10 \cdot 1/10)/(63/500) = 5/7$.

814. Answer A Bayesian statistics · Easy

Bayes: $P(D|+) = P(+|D)P(D)/P(+) = (9/10 \cdot 1/20)/(37/400) = 18/37$.

815. Answer D Bayesian statistics · Easy

Bayes: $P(D|+) = P(+|D)P(D)/P(+) = (19/20 \cdot 1/10)/(131/1000) = 95/131$.

816. Answer C Bayesian statistics · Easy

Bayes: $P(D|+) = P(+|D)P(D)/P(+) = (9/10 \cdot 1/100)/(117/2000) = 2/13$.

817. Answer C Bayesian statistics · Easy

Bayes: $P(D|+) = P(+|D)P(D)/P(+) = (4/5 \cdot 1/20)/(27/200) = 8/27$.

818. Answer D Bayesian statistics · Easy

Bayes: $P(D|+) = P(+|D)P(D)/P(+) = (19/20 \cdot 1/50)/(17/250) = 19/68$.

819. Answer C Bayesian statistics · Medium

The posterior is Beta(4, 8), with mean $4/(4+8) = 1/3$.

820. Answer C Bayesian statistics · Medium

The posterior is Beta(11, 17), with mean $11/(11+17) = 11/28$.

821. Answer A Bayesian statistics · Medium

The posterior is Beta(14, 9), with mean $14/(14+9) = 14/23$.

822. Answer C Bayesian statistics · Medium

The posterior is Beta(14, 4), with mean $14/(14+4) = 7/9$.

823. Answer A Bayesian statistics · Medium

The posterior is Beta(12, 4), with mean 12/(12+4) = 3/4.

824. Answer B Bayesian statistics · Medium

The posterior is Beta(11, 13), with mean 11/(11+13) = 11/24.

825. Answer A Bayesian statistics · Medium

The posterior is Beta(4, 9), with mean 4/(4+9) = 4/13.

826. Answer B Bayesian statistics · Hard

Posterior precision = 1/4 + 4/4 = 5/4. Posterior mean = $(\mu_0/\tau_0^2 + n\cdot\bar{x}/\sigma^2)$/precision = 5/2/5/4 = 2.

827. Answer C Bayesian statistics · Hard

Posterior precision = 1/4 + 4/16 = 1/2. Posterior mean = $(\mu_0/\tau_0^2 + n\cdot\bar{x}/\sigma^2)$/precision = 3/1/2 = 6.

828. Answer A Bayesian statistics · Hard

Posterior precision = 1/9 + 9/16 = 97/144. Posterior mean = $(\mu_0/\tau_0^2 + n\cdot\bar{x}/\sigma^2)$/precision = 241/144/97/144 = 241/97.

829. Answer B Bayesian statistics · Hard

Posterior precision = 1/4 + 8/1 = 33/4. Posterior mean = $(\mu_0/\tau_0^2 + n\cdot\bar{x}/\sigma^2)$/precision = 353/4/33/4 = 353/33.

830. Answer A Bayesian statistics · Hard

Posterior precision = 1/9 + 4/16 = 13/36. Posterior mean = $(\mu_0/\tau_0^2 + n\cdot\bar{x}/\sigma^2)$/precision = 67/18/13/36 = 134/13.

831. Answer D Bayesian statistics · Hard

Posterior precision = 1/4 + 8/16 = 3/4. Posterior mean = $(\mu_0/\tau_0^2 + n\cdot\bar{x}/\sigma^2)$/precision = 23/4/3/4 = 23/3.

832. Answer B Bayesian statistics · Hard

Posterior precision = 1/9 + 5/1 = 46/9. Posterior mean = $(\mu_0/\tau_0^2 + n\cdot\bar{x}/\sigma^2)$/precision = 142/9/46/9 = 71/23.

833. Answer C Numerical analysis · Easy

$x_1 = x_0 - f(x_0)/f'(x_0) = 4 - (6)/8 = 13/4$.

834. Answer D Numerical analysis · Easy

$x_1 = x_0 - f(x_0)/f'(x_0) = 2 - (-6)/4 = 7/2$.

835. Answer C Numerical analysis · Easy

$x_1 = x_0 - f(x_0)/f'(x_0) = 1 - (-4)/2 = 3$.

836. Answer D Numerical analysis · Easy

$x_1 = x_0 - f(x_0)/f'(x_0) = 1 - (-6)/2 = 4$.

837. Answer D Numerical analysis · Easy

$x_1 = x_0 - f(x_0)/f'(x_0) = 2 - (-1)/4 = 9/4$.

838. Answer A Numerical analysis · Easy

$x_1 = x_0 - f(x_0)/f'(x_0) = 2 - (-7)/4 = 15/4$.

839. Answer D Numerical analysis · Easy

$x_1 = x_0 - f(x_0)/f'(x_0) = 3 - (7)/6 = 11/6$.

840. Answer B Numerical analysis · Medium

With h = 1/2, T = h[f(0)/2 + f(h) + ... + f(2)/2] = 11/4. (The exact value 8/3 is slightly smaller, as expected for a convex function.)

841. Answer A Numerical analysis · Medium

With h = 4/5, T = h[f(0)/2 + f(h) + ... + f(4)/2] = 544/25. (The exact value 64/3 is slightly smaller, as expected for a convex function.)

842. Answer B Numerical analysis · Medium

With h = 1, T = h[f(0)/2 + f(h) + ... + f(2)/2] = 3. (The exact value 8/3 is slightly smaller, as expected for a convex function.)

843. Answer B Numerical analysis · Medium

With h = 3/2, T = h[f(0)/2 + f(h) + ... + f(3)/2] = 81/8. (The exact value 9 is slightly smaller, as expected for a convex function.)

844. Answer B Numerical analysis · Medium

With h = 5/4, T = h[f(0)/2 + f(h) + ... + f(5)/2] = 1375/32. (The exact value 125/3 is slightly smaller, as expected for a convex function.)

845. Answer A Numerical analysis · Medium

With h = 2, T = h[f(0)/2 + f(h) + ... + f(4)/2] = 24. (The exact value 64/3 is slightly smaller, as expected for a convex function.)

846. Answer D Numerical analysis · Medium

With h = 1, T = h[f(0)/2 + f(h) + ... + f(4)/2] = 22. (The exact value 64/3 is slightly smaller, as expected for a convex function.)

847. Answer C Numerical analysis · Hard

For $y' = \lambda y$, one RK4 step multiplies y by $1 + z + z^2/2 + z^3/6 + z^4/24$ with $z = \lambda h = -1$. Result: 9/8.

848. Answer D Numerical analysis · Hard

For $y' = \lambda y$, one RK4 step multiplies y by $1 + z + z^2/2 + z^3/6 + z^4/24$ with $z = \lambda h = -3/2$. Result: 105/128.

849. Answer B Numerical analysis · Hard

For $y' = \lambda y$, one RK4 step multiplies y by $1 + z + z^2/2 + z^3/6 + z^4/24$ with $z = \lambda h = -1/4$. Result: 1595/512.

850. Answer D Numerical analysis · Hard

For $y' = \lambda y$, one RK4 step multiplies y by $1 + z + z^2/2 + z^3/6 + z^4/24$ with $z = \lambda h = -4/5$. Result: 847/375.

851. Answer C Numerical analysis · Hard

For $y' = \lambda y$, one RK4 step multiplies y by $1 + z + z^2/2 + z^3/6 + z^4/24$ with $z = \lambda h = -3/4$. Result: 971/2048.

852. Answer B Numerical analysis · Hard

For $y' = \lambda y$, one RK4 step multiplies y by $1 + z + z^2/2 + z^3/6 + z^4/24$ with $z = \lambda h = -1$. Result: 3/8.

853. Answer A Numerical analysis · Hard

For $y' = \lambda y$, one RK4 step multiplies y by $1 + z + z^2/2 + z^3/6 + z^4/24$ with $z = \lambda h = -1/5$. Result: 12281/7500.

854. Answer C Optimization · Easy

$f'(x) = 6x - 7 = 0$ gives $x = 7/6$; $f'' = 6 > 0$, so this is a minimum.

855. Answer B Optimization · Easy

$f'(x) = 2x + 8 = 0$ gives $x = -4$; $f'' = 2 > 0$, so this is a minimum.

856. Answer A Optimization · Easy

$f'(x) = 8x - 2 = 0$ gives $x = 1/4$; $f'' = 8 > 0$, so this is a minimum.

857. Answer C Optimization · Easy

$f'(x) = 10x - 1 = 0$ gives $x = 1/10$; $f'' = 10 > 0$, so this is a minimum.

858. Answer C Optimization · Easy

$f'(x) = 8x + 7 = 0$ gives $x = -7/8$; $f'' = 8 > 0$, so this is a minimum.

859. Answer B Optimization · Easy

$f'(x) = 6x + 5 = 0$ gives $x = -5/6$; $f'' = 6 > 0$, so this is a minimum.

860. Answer D Optimization · Easy

$f'(x) = 8x + 0 = 0$ gives $x = 0$; $f'' = 8 > 0$, so this is a minimum.

861. Answer C Optimization · Medium

Lagrange conditions: $(x,y) = \lambda(4,2)/2$ scaled onto the line, giving minimum $c^2/(a^2+b^2) = 16/20 = 4/5$.

862. Answer C Optimization · Medium

Lagrange conditions: $(x,y) = \lambda(3,3)/2$ scaled onto the line, giving minimum $c^2/(a^2+b^2) = 9/18 = 1/2$.

863. Answer D Optimization · Medium

Lagrange conditions: (x,y) = λ(5,1)/2 scaled onto the line, giving minimum $c^2/(a^2+b^2) = 196/26 = 98/13$.

864. Answer B Optimization · Medium

Lagrange conditions: (x,y) = λ(5,2)/2 scaled onto the line, giving minimum $c^2/(a^2+b^2) = 64/29 = 64/29$.

865. Answer A Optimization · Medium

Lagrange conditions: (x,y) = λ(1,4)/2 scaled onto the line, giving minimum $c^2/(a^2+b^2) = 9/17 = 9/17$.

866. Answer B Optimization · Medium

Lagrange conditions: (x,y) = λ(2,5)/2 scaled onto the line, giving minimum $c^2/(a^2+b^2) = 324/29 = 324/29$.

867. Answer C Optimization · Medium

Lagrange conditions: (x,y) = λ(2,3)/2 scaled onto the line, giving minimum $c^2/(a^2+b^2) = 324/13 = 324/13$.

868. Answer C Optimization · Hard

The maximum occurs at a vertex of the feasible region. Enumerating all intersection points of constraint lines and keeping feasible ones, the best is at (27/5, 21/5) with value 144/5.

869. Answer B Optimization · Hard

The maximum occurs at a vertex of the feasible region. Enumerating all intersection points of constraint lines and keeping feasible ones, the best is at (0, 13/2) with value 39/2.

870. Answer A Optimization · Hard

The maximum occurs at a vertex of the feasible region. Enumerating all intersection points of constraint lines and keeping feasible ones, the best is at (7, 0) with value 21.

871. Answer C Optimization · Hard

The maximum occurs at a vertex of the feasible region. Enumerating all intersection points of constraint lines and keeping feasible ones, the best is at (19/3, 0) with value 19.

872. Answer B Optimization · Hard

The maximum occurs at a vertex of the feasible region. Enumerating all intersection points of constraint lines and keeping feasible ones, the best is at (0, 7) with value 35.

873. Answer A Optimization · Hard

The maximum occurs at a vertex of the feasible region. Enumerating all intersection points of constraint lines and keeping feasible ones, the best is at (7/5, 39/5) with value 191/5.

874. Answer A Optimization · Hard

The maximum occurs at a vertex of the feasible region. Enumerating all intersection points of constraint lines and keeping feasible ones, the best is at (2, 8) with value 34.

875. Answer B Operations research · Easy

EOQ = $\sqrt{(2\cdot1000\cdot25/5)} = \sqrt{10000} = 100$.

876. Answer D Operations research · Easy

EOQ = $\sqrt{(2\cdot250\cdot25/5)} = \sqrt{2500} = 50$.

877. Answer B Operations research · Easy

EOQ = $\sqrt{(2\cdot128\cdot25/4)} = \sqrt{1600} = 40$.

878. Answer D Operations research · Easy

EOQ = $\sqrt{(2\cdot40\cdot20/1)} = \sqrt{1600} = 40$.

879. Answer A Operations research · Easy

EOQ = $\sqrt{(2\cdot200\cdot50/2)} = \sqrt{10000} = 100$.

880. Answer B Operations research · Easy

EOQ = $\sqrt{(2\cdot1250\cdot20/5)} = \sqrt{10000} = 100$.

881. Answer A Operations research · Easy

EOQ = $\sqrt{(2\cdot80\cdot10/1)} = \sqrt{1600} = 40$.

882. Answer C Operations research · Medium

$\rho = \lambda/\mu = 10/11$. $L = \rho/(1-\rho) = \lambda/(\mu-\lambda) = 10/1 = 10$.

883. Answer C Operations research · Medium

$\rho = \lambda/\mu = 1/4$. $L = \rho/(1-\rho) = \lambda/(\mu-\lambda) = 1/3 = 1/3$.

884. Answer C Operations research · Medium

$\rho = \lambda/\mu = 9/10$. $L = \rho/(1-\rho) = \lambda/(\mu-\lambda) = 9/1 = 9$.

885. Answer D Operations research · Medium

$\rho = \lambda/\mu = 1/5$. $L = \rho/(1-\rho) = \lambda/(\mu-\lambda) = 1/4 = 1/4$.

886. Answer D Operations research · Medium

$\rho = \lambda/\mu = 4/11$. $L = \rho/(1-\rho) = \lambda/(\mu-\lambda) = 4/7 = 4/7$.

887. Answer C Operations research · Medium

$\rho = \lambda/\mu = 5/6$. $L = \rho/(1-\rho) = \lambda/(\mu-\lambda) = 5/1 = 5$.

888. Answer D Operations research · Medium

$\rho = \lambda/\mu = 5/8$. $L = \rho/(1-\rho) = \lambda/(\mu-\lambda) = 5/3 = 5/3$.

889. Answer B Operations research · Hard

Enumerating all 24 assignments (equivalently, running the Hungarian method) gives a minimum total cost of 27. A greedy row-by-row choice would give 27, which need not be optimal.

890. Answer C Operations research · Hard

Enumerating all 24 assignments (equivalently, running the Hungarian method) gives a minimum total cost of 31. A greedy row-by-row choice would give 32, which need not be optimal.

891. Answer D Operations research · Hard

Enumerating all 24 assignments (equivalently, running the Hungarian method) gives a minimum total cost of 35. A greedy row-by-row choice would give 35, which need not be optimal.

892. Answer B Operations research · Hard

Enumerating all 24 assignments (equivalently, running the Hungarian method) gives a minimum total cost of 19. A greedy row-by-row choice would give 19, which need not be optimal.

893. Answer D Operations research · Hard

Enumerating all 24 assignments (equivalently, running the Hungarian method) gives a minimum total cost of 30. A greedy row-by-row choice would give 30, which need not be optimal.

894. Answer D Operations research · Hard

Enumerating all 24 assignments (equivalently, running the Hungarian method) gives a minimum total cost of 22. A greedy row-by-row choice would give 29, which need not be optimal.

895. Answer D Operations research · Hard

Enumerating all 24 assignments (equivalently, running the Hungarian method) gives a minimum total cost of 21. A greedy row-by-row choice would give 26, which need not be optimal.

896. Answer A Mathematical modeling · Easy

12 years is 3 half-lives: $320/2^3 = 40$ g.

897. Answer B Mathematical modeling · Easy

15 hours is 3 doubling periods: $720 \cdot 2^3 = 5760$.

898. Answer C Mathematical modeling · Easy

15 hours is 3 doubling periods: $530 \cdot 2^3 = 4240$.

899. Answer D Mathematical modeling · Easy

20 hours is 5 doubling periods: $486 \cdot 2^5 = 15552$.

900. Answer D Mathematical modeling · Easy

9 hours is 3 doubling periods: $418 \cdot 2^3 = 3344$.

901. Answer C Mathematical modeling · Easy

8 hours is 2 doubling periods: $581 \cdot 2^2 = 2324$.

902. Answer C Mathematical modeling · Easy

8 hours is 4 doubling periods: $245 \cdot 2^4 = 3920$.

903. Answer B Mathematical modeling · Medium

$dP/dt = 4P(1 - P/400)$ is a downward parabola in P with maximum at $P = 400/2$, where it equals $4 \cdot (400/2)(1/2) = 400$.

904. Answer A Mathematical modeling · Medium

$dP/dt = 3P(1 - P/400)$ is a downward parabola in P with maximum at $P = 400/2$, where it equals $3 \cdot (400/2)(1/2) = 300$.

905. Answer B Mathematical modeling · Medium

$dP/dt = 6P(1 - P/400)$ is a downward parabola in P with maximum at $P = 400/2$, where it equals $6 \cdot (400/2)(1/2) = 600$.

906. Answer A Mathematical modeling · Medium

$dP/dt = 6P(1 - P/800)$ is a downward parabola in P with maximum at $P = 800/2$, where it equals $6 \cdot (800/2)(1/2) = 1200$.

907. Answer A Mathematical modeling · Medium

$dP/dt = 6P(1 - P/1000)$ is a downward parabola in P with maximum at $P = 1000/2$, where it equals $6 \cdot (1000/2)(1/2) = 1500$.

908. Answer B Mathematical modeling · Medium

$dP/dt = 1P(1 - P/100)$ is a downward parabola in P with maximum at $P = 100/2$, where it equals $1 \cdot (100/2)(1/2) = 25$.

909. Answer D Mathematical modeling · Medium

$dP/dt = 2P(1 - P/1000)$ is a downward parabola in P with maximum at $P = 1000/2$, where it equals $2 \cdot (1000/2)(1/2) = 500$.

910. Answer B Mathematical modeling · Hard

$R_0 = \beta/\gamma = 3$. Threshold $= 1 - 1/R_0 = 2/3$.

911. Answer B Mathematical modeling · Hard

$R_0 = \beta/\gamma = 5/2$. Threshold $= 1 - 1/R_0 = 3/5$.

912. Answer D Mathematical modeling · Hard

$R_0 = \beta/\gamma = 4$. Threshold $= 1 - 1/R_0 = 3/4$.

913. Answer A Mathematical modeling · Hard

$R_0 = \beta/\gamma = 3$. Threshold $= 1 - 1/R_0 = 2/3$.

914. Answer C Mathematical modeling · Hard

$R_0 = \beta/\gamma = 8/5$. Threshold $= 1 - 1/R_0 = 3/8$.

915. Answer A Mathematical modeling · Hard

$R_0 = \beta/\gamma = 3$. Threshold $= 1 - 1/R_0 = 2/3$.

916. Answer C Mathematical modeling · Hard

$R_0 = \beta/\gamma = 7/2$. Threshold $= 1 - 1/R_0 = 5/7$.

917. Answer D Game theory · Easy

Check each cell for profitable unilateral deviations. Only (Row 2, Col 2) has neither player wanting to deviate.

918. Answer A Game theory · Easy

Check each cell for profitable unilateral deviations. Only (Row 1, Col 1) has neither player wanting to deviate.

919. Answer C Game theory · Easy

Check each cell for profitable unilateral deviations. Only (Row 2, Col 2) has neither player wanting to deviate.

920. Answer D Game theory · Easy

Check each cell for profitable unilateral deviations. Only (Row 2, Col 1) has neither player wanting to deviate.

921. Answer A Game theory · Easy

Check each cell for profitable unilateral deviations. Only (Row 2, Col 2) has neither player wanting to deviate.

922. Answer D Game theory · Easy

Check each cell for profitable unilateral deviations. Only (Row 1, Col 2) has neither player wanting to deviate.

923. Answer A Game theory · Easy

Check each cell for profitable unilateral deviations. Only (Row 2, Col 2) has neither player wanting to deviate.

924. Answer D Game theory · Medium

Equate the two expected payoffs: 5p + 8(1−p) = 9p + 3(1−p). Hence p(a − b − c + d) = d − c, i.e. p = -5/-9 = 5/9.

925. Answer A Game theory · Medium

Equate the two expected payoffs: 7p + 4(1−p) = 9p + 3(1−p). Hence p(a − b − c + d) = d − c, i.e. p = -1/-3 = 1/3.

926. Answer B Game theory · Medium

Equate the two expected payoffs: 7p + 5(1−p) = 9p + 3(1−p). Hence p(a − b − c + d) = d − c, i.e. p = -2/-4 = 1/2.

927. Answer A Game theory · Medium

Equate the two expected payoffs: 0p + 9(1−p) = 6p + 3(1−p). Hence p(a − b − c + d) = d − c, i.e. p = -6/-12 = 1/2.

928. Answer A Game theory · Medium

Equate the two expected payoffs: 6p + 2(1−p) = 4p + 7(1−p). Hence p(a − b − c + d) = d − c, i.e. p = 5/7 = 5/7.

929. Answer D Game theory · Medium

Equate the two expected payoffs: 2p + 2(1−p) = 8p + 0(1−p). Hence p(a − b − c + d) = d − c, i.e. p = -2/-8 = 1/4.

930. Answer D Game theory · Medium

Equate the two expected payoffs: 2p + 9(1−p) = 8p + 4(1−p). Hence p(a − b − c + d) = d − c, i.e. p = -5/-11 = 5/11.

931. Answer D Game theory · Hard

With no saddle point both players mix. For a 2×2 game the value is (ad − bc)/(a + d − b − c) = (4 − 72)/-13 = 68/13.

932. Answer B Game theory · Hard

With no saddle point both players mix. For a 2×2 game the value is (ad − bc)/(a + d − b − c) = (84 − 8)/13 = 76/13.

933. Answer B Game theory · Hard

With no saddle point both players mix. For a 2×2 game the value is (ad − bc)/(a + d − b − c) = (20 − 1)/10 = 19/10.

934. Answer D Game theory · Hard

With no saddle point both players mix. For a 2×2 game the value is (ad − bc)/(a + d − b − c) = (144 − 88)/5 = 56/5.

935. Answer B Game theory · Hard

With no saddle point both players mix. For a 2×2 game the value is (ad − bc)/(a + d − b − c) = (99 − 0)/12 = 33/4.

936. Answer A Game theory · Hard

With no saddle point both players mix. For a 2×2 game the value is (ad − bc)/(a + d − b − c) = (4 − 77)/-14 = 73/14.

937. Answer D Game theory · Hard

With no saddle point both players mix. For a 2×2 game the value is (ad − bc)/(a + d − b − c) = (36 − 0)/12 = 3.

938. Answer B Dynamical systems and chaos theory · Easy

Solve x = 5x(1 − x): 1 = 5(1 − x), so x = 1 − 1/5 = 4/5.

939. Answer C Dynamical systems and chaos theory · Easy

Solve x = 10x(1 − x): 1 = 10(1 − x), so x = 1 − 1/10 = 9/10.

940. Answer D Dynamical systems and chaos theory · Easy

Solve x = 8x(1 − x): 1 = 8(1 − x), so x = 1 − 1/8 = 7/8.

941. Answer D Dynamical systems and chaos theory · Easy

Solve x = 11x(1 − x): 1 = 11(1 − x), so x = 1 − 1/11 = 10/11.

942. Answer C Dynamical systems and chaos theory · Easy

Solve x = 6x(1 − x): 1 = 6(1 − x), so x = 1 − 1/6 = 5/6.

943. Answer B Dynamical systems and chaos theory · Easy

Solve x = 7x(1 − x): 1 = 7(1 − x), so x = 1 − 1/7 = 6/7.

944. Answer C Dynamical systems and chaos theory · Easy

Solve x = 13x(1 − x): 1 = 13(1 − x), so x = 1 − 1/13 = 12/13.

945. Answer B Dynamical systems and chaos theory · Medium

trace = -6, det = 16, discriminant = -28. det > 0; trace < 0 gives stable; discriminant < 0 → spiral.

946. Answer A Dynamical systems and chaos theory · Medium

trace = -1, det = 3, discriminant = -11. det > 0; trace < 0 gives stable; discriminant < 0 → spiral.

947. Answer A Dynamical systems and chaos theory · Medium

trace = 2, det = -1, discriminant = 8. det < 0 means a saddle.

948. Answer A Dynamical systems and chaos theory · Medium

trace = -1, det = -12, discriminant = 49. det < 0 means a saddle.

949. Answer B Dynamical systems and chaos theory · Medium

trace = 6, det = -11, discriminant = 80. det < 0 means a saddle.

950. Answer C Dynamical systems and chaos theory · Medium

trace = -4, det = -15, discriminant = 76. det < 0 means a saddle.

951. Answer C Dynamical systems and chaos theory · Medium

trace = -3, det = -5, discriminant = 29. det < 0 means a saddle.

952. Answer D Dynamical systems and chaos theory · Hard

Factor f(f(x)) − x by removing the fixed-point factor f(x) − x. The remaining quadratic has root sum (r+1)/r = 14/11 (verified symbolically).

953. Answer D Dynamical systems and chaos theory · Hard

Factor f(f(x)) − x by removing the fixed-point factor f(x) − x. The remaining quadratic has root sum (r+1)/r = 13/10 (verified symbolically).

954. Answer A Dynamical systems and chaos theory · Hard

Factor f(f(x)) − x by removing the fixed-point factor f(x) − x. The remaining quadratic has root sum (r+1)/r = 22/17 (verified symbolically).

955. Answer C Dynamical systems and chaos theory · Hard

Factor f(f(x)) − x by removing the fixed-point factor f(x) − x. The remaining quadratic has root sum (r+1)/r = 5/4 (verified symbolically).

956. Answer D Dynamical systems and chaos theory · Hard

Factor f(f(x)) − x by removing the fixed-point factor f(x) − x. The remaining quadratic has root sum (r+1)/r = 21/16 (verified symbolically).

957. Answer C Dynamical systems and chaos theory · Hard

Factor f(f(x)) − x by removing the fixed-point factor f(x) − x. The remaining quadratic has root sum (r+1)/r = 24/19 (verified symbolically).

958. Answer A Dynamical systems and chaos theory · Hard

Factor f(f(x)) − x by removing the fixed-point factor f(x) − x. The remaining quadratic has root sum (r+1)/r = 17/13 (verified symbolically).

959. Answer A Information theory · Easy

$H = \Sigma\, p_i \log_2(1/p_i) = \Sigma\, k_i/2^{k_i} = 2$ bits.

960. Answer B Information theory · Easy

$H = \Sigma\, p_i \log_2(1/p_i) = \Sigma\, k_i/2^{k_i} = 17/8$ bits.

961. Answer B Information theory · Easy

$H = \Sigma\, p_i \log_2(1/p_i) = \Sigma\, k_i/2^{k_i} = 3/2$ bits.

962. Answer C Information theory · Easy

$H = \Sigma\, p_i \log_2(1/p_i) = \Sigma\, k_i/2^{k_i} = 15/8$ bits.

963. Answer A Information theory · Easy

$H = \Sigma\, p_i \log_2(1/p_i) = \Sigma\, k_i/2^{k_i} = 19/8$ bits.

964. Answer A Information theory · Easy

$H = \Sigma\, p_i \log_2(1/p_i) = \Sigma\, k_i/2^{k_i} = 5/2$ bits.

965. Answer C Information theory · Easy

$H = \Sigma\, p_i \log_2(1/p_i) = \Sigma\, k_i/2^{k_i} = 7/4$ bits.

966. Answer D Information theory · Medium

Repeatedly merge the two least probable nodes; the expected length equals the sum of all merged probabilities, which is 123/53.

967. Answer A Information theory · Medium

Repeatedly merge the two least probable nodes; the expected length equals the sum of all merged probabilities, which is 85/36.

968. Answer D Information theory · Medium

Repeatedly merge the two least probable nodes; the expected length equals the sum of all merged probabilities, which is 64/33.

969. Answer A Information theory · Medium

Repeatedly merge the two least probable nodes; the expected length equals the sum of all merged probabilities, which is 5/2.

970. Answer A Information theory · Medium

Repeatedly merge the two least probable nodes; the expected length equals the sum of all merged probabilities, which is 31/14.

971. Answer B Information theory · Medium

Repeatedly merge the two least probable nodes; the expected length equals the sum of all merged probabilities, which is 96/41.

972. Answer D Information theory · Medium

Repeatedly merge the two least probable nodes; the expected length equals the sum of all merged probabilities, which is 44/21.

973. Answer A Information theory · Hard

We need $2^r \geq k + r + 1$. For $k = 26$: $r = 4$ gives $16 < 31$, while $r = 5$ gives $32 \geq 32$. So $r = 5$.

974. Answer A Information theory · Hard

We need $2^r \geq k + r + 1$. For k = 57: r = 5 gives 32 < 63, while r = 6 gives 64 ≥ 64. So r = 6.

975. Answer B Information theory · Hard

We need $2^r \geq k + r + 1$. For k = 8: r = 3 gives 8 < 12, while r = 4 gives 16 ≥ 13. So r = 4.

976. Answer B Information theory · Hard

We need $2^r \geq k + r + 1$. For k = 30: r = 5 gives 32 < 36, while r = 6 gives 64 ≥ 37. So r = 6.

977. Answer C Information theory · Hard

We need $2^r \geq k + r + 1$. For k = 4: r = 2 gives 4 < 7, while r = 3 gives 8 ≥ 8. So r = 3.

978. Answer A Information theory · Hard

We need $2^r \geq k + r + 1$. For k = 20: r = 4 gives 16 < 25, while r = 5 gives 32 ≥ 26. So r = 5.

979. Answer D Information theory · Hard

We need $2^r \geq k + r + 1$. For k = 11: r = 3 gives 8 < 15, while r = 4 gives 16 ≥ 16. So r = 4.

980. Answer D Financial mathematics · Easy

$A = P(1 + r)^n = 500 \cdot (1 + 0.08)^3 = 629.86$.

981. Answer B Financial mathematics · Easy

$A = P(1 + r)^n = 1000 \cdot (1 + 0.08)^2 = 1166.40$.

982. Answer B Financial mathematics · Easy

$A = P(1 + r)^n = 5000 \cdot (1 + 0.04)^3 = 5624.32$.

983. Answer C Financial mathematics · Easy

$A = P(1 + r)^n = 1000 \cdot (1 + 0.04)^5 = 1216.65$.

984. Answer D Financial mathematics · Easy

$A = P(1 + r)^n = 1000 \cdot (1 + 0.1)^2 = 1210.00$.

985. Answer C Financial mathematics · Easy

$A = P(1 + r)^n = 5000 \cdot (1 + 0.08)^5 = 7346.64$.

986. Answer B Financial mathematics · Easy

$A = P(1 + r)^n = 1000 \cdot (1 + 0.1)^5 = 1610.51$.

987. Answer B Financial mathematics · Medium

PV = C·(1 − (1+i)^(−n))/i = 200·(1 − 1.05^(−11))/0.05 = 1661.28.

988. Answer C Financial mathematics · Medium

PV = C·(1 − (1+i)^(−n))/i = 100·(1 − 1.05^(−5))/0.05 = 432.95.

989. Answer B Financial mathematics · Medium

PV = C·(1 − (1+i)^(−n))/i = 1000·(1 − 1.08^(−8))/0.08 = 5746.64.

990. Answer D Financial mathematics · Medium

PV = C·(1 − (1+i)^(−n))/i = 200·(1 − 1.10^(−12))/0.1 = 1362.74.

991. Answer B Financial mathematics · Medium

PV = C·(1 − (1+i)^(−n))/i = 500·(1 − 1.08^(−12))/0.08 = 3768.04.

992. Answer D Financial mathematics · Medium

PV = C·(1 − (1+i)^(−n))/i = 200·(1 − 1.08^(−7))/0.08 = 1041.27.

993. Answer B Financial mathematics · Medium

PV = C·(1 − (1+i)^(−n))/i = 100·(1 − 1.05^(−6))/0.05 = 507.57.

994. Answer D Financial mathematics · Hard

Risk-neutral probability q = (1 + r − d)/(u − d) = 3/5. Payoffs: up 8, down 0. Price = [q·8 + (1−q)·0]/(1 + r) = 4.62.

995. Answer B Financial mathematics · Hard

Risk-neutral probability q = (1 + r − d)/(u − d) = 5/9. Payoffs: up 16, down 0. Price = [q·16 + (1−q)·0]/(1 + r) = 8.47.

996. Answer C Financial mathematics · Hard

Risk-neutral probability q = (1 + r − d)/(u − d) = 3/5. Payoffs: up 15, down 0. Price = [q·15 + (1−q)·0]/(1 + r) = 8.65.

997. Answer B Financial mathematics · Hard

Risk-neutral probability q = (1 + r − d)/(u − d) = 1/2. Payoffs: up 20, down 0. Price = [q·20 + (1−q)·0]/(1 + r) = 9.52.

998. Answer C Financial mathematics · Hard

Risk-neutral probability q = (1 + r − d)/(u − d) = 4/7. Payoffs: up 30, down 0. Price = [q·30 + (1−q)·0]/(1 + r) = 15.58.

999. Answer D Financial mathematics · Hard

Risk-neutral probability q = (1 + r − d)/(u − d) = 4/7. Payoffs: up 24, down 0. Price = [q·24 + (1−q)·0]/(1 + r) = 12.47.

1000. Answer A Financial mathematics · Hard

Risk-neutral probability q = (1 + r − d)/(u − d) = 5/9. Payoffs: up 28, down 0. Price = [q·28 + (1−q)·0]/(1 + r) = 14.81.